a machine a load of 400 N can be raised by a force of 80 N applied to the rim of the wheel. The radii of the wheel and axle are 32 cm and 4 cm respectively. Calculate the force ratio (Mechanical advantage), the velocity ratio, and the efficiency of the machine.

13 What type of lever is the crowbar?
A 2 m crowbar is employed to lift a load of 300 kg. What force must be applied to one end of the crowbar when a stump acting as a fulcrum is placed under the crowbar 0.5 m away from the end on which the load rests? Neglect the weight of the crowbar.

14 Draw a block and tackle system of pulleys with a velocity ratio of 5. A block and tackle with a velocity ratio of 5 is used to raise a mass of 25 kg through a vertical distance of 40 cm at a steady rate. If the effort is equal to 60 N, determine
   a. the distance moved by the effort;
   b. the work done by the effort in lifting the load;
   c. the loss in energy involved in operating the machine.

15 With the aid of labelled diagrams describe and explain one application in each case of (a) an inclined plane (b) a simple lever.
A lift is used to carry boxes to the top floor of a hotel 20 m high. A total weight of 100 kg of boxes was carried up in 20 seconds. If the useful output of the engine driving the lift mechanism is 1.2 kilowatt, calculate the efficiency of the engine of the lift mechanism.

16 (a) (i) What is meant by a *machine*?
   (ii) List two examples of a simple machine.
   (iii) Explain the statement "The velocity ratio of a machine is 5"
   (b) (i) Define the efficiency of a machine.
   (ii) Explain why a machine can never be 100% efficient.
   (c) A screw jack, 25% efficient and having a screw of pitch 0.4cm is used to raise a load through a certain height. If in the process the handle turns through a circle of radius 40cm, calculate the
   (i) Velocity ratio of the machine.
   (ii) The Mechanical advantage of the machine.
   (iii) Effort required to raise a load of 1000N with the machine. (take $\pi = 3.14$)
(SSCE)

Scanned with CamScanner

8 HEAT ENERGY 1: TEMPERATURE AND ITS MEASUREMENT

8.1 Heat and Temperature

In chapter 4 of Book 1, we defined temperature as the degree of hotness or coldness of a body. It is the property of an object which determines which way heat energy will flow when it is placed in contact with another object. Heat always flows from a body at a higher temperature to another at a lower temperature. We also distinguished between heat and temperature. We noted that heat is a form of energy - thermal energy. When a body absorbs heat without changing its state, its temperature rises. Heat depends on the mass of a body and its temperature. A fixed quantity of heat given to a small body makes it hotter than it would a large body.

Heat is a measure of the total internal energy of a body. It is a form of energy due to a temperature difference. **Temperature** is the degree of hotness or coldness of the body and is related to the energy of movement. It is a measure of the average kinetic energy of the molecules in a body. Unit of Heat is the Joule, unit of temperature is degree Celsius (°C) or Kelvin (K).

In the present chapter we will discuss the various methods of measuring temperature, the relationship between pressure and temperature of a gas, and types of thermometers. We shall also give a molecular explanation of temperature.

At the end of the chapter, the student should be able to:

1 Construct a device for measuring the temperature of a body.
2 Use the variation of
   a. pressure of a gas with temperature
   b. the expansion of solid, liquid or gas with temperature
   c. electrical resistance of a material, to measure the temperature of a body.
3 Distinguish between heat and temperature and between temperature points and temperature intervals.
4 Select those liquids which are suitable for use in liquid-in-glass thermometers from a given list of liquids and their properties.
5 Describe the absolute scale of temperature and explain the meaning of the absolute zero of temperature.
6 Convert a given temperature in the Celsius scale to a temperature on the Kelvin scale.
7 Describe the kinetic molecular model of temperature.

8.2 Methods of Measuring Temperature

Our sense of touch can give us a general impression of the degree of hotness or coldness of a body. This is however not a reliable method of estimating or measuring a temperature, because the response of the human sense of touch to a temperature change tends to be influenced by its previous experience. Thus warm water will feel cool if a hand initially dipped in hot water is transferred to it. Hence in order to gauge accurately the exact degree of hotness, an instrument called the *thermometer* is used. Thermometers are much more reliable instruments for measuring temperatures.

Table 8.1

Type of Thermometer	Thermometric Substance	Physical Property
1 Liquid-in-glass thermometer	Mercury or Alcohol	Change in volume of liquid with temperature.
2 Gas thermometer	Gas	Change of gas pressure at constant volume with temperature.
3 Resistance thermometer	Resistance wire	Change in the electrical resistance of wire with temperature.
4 Thermocouple	Two dissimilar metals (e.g. copper and constantan)	Change in electric potential difference (or current) between two metal junctions at different temperatures.
5 Bimetallic thermometer	Two dissimilar metals (e.g. iron and copper)	The differential expansion of the two metals of the bimetallic strip.

Scanned with CamScanner

Thermometers use any physical property of a substance which varies in a known way with temperature, and is easily measurable as a means of gauging temperature. The substance of whose physical property is so used is known as a *thermometric substance.*

Types of thermometers and their thermometric substances are summarised in table 8.1.

8.3 Fixed Temperature and Temperature Scales of Thermometer

Each thermometer has two reference temperatures or *fixed points* called the *Upper fixed temperature point* and the *Lower fixed temperature point.*

The **Upper fixed point** is the temperature of steam from pure water boiling at standard atmospheric pressure of 760 mm of mercury.

The **Lower fixed point** is the temperature of pure melting ice at the standard atmospheric pressure of 760 mm of mercury.

The difference in temperature between the two temperature points is called the *Fundamental Interval* (or temperature interval) of a thermometer. The calibration of this interval depends on the temperature scale chosen. There are three types of scales in current use

1 the *Celsius scale,*
2 the *Fahrenheit scale* and
3 The *Absolute (or thermodynamic or Kelvin) scale.*

The lower and upper fixed points are 0°C and 100°C for the Celsius scale; 32°F and 212°F for the Fahrenheit scale. The fundamental interval in the Celsius scale is divided into 100 equal parts, each part of which defines 1°C in this scale. For the Fahrenheit scale, the fundamental interval is divided into 180 units or degrees (°F).

The S.I. unit of temperature is the Kelvin (K) and its scale is called the *Absolute* or *Thermodynamic temperature* scale. The Fundamental interval for the Kelvin scale goes from a lower fixed point of 273 K to an upper fixed point of 373 K, i.e a difference of 100 K This interval is divided into 100 equal parts each of which is equal to 1 K. Temperatures on this scale are not measured in degrees but in units called Kelvin (K). Hence the unit symbol K is written without the degree sign. The lower fixed point or the zero on the Kelvin scale is equal to –273°C. It is called the absolute zero.

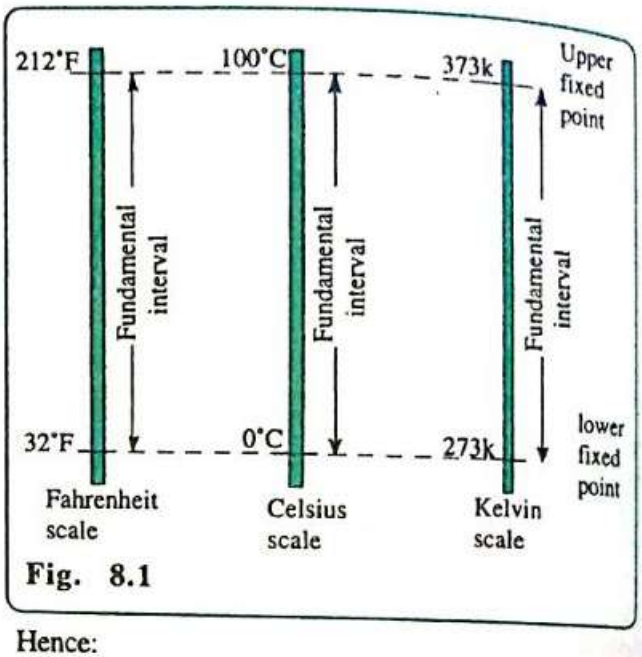


Fig. 8.1

Hence:

$-273°C = 0\ K$ or $0°C = 273\ K$

A temperature of θ°C in the Celsius scale is related to T of the Kelvin scale by

$$T = \theta + 273 \qquad 8.1$$

The Upper fixed point on the Kelvin scale is 373 K or 100°C. Nevertheless a temperature change of 1°C is equal to a temperature change of 1 K. Thus we can for example say that the thermal expansivity of brass is 18×10^{-6}/K or 18×10^{-6}/°C or that a heat quantity which has a value of 100 Joules per °C (or 100 J/°C) has a value of 100 J/K in S.I. units.

Example 8.1

The lower and upper fixed points of a certain thermometer are 30 cm apart. At a certain day, the length of mercury thread in the thermometer is 9 cm above the ice point (0°C). What is the temperature recorded by the thermometer in (a) Celsius scale (b) Kelvin scale.

Solution

Let the temperature recorded in Celsius be θ°C. The fixed points are 30 cm apart. We assume the temperature increases in a linear scale. Then from fig 8.2, by taking proportions

$$\frac{AB}{AC} = \frac{DE}{DF} = \frac{LM}{LN}$$

$$\frac{\theta - 0}{100 - 0} = \frac{9\ \text{cm}}{30\ \text{cm}}$$

Scanned with CamScanner

$$\frac{\theta}{100} = \frac{9}{30} = \frac{3}{10}$$

Hence $\theta = \underline{30^\circ C}$

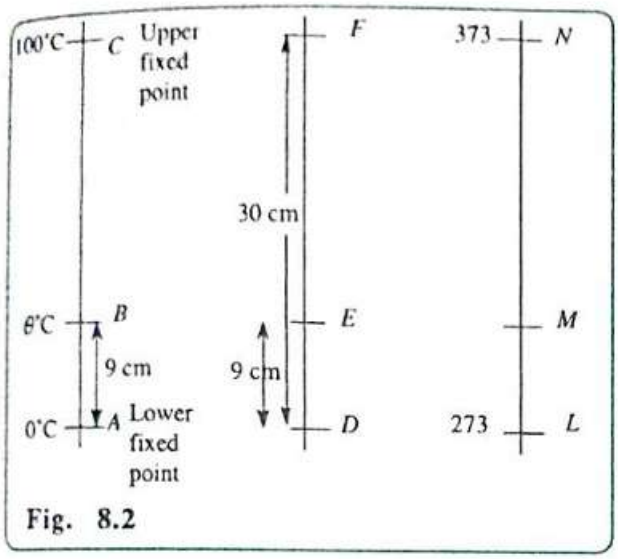


Fig. 8.2

From equation 8.1, $T = \theta + 273 = 30 + 273$

$= \underline{303\ K}$

The Upper Fixed Points

The upper fixed point of an unmarked thermometer can be determined using a hypsometer, a double walled copper vessel constructed as shown in fig 8.3.

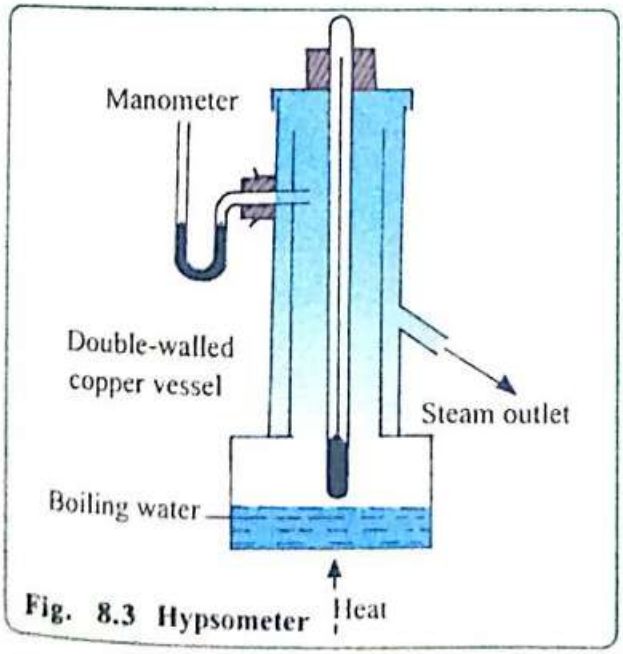


Fig. 8.3 Hypsometer

Almost the whole length of the thermometer hangs freely in the steam in the inner chamber well above the boiling water and is placed in such a way that its mercury thread is just visible above the top of the cork. A manometer is attached to the inner chamber of the hypsometer to ensure that the pressure within it is 760 mm of mercury. This is so when the mercury levels in both arms of the manometer are the same. Also attached to the vessel is a steam-outlet to ensure that no steam condenses on the thermometer. The water is heated until it starts boiling.

When the thermometer has been in the steam for some time and the position of the mercury thread has remained steady, its level is marked on the stem by a light scratch. This mark is the upper fixed point or the upper temperature point.

An important precaution is to ensure that the bulb of the thermometer does not come in contact with the boiling water. The thermometer should be suspended only in the steam. Also we must avoid parallax error in locating the boiling point.

The Lower Fixed Point

The lower fixed point is determined by placing the thermometer upright in pure melting ice contained in a glass funnel as shown in fig 8.4. When the mercury level remains steady for some time, a mark is made on the stem of the thermometer to indicate this level, which represents the lower fixed temperature point.

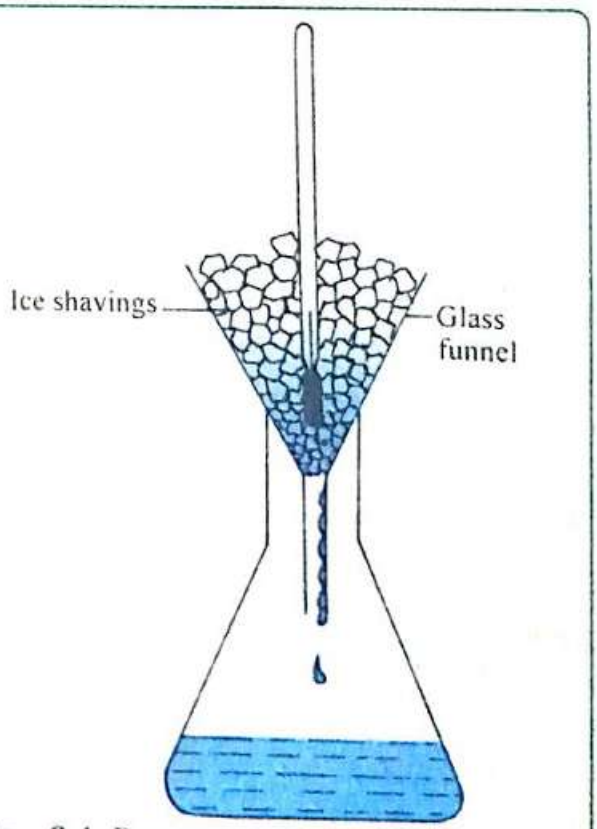


Fig. 8.4 Determination of lower fixed point

Scanned with CamScanner

Example 8.2

A faulty Celsius thermometer reads 0.7°C at the melting point of pure ice and 99.5°C at the boiling point of water at normal pressure.

a. What is the correct temperature when it reads 30 C?

b. At what temperature will its reading be exactly correct?

Solution

a. Let the correct temperature when the faulty thermometer reads 30°C be t°C. Fundamental interval of faulty thermometer is 99.5°C – 0.7°C = 98.8°C. Fundamental interval of accurate thermometer is 100°C – 0°C.

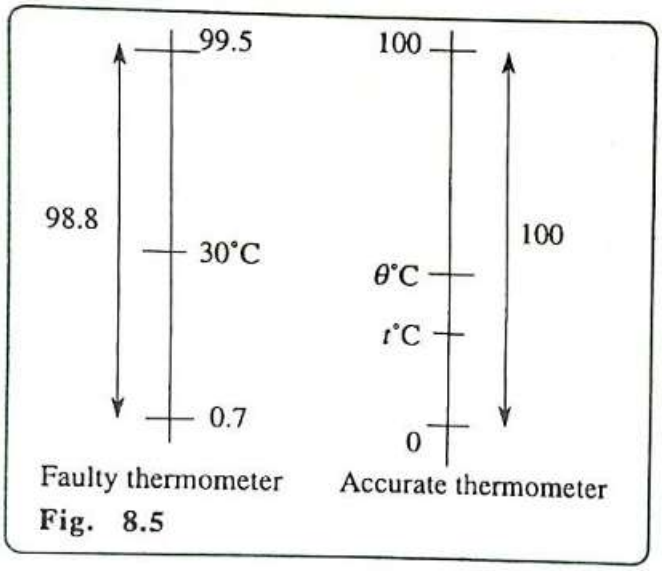


Fig. 8.5

By taking proportions we have

$$\frac{t-0}{100-0}=\frac{30-0.7}{99.5-0.7}=\frac{29.3}{98.8}$$

$$\frac{t}{100}=\frac{29.3}{98.8}$$

$$t=100\times\frac{29.3}{98.8}$$

$$=\underline{29.6°C}$$

b. Let the faulty thermometer record the true temperature at θ°C.

$$\therefore \quad \frac{\theta}{100}=\frac{\theta-0.7}{98.8}$$

$$98.8\theta=100\theta-70$$

$$\theta=\frac{70}{1.2}$$

$$=\underline{58.33°C}$$

8.4 Types of Thermometers

1 Liquid-in-Glass thermometers

The most common liquids used in thermometers are mercury and alcohol. The thermometer measures temperature by measuring the change in volume of a fixed mass of liquid due to a change in temperature.

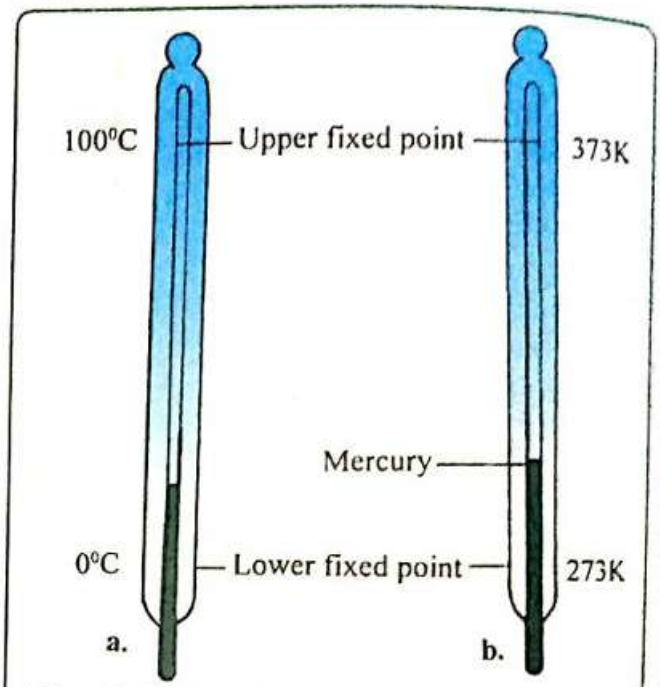


Fig. 8.6 Liquid-in-glass thermometer

For high sensitivity, liquid-in-glass thermometers should have

(i) a bulb made of thin glass; this enables the liquid in the bulb to assume the temperature of its surrounding quickly

(ii) A narrow capillary tube with uniform bore. This makes it possible for small temperature changes to cause large changes in the length of the mercury column.

(iii) A liquid with a high expansivity.

To be used as a thermometric liquid, such a liquid should

(i) Expand or contract uniformly with temperature.

(ii) Have a high boiling point and a low melting point.

(iii) Be easily seen in glass.

Water has none of the above properties. It has a small range of expansion; it freezes at 0°C and boils at 100°C. It does not expand uniformly (it also contracts from 0°C to 4°C). It wets glass, and is colourless, making the meniscus in glass difficult to read. The above reasons make water unsuitable as a thermometric liquid.

Mercury-in-glass thermometers are more commonly used in school laboratories and hospitals

Scanned with CamScanner

than *Alcohol-in-glass*. Mercury-in-glass thermometer is cheap and simple, but it is not reliable enough for accurate work.

Comparison of Mercury and Alcohol as Thermometric Liquids

Mercury has the following advantages over alcohol as a thermometric liquid.

1 Mercury has a much greater conductivity than alcohol and expands rapidly, and thus indicates a temperature change quickly. Alcohol being a relatively poor conductor, expands slowly and thus responds to temperature changes slowly.
2 Its silvery surface makes mercury opaque, and hence it can be easily seen. Alcohol has to be coloured to be easily seen.
3 Mercury does not wet glass but alcohol because of its concave meniscus tends to cling to the walls of stem of the thermometer. This leads to an inaccurate reading when the thread is falling.
4 Mercury is not easily vaporised whereas alcohol is very easily vaporised even at low temperatures.
5 Mercury boils at 357°C while alcohol boils at 78°C. Mercury can therefore be used to measure higher temperatures than alcohol.

Alcohol has the following advantages over mercury as a thermometric liquid.

1 It freezes at −115°C whereas mercury freezes at −39°C. Alcohol can therefore be used for measuring very low temperatures.
2 The expansivity of alcohol is about six times that of mercury for the same temperature rise.

Clinical Thermometer

This is a form of mercury-in-glass thermometer used in the hospitals for measuring the temperature of the human body. The temperature of a normal healthy person is about 37°C but it may rise to about 41°C in cases of high fever. The temperature range of the clinical thermometer is therefore between 35°C to 43°C.

The thermometer has a small bulb made of thin glass. The bulb has a narrow bore through which a fine mercury thread moves. The fine bore of the tube makes it possible for small temperature changes to cause large changes in the length of mercury column, making the thermometer very sensitive to temperature changes. The most prominent feature of the thermometer is the *kink* or *constriction* of bore near the bulb. The kink allows the expanding mercury to flow past it when the thermometer is placed in contact with the human body. The mercury thread however remains in position in the upper part of the capillary tube when the thermometer is taken away from the patient. The body temperature can thus be read at leisure. Mercury can be returned to the bulb past the kink by vigorously shaking the thermometer. Some clinical thermometers have a thick stem which acts as a cylindrical lens, magnifying the thread of mercury for easy viewing.

It is not advisable to sterilize the clinical thermometer in boiling water at normal pressure because the thermometer has a small temperature range which does not extend up to 100°C. At such a high temperature, the glass will break due to excessive expansion of the mercury.

Combined Maximum and Minimum Thermometer

This is a type of liquid-in-glass thermometer used by weather-men in meteorological stations. It records both the maximum and minimum temperatures for the day. The one most commonly in use is the *Six's Combined Maximum and Minimum Thermometer* invented by James Six. Its most prominent features are as shown in fig 8.8.

It consists of a U-shaped stem Y connected to two large bulbs P and Q. Both bulbs contain alcohol which are separated by a column of mercury in the bend of the U stem. While the alcohol in P completely fills that bulb, in Q there is some air space near the top of the bulb. Two steel indexes, I_1 and I_2 float in the alcohol in both sides of the tube. Temperature scales are provided against each limb of the U-tube.



Fig. 8.7 Clinical thermometer

Scanned with CamScanner

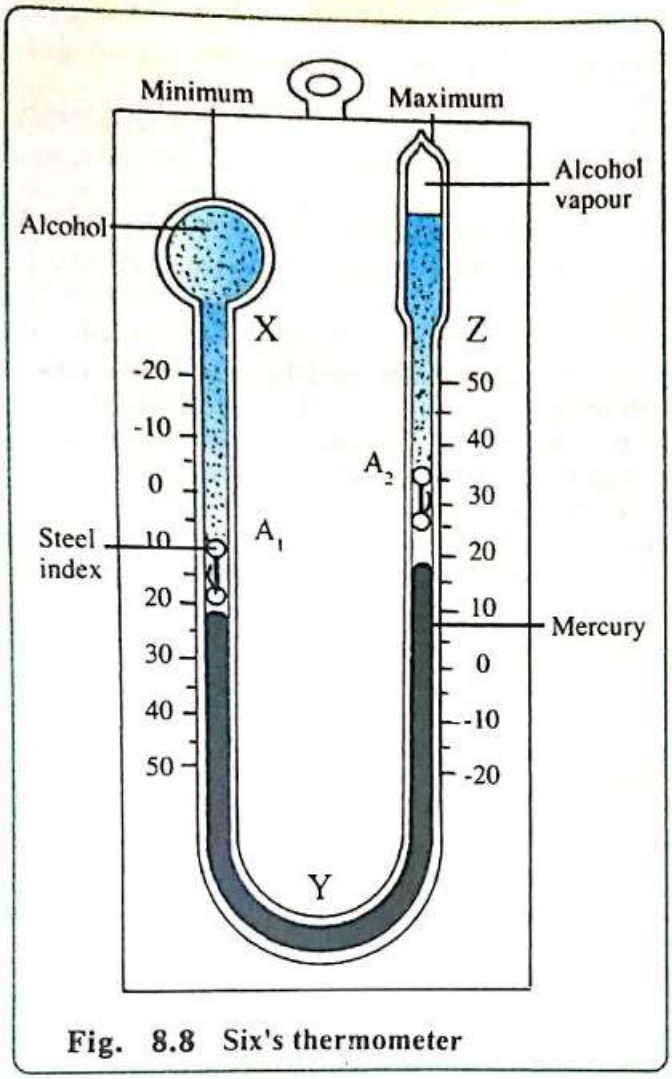


Fig. 8.8 Six's thermometer

An increase in temperature causes the alcohol to expand, pushing the mercury round the thermometer. As a result, the steel index I_1 is pushed upwards. Its lower end indicates the maximum temperature, this being read from the scale. Decrease in temperature causes the alcohol to contract. Mercury moves up the minimum arm pushing index I_2 upwards. The lower end of this index indicates the minimum temperature reached. Tiny springs in I_1 and I_2 help them to remain in positions to which they have been pushed. A magnet is used to reset the indices by bringing them in contact with the mercury column once more.

2 The Gas Thermometers

Mercury-in-glass thermometers are not very suitable for extremely accurate measurements of temperature. Their inaccuracy is due to the fact that the glass also expands with temperature in an irregular manner. Also, such thermometers have a relatively small range determined by the freezing and boiling points of mercury (–39°C to 358°C).

Gas thermometers are therefore used for accurate temperature measurements. The principle of the gas thermometer is based on the fact that at constant volume, the pressure of a gas increases linearly with increase in temperature. A constant volume gas thermometer consists of a large bulb which contains a gas (e.g hydrogen or helium). the bulb is connected to a narrow glass tube attached to a mercury manometer.

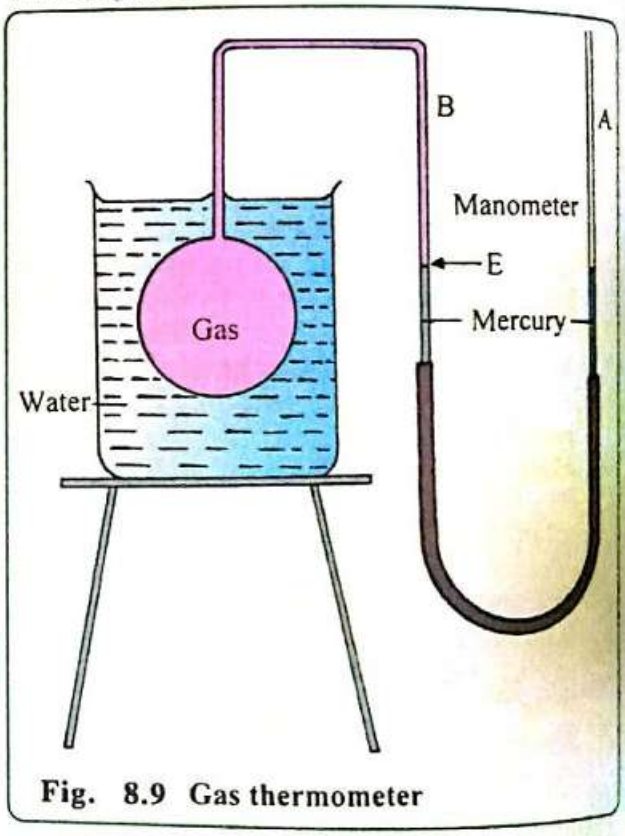


Fig. 8.9 Gas thermometer

When the gas in the bulb is heated to a particular temperature, the gas expands and pushes the mercury down in tube B and up in tube A. The expansion of the glass is negligible compared to the gas expansion. The gas is kept at constant volume by moving the manometer up or down to bring the mercury level on the B side to a fixed mark E. The pressure of the gas is then read from the manometer. When the level of mercury in A is above that in B, the gas pressure is atmospheric pressure (H) plus the difference (h) in the levels of mercury in A and B, i.e ($H + h$). When the level of mercury in A is below that in B, the gas pressure is $H - h$. Thus the pressure of the gas in the bulb is related to the difference in the mercury levels in the manometer.

Scanned with CamScanner

To calibrate the thermometer, the gas pressure is measured when the bulb is placed in pure melting ice and again when it is placed in steam. A straight line graph of pressure versus temperature is then plotted using the two known values (fixed points). This graph can then be used to read off accurate temperature values from the measured pressure readings.

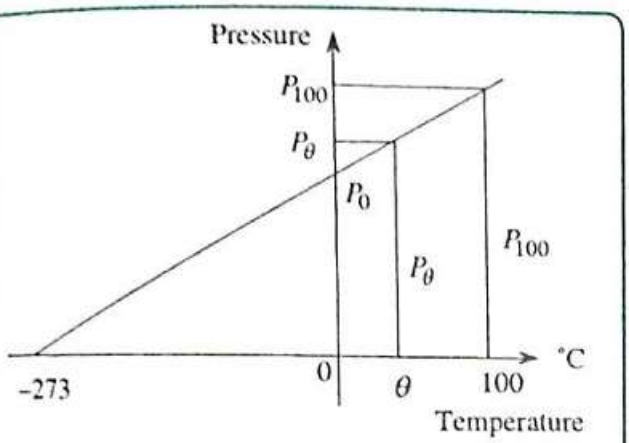


Fig. 8.10 Variation of pressure with temperature

From fig 8.10, we see that the pressure of a gas at constant volume varies directly as the temperature. If the graph is produced to cut the temperature axis, it will do so at –273°C. This temperature is known as the *Absolute zero* temperature. At this temperature the pressure of the gas is zero and the gas molecules have zero kinetic energy.

The advantage of this type of thermometer over other thermometers is that it is very sensitive, accurate, consistent and has a wide range. Its major disadvantage is that it is large and cumbersome and therefore not suitable for measuring the temperatures of small volumes of liquids or small enclosures. It also requires a knowledge of the pressure of the gas at the fixed points any time we require to use the instrument.

Example 8.3

A constant volume gas thermometer records a pressure of 250 mm Hg at 0°C and 290 mm Hg at 100°C. Calculate the room temperature when the gas pressure is 262 mm Hg.

Solution

Difference in the pressures of fixed points
= 290 – 250 = 40 mm Hg
Difference in temperatures of fixed points
= 100 – 0 = 100°C
Pressure difference between room temperature and ice point
= 262 – 250 = 12 mm Hg.
If the room temperature is t°C, we have by taking proportions that:

$$\frac{t}{100} = \frac{12}{40}$$

$t = \underline{30°C}$

3 Thermoelectric Thermometers

Such thermometers are used for measuring very high temperatures. They work on the principle of the *thermocouple*. When two different metals (e.g. copper and constantan) are joined at the ends and one end (the hot junction) is heated, while the other end (the cold junction) is kept constant in melting ice, an electric current flows along the metals. This is the *thermoelectric effect* and the set up is called a *thermocouple*.

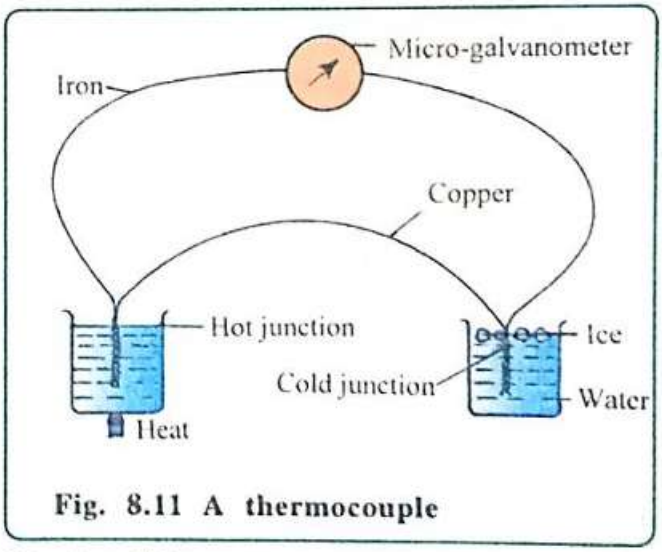


Fig. 8.11 A thermocouple

The size of the current is measured by a sensitive galvanometer which is connected in series with the hot and cold junctions. This current is dependent on the difference between the temperatures of the hot and cold junctions. The galvanometer is usually calibrated to read the temperature of the hot junction directly.

An instrument which uses the thermoelectric effect to measure temperatures is known as a *Thermoelectric thermometer*. Such thermometers are used in industry for measuring high temperatures. The advantages of the thermometer are

Scanned with CamScanner

1 It is very sensitive and can measure rapidly changing or momentary temperatures.
2 It can also measure high temperatures up to 500°C.
3 It is very small in size and can therefore measure the temperature almost at a point.

4 Resistance Thermometers

This thermometer uses the fact that the electrical resistance of a metallic conductor changes proportionally with its temperature. The higher the temperature the greater is the resistance. It consists of a long thin platinum wire wound round a small spool made of mica or silica.

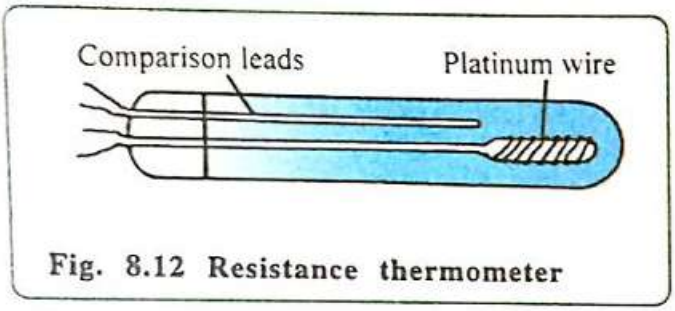


Fig. 8.12 Resistance thermometer

The ends of the wire are connected to a resistance measuring device such as a wheatstone bridge. If for example, a given platinum wire has a resistance R_0 at 0°C and R_{100} at 100°C, then the temperature t, when its resistance is R is given by

$$t = \frac{R - R_0}{R_{100} - R_0} \times 100\ (^\circ\text{C}) \qquad 8.2$$

In many cases the resistance scale is calibrated to read the temperature directly in degrees.

Resistance thermometers are useful in the accurate measurement of very low or very high temperatures.

Example 8.4

The electrical resistance of a platinum wire at 0°C is 9.8 ohms and 11.0 ohm at 100°C. Calculate the temperature when the resistance is 10.4 ohms assuming that the change in the electrical resistance of conductor is proportional to the change in temperature.

Solution

From equation 8.2 above we have that

$$t = \frac{10.4 - 9.8}{11.0 - 9.8} \times 100$$
$$= \frac{0.6}{1.2} \times 100$$
$$= \underline{50^\circ\text{C}}$$

8.5 Molecular Explanation of Temperature

According to the kinetic theory of matter, all matter is composed of tiny particles called molecules. These molecules are always in motion with varying velocities. When a substance is heated, the velocity of the molecules and hence their kinetic energy increases. This increase in the kinetic energy of the molecules appears as an increase in the temperature of the body. The temperature of a body is thus proportional to the average kinetic energy of its molecules.

If the same quantity of heat is given to a body of mass m_1, and to another body of similar material but of a smaller mass m_2, the body of smaller mass becomes hotter than that of larger mass. This is because the smaller mass contains fewer molecules to share the heat than the larger mass.

The average kinetic energy of the molecules of a body becomes zero at absolute zero temperature (−273°C). At this temperature the pressure of a gas is also zero.

Summary

1 Temperature is the degree of hotness or coldness of a body. It is the property of a body which determines the direction in which heat will flow when in contact with another body. Heat is a form of energy (thermal energy) which flows as a result of temperature difference. Temperature is a scalar quantity.
2 Thermometers use any physical property of a substance which varies in a known way with temperature and is easily measurable. Liquid-in-glass thermometers use the variation of volume of liquid with temperature; for Gas thermometers it is the variation of gas pressure at constant volume with temperature; for thermoelectric thermometers, it is the variation of electric potential difference (or current) between two metal junctions at different temperatures; Resistance thermometers use the variation of electrical resistance of platinum wire with temperature.
3 Clinical thermometer is a kind of mercury-in-glass thermometer used to measure the temperature of the human body. It has a short range of temperature 35°C – 43°C and a constriction or kink near the bulb.
4 The Kelvin scale of temperature T(K) is related to the Celsius (θ°C) by $T(\text{K}) = 273 + \theta^\circ\text{C}$. Change of 1°C = change of 1 K. The absolute

Scanned with CamScanner

zero of temperature is a temperature of 0(K) or –273°C.

5 The temperature of a body is proportional to the average kinetic energy of its molecules.

Exercise 8

1 A thermometer with an arbitrary scale, *S*, of equal divisions registers –30°S at the ice point and +90°S at the steam point. Calculate the Celsius temperature corresponding to 60°S.
A 25.0°C B 50.0°C C 66.7°C
D 75.0°C

(JAMB)

2 Water is not suitable for use as a thermometric liquid because (i) It wets glass (ii) It needs to be coloured (iii) It expands abnormally (iv) It has a low density.
A I only B I and II only
C I, II, III only D II and III only
E III and IV only.

3 A thermometer has its stem marked in millimeter instead of degree Celsius. The lower fixed point is 30 mm and the upper fixed point is 180 mm. Calculate the temperature in degree Celsius when the thermometer reads 45 mm.
A 67.5°C B 30.0° C 25.0°C
D 15.0°C E 10.0°C.

(SSCE)

4 Which of the following *cannot* be used to measure the temperature of a substance?
A Variation of pressure with temperature
B Expansivity of liquid
C Change in resistance of a conductor
D Thermo-electric effect
E Change in colour with temperature

(SC/GCE)

5 Which of the following thermometers can be used to measure a range of temperatures from –50 to –80°C?
I. A clinical thermometer
II. A mercury thermometer
III. An alcohol thermometer
A I only. B II only.
C III only. D I and II only.
E II and III only.

(SC/GCE)

6 Which of the following thermometers responds best to changing temperature?
A Mercury thermometer.
B Alcohol thermometer.
C Resistance thermometer.
D Thermoelectric thermometer
E Gas thermometer.

7 State three desirable properties of a thermometric liquid. List four advantages and four disadvantages of (a) Mercury (b) Alcohol, as thermometric liquids.
Why is water considered as an unsuitable liquid for a thermometer.

8 A platinum resistance thermometer measures temperature on the assumption that the change in electrical resistance of platinum is proportional to the change in temperature. The resistance of the platinum wire at 0°C is 10.4 ohms and 10.71 ohms at 100°C. Calculate the temperature when the resistance is 9.61 ohms. What will be the resistance of the wire when the temperature is 45°C.

(SC/GCE)

9 Write down the important features of (i) a clinical thermometer (ii) Six's maximum and minimum thermometer. What are the functions of any three named components of these thermometers?

Why is it not advisable to sterilize a clinical thermometer in boiling water at normal pressure?

10 a. (i) Distinguish between *temperature* and *heat*. State the units in which they are measured.
(ii) State two physical properties used for measuring temperature.
b. (i) Describe with the aid of a diagram, how the upper fixed point is determined for a mercury thermometer. State *one* precaution to ensure accurate results.
(ii) State *one* advantage which a constant-volume gas thermometer has over other thermometers and *one* reason why it is seldom used as an everyday laboratory instrument.

(SSCE)

11 With the aid of a labelled diagram describe the action of a thermometer suitable for recording the highest and lowest temperatures recorded in a period of time.

12 What do you understand by the
a. *fixed points* of a thermometer,
b. the *fundamental interval* of a thermometer?

13 A thermometer which was not accurately calibrated indicates –0.5°C at the lower fixed point, and 106°C at the upper fixed point. What temperature does the thermometer register when the true temperature is 60°C?

Scanned with CamScanner

14 Draw a labelled diagram of a clinical thermometer and explain how it works. Give the reasons for the following features of the thermometer
   (i) a bulb of thin glass
   (ii) the constriction near the bulb
   (iii) a tube of fine bore.

15 The ice and steam points on a thermometer are found to be 96.0 mm apart. What temperature is recorded when the length of mercury thread is 33.6 mm above the ice point mark?

(SC)

16 With the aid of diagrams describe the action of
   a. Thermoelectric thermometer.
   b. Resistance thermometer. State one advantage of each thermometer.

17 Describe the kinetic molecular model of temperature.

18 (a) Define upper fixed point and lower fixed point as used in thermometry.
   (b) The electric resistance of the element in a platinum resistance thermometer at 100° 0°C and room temperature are 75.000; 63.000 and 64.992 respectively. Use these data to determine the room temperature.

(SSCE)

Scanned with CamScanner

9 MEASUREMENT OF HEAT ENERGY

Heat energy is the energy that is transferred from a hot object to a cooler one as a result of their temperature difference. Such energy always flows from the hot object to the cold object. Heat energy is only one form of energy and is called *thermal energy*. Like other forms of energy, it is measured in *joules*.

This chapter will discuss two important concepts of heat energy – *specific heat* and *latent heat*. We shall also treat other related topics such as *evaporation*, *boiling* and *sublimation*.

At the end of this chapter the student should be able to:

1 Explain the relationship between the heat supplied to a substance and (i) its temperature change at constant mass (ii) its mass at constant temperature change.
2 Explain the terms specific heat capacity and thermal capacity.
3 Explain why there is unequal rise in temperature for different substances of the same mass supplied with the same quantity of heat.
4 State the assumptions underlying the experiments involving the measurements of heat quantities.
5 Calculate unknown quantities using the relation $H = mc\Delta T$ when no change of state is involved.
6 Determine the melting point of a solid.
7 Determine the boiling point of a given liquid.
8 List the effects of impurities and pressure on
(i) the melting point of solid (ice)
(ii) the boiling point of a liquid.
9 Solve simple problems involving latent heat.
10 Distinguish between evaporation and boiling and explain sublimation.
11 Explain the working principles of such common devices as (a) refrigerator (b) air conditioner (c) pressure cooker.
12 Explain the effects of humidity on personal comfort.

9.1 Specific Heat

Experiments show that the rise in temperature of a body is proportional to the quantity of heat supplied to the body, the mass of the body and the nature of the substance of the body. We can in general write that

$$H = mc\theta \qquad 9.1$$

where H is the quantity of heat, m, is the mass of the body, θ is the change in its temperature and c is a constant known as the *specific heat capacity* of the body. This constant is dependent on the nature of the body. From equation 9.1, we can write that:

$$c = \frac{H}{m\theta} \qquad 9.2$$

Hence we can define c thus:

The *specific heat capacity* (c) of a substance is the quantity of heat required to raise the temperature of unit mass of the substance by 1°C (or 1 K).

When H is in Joules, m in kg and θ in °C or K then the unit of specific heat capacity is Joules per kilogram per Kelvin (J/kg.K). Water has a specific heat capacity of 4200 J/kg.K, (or 42 J/g.k). Thus, to raise the temperature of 1 kg of water by 1 K, we require 4200 J of heat. Table 9.1 gives the specific heat capacity of various substances.

Table 9.1

Substance	Specific heat capacity in $J.kg^{-1}.K^{-1}$
Aluminium	900
Brass	380
Copper	390
Glass	670
Iron	460
Lead	129
Platinum	135
Silver	234
Tin	226
Zinc	384
Alcohol	2520
Glycerine	2430
Ice	2100
Mercury	140
Paraffin oil	2130
Turpentine	1760
Water	4200

If we consider the quantity of heat required to raise the entire mass of the body by 1 K instead of just 1 kg of the mass, the heat involved is known as *heat capacity* or *thermal capacity*.

Scanned with CamScanner

The Thermal or Heat capacity (C_p) of a body is the quantity of heat required to raise the temperature of the body by 1 K.

The unit of thermal or heat capacity is joules per kelvin (J/K). It is easy to see that

Heat capacity (C_p) = mass (m) × specific heat capacity (c)

Thus from equation 9.1 we see that

$$H = mc\theta = C_p\theta \qquad 9.3$$

where $C_p = mc$ = thermal capacity.

Quantity of Heat = Thermal capacity × temperature rise

Example 9.1

A copper rod with heat capacity 585 J.K^{-1} is heated until its temperature changes from 35°C to 80°C. Calculate the quantity of heat supplied to the rod. If the specific heat capacity of copper is 390 J.kgK^{-1}, find the mass of the rod.

Solution

H = Heat capacity × temperature rise

$H = C_p\theta$

C_p = heat capacity = 585 J.K^{-1}

$H = 585(80 - 35)$ J = (585×45) J

= 26325 J = 26.325 kJ (kilojoules)

Let mass of rod be m (kg)

$C_p = mc$

$585 = 390m$

$m = \frac{585}{390} = 1.5$ kg

Example 9.2

What is the quantity of heat required to raise the temperature of 300 g of aluminium cube from 30°C to 70°C? (Specific heat capacity of aluminium = 900 Jkg^{-1}K^{-1}).

Solution

Rise in temperature = 70 – 30 = 40°C (or 40 K)

Mass of Aluminium = 300 g = 0.3 kg

$H = mc\theta$

$= (0.3 \times 900 \times 40)$ J

= 10800 J = 10.8 kJ

Example 9.3

a. What is meant by the statement that the specific heat capacity of water is 4200 JkgK^{-1}?

b. Calculate the temperature change when 1000 J of heat is supplied to 100 g of water.

Solution

a. The statement means that 4200 joules of heat energy is required to raise the temperature of 1 kg of water by 1 K.

b. Let the temperature change be θ°C.

From $H = mc\theta$ we have

$1000 \text{ J} = 0.1 \times 4200 \times \theta$

$\theta = \frac{1000}{0.1 \times 4200}$

= 2.38°C

Experiment to Measure the Specific Heat Capacity of a Solid by the Electrical Method

The specific heat capacity of a solid in the form of a metal block can be determined by an electrical method.

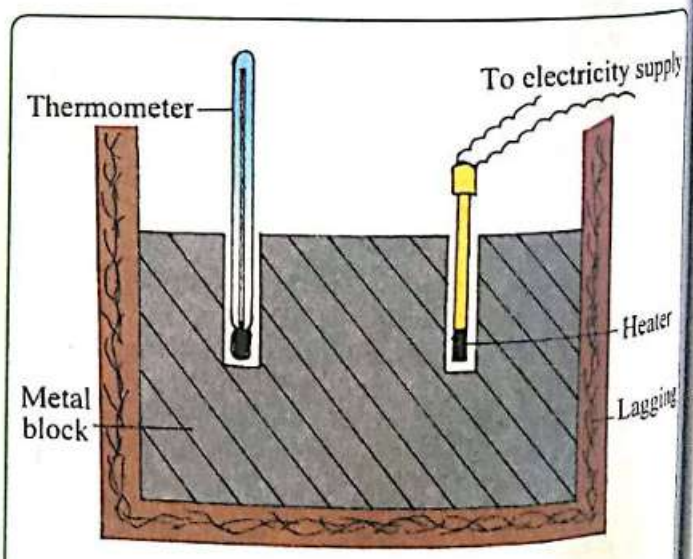


Fig. 9.1 Measurement of spcific heat capacity of solid by electrical method

Scanned with CamScanner

Two holes are bored in the block, after which the block is weighed. A thermometer and an electrical heater are inserted in each of the holes. A little oil in each hole helps to establish a good thermal contact with the block. The metal block is surrounded with a lagged jacket to reduce heat losses.

The initial temperature of the block is noted and the electrical heater is switched on and current allowed to flow for some time until the temperature rises by about 15°C. The exact time of flow of a known current is measured with a stop-clock, and the initial and final temperatures of the metal block are measured by the thermometer.

The specific heat capacity is calculated as follows:

Let mass of block be m (kg)

Let value of steady current be I (A)

Let value of p.d. across heater be V (v)

Let temperature rise of block be θ°C

Let time of flow of current be t (secs)

Let the specific heat capacity of block be c ($Jkg^{-1}K^{-1}$)

Heat supplied by current = $H = IVt$ joules.

Heat required to raise the temperature of block by θ°C = $mc\theta$

Assuming no heat loss to the surroundings, we have

$$IVt = mc\theta$$

$$c = \frac{IVt}{m\theta}\ Jkg^{-1}K^{-1} \qquad 9.4$$

In this calculation, we have ignored errors due to loss of heat to the surroundings, lack of thermal contact between metal and thermometer, and between metal and heater and also the small specific heat capacity of the heater.

Experiment to determine the Specific Heat Capacity of a Liquid by the Electrical Method.

The apparatus for this experiment is as shown in fig 9.2. A plastic container is first weighed and reweighed when about two-thirds full of the liquid (e.g. water). The ammeter, voltmeter and heating coil are connected in the circuit as shown. A plastic stirrer and a thermometer are also fitted into the container through holes in the wooden lid. The current is switched on after the initial temperature of the liquid has been recorded. The rheostat is adjusted to give a suitable steady current for about 3 minutes.

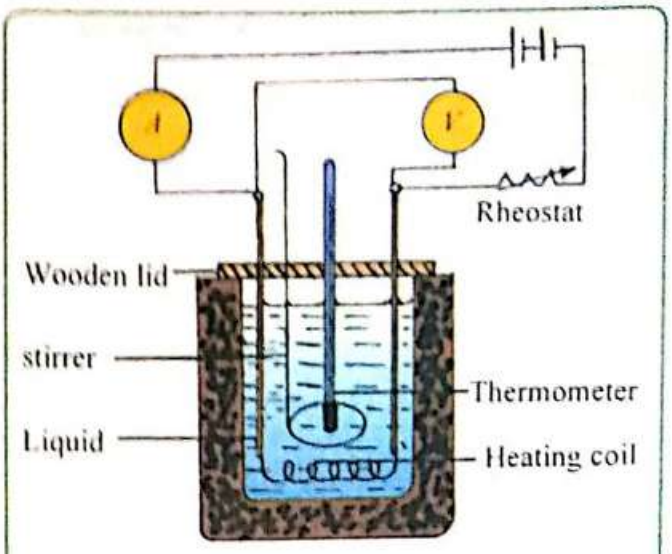


Fig. 9.2 Determination of specific heat capavity

The liquid is stirred gently while being heated. The final temperature is read after the current is switched off. The specific heat capacity of the liquid is calculated as follows:

Mass of liquid = m kg

Initial temperature of liquid = θ_1°C

Final temperature of liquid = θ_2°C

Current flowing in circuit = I A

Voltage across heating coil = V volts

Time of flow of current = t seconds

Let the specific heat capacity of liquid = c $J.kg^{-1}K^{-1}$

Rise in temperature = $(\theta_2 - \theta_1)$°C

Heat gained by liquid = $mc(\theta_2 - \theta_1)$ joules .

Electrical energy supplied to liquid = IVt joules .

Since heat gain by liquid = electrical energy supplied, we can write

$$mc(\theta_2 - \theta_1) = IVt$$

$$c = \frac{IVt}{m(\theta_2 - \theta_1)} \qquad 9.5$$

Errors can arise in this experiment as a result of heat lost to the surroundings. This is reduced by cooling the water with ice to about 10°C below room temperature before the start of the experiment, and heating the water until its final temperature is about 10°C above the room temperature. Another form of error can arise due to the neglect of the heat capacity of the plastic container in our calculation.

Scanned with CamScanner

Specific Heat Capacity by Method of Mixtures

The *calorimeter* is an apparatus used in experiments that involve the exchange of heat between two bodies at different temperatures. The apparatus is usually made of copper or aluminium which are good conductors of heat. The most common method employed in calorimeter experiments is known as the *method of mixtures*. In this method, a hot substance is mixed with a cold substance, during which heat flows from the hot to the cold substance until both reach the same temperature. When this happens, it follows that *the quantity of heat lost by the hot substance must be equal to the quantity of heat gained by the cold substance*. This is the *principle of mixtures* which is applicable in all problems involving a heat transfer. This assumes that no heat is lost to the surroundings. Losses are however often possible through the processes of *conduction*, *convection*, *radiation* and *evaporation*. To reduce losses due to conduction the calorimeter is surrounded with a poor conductor, e.g, cotton wool or felt. This is called *lagging the calorimeter*. Losses due to convection and evaporations are reduced by providing the calorimeter with a lid (containing holes to take a thermometer and a stirrer). Heat lost by radiation is reduced by polishing the outside of the calorimeter.

Experiment to determine the Specific Heat Capacity of a Solid by Method of Mixtures

The apparatus for this experiment is set up as shown in fig 9.3.

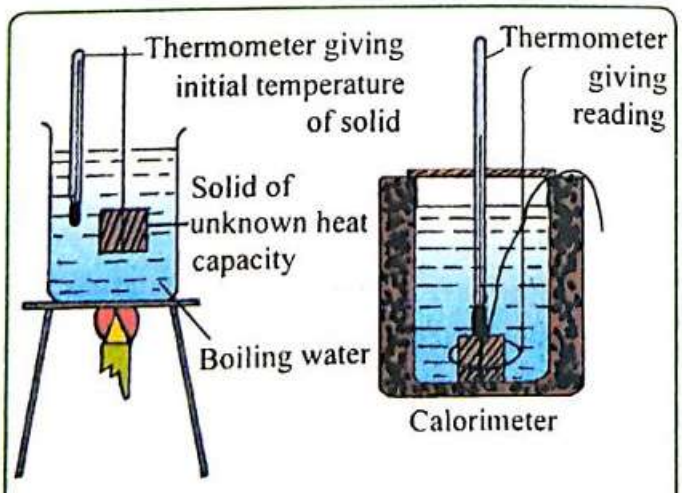


Fig. 9.3 Determination of specific heat capacity by the principle od mixtures

The solid in the form of a brass or copper block is weighed, tied to a string and left for some minutes in a beaker of boiling water. The calorimeter together with the stirrer is first weighed empty and reweighed when it is about two-thirds full of water. The initial temperature of the water is read with the thermometer. After the solid has stayed in the boiling water for about 15 minutes, it is quickly transferred into the water in the calorimeter. The calorimeter is covered with the lid, and the water is gently stirred to ensure a uniform distribution of temperature. The highest steady temperature of the mixture is read and recorded. The specific heat capacity of the solid is calculated as follows:

Mass of solid = m_1 kg

Mass of calorimeter + stirrer = m_2 kg

Mass of calorimeter + stirrer + water = m_3 kg

Initial temperature of water + calorimeter = θ_1

Final temperature of mixture = θ_2

Temperature of boiling water = 100°C

Specific heat capacity of solid = c_1

Specific heat capacity of water = 4200 $J.kg^{-1}.k^{-1}$

Let specific heat capacity of calorimeter material = c_2

Heat lost by solid in cooling from 100°C to final temperature of mixture $(\theta_2) = m_1 \times c_1(100-\theta_2)$J

Heat gained by water $(m_3 - m_2)\cdot 4200\cdot(\theta_2 - \theta_1)$J

Neglecting any heat losses to the surroundings we write:

Heat lost by solid = heat gained by water + heat gained by calorimeter and stirrer.

$$\therefore \quad m_1 \times c_1 \times (100-\theta_2) = (m_3 - m_2)\times 4200\times(\theta_2-\theta_1)\text{J} + m_2 \times c_2 \times (\theta_2 - \theta_1)$$

Since $m_1, m_2, \theta_2, \theta_1$ and c_2 are known, c_1 can be calculated from the above equation.

Precautions

1. The calorimeter should be lagged to prevent heat exchanges with the surroundings.
2. The hot metal solid should be quickly transferred to the calorimeter.
3. The mixture should be gently stirred to obtain an even temperature.

Experiment to determine the Specific Heat Capacity of a liquid by Method of Mixtures.

The experiment is carried out exactly as the previous experiment above, but this time we use the liquid of unknown specific heat capacity instead of water in the calorimeter, and we use a solid

Scanned with CamScanner

whose specific heat capacity is known. The specific heat capacity of the liquid is calculated as below:

Let specific heat capacity of solid = c_1

Let specific heat capacity of calorimeter = c_2

Let specific heat capacity of liquid = c_3

Mass of solid = m_1 kg

Mass of calorimeter + stirrer = m_2 kg

Mass of calorimeter + stirrer + liquid = m_3 kg

$\therefore$ Mass of liquid = $(m_3 - m_2)$ kg

Initial temperature of liquid and calorimeter = θ_1

Initial temperature of solid in boiling water = 100°C

Final temperature of mixture = θ_2

Heat lost by solid in cooling from 100°C to θ_2 = $m_1 \times c_1(100 - \theta_2)$

Heat gained by liquid + heat gained by calorimeter + stirrer

$= (m_3 - m_2) \times c_3 \times (\theta_2 - \theta_1) + m_2 \times c_2 \times (\theta_2 - \theta_1)$

Neglecting heat lost to the surroundings we can write:

Heat lost by solid = total heat gained by liquid and calorimeter + stirrer.

$$m_1 \times c_1(100 - \theta_2) = (m_3 - m_2) \times c_3 \times (\theta_2 - \theta_1) + m_2 \times c_2 \times (\theta_2 - \theta_1)$$

Since $m_1, m_2, m_3, \theta_1, \theta_2, c_1, c_2$ are known, c_3 can be calculated from the above equation.

The student does not need to memorize any of the above equations for the specific heat of solid or liquid. What is necessary is just the understanding that heat gained or heat lost by material is equal to the mass of material × specific heat capacity of material × temperature change. Heat is gained by a material when its temperature rises from a lower to a higher value, and heat is lost when the temperature of the material falls from a higher to a lower value.

Example 9.4

The hot water tap of a bath delivers water at 85°C at a rate of 8 kg/min. The cold water tap of the bath delivers water at 30°C at the rate of 20 kg.min^{-1}. If both taps are left on for 5 minutes, calculate the final temperature of the bath water, ignoring heat losses to the surroundings.

Solution

Let the final temperature be θ°C.

Mass of hot liquid delivered in 5 minutes = 8 × 5 = 40 kg.

Mass of cold liquid delivered in 5 minutes = 20 × 5 = 100 kg.

Heat lost by hot water in cooling from 85°C to θ°C

$= 40 \times c \times (85 - \theta)$ J

where c is the specific heat capacity of water.

Heat gained by the cold water in warming from 30°C to θ = $100 \times c \times (\theta - 30)$ J

Since heat gained = heat lost, we have:

$100 \times c \times (\theta - 30) = 40 \times c \times (85 - \theta)$

$\theta = \underline{45.7°C}$

Example 9.5

A piece of copper weighing 400 g is heated to 100°C and then quickly transferred to a copper calorimeter of mass 10 g containing 100 g of liquid of unknown specific heat capacity at 30°C. If the final temperature of the mixture is 50°C, calculate the specific heat capacity of the liquid. (Specific heat capacity of copper = 390 J.kg^{-1}k^{-1})

Solution

Mass of copper = 400 g = 0.4 kg.

Mass of copper calorimeter = 10 g = 0.01 kg.

Mass of liquid = 100 g = 0.1 kg.

Initial temperature of copper piece = 100°C.

Initial temperature of liquid and calorimeter = 30°C.

Final temperature of mixture = 50°C ,

Heat lost by copper = 0.4 × 390 × (100 − 50) = 7800 J.

Heat gained by liquid = $0.1 \times c \times (50 - 30) = 2c$ J

Heat gained by calorimeter

$= 0.01 \times 390 \times (50 - 30) = 78$ J,

Total heat gained = heat lost.

$2c + 78 = 7800$

$c = \underline{3861 \text{ J.kg}^{-1}\text{K}^{-1}}$

Example 9.6

An electric heater, rated 15 V 40 W , fitted into a metal block supplies heat to the block of mass 1.5 kg and specific heat capacity of 460 Jkg^{-1}K^{-1}. Calculate the temperature rise in the block if the current flows for 10 minutes.

Solution

Power of heater = IV = 40 W

Heat produced by heater in 10 minutes =

Power × time (in sec) = 40 × 10 × 60 J

= 24000 J.

Let temperature rise in metal block = θ K.

Heat gained by metal block = $1.5 \times 460 \times \theta$ J ,

$\therefore \quad 24000 = 1.5 \times 460 \times \theta$

$$\theta = \frac{24000}{1.5 \times 460}$$

Scanned with CamScanner

= 34.78 K or 34.78°C

Example 9.7

A waterfall is 70 m high. Calculate the difference between the temperature of the water at the top and at the foot of the waterfall. (g = 10 ms^{-2}), specific heat capacity of water = 4200 $J.kg^{-1}K^{-1}$. What important assumption have you made in your calculation?

Solution

Potential energy of the water at the top of the waterfall = $mgh = m \times 10 \times 70$ J

where m is certain mass of water in kg.

Let the rise in temperature of water when it falls through 70 metres, be θ K

∴ Heat gained by water = $m \times 4200 \times \theta$

Assuming that all the potential energy of the water is eventually converted to heat energy, we have

$$m \times 10 \times 70 = m \times 4200 \times \theta$$

$$\theta = \frac{m \times 10 \times 70}{m \times 4200} = \frac{700}{4200} \text{ K}$$

$$= 0.17 \text{ K}$$

The important assumption we have made in the calculation is that the law of conservation of energy holds true and all the potential energy has been converted to heat energy.

9.2 Change of State: Latent Heat

Matter can exist in three states – solid, liquid, and gas. Supply of heat to a solid will not only cause its temperature to rise but can also cause the solid to change to a liquid in a process known as *Fusion*. Similarly a liquid can change to a gas or vapour through the addition of heat by the process of *vaporization* or *evaporation*. The reverse processes are also possible. Removal of heat from a gas can cause it to revert to a liquid and the removal of heat from a liquid can cause it to revert to a solid. The change of state of any substance under normal atmospheric pressure takes place at known temperatures. *For pure crystalline substances the change of state occurs at fixed and definite temperatures.*

Activity 9.1: To determine the Melting Point of Naphthalene

Put some solid naphthalene into a test tube placed in a beaker of hot water. Insert a thermometer into the test tube. Read the temperature of the melting naphthalene at one minute interval as the water in the beaker is heated more and more.

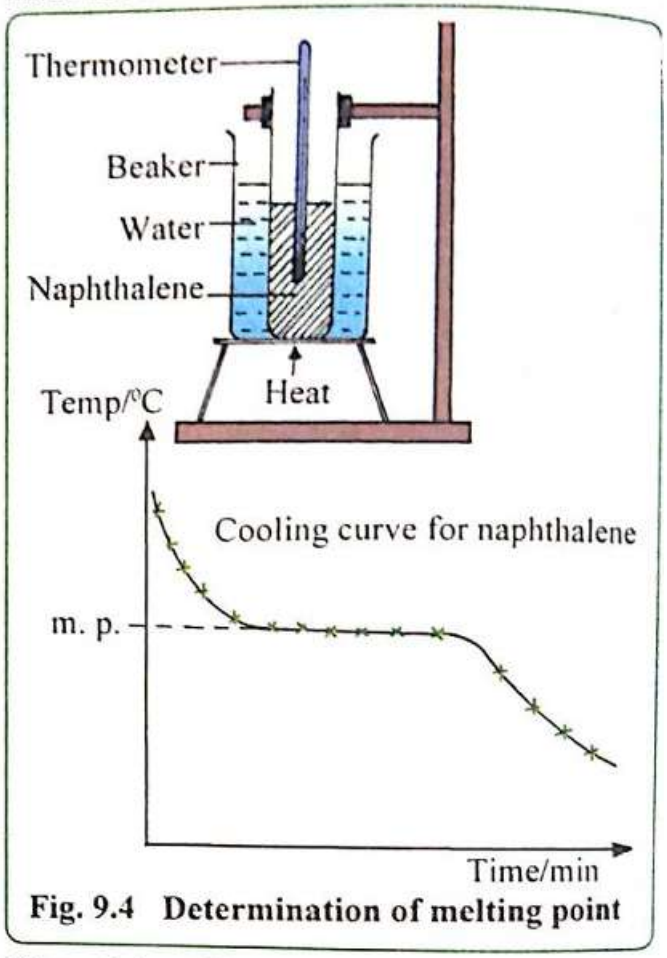
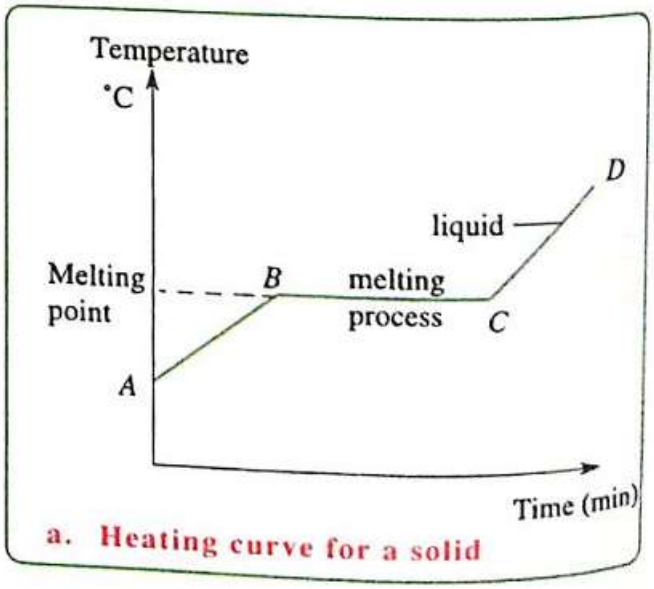


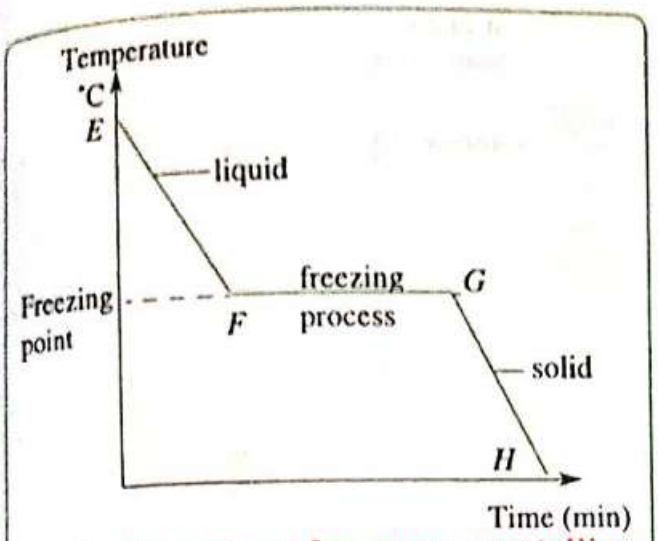
Fig. 9.4 Determination of melting point

When all the solid naphthalene has melted, remove the beaker of hot water and allow the molten naphthalene to cool slowly undisturbed. Again record the temperature at intervals of one minute. Plot the graph of temperature versus time in the two cases.

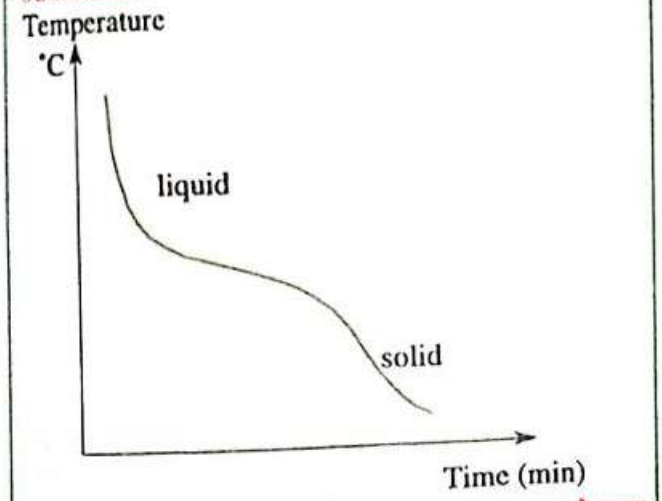


a. Heating curve for a solid

Scanned with CamScanner



b. Cooling curve for pure crystalline substance



c. Cooling curve for wax (amorphous substance)

Fig. 9.5

Repeat the same experiment using ice and wax and plot their temperature versus time graphs.

From fig. 9.5(a) we see that the temperature of the naphthalene rises along AB until the naphthalene begins to melt. During melting the temperature remains constant along BC, even though heat is being supplied to the substance. The constant temperature persists until all the naphthalene has melted. Beyond this point and along CD the temperature rises once more.

For the cooling curve, (fig. 9.5(b),) the temperature is seen to drop steadily with time along EF until the molten substance begins to solidify at F. During solidification the temperature remains constant along FG until after all the naphthalene has solidified at G? Beyond G, the temperature again begins to decrease with time along GH.

An important observation is that the constant temperature represented by BC for melting is the same as the constant temperature represented by FG for freezing (or solidification). Thus the temperature at which a solid melts to its liquid form (melting point) is the same as the temperature at which the liquid solidifies or freezes to its solid state (freezing point). Also while the change of state is taking place, the temperature of the substance remains constant. Crystalline substances have definite melting or freezing points but amorphous substances (e.g wax, glass etc) do not have such definite temperatures (fig. 9.5(c)) This is an important distinguishing feature between crystalline and amorphous substances. Such characteristic property of amorphous substances is used in the blowing of molten glass to any desired shape.

Latent Heat of Fusion

Our experiment above shows that during the melting process the temperature of the crystalline substance remains constant despite the fact that heat is being supplied. For instance, when heat is supplied to ice at its melting point (0°C), it absorbs the heat and melts to form water at the same temperature of 0°C.

Conversely, when a liquid such as water or molten naphthalene at the freezing point is cooled, it changes from the liquid to a solid state at the same temperature. This heat which does not show itself as a rise (or fall) in the temperature of the substance but rather remains hidden in the liquid is known as *latent heat*.

Latent heat is the heat supplied or removed which causes a change of state without a change in temperature

Latent heat is an invisible heat, hence the thermometer does not detect it. It depends on the mass and nature of a substance.

The Latent Heat of Fusion is the quantity of heat required to convert a substance from its solid to its liquid state without a change in temperature.

When only unit mass of the substance is considered, the heat involved is known as *specific latent heat.*

Specific Latent Heat of Fusion (l) of a substance is the quantity of heat required to convert unit mass of the solid at the melting point to its liquid form without a change of temperature.

Scanned with CamScanner

In S.I. units, specific latent heat of fusion is expressed in Joules per kilogram ($J.kg^{-1}$). The heat involved (H), the mass (m), and the specific latent heat of fusion (l) are related by the formula:

$$H = ml \qquad 9.6$$

While the substance is freezing (fig. 9.5b) it is evolving its latent heat of fusion, which compensates for the heat lost by cooling. Hence the temperature does not fall but remains constant.

Example 9.8

How much heat is required to change 3 kg of ice at 0°C to water at the same temperature. Specific latent heat of fusion of ice = 336000 Jkg^{-1}.

Solution

$H = ml$

$= 3 \times 33600$ joules

$= \underline{10.08 \times 10^5 \text{ joules}}$

Example 9.9

What quantity of heat will be necessary to change 2 kg of ice at 0°C to water at 10°C. Specific latent heat of fusion of ice = 3.36×10^5 Jkg^{-1}, and specific heat capacity of water = 4200 $JkgK^{-1}$.

Solution

In this example, we require heat to first melt the ice to water at 0°C and heat to raise the temperature of water from 0°C to 10°C.

Heat required to change 2 kg of ice at 0°C to water at 0°C = $2 \times 3.36 \times 10^5$ J.

$= \underline{6.72 \times 10^5 \text{ J}}$

Heat required to raise the temperature of 2 kg of water from 0°C to 10°C

$= 2 \times 4200 \times (10 - 0)$ J

$= 8.4 \times 10^4 = 0.84 \times 10^5$ J

$\therefore$ total heat needed = $(6.772 + 0.84)10^5$ J

$= \underline{7.56 \times 10^5 \text{ joules}}$

Experiment to determine the specific latent heat of fusion of ice

The apparatus for this experiment is as shown in fig. 9.6.

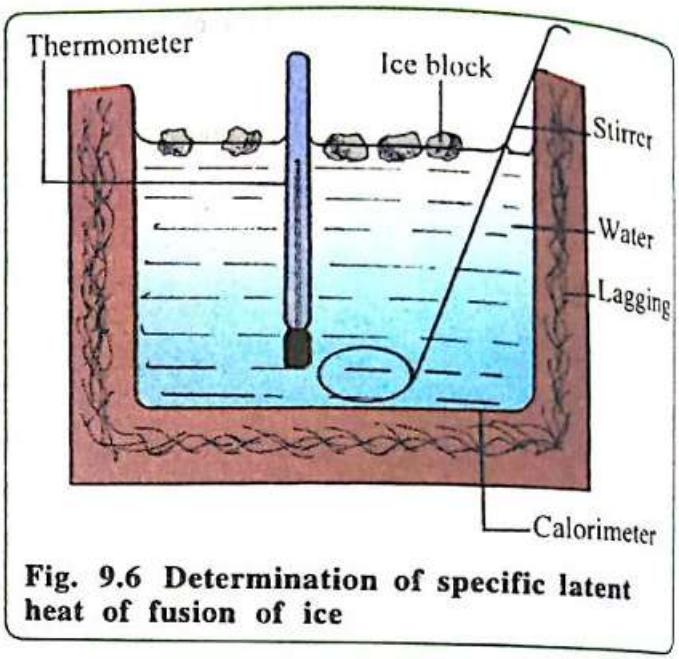


Fig. 9.6 Determination of specific latent heat of fusion of ice

The copper calorimeter with a stirrer is first weighed empty and reweighed when about half-full of water warmed to about 5°C above room temperature. The exact temperature is read and recorded. Some small ice blocks are dried with blotting paper and then added to the water in the calorimeter. The mixture is gently and continuously stirred. Each piece of ice is allowed to melt completely before the next is added. When the temperature of the mixture is about 5°C below room temperature, the exact temperature is read with a thermometer. The calorimeter with its content is reweighed to find the mass of ice added. The specific latent heat capacity of ice is then calculated as follows:

Mass of calorimeter and stirrer = m_1 (kg).

Mass of calorimeter + stirrer + water = m_2 (kg).

Mass of calorimeter + stirrer + water + ice = m_3 (kg),

Initial temperature of water = θ_1°C

Final temperature of mixture = θ_2°C.

Specific heat capacity of calorimeter

$= c$ ($Jkg^{-1}K^{-1}$)

Specific heat capacity of water = 4200 ($J.kg^{-1}K^{-1}$)

Specific latent heat of fusion of ice = l ($J.kg^{-1}$).

Mass of water = $m_2 - m_1$ (kg).

Mass of ice added = $m_3 - m_2$ (kg).

Heat required to melt the ice = $(m_3 - m_2)l$ (J).

Scanned with CamScanner

Heat required to raise the temperature of melted ice from 0°C to θ_2°C

$= (m_3 - m_2) \times 4200 \times \theta_2$ (J)

Heat lost by the calorimeter and water in cooling from θ_1°C to θ_2°C

$= [m_1 c + (m_2 - m_1) \times 4200](\theta_1 - \theta_2)$ J

Since heat gained by ice = heat lost by calorimeter and water, we can write

$(m_3 - m_2)l + (m_3 - m_2) \times 4200 \times \theta_2$

$= [m_1 c + (m_2 - m_1) \times 4200] \times (\theta_1 - \theta_2)$

Since $m_1, m_2, m_3, \theta_1, \theta_2, c$ are known, l can be calculated from the above equation.

Precautions

1 Only dry ice pieces should be used.
2 Ice should be added in small quantities at a time.
3 The mixture should be gently and continuously stirred to obtain an even temperature of the mixture.
4 The final temperature should be about the same amount below room temperature as the initial temperature is above room temperature. In this way we can largely compensate for the heat exchange between the calorimeter and its contents and the surrounding air.
5 The calorimeter should be lagged.

Expansion and contraction on Fusion

Water expands on freezing to ice. Because of this expansion,

(i) a glass bottle containing water, cracks when the water freezes
(ii) ice is less dense than water and so floats on water with nine-tenths of its volume submerged in water.

Similarly ice contracts on melting to water.

Effects of pressure and impurities on freezing point

1. An ice block is seen to melt if sufficient pressure is applied to it. This is because *an increase in pressure lowers the melting point of ice or the freezing point of water*. Usually an increase in pressure lowers the freezing point of any liquid which expands on solidifying. But for substances which contract on solidifying (e.g paraffin-wax) an increase in pressure results in higher freezing point.

When we subject ice to high pressure it melts, but when this pressure is removed, the ice re-freezes. This phenomenon is known as *regelation*. Hence a piece of wire with heavy weights attached to both ends can cut through a block of ice but the block remains solid behind the wire (fig. 9.7).

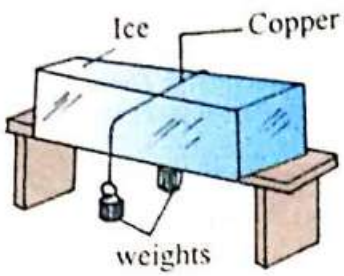


Fig. 9.7 Regelation

The ice directly below the wire melts because the increased pressure lowers the melting point of ice, hence the wire falls through the water formed. Above the wire, decrease in pressure raises the freezing point of the melted ice and the water freezes again. On re-freezing, it liberates latent heat which is conducted by the wire to melt more ice beneath it. The process is repeated till the wire passes right through the ice block without cutting it into two. When we press two pieces of ice blocks one against the other, they stick together. The increased pressure lowers the melting point of ice. The melted ice flows away as water to places of lower pressure. There, it re-freezes, binding the two pieces of ice together.

2. Impurities lower the melting point of a pure solid. Hence a *freezing mixture of ice and salt* has a lower temperature than 0°C, the freezing point of pure water (or the melting point of ice).

Example 9.10

A copper calorimeter weighing 40 g contains 200 g of water at 35°C. 20g of dry ice is added to the water and this reduces the temperature to 25°C after all the ice has melted. What value of the specific latent heat of fusion of ice is given by this experiment? (Specific heat capacity of water = 4200 $Jkg^{-1}k^{-1}$, Specific heat capacity of copper = 400 $Jkg^{-1}k^{-1}$)

Solution

Let specific latent heat of fusion of ice = l (Jkg^{-1})
Mass of copper calorimeter = 40 g = 0.04 kg
Mass of water = 200 g = 0.2 kg
Mass of ice added = 20 g = 0.02 kg
Heat gained by ice in melting = 0.02 l (J)

Scanned with CamScanner

Heat required to raise temperature of melted ice from 0°C to 25°C
$= 0.02 \times 4200 \times 25 = 2100$ J.
Total heat gained $= (0.02l + 2100)$ J.
Heat lost by water in cooling from 35°C to 25°C
$= 0.2 \times 4200 \times 10 = 8400$ J
Heat lost by calorimeter in cooling from 35°C to 25°C $= 0.04 \times 400 \times 10 = 160$ J.
Total heat lost $= (8400 + 160) = 8560$ J.
Since heat gained = heat lost, we have:
$0.02l + 2100 = 8560$
$0.02l = 6460$
$l = 3.23 \times 10^5$ J.kg^{-1}

Latent Heat of Vaporization

Vaporization is the process in which a substance changes from the liquid to the gaseous state. Like fusion, it occurs at a definite temperature for a given substance. This temperature is known as the *boiling point* of the liquid.

The specific latent heat of vaporization (L) of a substance is the quantity of heat required to change unit mass of the substance from its liquid at the boiling point to vapour without a change in temperature

The unit for specific latent heat of vaporization is Jkg^{-1}. In general, the quantity of heat, H, required for vaporization of mass m is given by

$$H = mL \qquad 9.7$$

Experiment to determine the Specific Latent Heat of Vaporization of Steam.

The specific latent heat of vaporization of steam can be found by passing steam into water in a calorimeter, where it condenses and releases its latent heat. The apparatus is as shown in fig 9.8. The calorimeter is first weighed empty and then reweighed when about half-full of water. The temperature of the water is read and recorded. The calorimeter is lagged to insulate it from the heat of surrounding. Steam from the boiler is passed through a 'steam trap' to remove any condensed moisture. Dry steam only therefore passes into the water in the calorimeter. A screen is usually placed between the boiler and the calorimeter to prevent the latter from receiving heat directly from the boiler. The passing of dry steam into the calorimeter is stopped when the temperature of the water rises by about 30°C above room temperature.

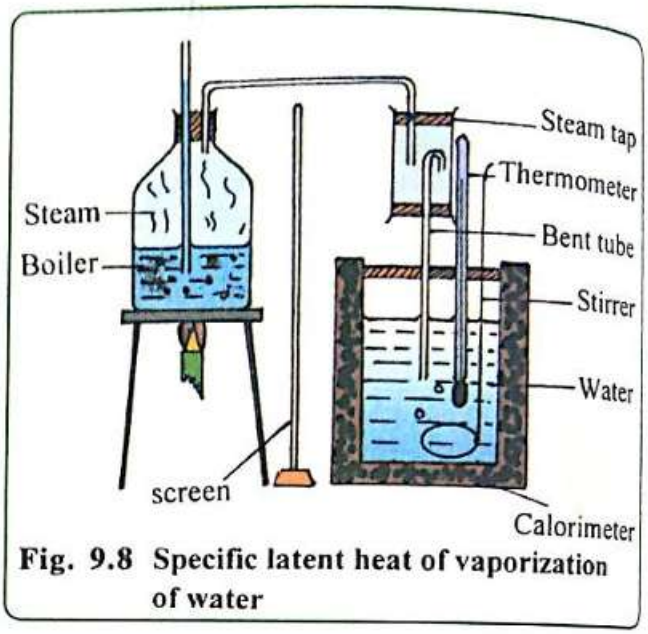


Fig. 9.8 Specific latent heat of vaporization of water

The water is continuously and gently stirred until a steady final temperature is obtained. The calorimeter and contents are then cooled and reweighed to find the mass of condensed steam. The specific latent heat of vaporization of steam is then calculated as follows:
Mass of calorimeter $= m_1$ (g)
Mass of calorimeter + water $= m_2$ (kg)
Mass of calorimeter + water + condensed steam $= m_3$ (kg)
Initial temperature of water + calorimeter $= \theta_1$°C
Final temperature of mixture $= \theta_2$°C
Temperature of steam = 100°C
Let specific latent heat of steam $= L$ (Jkg^{-1})
Mass of water $= m_2 - m_1$ (kg)
Mass of condensed steam $= m_3 - m_2$ (kg)
Specific heat capacity of water = 4200 Jkg^{-1}k^{-1}
Specific heat capacity of copper calorimeter $= 400$ Jkg^{-1}k^{-1}
Heat lost by steam in condensing to water at 100°C $= (m_3 - m_2)L$ (J)
Heat lost by condensed steam in cooling from 100°C to θ_2°C
$= (m_3 - m_2) \times 4200 \times (100 - \theta_2)$ (J)
Total heat lost
$= (m_3 - m_2)L + (m_3 - m_2) \times 4200 \times (100 - \theta_2)$ J
Heat gained by calorimeter $= m_1 \times 400 \times (\theta_2 - \theta_1)$
Heat gained by water $= (m_2 - m_1) \times 400 \times (\theta_2 - \theta_1)$

Scanned with CamScanner

Total heat gained:
$m_1 \times 400 \times (\theta_2 - \theta_1) + (m_2 - m_1) \times 400 \times (\theta_2 - \theta_1)$
By equating total heat gained to total heat lost, we can obtain the value of L as is illustrated in example 9.11

Precautions

1 Only dry steam should be used.
2 The calorimeter should be lagged and should be further prevented from receiving extra heat directly from the boiler.
3 The water in the calorimeter should be cooled to about 10°C below room temperature before steam is passed and the passing of steam should be continued until the temperature of mixture is about 10°C above room temperature.
4 The mixture should be gently and continuously stirred to ensure an even final temperature.

Example 9.11

In an experiment to find the specific latent heat of vaporization of steam, the mass of calorimeter was 50 g. The total mass of calorimeter and water was 300 g. The initial temperature of the water in the calorimeter before steam was passed was 20°C. The temperature increased to 65°C after steam of total mass 16 g was passed. Assuming that the steam was at 100°C, estimate the value of the latent heat of vaporization of steam. (Specific heat capacity of calorimeter = 400 $Jkg^{-1}k^{-1}$, Specific heat capacity of water = 4200 $Jkg^{-1}k^{-1}$)

Solution

Mass of calorimeter = 50 g = 0.05 kg.
Mass of water = 300 – 50 = 250 g = 0.25 kg.
Mass of steam = 16 g = 0.016 kg .
Heat lost by steam in condensing to water at 100°C
$= 0.016L$ (J)
Heat lost by condensed water in cooling from 100°C to 65°C
$= 0.016 \times 4200 \times 35 = 2352$ (J)
Total heat lost = $(0.016L + 2352)$ J
where L is the specific latent heat of vaporization of steam.
Heat gained by calorimeter in warming from 20°C to 65°C
$= 0.05 \times 400 \times 45$ J
$= 900$ J
Heat gained by water in warming from 20°C to 65°C = $0.25 \times 4200 \times 45 = 47250$ J
Total heat gained = $47250 + 900 = 48150$ J
Total heat gained = total heat lost

$48150 = 0.016L + 2352$
$L = \underline{2.86 \times 10^6 \text{ Jkg}^{-1}}$

Example 9.12

Estimate the quantity of heat needed to melt 150 g of ice at –10°C to water at 15°C.
(Specific heat capacity of water = 4200 $JkgK^{-1}$
Specific heat capacity of ice = 2100 $JkgK^{-1}$
Specific latent heat of fusion of ice = 3.36×10^5 Jkg^{-1}).

Solution

Heat required to raise the temperature of ice from –10°C to 0°C = $0.15 \times 2100 \times 10 = 3150$ J
Heat required to melt the ice at 0°C
$= 0.15 \times 3.36 \times 10^5$ J $= 5.04 \times 10^4$ J $= 50400$ J
Heat required to raise the temperature of water from 0°C to 15°C $= 0.15 \times 4200 \times 15 = 9450$ J
∴ total heat = 3150 + 50400 + 9450
$= \underline{6.30 \times 10^4 \text{ J} = 63 \text{ kJ}}$

Example 9.13

Explain the terms specific latent heat and heat capacity of a substance.

Heat is supplied to a liquid of mass 500 g contained in a can by passing a current of 4 A through a heating coil of resistance 12.5 ohm immersed in the liquid. The initial temperature of the liquid is 24°C. The liquid reaches its boiling point in 10 minutes after the current is switched on. It takes a further 2 minutes after the liquid starts to boil for all the liquid to boil away. Calculate

a. the specific heat capacity of the liquid
b. the specific latent heat of vaporization of the liquid.

(Boiling point of liquid = 84°C, thermal capacity of the can = 400 J/K

(SC/GCE)

Solution

a. Mass of liquid = 500 g = 0.5 kg
Heat required to raise temperature of liquid from 24°C to 84°C (the boiling point of liquid)
$= 0.5 \times c \times (84 - 24)$ J $= 30c$ (J)
where c = specific heat capacity of the liquid.
Heat required to raise temperature of can from 24°C to 84°C = $400 \times 60 = 24000$ J
(i.e thermal capacity × change in temperature)
Heat supplied by heating coil = $IVt = I^2Rt$
$= 4 \times 4 \times 12.5 \times 10 \times 60 = 120000$ J

Scanned with CamScanner

Since this heat is used to raise the temperature of the can and the liquid to boiling point, we have

$30c + 2400 = 120000$

$c = \underline{3200\ \text{Jkg}^{-1}k^{-1}}$

b. Let specific latent heat of vaporization of the liquid = L (Jkg^{-1})
Heat required to vaporize liquid = $0.5L$ (J)
Heat supplied by current
$= I^2Rt = 16 \times 12.5 \times 2 \times 60 = 24000\ \text{J}$
Since this heat is required to boil away the liquid at 84°C, we have, neglecting any heat losses to the surrounding,

$0.5L = 24000$

$L = \underline{4.8 \times 10^4\ \text{Jkg}^{-1}}$

9.3 Evaporation

When water or any other liquid is exposed in an open dish, it slowly evaporates, that is, it goes spontaneously into the gaseous state.

Evaporation is a process where a liquid turns spontaneously into vapour below its boiling point.

It takes place at all temperatures. Various factors affect the rate of evaporation. Amongst these are:

1 *Temperature*. The rate of evaporation increases with increasing temperature.
2 *Pressure*. The greater the pressure, the slower the evaporation and vice versa.
3 *Wind and dryness of the air*. The drier the air (i.e. the less water vapour it contains) the faster the evaporation. Wet clothes dry faster in dry air. We observe this very regularly during the dry Harmattan season. The dryness of the air around the clothes causes rapid evaporation of the moisture from the wet material. Also wet clothes placed in an air draft (e.g. draft from a fan) will dry up faster than if placed in still air. The draft or wind blows away the water vapour around the material causing more evaporation to take place by bringing drier air into contact with the wet material.
4 *Nature of the liquid*. Liquids with low boiling points evaporate more rapidly than those with higher boiling points. Hence ether (b.p. 35°C) evaporates more rapidly than mercury (b.p. 357°C).
5 *Area of liquid exposed*. The greater the surface area of liquid exposed, the faster the evaporation.

Molecular Explanation of Evaporation

According to the kinetic molecular theory of matter, a liquid consists of molecules that are in constant motion. The average velocity and hence the average kinetic energy of the molecules is related to the temperature of the liquid. When the temperature increases, the molecules gain more kinetic energy.

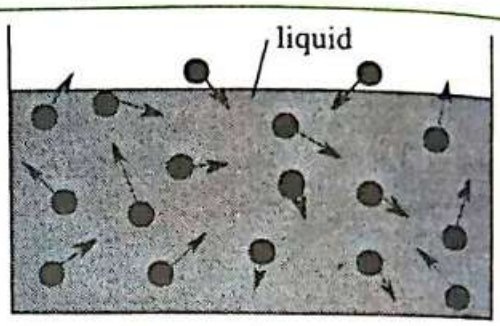


Fig. 9.9 Molecular explanation of evaporation

Molecules with high speeds near the surface of the liquid may have enough kinetic energy to break away from the attraction of other molecules and move outside the liquid surface as molecules of vapour. Some of these molecules stay outside the liquid, some however return to the liquid due to the attractive forces from the molecules in the liquid. As more and more molecules stay outside the liquid surface, the liquid evaporates more and more. High temperatures increase the velocities of all the molecules, thus allowing more molecules to escape from the surface, hence increasing the rate of evaporation. The presence of a draft of air or wind increases the rate of evaporation by sweeping away the molecules of vapour above the liquid surface, thus making way for fresh supply of escaping molecules.

Cooling by Evaporation

Whenever methylated spirit or petrol is spilled over any part of our body, we usually feel a cooling effect as the liquid evaporates. The body becomes cooler because the latent heat needed to convert the liquid to the vapour state is extracted from the liquid or any object in contact with it. This extraction of the latent heat from the liquid leads to a fall in its temperature. The faster the evaporation the greater is the fall in temperature.

The human body utilizes the effect of evaporation for cooling. Perspiration cools the body as sweat evaporates from the surface of the

Scanned with CamScanner

skin. As the latent heat of vaporization is extracted from the body, it is cooled in the process. Cooling by evaporation is also utilized by doctors for numbing pains from needle points. Volatile ether or methylated spirit is usually dabbed on the skin before the body is injected with some drug. As the methylated spirit evaporates, it cools the part of the skin and numbs it so that the pain from the needle prick during an injection process is not much felt.

Activity 9.2: To show that rapid evaporation causes cooling

1 Spread some water on a wooden block and place a small beaker half-filled with ether on the block.

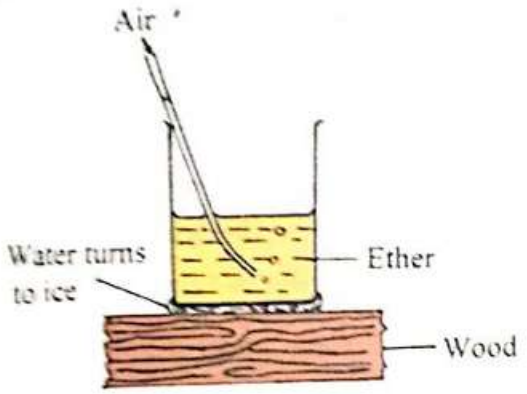


Fig. 9.10 Cooling by rapid evaporation

2 Blow air rapidly into the ether through a small straw pipe so that the ether evaporates.
3 Notice that the water beneath the beaker eventually turns to ice, fusing the beaker to the block of wood.

Explanation The latent heat of vaporization of the ether is taken from the surroundings, including the water beneath the beaker. As the water is cooled down to 0°C, ice is formed.

How does the kinetic molecular theory explain cooling by evaporation?

The molecules of a liquid have an average velocity or average kinetic energy which increases with temperature. Molecules near the surface which happen to be moving faster than the average escape from the attraction of their neighbours and out of the liquid. Thus, the liquid loses its most energetic molecules while the less energetic ones are left behind. The average kinetic energy of the remaining molecules is therefore reduced. This results in the lowering of the temperature of the liquid, since temperature is a measure of the average kinetic energy. Hence cooling of the liquid is brought about.

The Refrigerator: Working Principles

The working principle of the refrigerator utilizes the cooling effect of evaporation.

The volatile liquid used is called the *refrigerant*. Examples of such liquids or liquefied gas are ammonia, and ethyl chloride (Freon).

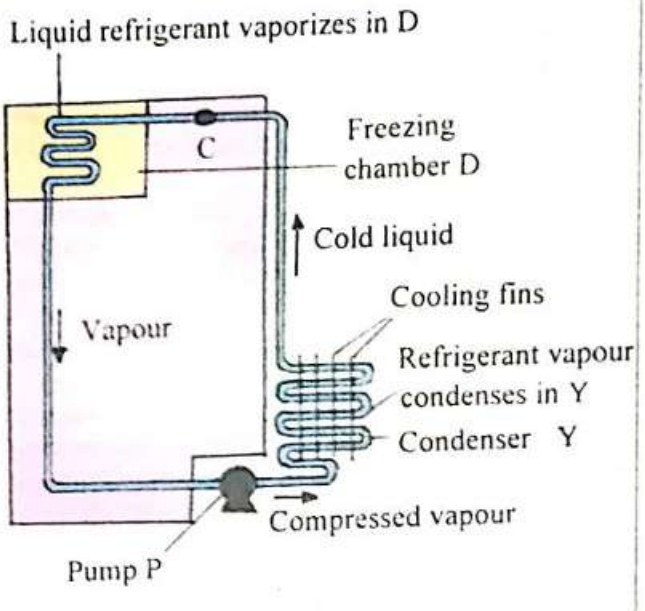


Fig. 9.11 Refrigerator

The volatile liquid is contained inside coiled copper pipes or tubes which surround the freezing chamber. As the liquid evaporates, it absorbs the necessary latent heat of vaporization from its surrounding thereby cooling the inside of the refrigerator and its contents. The vapour is removed by the pump which compresses it into a condenser attached outside the refrigerator and fixed with metal cooling fins. The compressed vapour condenses and gives out latent heat. This heat is removed by conduction into the cooling fins and thence by convection and radiation to the surrounding. The condensed vapour goes back into circulation around the freezing chamber, where it again evaporates, thus setting up a continuous circulation of liquid and vapour.

A thermostat inside the refrigerator switches the pump motor on and off at intervals. In this way it controls the rate of vaporization and the consequent degree of cooling.

Scanned with CamScanner

9.4 Boiling

As we heat water in a container, its temperature rises and the liquid evaporates. As the heating is continued, the temperature is found to rise until the water begins to boil. During boiling bubbles of air formed inside the liquid rise to the surface. The temperature of the liquid remains steady during boiling. This constant temperature is called the *boiling point* of the liquid.

Evaporation and Boiling compared

	Evaporation	Boiling
1	This is the change from liquid to vapour at temperatures below normal boiling point.	This is the change from liquid to vapour at the boiling point.
2	Takes place slowly at the liquid surface.	Occurs throughout the entire volume of the liquid
3	Takes place at all temperatures	Takes place at a particular temperature for a given pressure.
4	Temperature need not be steady during evaporation.	Temperature remains steady during boiling.
5	Wind assists evaporation	Wind has no effect on boiling..

Effect of Pressure on Boiling Point

An increase in pressure at the surface of a liquid raises the boiling point of the liquid and conversely, a decrease in pressure lowers the boiling point of the liquid. We can demonstrate this by a simple experiment.

Activity 9.3

1 Heat some water in a round bottomed flask until the water boils. Allow the water to boil for about five minutes.

2 Cork the flask tightly and invert it over a sink as shown in fig 9.12.
Allow the flask to cool until the water completely stops boiling.

3 Pour some water over the flask, and notice that the boiling recommences once more even though the temperature is below boiling point.

4 Stop the flow of water over the flask and notice that the boiling again ceases.

The cold running water over the flask causes some of the vapour inside the flask to condense. This reduces the pressure of the warm water inside the flask. This reduction of pressure lowers the boiling point of the liquid, thus making the liquid to boil at a lower temperature.

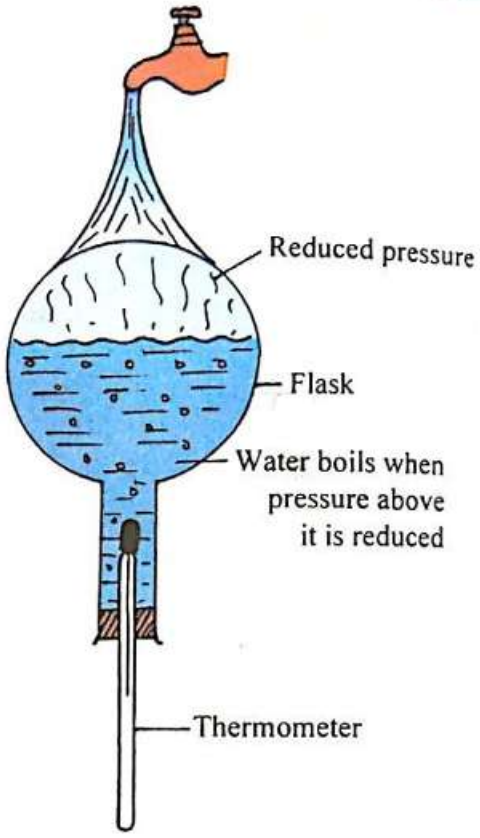


Fig. 9.12 Effect of pressure on boiling point

The *Pressure Cooker* is a practical application of the effect of pressure on the boiling point of a liquid. The pressure cooker is basically a strong metal saucepan with a lid that can be held down. There is a loaded pin valve on this lid which controls the flow of steam produced when the water in the container is heated.

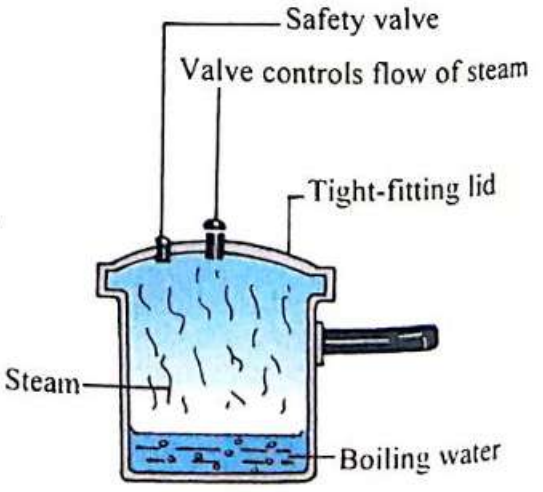


Fig. 9.13 Pressure cooker

Scanned with CamScanner

Food is cooked much faster in the pressure cooker because the increased pressure of the trapped gas above the liquid raises the boiling point of the liquid inside the cooker to about 120°C or more. Thus a high cooking temperature is reached very fast, thereby reducing the cooking time and saving fuel.

Pressure cookers are very indispensable to mountaineers and other people who go where atmospheric pressure is low. The low pressure reduces the boiling point of water to below 100°C. So it will take a long time to cook food if pressure cookers are not used.

Effect of Impurities on Boiling Point

The presence of impurities or of dissolved substance in a liquid raises the boiling point of the liquid but lowers its freezing point.

The effect of dissolved substance on the boiling point of a liquid is illustrated in the following activity.

Activity 9.4

1 Fill a flask half-full with a salt solution. Fit a thermometer, stirrer and a bent tube to the flask as shown in fig. 9.14.

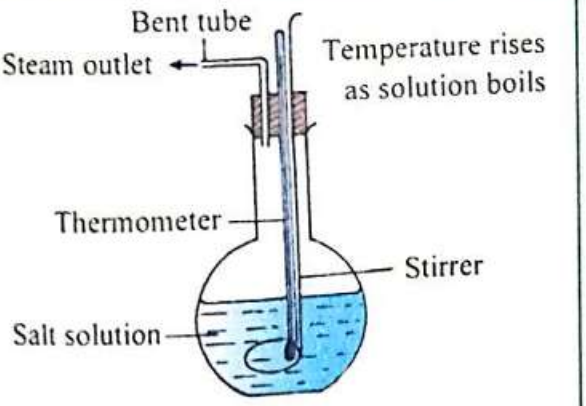


Fig. 9.14 Effect of dissolved substances on boiling point

2 Ensure the bulb of the thermometer is well inside the salt solution.

3 Heat the flask with a bunsen burner until the solution boils.

4 Read the temperature as shown by the thermometer while the solution boils. The boiling point will be seen to be above 100°C and to continue to rise as the solution is further heated.

Why is the boiling point not constant in the above experiment? As the boiling continues, more and more water evaporates leaving the solution more and more concentrated with salt. This increase in the concentration of the salt raises the boiling point of the solution higher and higher until a saturated solution is obtained.

Sublimation

Under certain conditions, some substances (e.g. iodine crystals, dry ice) can go from solid to vapour or gaseous state without passing through the liquid state. This process is called *sublimation*.

Molecular Explanation of 'Change of State'

A 'change of state' may involve the change of a substance from its solid to its liquid form or from liquid to gaseous form or vice versa.

According to the kinetic molecular theory, the molecules of a solid vibrate about a mean position. As the temperature of the solid increases the molecules acquire greater kinetic energy and are able to vibrate farther away from their mean position, thus weakening the intermolecular forces binding them together. At a certain maximum temperature the molecules have maximum kinetic energy. This maximum temperature is the *melting point* of the solid. Any further heat supplied to the substance does not increase the kinetic energy of the molecules. Instead the heat energy is used to break down the inter-molecular forces binding the molecules together in a solid. The molecules are now free and are able to slide over one another. The solid is said to *melt* at this maximum temperature or melting point.

As any further heating does not increase the kinetic energy of the molecules, the temperature of the substance remains constant. The heat required to break down the intermolecular forces of attraction in a solid is the *latent heat of fusion*.

The change from liquid to gas takes place in a similar fashion. As heat is added to the liquid, its molecules acquire higher kinetic energy and those at the surface are able to overcome the weak forces binding them to other molecules and escape in a process of evaporation. At a certain maximum temperature the average speed or kinetic energy of all the molecules is great enough for them all to break loose and to be practically independent of each other and exist as a gas. The *latent heat of vaporization* is the heat needed to overcome the forces of attraction between the molecules of a

Scanned with CamScanner

liquid. This energy is much greater than that needed to convert the solid to the liquid.

Saturated and Unsaturated Vapour

When a liquid evaporates in a closed container, the vapour formed above the liquid exerts a pressure. According to the kinetic molecular theory, the molecules of the vapour are in constant motion and will hence exert a pressure just like the molecules of a gas. This pressure is called the *vapour pressure* of the liquid. When the enclosed space above a liquid is saturated with vapour molecules and can hold no more molecules, the pressure exerted by this *saturated vapour* is said to be the *saturated vapour pressure* (s.v.p) of the liquid.

A saturated vapour is a vapour that is in contact with its own liquid within a confined space.

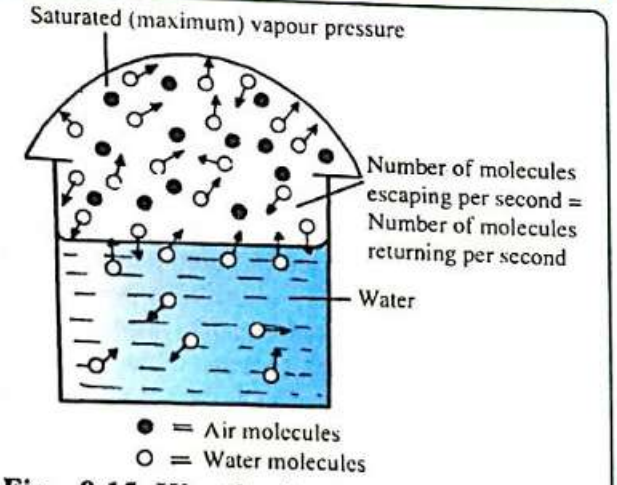


Fig. 9.15 Kinetic theory explanation of s.v.p.

The vapour is said to be saturated when the number of molecules escaping from the liquid per unit time is equal to the number returning to the liquid per unit time. The saturated vapour is thus said to be in a state of *dynamic equilibrium* with its own liquid.

Saturated vapour pressure (s.v.p) increases with temperature as we show by the following experiment.

Experiment to show the Variation of s.v.p. with Temperature

The apparatus for this experiment is as shown in fig 9.16. Two long barometer tubes are filled with mercury and inverted over a mercury trough. The mercury level falls in both leaving a vacuum above each tube. The tubes are surrounded with a water jacket. A few drops of water are introduced into one of the tubes by means of a bent pipette until the space above the mercury is saturated with water vapour, leaving some water on top of the mercury.

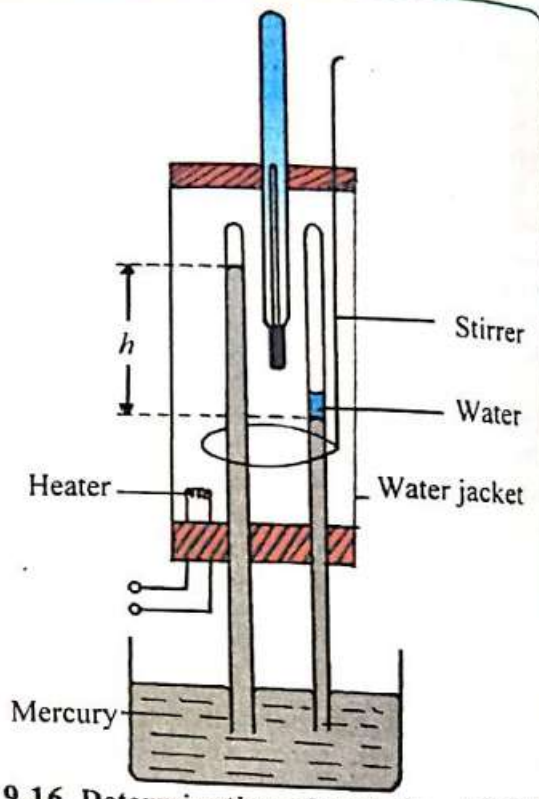


Fig. 9.16 Determination of variation of S.V.P. of water with temperature

The mercury level in this tube is found to be depressed more than that in the other tube. The temperature of the vapour is varied by heating the water in the water jacket by means of a heating coil. At each temperature reading, the saturated vapour pressure is the difference between the heights of the mercury columns in the two tubes. By recording this s.v.p. at each temperature, it will be seen that the s.v.p. increases with temperature as shown in fig 9.17.

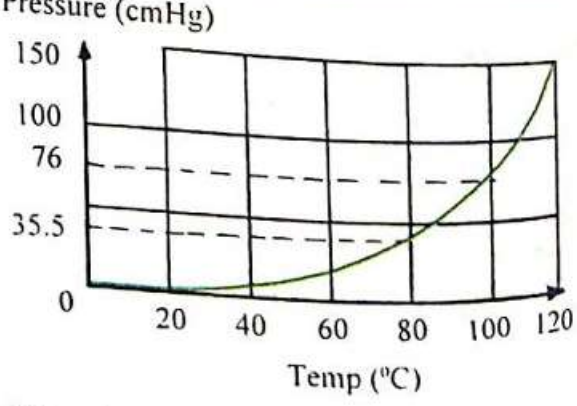


Fig. 9.17 Variation of S.V.P. of water with temperature

Scanned with CamScanner

At 100°C, the boiling point of water, the s.v.p. will be seen to be equal to the external atmospheric pressure. Above 100°C, there is a very rapid increase of s.v.p. with temperature.

Although the s.v.p. increases with temperature, a saturated vapour does not obey Boyle's Law or Charles Law (see next chapter), but an unsaturated vapour obeys both gas laws approximately. Therefore s.v.p at any temperature is independent of volume of the vapour.

Determination of Boiling Point of a Liquid

The fact that a liquid boils when its s.v.p. becomes equal to the external atmospheric pressure can be used to determine the boiling point of a liquid when only a small quantity of the liquid is available. The apparatus for this experiment is as shown in fig. 9.18.

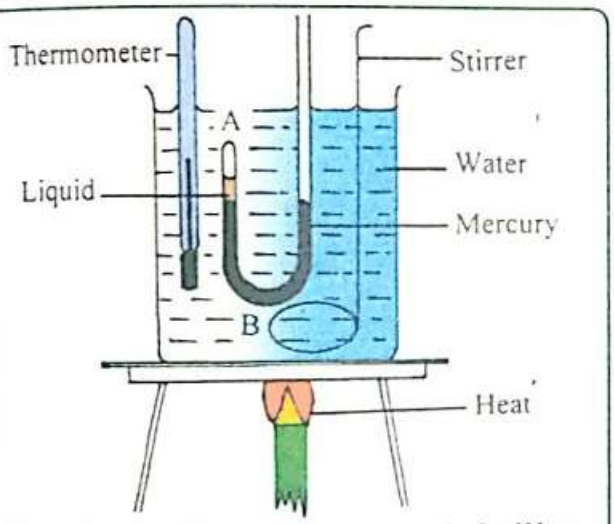


Fig. 9.18 Determination of boiling point of a liquid

It consists of a J-shaped tube with the end of the shorter arm sealed. The tube is filled with mercury such that the entire short arm from the closed end A and the bend at B are completely filled. A drop of the liquid is introduced into the space on top of the mercury in the shorter arm of the tube. The liquid evaporates and the pressure of the vapour depresses the mercury level slightly.

The tube is then placed in a beaker of water and heated. If all the liquid in the short arm of the J-tube evaporates, more drops of liquid are introduced. As the temperature of the water bath increases, the s.v.p. of the liquid increases. The temperature at which the mercury levels in both arms of the tube become equal is read from the thermometer. This temperature is the boiling point of the liquid because at this temperature the s.v.p. of the liquid is equal to the atmospheric pressure, since the longer arm of the tube is open to the atmosphere. At this temperature also the liquid in contact with the vapour is seen to be boiling.

Precautions

(i) The water in the beaker must be continuously stirred to ensure even temperature.

(ii) The thermometer readings of the boiling point are taken, one while the temperature is rising and the second, while the temperature is falling after the heating has been stopped. The mean temperature gives the more accurate boiling point of the liquid.

(iii) A pure sample of the liquid must be used.

With the above experiment we can define the boiling point of a liquid as follows:

The **boiling point** of a liquid is that temperature at which its s.v.p. is equal to the atmospheric pressure.

9.5 Relative Humidity and Dew Point

There is always some water vapour present in the atmosphere. This is due to evaporation from oceans, rivers, lakes etc. The water vapour content of the atmosphere is known as its *Humidity*. If there is a high percentage of water in the air, we say that the air is very *humid*. *Relative humidity* is a term to describe how humid or moist the air is.

Relative Humidity is the ratio of the mass of water vapour present in a certain volume of air to the mass of water vapour required to saturate the same volume of air at the same temperature.

Relative Humidity is usually expressed in percentage. A relative humidity of 100% means that the air is completely saturated with water vapour. A relative humidity of 50% means that the air is only half saturated. Relative Humidity values are used by meteorologists in weather forecasts.

Effects of humidity on personal comforts. Air Conditioning.

Very high humidities or very low ones make us feel uncomfortable. In a hot and humid weather, as we usually experience during the rainy season, the body sweats but cannot dispose of the moisture. The skin therefore feels damp and unpleasant. Some people may experience some difficulty in breathing. A good air conditioning system helps to maintain

Scanned with CamScanner

the correct humidity and temperature in a building. In a hot and humid room, the air conditioning system both cools and removes moisture from the room. During the Harmattan or dry season, the humidity is low and the air is dry. Our throat can become uncomfortably dry and rough. A good air-conditioning system warms and adds moisture to the air in a room through special humidifiers.

Air conditioning aims to maintain the room at about 25°C and the relative humidity at about 50% for the best bodily comfort.

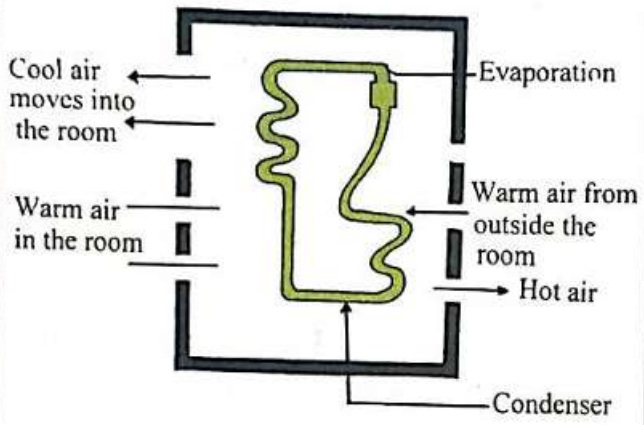


Fig. 9.19 An air-conditioning system

The air-conditioning system will not only cool or warm a room, but will also clean, filter and circulate the air in a room. As a result of such control of the temperature and moisture, the conditions in a building become more comfortable and healthy.

Dew Point

Water vapour is always present in the atmosphere. If a sample of air is cooled, a temperature will be reached when the air becomes saturated with the water vapour present. This temperature is known as the *Dew Point*, and any further cooling will cause the excess water vapour to condense into tiny visible droplets called *Dew*.

Dew Point is the temperature at which the water vapour present in the air is just sufficient to saturate it.

Another way of defining relative humidity is therefore related to dew point.

$$\text{Relative humidity} = \frac{\text{s.v.p. at dew point}}{\text{s.v.p. at air temperature}} \times 100\%$$

Dew point is dependent upon prevalent atmospheric conditions like temperature, wind and the amount of water vapour in the atmosphere.

When a bottle of ice-cold water is brought out from a refrigerator into a warm room, it quickly cools the air immediately around it to below the dew point. Hence moisture or dew forms on the outside of such a bottle due to the condensation of water vapour on the cold surface.

This kind of condensation process is responsible for dew, mist, fog, clouds, rain etc.

Mist and Fog

When moist air near the surface of the land cools until it reaches dew point, water vapour originally present in the air begins to condense around tiny suspended dust particles in the air. They form suspended water droplets in the atmosphere. A collection of these suspended water droplets constitute a mist. Mist reduces visibility. In severe mist situation, we have a *fog*. Under *foggy* situations motorists have to drive slowly and visibility can be so reduced that it becomes necessary to drive with the vehicle head-lamps switched on even in the day.

Clouds and Rain

A thick mist when formed high up in the atmosphere is known as a *cloud*. As warm moist air rises over the land or sea during the day, it becomes cooler as it moves higher and higher in the atmosphere. If this cooling process continues long enough the air may become saturated, causing the water vapour to condense as tiny water droplets. A combination of several such water droplets constitute a cloud. Instability in the condition of the cloud may cause rain to fall. Certain types of clouds known as *cirrus clouds* form at very great heights. They are composed of ice crystals due to the very low temperatures prevalent at such heights.

Measurement of Relative Humidity – the Hygrometer

An instrument known as the *Hygrometer* is used to measure the relative humidity of air. A simple and common form of the hygrometer is the *Wet-and-Dry-bulb hygrometer*. It consists of two thermometers, the wet bulb and the dry bulb thermometers, mounted side by side. The wet bulb thermometer has its bulb covered with a piece of moist cloth, the lower end of which dips into a small water pot containing some water.

Scanned with CamScanner

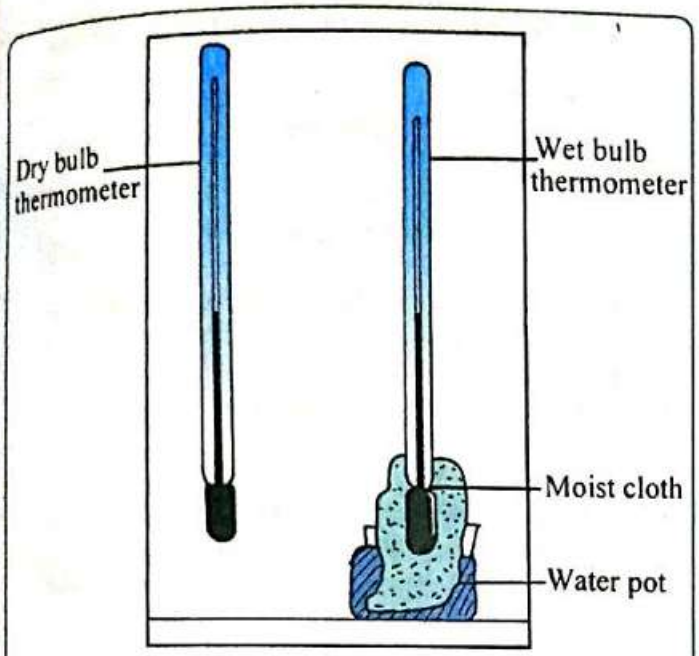


Fig. 9.20 Wet and dry bulb hygrometer

The dry bulb thermometer reads the temperature of the dry air. The wet bulb thermometer records a lower temperature because the evaporation of water from the moist cloth cools the bulb. In a dry atmosphere (low Relative Humidity), the water evaporates rapidly from the wet cloth. There is therefore a big temperature difference between the dry and the wet thermometer. This difference in the readings of the thermometers is a measure of the relative humidity.

If the air is humid (high relative humidity) water from the wet cloth evaporates only very slowly. The difference in the temperatures recorded by the two thermometers is therefore low.

The difference of temperatures between the wet and the dry bulb thermometers is greatest for very dry air and zero for saturated air.

Example 9.14

On a certain day, the average air temperature was 30°C and the dew point was 8°C. If the s.v.p. of water vapour was 15.6 mmHg at 30°C and 9.5 mmHg at 8°C, find the percentage relative humidity of the air.

Solution

Rel. humidity

$$= \frac{\text{s.v.p. at dew point}}{\text{s.v.p. at room temperature}} \times 100\%$$

$$= \frac{9.5}{15.6} \times 100\%$$

$$= \underline{60.9\%}$$

Summary

1 Thermal (or Heat) Capacity of a body is the quantity of heat required to raise its temperature by 1 K or 1°C.

2 Units of quantity of heat are Joules (J) or kilojoules (KJ) or MegaJoule (MJ). 1 KJ = 10^3J, 1 MJ = 10^6J.

3 Specific heat capacity is the quantity of heat required to raise the temperature of unit mass of a body by 1 K or 1°C. Unit is $Jkg^{-1}k^{-1}$ or $Jg^{-1}k^{-1}$.

4 Quantity of heat, $H = mc\theta$ where m is mass, c is specific heat capacity and θ is temperature change.

5 Thermal (or Heat) Capacity = mass × specific heat capacity.

6 We can find the specific heat capacities of solids and liquids by methods of mixtures or by the electrical method. (Heat gained = Heat lost).

7 Latent heat is the heat supplied or removed from a body, which causes a change of state without a change of temperature. It is a 'hidden' heat which cannot be detected by a thermometer.

8 Specific latent heat of fusion (l) of a substance is the quantity of heat required to convert unit mass of the substance from solid to liquid state without change in temperature. SI unit is Jkg^{-1}.

9 Specific latent heat of vaporization (L) is the quantity of heat required to convert unit mass of the substance from liquid at its boiling point to vapour at the same temperature. S.I. unit is Jkg^{-1}.

10 We can determine melting point of a substance by plotting its cooling curve. As the molten substance solidifies its temperature remains constant. This is because its latent heat is given out and balances the heat lost to the surroundings.

11 Increased pressure lowers the freezing point and raises the boiling point of water.

12 The presence of impurities in a liquid lowers the freezing point and raises the boiling point of the liquid.

13 Water expands on freezing, and ice contracts on melting.

14 The factors which affect evaporation are temperature, surface areas, dryness of surrounding air, presence of draught or wind, boiling point of liquid.

Scanned with CamScanner

15 Evaporation leads to cooling. The human body uses perspiration to control its temperature. Evaporation is a surface phenomenon but boiling is a bulk phenomenon.

16 A saturated vapour is one that is in contact with its own liquid. Saturated vapour pressure is the pressure exerted by a vapour that is in contact with its own liquid. When a vapour is saturated, the number of molecules escaping from the liquid per unit time is equal to the number returning to the liquid per unit time. s.v.p. depends on temperature but not on volume.

17 A liquid boils when its s.v.p. is equal to the external atmospheric pressure.

18 Pressure cookers use high pressures to increase the boiling point of water to about 120°C or more. Under such high temperature, food cooks faster and fuel is saved.

19 Dew Point is the temperature at which the water vapour present in the air is just sufficient to saturate it. Water vapour will condense on surfaces which have a temperature lower than that of dew point.

20 Relative humidity

$$= \frac{\text{mass of water vapour in a given volume of air}}{\text{mass of water vapour required to saturate the same volume of air at the same temperature}}$$

$$= \frac{\text{s.v.p. at dew point}}{\text{s.v.p. at room temperature}}$$

Exercise 4

1 The quantity of heat required to change the temperature of a unit mass of a substance by 1°C is called
- A Heat capacity of fusion
- B Specific heat capacity of vaporization
- C Specific latent heat
- D Specific heat capacity

(JAMB)

2 Why does melting ice cool an orange drink far better than the same mass of ice-cold water?
- A Because ice releases latent heat during melting.
- B Because the melting ice absorbs latent heat during melting.
- C Because the ice is at lower temperature than the ice-cold water.
- D Because melting ice has a lower specific heat than ice-cold water.
- E Because the drink mixes better with ice than with ice-cold water.

3 When drops of methylated spirit are applied to our body, it normally feels cool because the methylated spirit
- A easily evaporates
- B extracts its latent heat of evaporation from the body
- C gives away its latent heat of evaporation to the body
- D evaporates without boiling

4 The s.v.p. of a liquid depends on its
A Density B Temperature C Volume
D Pressure.

5 A liquid whose boiling point is 84°C is taken down to a deep well and its boiling point is measured. We expect the boiling point at the bottom of the well
- A Is still 84°C
- B Is less than 84°C
- C Is more than 84°C
- D Will depend on the temperature of the well.

6 A tap supplies water at 26°C while another supplies water at 82°C. If a man wishes to bath with water at 40°C the ratio of the mass of hot water to that of cold water is
A 1:3 B 3:1 C 3:7 D 7:3 E 15:8
(SSCE)

7 A heater marked 60W evaporates 6×10^{-3} kg of boiling water in 60 seconds. What is the specific latent heat of vaporization of water in Jkg^{-1}?
- A 6.0×10^{5}
- B 6.0×10^{6}
- C 3.0×10^{-5}
- D 6×10^{-5}
- E 3×10^{5}

8 The specific latent heat of fusion of a substance is the quantity of heat required to
- A heat unit mass of the substance through 1°C
- B change the state of unit mass of the substance at its melting point
- C melt all the substance present at the melting point
- D change the state of unit mass of substance at its boiling point.

9 When two objects P and Q are supplied with the same quantity of heat, the temperature change in P is observed to be twice that of Q. The mass of P is half that of Q. The ratio of the specific heat capacity of P to Q is
A 1:4 B 4:1 C 1:1 D 2:1
(JAMB)

Scanned with CamScanner

10 When an athlete perspires after running
A Evaporation occurs and helps to cool the body.
B Convection cools the body.
C The body absorbs cold from the surrounding air.
D Heat is conducted away from the body.

11 Define (i) *specific latent heat of fusion* (ii) *specific heat capacity* (iii) *thermal capacity.*
What mass of ice at −14°C will be needed to cool 200 cm^3 of an orange drink (essentially water) from 25°C to 10°C? (Specific latent heat of fusion of ice = 3.36×10^5 $Jkg^{-1}k^{-1}$, specific heat capacity of ice = 2100 $Jkg^{-1}k^{-1}$, specific heat capacity of water = 4200 $Jkg^{-1}k^{-1}$.

12 150 g of ice at 0°C is mixed with 300 g of water at 50°C. Calculate the temperature of the mixture.

13 Calculate the specific heat capacity of the materials for which the heat supply, mass, and temperature rise are given below:
(i) 2.52×10^4 J, 0.3 kg, 20 K
(ii) 8×10^3 J, 500 g, from 20°C to 60°C
(iii) 9.45×10^3 J, 150 g, from 15°C to 45°C

14 An electric heater raises the temperature of 120 g of water in a thin light vessel through 10 K in 2 min., when placed in 70 g of water contained in a metal vessel of mass 0.55 kg the temperature rises through 9 K in the same time. Calculate from the above
a. the heat supplied in 2 min by the heater
b. the power of the heater
c. the heat supplied to the 70 g of water
d. the heat supplied to the metal vessel
e. the heat capacity of the vessel
f. the specific heat capacity of its material (specific heat capacity of water = 4200 $J.kg^{-1}k^{-1}$)

15 With the aid of suitably labelled diagram describe the principle of operation of refrigerator.

16 Define the term *specific heat capacity.* How would you determine the specific heat capacity of a solid? What precautions would you take to obtain an accurate result.

17 a. Define the boiling point of a liquid.
b. Describe an experiment to determine the boiling point of a small quantity of a liquid.
c. A piece of copper of mass 300 g at a temperature of 950°C is quickly transferred to a vessel of negligible thermal capacity containing 250 g of water at 25°C. If the final steady temperature of the mixture is 100°C calculate the mass of the water that will boil away.
(Specific heat capacity of copper = 4.0×10^2 $Jkg^{-1}k^{-1}$. Specific heat capacity of water = 4.2×10^3 $Jkg^{-1}k^{-1}$. Specific Latent heat of vaporization of steam = 2.26×10^6 Jkg^{-1})
d. State four other effects of heat on a substance other than expansion.
(SSCE)

18 Explain the following
a. When methylated spirit is poured on the skin, it has cooling effect.
b. A piece of ice at 0°C cools a drink more effectively than the same mass of cold water at 0°C.
c. Ice floats on water.
d. Moisture is deposited on the outside of a glass bottle containing very cold water.
e. An earthen pot is better than a glass pot for storing cold drinking water in hot climates.
f. A thin wire with heavy weights attached to its ends can cut through an ice block without breaking it.
g. A pressure cooker boils food more quickly than an ordinary pot.

19. What are the differences between *evaporation* and *boiling*? What is the effect of
a. Pressure on the melting point of ice.
b. Impurities on the boiling point of water.

20 Define the *specific latent heat of vaporization of a substance.*
a. Calculate the total energy required to evaporate completely 1 kg of ice that is initially at −10°C.
(Specific heat capacity of ice = 2.2×10^3 $Jkg^{-1}k^{-1}$, Specific heat capacity of water = 4.18×10^3 $Jkg^{-1}k^{-1}$, Specific latent heat of fusion of ice = 3.34×10^5 Jkg^{-1} Specific latent heat of vaporization of water = 2.26×10^6 Jkg^{-1}.
b. Use the kinetic theory to explain cooling by evaporation.
(SC/GCE)

21 Distinguish between the *specific latent heat of fusion* and the *specific heat capacity* of a substance.

Scanned with CamScanner

A large beaker is filled with ice-cold water and is placed in a larger vessel containing a mixture of ice and water at 0°C. A lump of aluminium weighing 120 g, initially cooled to -70°C, is then suspended inside the beaker.

(i) Explain why ice is formed on the aluminium lumps and not on the walls of the beaker.

(ii) Calculate the maximum weight of ice which can form on the aluminium = 9.10×10^2 $Jkg^{-1}k^{-1}$, specific latent heat of fusion of ice = 3.34×104 Jkg^{-1})

(SC/GCE)

22 Explain why food can be cooked more rapidly by boiling it in salt water than by boiling it in pure water.

(SC/GCE)

23 What is meant by the statement that the *specific latent heat of steam* is 2.26×10^6 Jkg^{-1}

How would you determine the value of the specific latent heat of steam in the laboratory? What precautions are necessary to obtain an accurate result?

Dry steam at 100°C is bubbled into 500 g of water originally at 20°C. What will be the temperature of the water after 30 g of steam has condensed?

(Specific Latent heat of steam = 2.26×10^6 Jkg^{-1}, Specific heat capacity of water = 4200 $Jkg^{-1}k^{-1}$).

24 a. Explain the statement: *the saturation vapour pressure of water at 60°C is 149.4 mm of mercury.*

b. Describe an experiment to determine the saturation vapour pressure of water in the temperature range 25°C - 70°C. Sketch the curve you would expect to obtion.

c. Define *dew point.* Compare and contrast a mist and a cloud.

(SC/GCE)

25 a. Distinguish between saturated vapour and unsaturated vapour. Use the kinetic theory to explain saturation vapour pressure.

b. Describe an experiment to show that the saturation vapour pressure of water at its boiling point is equal to the air pressure on its surface.

(SC/GCE)

26 a. Explain what is meant by the following statement: *The specific latent heat of fusion of ice* 3.4×10^5 Jkg^{-1}.

b. Describe an experiment to determine the specific latent heat of fusion of ice. State two precautions necessary to obtain an accurate result.

c. Using the kinetic theory of matter, explain why ice can change to water at 0°C without any change in temperature.

(SSCE)

27 a. What is meant by the statement: The specific heat capacity of copper is 400 $Jkg^{-1}k^{-1}$?

b. (i) Describe an experiment to determine the specific heat capacity of copper using a copper ball.

(ii) State two precautions necessary to obtain an accurate result.

(iii) A piece of copper ball of mass 20 g at 200°C is placed in a copper calorimeter of mass 60 g containing 50 g of water at 30°C, ignoring heat losses, calculate the final steady temperature of the mixture. (Specific heat capacity of water = 4.2 $Jg^{-1}k^{-1}$)

(SSCE)

28 A room with floor measurements 7 m x 10 m contains air of mass 250 kg at a temperature of 34°C. The air is cooled until the temperature falls to 24°C. Calculate

(i) the height of the room

(ii) the quantity of energy extracted to cool the room

(iii) which is higher: the calculated value or the actual energy needed to cool the room? Give a reason for your answer.

(Specific heat capacity of air = 1010 $Jkg^{-1}k^{-1}$, density of air = 1.25 kgm^{-3})

Scanned with CamScanner

29 a. Define heat capacity and state its unit.

b. List two effects of heat on a substance.

c. Explain how a tightly fitted glass stopper could be removed from a reagent bottle.

d. A quantity of pepper soup of mass 800 g poured into a plastic container with a tight fitting lid has a temperature of 30°C. The container is then placed in a microwave oven, rated 1200W and operated for 3 mins.

(i) Calculate the final temperature attained by the soup. (Assuming no heat losses)

(ii) Explain why containers with tight fitting lids are not suitable for use in microwave cooking.

(iii) When the soup is brought out and allowed to cool, a dent is observed on the container. Explain. (Take specific heat capacity of the soup = $4000 Jkg^{-1}k^{-1}$) (SSCE)

30 Using an electric drill, it takes 150 s to make a hole in a piece of brass of mass 0.45 kg. During this process, the average power delivered to the drill from the electricity mains is 300 W. How much electric energy is used in drilling the hole? If 70% of the energy supplied to the drill appears as heat in the brass, what is the initial rate of rise of temperature of the metal?
(Specific heat capacity of brass is $390 Jkg^{-1}k^{-1}$)

Scanned with CamScanner

10 THE GAS LAWS

The three properties of a gas that are necessary to describe the state of the gas are the pressure, the volume and the temperature. For a given mass of a gas, the above quantities are found to be related through the *Gas Laws*. This chapter discusses the measurement of gas pressure, Barometers in practical use, Boyle's law and its application, Charles law and its application and the General gas law.

At the end of the chapter the student should be able to:

1 Explain, using the ideas of the kinetic theory of gases
   a. the variation of volume with temperature of a gas when the pressure is kept constant.
   b. the variation of pressure with volume of a gas when the temperature is kept constant.
2 Explain Boyle's and Charles' laws of gases.
3 Deduce the general gas law from a given mass of gas which obeys Charles' law.
4 Identify the limitations of the gas laws.
5 Solve simple problems involving the gas laws.
6 Identify and use instruments for measuring pressure.

10.1 Measurement of Gas Pressure

Gas pressure is the force per unit area exerted by a gas. The *Manometer* is an instrument for measuring gas pressure.

It consists of a U-tube containing water or mercury.

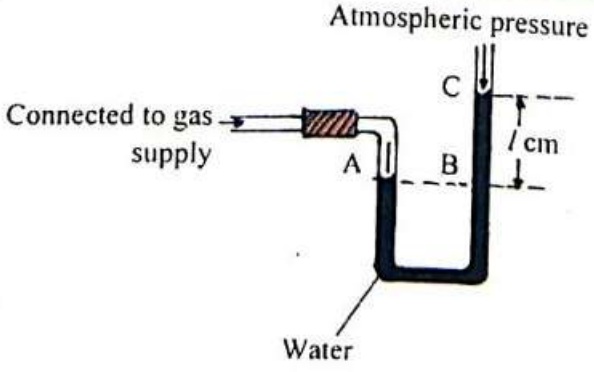


Fig. 10.1

One end of the U-tube is open to the atmosphere, the other end is joined to the gas supply. When the tap from the gas supply is opened, the gas exerts a pressure at A and forces the water in the open tube to rise to a certain height, h cm, above the point B. When the water levels are steady, the pressure P of the gas at A = the pressure in the liquid at B. Since the right side of the U-tube is open to the atmosphere, the pressure at B = Atmospheric pressure (H) plus the pressure due to a column of water of height (h)

$\therefore$ Pressure of gas at $A = H + h$

If the pressure due to h cm of water is equal to the pressure due to y cm of mercury, then using the relation $P = h\rho g$, we can write

$$h\rho_1 g = y\rho_2 g$$

where ρ_1 = density of water, ρ_2 = density of mercury.

Using $\rho_1 = 1$ g cm^{-3} and $\rho_2 = 13.6$ g cm^{-3}, we can write:

$$y = \frac{h\rho_1}{\rho_2} = \frac{h}{13.6}$$

Hence h cm of water exerts the same pressure as y ($= h/13.6$) cm of mercury.

Hence gas pressure (P) at A is given by

$$P = \left(H + \frac{h}{13.6}\right) \text{ cm of mercury}$$

$$P = 76 + \frac{h}{13.6} \text{ cm Hg}$$

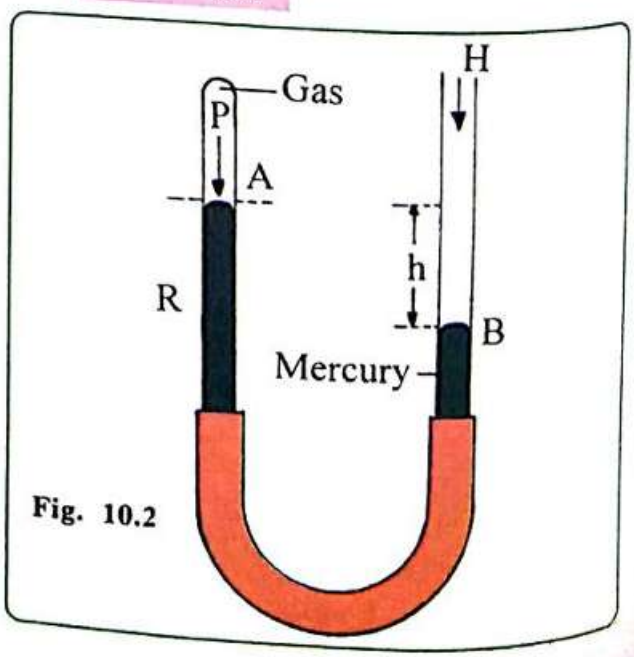


Fig. 10.2

Scanned with CamScanner

where the atmospheric pressure is taken to be equivalent to 76 cm of mercury.

Suppose some dry gas is present above the mercury in the enclosed space of a U-tube manometer, using mercury instead of water. The pressure at B of the open end of the manometer is atmospheric pressure, H. The pressure (P) at A is less than atmospheric pressure. If the difference in the heights of mercury in the two arms of the U-tube is h cm, then the pressure of gas is given by

$P = (H - h)$ cm of mercury.

The Barometer

The *Barometer* is an instrument for measuring atmospheric pressure. We can construct a Simple Barometer by the following activity:

Activity 10.1 Construction of a Simple Barometer

1 Take a clean, dry, thick walled glass tube about one metre long, and fill it completely with mercury.

2 Cover the top of the tube with a finger and invert it so that all the tiny air bubbles collect together.

3 Turn the tube round again and remove the finger. Fill up the tube again completely with mercury to exclude all the air.

4 Close the mouth of the tube with a finger and invert it into a dish of mercury.

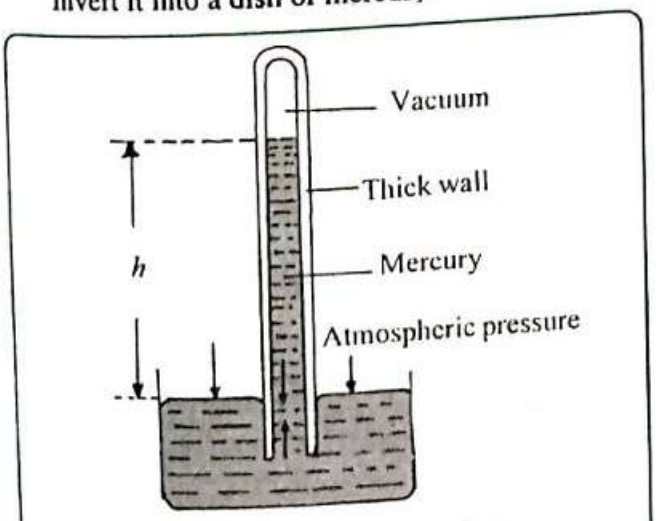


Fig. 10.3 Simple barometer

5 Remove the finger and observe that the level of the mercury immediately falls in the tube to a height, h, as shown in fig. 10.3. Measure this height above the mercury level in the open dish. Since air was completely excluded, the space above the mercury level in the tube is a vacuum, often called the 'Torricellian vacuum', in honour of Torricelli, who first made a simple barometer using mercury.

The height of the mercury level in the tube above the mercury level in the dish is found to be about 76 cm or 760 mm.

The simple barometer is not an accurate instrument as it can contain air or water vapour instead of vacuum. It is also bulky and not convenient for carrying about.

Faulty Barometer

A simple barometer constructed as above may contain some air or water vapour above the mercury column instead of a vacuum. Such a barometer is then said to be faulty. To test the vacuum, the tube is tilted; if the vertical height of the mercury column remains constant (at 76 cm) until the tube is completely filled, then the space is a perfect vacuum.

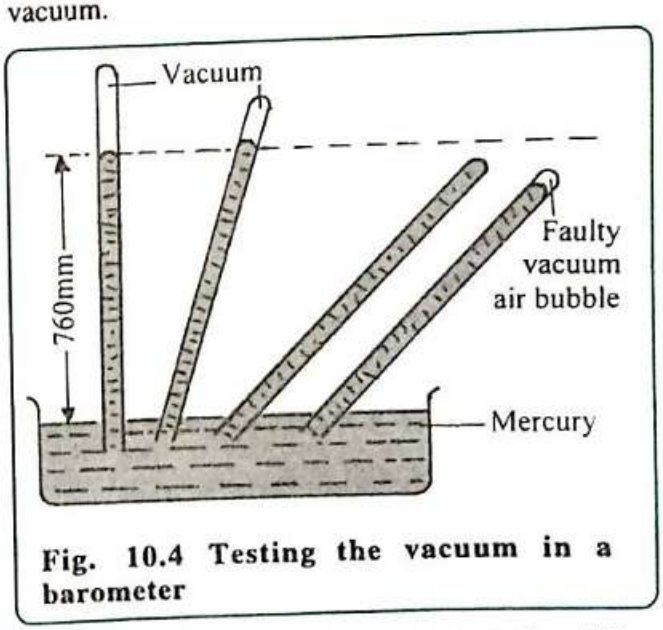


Fig. 10.4 Testing the vacuum in a barometer

If the space is not vacuum, an air bubble will be seen at the top of the tube when the tube is tilted to certain height below 76 cm.

Barometers in Practical Use

There are two types of barometers in practical use - the *Fortin barometer* and the *Aneroid barometer*.

The Fortin Barometer

This type of barometer is used for measuring atmospheric pressure in the laboratory. It gives a more accurate reading than the simple barometer.

The Fortin barometer is basically a simple barometer enclosed in a metal case with glass windows through which the mercury level can be seen at the top and the bottom of the metal case.

Scanned with CamScanner

Instead of the dish of the simple barometer, we have a leather bag to contain the mercury. Before taking the barometer reading, a screw at the lower end of the bag is used to lower or raise the level of mercury in this bag until it touches the tip of an ivory pointer. The height of the mercury level is then measured from the tip of this pointer to the level of mercury in the barometer tube.

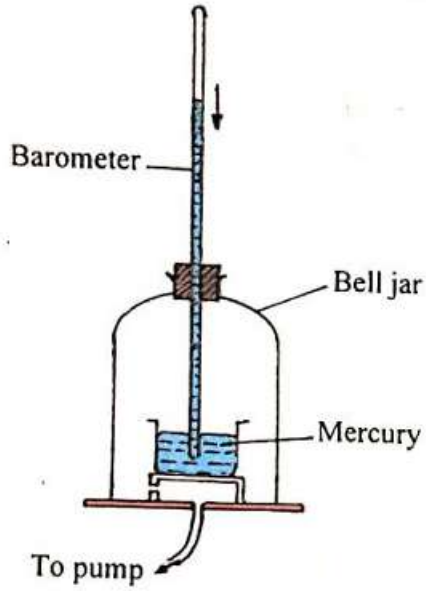


Fig. 10.5 Height of mercury column depends on air pressure

The graduation of the instrument has a main scale and a vernier scale which moves over the main scale.

The Aneroid Barometer

This type of barometer contains no liquid and is therefore more convenient to carry about.

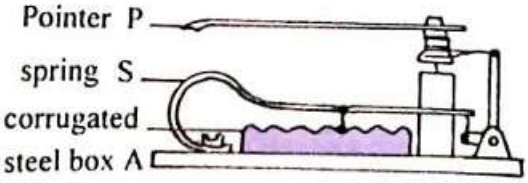


Fig. 10.6 Aneroid barometer

Its basic structure consists of a corrugated steel box which is partially evacuated of air. The top surface of the box is held by a strong spring, S, which prevents it from collapsing. When the air pressure rises, the top of the box is pressed inwards; when the air pressure falls, the top of the box moves upwards. This up and down movement of the box is magnified by a system of levers connected to a small rod fixed on the top of the box. As a result, a pointer moves over a suitably calibrated scale.

Atmospheric pressure decreases with height. So the aneroid barometer can be used as an *altimeter* to measure the height of a place. Such an instrument is used in aeroplanes by pilots to know the height of the aeroplanes above sea-level. Though the aneroid barometer is not very accurate, it is often used because it is small and convenient to carry about.

Magnitude of atmospheric pressure

The pressure of the atmosphere at sea level corresponds to a barometric height of 76 cm of mercury (or 76 cm Hg.) Hence 76 cm Hg is taken as the standard (or normal) pressure. Since we have defined pressure as force per unit area, let us derive the equivalent of 76 cm Hg in Newtons per square metre, the S.I. unit.

We recall that $P = h\rho g$

h = 76 cm Hg = 0.76 m Hg

density of mercury, $\rho = 13600\ \text{kgm}^{-3}$

$g = 9.8\ \text{ms}^{-2}$

Thus

$$P = 0.76 \times 13600 \times 9.8\ \text{Nm}^{-2}$$
$$= 101293\ \text{Nm}^{-2}$$
$$= 1.013 \times 10^5\ \text{Nm}^{-2}$$

Hence standard atmospheric pressure (or 1 Atmosphere, Atm) equals $1.013 \times 10^5\ \text{Nm}^{-2}$. Other units of pressure are the *Pascal* (Pa), and the *bar* (b).

1 Pascal (Pa) = 1 Nm^{-2}

1 bar = $10^5\ \text{Nm}^{-2}$ or 10^5 Pascals

A small unit of pressure is the *millibar*.

$$1\ \text{millibar (mb)} = \frac{1}{1000}\ \text{bar} = \frac{10^5}{1000}\ \text{Nm}^{-2}$$
$$= 100\ \text{Nm}^{-2}$$

Example 10.1

What is the height of water that will support 1 atmosphere of pressure?

Solution

Use $P = h\rho g$

density of water = $\rho = 1000\ \text{kg m}^{-3}$

$g = 9.8\ \text{ms}^{-2}$

Scanned with CamScanner

1 (Atm) = 76 cm (or 0.76 m) of mercury,
density of mercury = 13600 kg m^{-3}
Thus
1 Atm = $0.76 \times 13600 \times 9.8 = h \times 1000 \times 9.8$

$$\therefore \quad h = \frac{0.76 \times 13600 \times 9.8}{1000 \times 9.8}$$

$$= \underline{10.34 \text{ m}}$$

Hence one atmosphere of pressure supports a height of 10.34 m of water or a height of 0.76 m of mercury.

10.2 Boyle's Law and Its Application

The relationship between the pressure and the volume of a fixed mass of a gas is given by Boyle's law.

Boyle's law states that the pressure of a fixed mass of a gas is inversely proportional to its volume, provided that the temperature is kept constant.

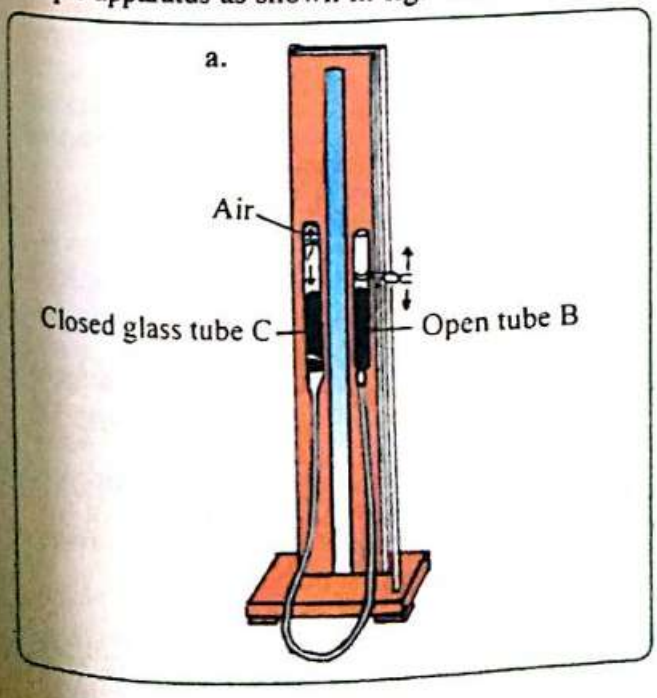
In symbols, Boyle's law is written as

$$P \propto \frac{1}{V} \text{ or } PV = \text{constant}$$

where P is the pressure and V is the volume of a fixed mass of gas.

Experimental Verification of Boyle's Law

We can verify Boyle's law in the laboratory using a simple apparatus as shown in fig. 10.7.



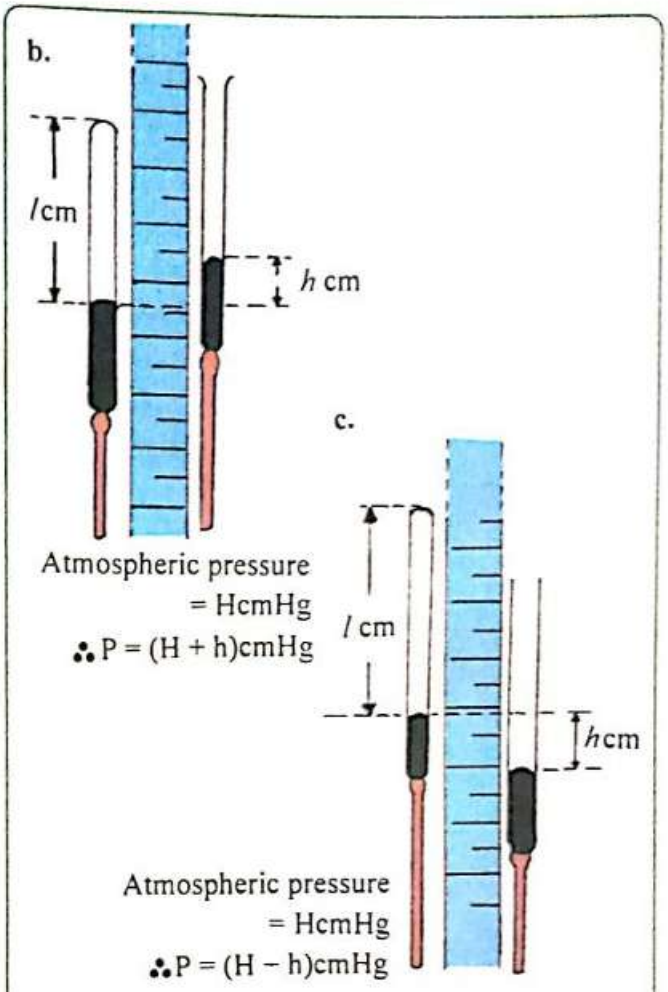


Fig. 10.7 Verification of Boyle's law

Some dry air is enclosed in a glass tube of uniform cross-section attached by rubber tubing to a mercury reservoir. By raising or lowering the open tube B, the length of the air column in C can be varied. This length which can be read against the metre scale is taken as a measure of the volume of the gas, since the tube has a uniform cross-section. The difference (h) in the levels of mercury at C and B is read for each value of the length of gas. The atmospheric pressure (H) is read from a Fortin barometer. When the mercury surface in B is lower than that in C, the pressure of gas is given by

$$P = (H - h) \text{ cm Hg}$$

When the mercury surface in B is higher than that in C, the gas pressure P is given by:

$$P = (H + h) \text{ cm Hg}$$

By lowering or raising the tube B, we obtain a set of readings of l and h and the result is recorded as shown in Table 10.1

Scanned with CamScanner

Table 10.1

Atmospheric Pressure H = ____ mm Hg

h (mm Hg)	P = (H ±h) mm Hg	l (∝ V) l in mm	$\frac{1}{l} \propto \frac{1}{V}$

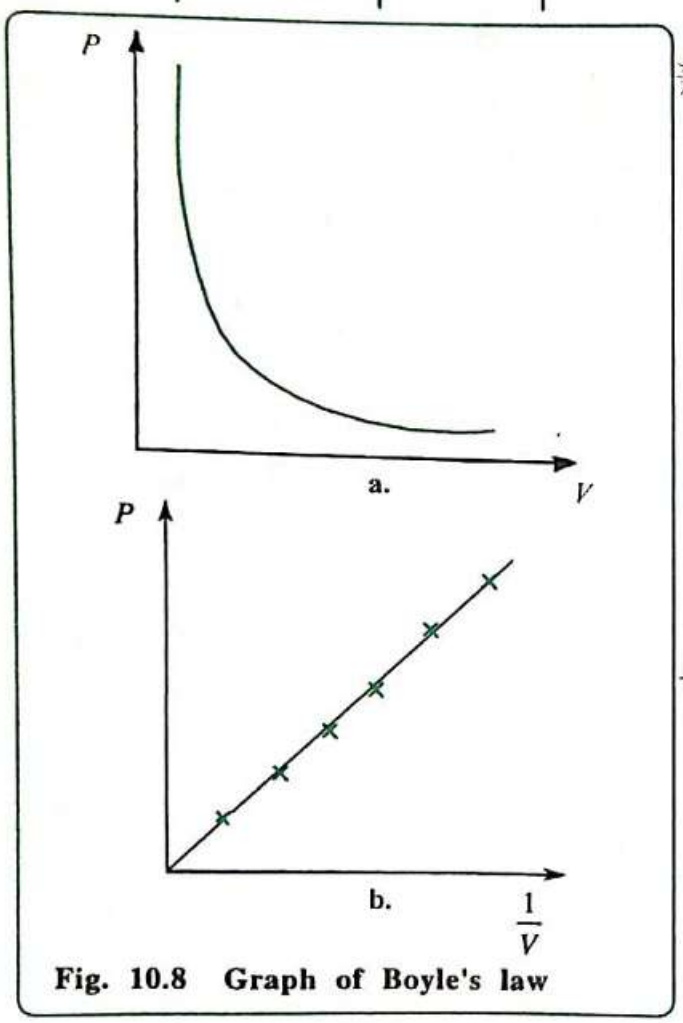


Fig. 10.8 Graph of Boyle's law

A graph of P against V (or l) is then plotted, and we obtain a curve as shown in fig. 10.8a. If P is plotted against $\frac{1}{V}$ (or $\frac{1}{l}$) a straight line graph passing through the origin is obtained (fig. 10.8b). This shows that the pressure and volume of air or any gas, are inversely proportional, thus verifying Boyle's law.

Precautions

1 There should be no air bubbles in the mercury.
2 Only dry air should be used in the experiment.
3 Readings should be taken when the mercury levels are steady. Top of mercury meniscus is to be read.
4 Time must be allowed for the air to assume room temperature, the constant temperature of the experiment.

Example 10.2

The pressure of a fixed mass of gas is 960 mm Hg when the volume of the gas is 3.0 m^3. Find the pressure of the gas when its volume is 4.5 m^3. Assume that the temperature is constant.

Solution

By Boyle's law PV = constant at constant temperature

$\therefore \quad P_1V_1 = P_2V_2$

$960 \times 3 = P_2 \times 4.5$

$$P_2 = \frac{960 \times 3}{4.5}$$

$= \underline{640 \text{ mm Hg}}$

Example 10.3

An air bubble of radius 3 cm rises from a depth of 7.0 m in a swimming pool. What will be the radius of the bubble just before it reaches the water surface? Assume that the water temperature is uniform. Take approximate height of water barometer as 10.34 m.

Solution

At the depth of 7.0 m of water, total pressure P_1 = 10.34 + 7.0 = 17.34 m of water

V_1 = Volume of air bubble $= \frac{4}{3}\pi \times r^3$

$$= \frac{4}{3}\pi \times 27 \text{ cm}^3$$

At water surface, pressure = 10.34 m of water = P_2

Let radius of bubble be r_1 (cm)

Then from Boyle's law

$P_1V_1 = P_2V_2$

$$17.34 \times \frac{4}{3} \times \pi \times 27 = 10.34 \times \frac{4}{3} \times \pi \times r_1^3$$

$r_1^3 = 45.28$

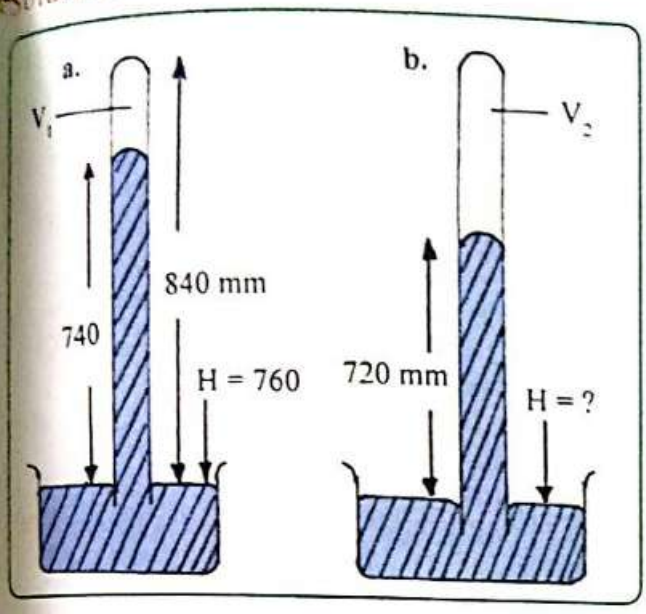
$r_1 = \underline{3.56 \text{ cm}}$

Example 10.4

A mercury barometer, with a tube of uniform diameter and length 840 mm above the mercury level in the reservoir, reads 740 mm on a day when the atmospheric pressure is 760 mm of mercury. Estimate the pressure of the atmosphere on a day when it reads 720 mm. Assume the average temperature is the same on both days.

Scanned with CamScanner

Solution



Since tube is of uniform diameter, $l \propto V$
Vol. of gas in (a) = 840 – 740 = 100 $(mm)^3 = V_1$
Pressure of gas in (a) = 760 – 740 = 20 mm Hg
Vol. of gas in (b) = 840 – 720 = 120 $(mm)^3$
Let atmospheric pressure be H mm Hg on the second day
∴ Pressure of gas = $(H - 720)$ mm Hg
Using Boyle's law, $P_1V_1 = P_2V_2$

$$20 \times 100 = (H - 720) \times 120$$

$$(H - 720) = \frac{20 \times 100}{120} = 16.7$$

$$H = 720 + 16.7 = \underline{736.70 \text{ mm Hg}}$$

Example 10.5

A tube of uniform cross-section closed at one end, contains some dry air which is sealed by a thread of mercury 15.5 cm long. When the tube is held vertically with the closed end at the bottom the air column is 22.0 cm long; but when it is held horizontally, the air column is 26.4 cm long. Calculate the atmospheric pressure.

Solution

Let Atmospheric Pressure = H (cm Hg)
Total pressure on gas in (a) = $(H + 15.5)$ cm Hg
Vol. of gas = $l \times A = 22$ cm^3
where A is cross-sectional area of tube
In (b) total pressure is only atmospheric pressure = H (cm Hg)
The column of mercury in the horizontal position has negligible pressure on gas.
Vol. of gas = $26.4 \times A$

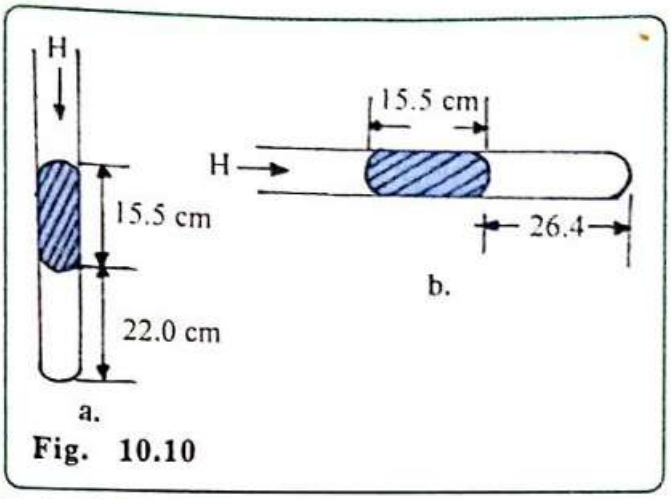


Fig. 10.10

Using $P_1V_1 = P_2V_2$

$$(H + 15.5) \times 22A = H \times 26.4A$$

$$H + 15.5 = H \times \frac{26.4}{22.0} = 1.2H$$

$$15.5 = 1.2H - H = 0.2H$$

$$H = \frac{15.5}{0.2}$$

$$= \underline{77.5 \text{ cm Hg}}$$

10.3 Charles' Law and Its Applications

Charles law gives the relationship of the volume of a gas to its temperature at constant pressure.

Charles' law states that the volume of a fixed mass of gas increases by 1/273 of its volume at 0°C per degree Celsius rise in temperature, provided its pressure remains constant.

Experimental Verification of Charles' Law

We can verify Charles' law in the laboratory using the apparatus shown in fig. 10.11.

A column of dry air is trapped by a pellet of mercury in a capillary tube of uniform diameter, which has been previously cleaned and dried. The tube is then sealed at one end. It is tied, together with a thermometer, to a half-metre rule by means of two rubber bands, and placed inside a beaker of water with enough water to cover most of the capillary tube.

Some ice shavings are added to the water in the beaker until the temperature of the mixture gets down to about 0°C. The exact temperature is read with the thermometer after the water has been continually stirred to ensure an even temperature.

Scanned with CamScanner

The length of the air column is now read over the metre scale and recorded against the temperature.

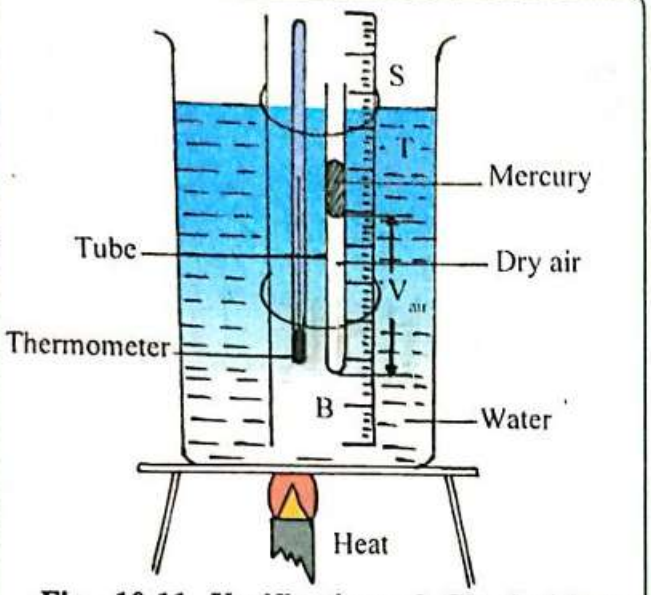


Fig. 10.11 Verification of Charles' law

The water is now heated gently and lengths of the air columns at different temperatures are read as the temperature of the water rises. Before each reading of temperature, the water should be stirred to obtain an even distribution. The results are recorded as shown in table 10.2.

Table 10.2

t° C	l ($\propto V$)

Because the tube is of uniform cross-section, we can take the volume of the gas as proportional to its length.

A graph of volume (or length) of gas is then plotted against the corresponding temperature. The graph obtained (fig 10.12) is a straight line whose slope is approximately $\frac{1}{273}$. Since the volume of gas was under a constant pressure (Atmospheric pressure plus that due to the mercury column), we conclude that Charles' law has been verified.

Precautions

1 Temperatures are taken when the thermometer reading is steady.
2 Water must be continuously stirred to ensure even temperature.

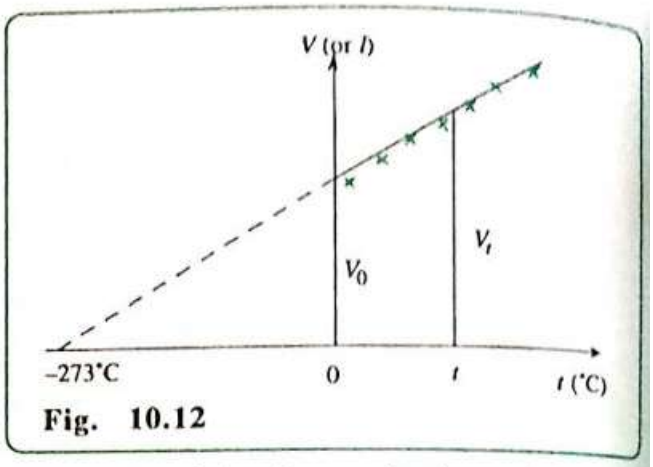


Fig. 10.12

3 The bore of the tube must be clean and dry.
4 Only dry air or gas should be used.
5 It is essential to carry out the experiment slowly, so that heat has time to pass through the thick walls of the capillary tube and bring the gas to the same temperature as the water in the beaker.

Cubic Expansivity of a Gas

We can use the above experiment to obtain the value of the cubic expansivity of a gas at constant pressure.

Cubic expansivity of a gas (γ) is defined as increase in volume per unit volume at 0°C per degree celsius rise in temperature.

$$\gamma = \frac{\text{Increase in volume from 0°C}}{\text{Volume at 0°C} \times \text{temperature rise}}$$

Thus if V_0 = Volume at 0°C, V_t = Volume at t°C, then

$$\gamma = \frac{V_t - V_0}{V_0 \times t} \qquad 10.1$$

Careful experiments show that the fraction γ in equation 10.1 is the same for all gases and is equal to $\frac{1}{273}$. Thus the cubic expansivity of a gas, unlike that of solids and liquids, is the same irrespective of the nature of the gas.

Another important point to note is that the cubic expansivity of a gas is calculated from the fractional increase of its volume at 0°C.

Scanned with CamScanner

Example 10.6

The volumes of a fixed mass of gas at constant pressure are 200 cm^3 at 0°C and 295 cm^3 at 130°C. Calculate the cubic expansivity of the gas.

Solution

$$\gamma = \frac{V_t - V_0}{V_0 \times t}$$

$$= \frac{295 - 200}{200 \times 130}$$

$$= \underline{0.00365/°C \text{ or } 0.00365/K}$$

Thermodynamic (or Absolute) Temperature

If the graph of fig. 10.12 is produced backwards it will cut the temperature axis at a temperature of -273°C. This implies that the volume of a gas is theoretically zero at -273°C, if the gas has not liquefied before that temperature. Because of this, the temperature -273°C is called the *absolute zero* of temperature. It is considered to be the lowest possible temperature of a gas. A temperature scale in which the value of the degree is the same as the Celsius degree but whose zero point is the absolute zero of temperature or -273°C is known as the *thermodynamic* or *Absolute* or *Kelvin* (K) *temperature scale*. A temperature t°C on the Celsius scale is related to an absolute or Kelvin temperature T by the equation

$$T\,(K) = 273 + t°C \qquad 10.2$$

Thus to convert the Celsius temperature t° to Kelvin scale, we add 273 to the Celsius temperature.

Using this scale, Charles' law can be stated as follows

The volume of a fixed mass of gas is directly proportional to its absolute temperature if its pressure remains constant.

In symbols, Charles' law states that

$$\frac{V}{T} = \text{Constant, if the pressure is kept constant or}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

where V_1 = volume of gas at T_1

V_2 = volume of gas at T_2

This new statement of Charles' law can be derived from the graph of volume of gas against temperature at constant pressure (fig. 10.13)

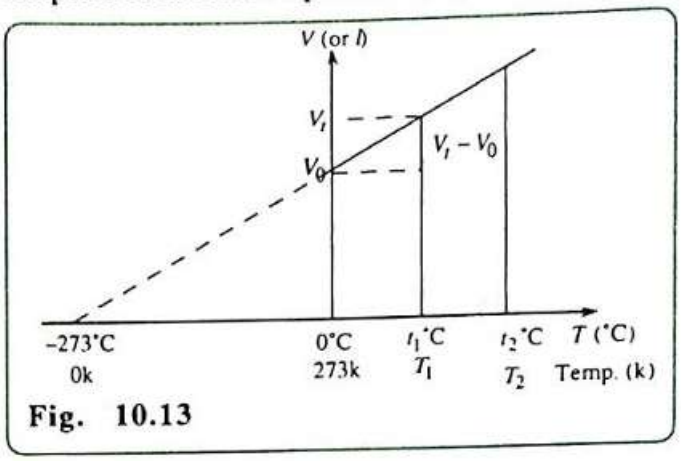


Fig. 10.13

Cubic expansivity of gas at constant pressure

$$\gamma = \frac{V_t - V_0}{V_0 t}$$

$$\frac{1}{273} = \frac{V_t - V_0}{V_0 t}$$

or $$V_t = V_0\left(1 + \frac{t}{273}\right) = V_0\left(\frac{273 + t}{273}\right)$$

$$= V_0\left(\frac{T}{273}\right) \qquad 10.4$$

Since $\frac{V_0}{273}$ is a constant, we can state that $V \propto T$ or *the volume of a given mass of gas at a constant pressure is proportional to its absolute or Kelvin temperature T*. Thus if in our experiment to verify Charles' law, we plot the volume of the gas against its absolute temperature we shall obtain a straight line graph passing through the origin (fig. 10.14).

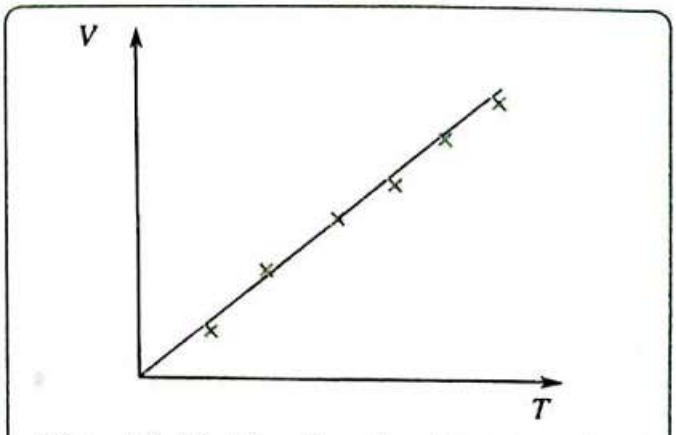


Fig. 10.14 Graph of volume against absolute temperature

Scanned with CamScanner

Example 10.7

A fixed mass of gas is heated at constant pressure from 10°C to 90°C. If its volume at 10°C is 200 cm^3, calculate the volume at 90°C.

Solution

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \text{(from Charles' law)}$$

$10°C = (273 + 10)\ K = 283\ K$

$90°C = (273 + 90)\ K = 363\ K$

$$\frac{200}{283} = \frac{V_2}{363}$$

$$V_2 = \frac{363 \times 200\ cm^3}{283}$$

$$= \underline{256.54\ cm^3}$$

10.4 The Pressure Law or Gay-Lussac's Law

The manner in which the pressure of a gas changes with its temperature, at constant volume is given by the *Gay-Lussac's law*. There are two alternative ways of stating this law:

1 The Gay-Lussac's law states that the pressure of a fixed mass of gas at constant volume increases by $\frac{1}{273}$ of its pressure at 0°C for every degree Celsius (or Kelvin) rise in temperature.

In symbols, we write

$$P_t = P_0\left(1 + \frac{t}{273}\right) \qquad 10.5$$

where P_0 = Pressure at 0°C and P_t = Pressure at temperature t°C.

2 The Gay-Lussac's law states that the pressure of a fixed mass of gas at constant volume is proportional to the absolute temperature of the gas.

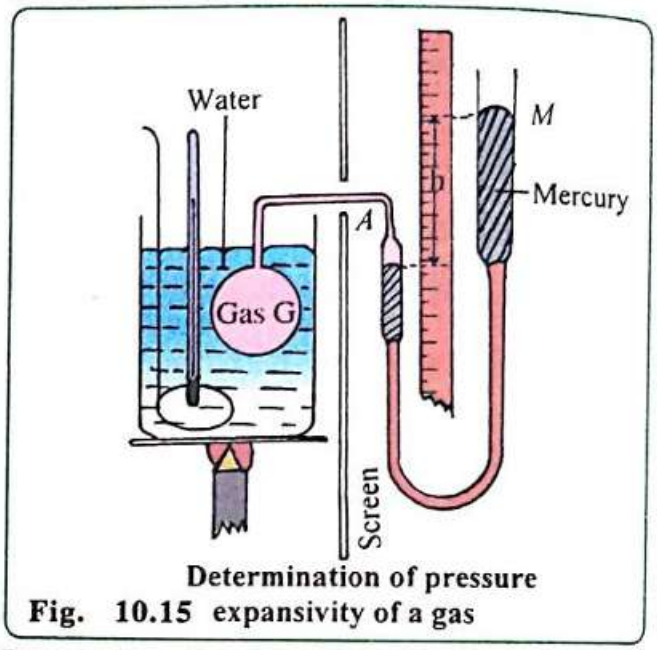
In symbols the law states that

$P \propto T$ or

$$\frac{P}{T} = \text{Constant or } \frac{P_1}{T_1} = \frac{P_2}{T_2} \qquad 10.6$$

Experiment to measure the Expansivity of a Gas at Constant Pressure

The experiment is performed using the simple constant-volume gas thermometer, as shown in fig. 10.15.



Determination of pressure

Fig. 10.15 expansivity of a gas

Dry air is enclosed by the glass bulb which is completely surrounded by water inside a large beaker. The water in the beaker is first cooled to about 0°C by the addition of some ice shavings. The manometer is adjusted until the mercury surface coincides with the fixed mark *A*. This ensures that the volume of the gas trapped in the glass bulb remains constant. The pressure of the gas at 0°C is obtained by reading the difference h between the mercury levels *M* and *A*. The temperature of the water is raised by heating with a bunsen burner. At each rise in temperature, at intervals of 20°C, and up to about 100°C, the readings of the difference in height of mercury levels in *M* and *A* are recorded as shown in table 10.3.

Table 10.3

t°	h (cm Hg)	$P = (H \pm h)$ cm Hg

When *A* is below *M*, the pressure of the gas is $H + h$, where H is the atmospheric pressure; when A is above *M*, the gas pressure is $H - h$. Before each reading of pressure is taken, care must be taken to ensure that the mercury level is maintained at the

Scanned with CamScanner

fixed mark A. The water should also be constantly stirred to maintain an even temperature.

The graph of pressure (P) against temperature, t, is then plotted (fig. 10.16).

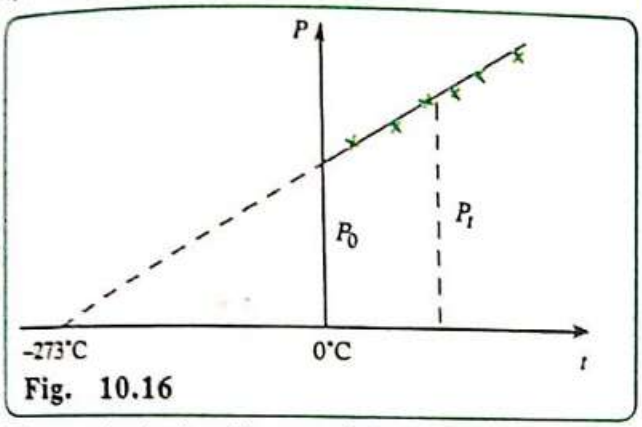


Fig. 10.16

The graph obtained is a straight line showing that at constant volume, the pressure of a gas varies linearly with the temperature.

If the graph is extrapolated to meet the temperature axis, it will be found to intersect it at -273°C.

The result of this experiment if accurately performed shows that for all permanent gases and dry air, the pressure increases by about $\frac{1}{273}$ of its pressure at 0°C for each degree Celsius rise in temperature. Thus the pressure expansivity of such gases is the same as the volume expansivity. So we can similarly define the pressure expansivity of the gas as

$$\gamma = \frac{\text{Increase in pressure from 0°C}}{\text{Pressure at 0°C} \times \text{temperature rise}}$$

In symbols, we can write:

$$\gamma = \frac{P_t - P_0}{P_0 \times t}$$

or $$P_t = P_0\left(1 + \frac{t}{273}\right) \qquad 10.7$$

(since $\gamma = \frac{1}{273}$).

From equation 10.6, we have that

$$P_t = \frac{P_0}{273}(273 + t) = \frac{P_0 T}{273}$$

Since P_0 and $\frac{1}{273}$ are constants, we can write that

$$P \propto T \text{ or } \frac{P}{T} = \text{constant.}$$

For a gas with pressure P_1 at T_1 and P_2 at T_2, at constant volume

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \qquad 10.8$$

Example 10.8

The pressure of a gas at constant volume is 90 cm Hg at 20°C. Calculate its pressure at 70°C.

Solution

$\frac{P_1}{T_1} = \frac{P_2}{T_2}$ (By pressure law)

$T_1 = 20 + 273 = 293\,K$

$T_2 = 70 + 273 = 343\,K$

$\frac{90}{293} = \frac{P_2}{343}$

$P_2 = \frac{90 \times 343}{293}$

$= \underline{105.3 \text{ cm Hg}}$

Example 10.9

Air pressure in an automobile tyre

An automobile tyre is filled to a gauge pressure of 220kpa early in the morning when the temperature is 30°c.After the car is driven all day over hot roads the tyre temperature is 85°c. Estimate the new gauge pressure. (Remember, gauge pressure is the pressure in excess of atmospheric pressure; Assume atmospheric pressure is 101kpa).

Solution

We assume that the volume of the tyre is approximately constant.

$\frac{P_1}{T_1} = \frac{P_2}{T_2}$ (By pressure law)

$$P_2 = \frac{P_1 T_2}{T_1} = \frac{(220+101) \times (273+85)}{(273+30)}$$

$$\frac{321 \times 358}{303} = 379.27 \text{ kpa}$$

But this is absolute pressure , so gauge pressure + (379.27 - 101) kpa =278.27 kpa

10.5 The General Gas Law

We can deduce the general gas law from Boyle's law, Charles' law and Pressure law. From Boyle's law we have that

PV = Constant if T is constant

From Charles' law, we have that

$\frac{V}{T}$ = Constant if P is constant

From the Gay-Lussac's or Pressure law we have that

$\frac{P}{T}$ = Constant at constant volume

Any two of the three equations can be combined into one equation to obtain:

$\frac{PV}{T}$ = Constant

or $$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \qquad 10.9$$

Scanned with CamScanner

where P_1, V_1 are the gas pressure and volume at temperature T_1 and P_2, V_2 are the gas pressure and volume at temperature T_2. Equation 10.9 is known as the *Ideal Gas Equation* or the *General Gas Equation.*

The above equation can be used to find the volume of a gas when both its pressure and temperature change. The law does not hold good when a gas is near its liquefying temperature. It also assumes that the attraction between the gas molecules and the volume they occupy are both negligible. This is not true for real gases.

Example 10.10

A mass of gas at 7°C and 70 cm of mercury has a volume of 1000 cm^3. Determine its volume at 27°C and pressure of 85 cm of mercury.

Solution

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$T_1 = 273 + 7 = 280\,K$; $T_2 = 273 + 27 = 300\,K$

$$\frac{70 \times 1000}{280} = \frac{85V_2}{300}$$

$$V_2 = \frac{70 \times 1000 \times 300}{280 \times 85}$$

$$= \underline{882.35\ cm^3}$$

Example 10.11

A vessel is filled with a gas at a temperature 30°C and a pressure of 76 cm Hg. Calculate the final pressure if the volume of the gas is doubled while it is heated to 80°C.

Solution

$P_1 = 76$ cm Hg; $V_1 = V$; $T_1 = 273 + 30 = 303$ K

$P_2 = ?\ V_2 = 2V$; $T_2 = 273 + 80 = 353$ K

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{76 \times V}{303} = \frac{P_2 \times 2V}{353}$$

$$P_2 = \frac{353 \times 76}{303 \times 2}\ \text{cm Hg}$$

$$= \underline{44.27\ \text{cm Hg}}$$

Standard Temperature and Pressure

The standard temperature and pressure (s.t.p.) are a temperature of 0°C (or 273 K) and a pressure of 76 cm Hg. By expressing the volume of any gas at s.t.p., it is possible to compare volumes of different masses of a gas.

It is easy to convert the volume of a gas at any temperature and pressure to its volume at s.t.p. and vice versa.

Example 10.12

1200 cm^3 of a gas is collected at 37°C and 600 mm Hg. Convert this volume to s.t.p.

Solution

$P_1 = 600$ mm Hg; $T_1 = 273 + 37 = 310$ K; $V_1 = 1200\ cm^3$

$P_2 = 760$ mm Hg; $T_2 = 273$, $V_2 = ?$

Using the general gas equation we have:

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{600 \times 1200}{310} = \frac{760 \times V_2}{273}$$

$$V_2 = \frac{600 \times 1200 \times 273}{760 \times 310}$$

$$= \underline{834.29\ cm^3}$$

10.6 Kinetic Molecular Theory of Gas and its Explanation of the Gas Laws

We can explain the effect of temperature on the volume and pressure of a gas using the kinetic molecular theory of gases. This theory makes the following assumptions:

1. A gas is considered to be made up of a large number of molecules acting like elastic spheres.
2. The molecules are in a state of constant motion knocking against each other and the walls of the containing vessel.
3. The particles exert attractive forces on one another.
4. The particles possess kinetic energy due to their motion.
5. The volume of the molecules of a gas is negligible compared with the volume of the container.

The molecules hit the walls of the container and rebound, thus changing the momentum of each molecule, in the process. According to Newton's 2nd law of motion, the rate of change of the momentum of the molecules is proportional to the forces they exert on the walls of the container. Since pressure is the force per unit area normal to a surface, the gas pressure due to all the molecules bombarding the walls is proportional to their average total momentum change per second.

Boyle's Law

Consider a fixed mass of gas at constant temperature. If the gas volume is decreased, the molecules take less time to travel between the piston and the wall opposite. The molecules

Scanned with CamScanner

therefore hit the walls of the container more often in unit time. Therefore the gas pressure increases as more impacts per second are made. Hence decrease in volume of gas at constant temperature leads to increase in pressure. Conversely, if the gas volumes is increased at constant temperature, the molecules take longer time to bombard the walls. There are therefore fewer impacts per second. The pressure of the gas thus decreases. Thus the pressure of the gas varies inversely with its volume at constant temperature. This is in agreement with Boyle's law.

Charles' law

When the gas is heated at constant pressure, the molecules gain kinetic energy and hit the walls of the container more often in a second. This leads to an increase in pressure. In order to keep the pressure constant, the volume of the containing vessel must be increased in order that the molecules would travel a longer distance before striking the walls of the container. Thus an increase in temperature at constant pressure would lead to an increase in the gas volume. This is a statement of Charles law.

Pressure law or Gay Lussac's law

If the temperature of a fixed mass of gas is increased at constant volume, the average kinetic energy of the molecules increases. They therefore gain more speed. Hence they strike the walls of the container harder and more often in a second. This brings about an increase in the pressure of the gas. Thus the pressure of a fixed mass of gas increases as its temperature increases at constant pressure. This is a statement of Gay-Lussac's law.

Summary

1 A *manometer* is used to measure the pressure of a gas.

2 A *Barometer*, e.g *Fortin barometer*, and *Aneroid barometer* can be used to measure Atmospheric pressure. The Aneroid barometer can also be used as an *Altimeter* to measure the height of a place.

3 Pressure can be measured in mm Hg or in Pascals (P) or in Newtons per square metre (Nm^{-2}) or in bars.
1 Pascal = 1.013×10^5 Nm
1 bar = 10^5 Nm^{-2} or 10 Pascals
1 millibar (mb) = 10^{-3} bar = 100 Pascals

4 *Boyle's law* states that the pressure of a fixed mass of gas is inversely proportional to its volume provided the temperature is kept constant

$$P \propto \frac{1}{V} \text{ or } PV = \text{constant}$$

5 Charles' law states that the volume of a fixed mass of gas increases by 1/273 of its volume at 0°C for each °C (or Kelvin) rise in temperature, provided its pressure remains constant. Hence the volume is proportional to the absolute temperature

$$V \propto T \text{ or } VT^{-1} = \text{constant or}$$

$$V_t = V_0\left(1+\frac{1}{273}t\right).$$

6 Gay-Lussac's law states that the pressure of a fixed mass of gas at constant volume is directly proportional to its absolute temperature.

$$P_t = P_0\left(1+\frac{1}{273}t\right) = KT \text{ or } \frac{P}{T} = \text{constant.}$$

7 General Gas Equation or Ideal Gas Equation:

$$\frac{PV}{T} = \text{Constant}.$$

The equation indicates that the energy term (PV) increases directly with the absolute temperatures T.

8 The cubic expansivity of a gas at constant pressure is the fractional change in the volume of the gas at 0°C per °C (or per Kelvin) rise in temperature. The cubic or volume expansivity is the same for all gases and is equal to $\frac{1}{273}$.

9 The temperature –273°C is called the *absolute zero of temperature*. According to the kinetic theory it is the temperature at which molecules of an ideal gas would cease to move. In practice all gases, on cooling, liquefy or solidify before –273°C is reached.

10 The kinetic (molecular) theory assumes that matter is made up of molecules which are in continual motion. The pressure exerted by a gas is due to the bombardment of molecules on the sides of the containing vessel.

Exercise 10

1 A mass of gas occupies 20 cm^3 at 5°C and 760 mm Hg pressure. What is its volume at 30°C and 800 mm Hg pressure?
A 41.4 cm^3 B 20.7 cm^3
C 50 cm^3 D 0.4 cm^3
E 25 cm^3.

Scanned with CamScanner

2

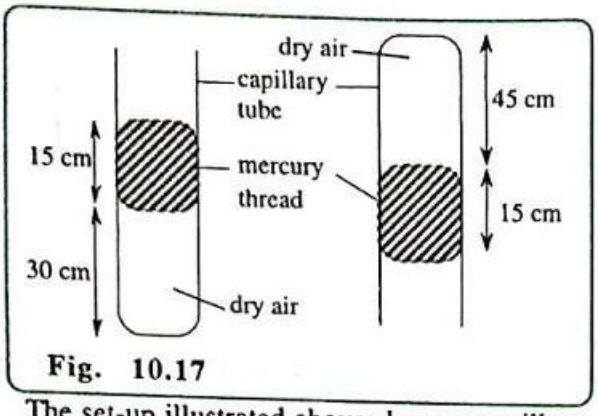


Fig. 10.17

The set-up illustrated above shows a capillary tube of uniform cross-sectional area in two different arrangements. Using the data in the diagrams, calculate the pressure of the atmosphere.

A 76 cm Hg B 75 cm Hg
C 74 cm Hg D 60 cm Hg
E 45 cm Hg

(SSCE)

3 A fixed mass of gas of volume 600 cm^3 at a temperature of 27°C is cooled at constant pressure to a temperature of 0°C. What is the change in volume?

A 54 cm^3 B 273 cm^3
C 300 cm^3 D 546 cm^3
E 600 cm^3.

(SSCE)

4

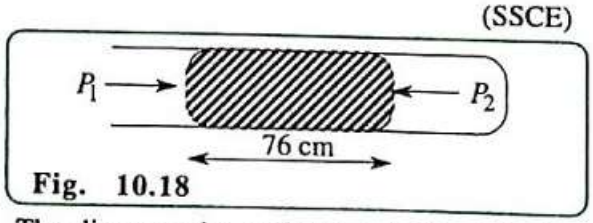


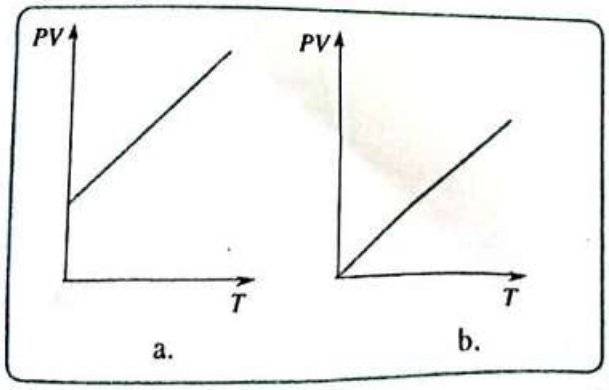
Fig. 10.18

The diagram above shows a horizontal tube containing a column of mercury 76 cm long. Calculate the ratio of the pressures P_2/P_1. (Atmospheric pressure is 76 cm of mercury)

A 0 B $\frac{1}{2}$ C 1 D 2

(JAMB)

5 Which of the following graphs illustrates the general gas law?



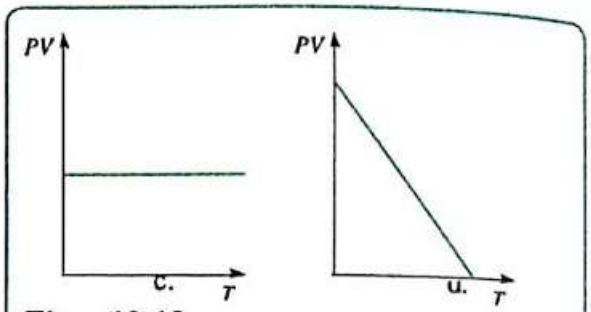


Fig. 10.19

6 A given mass of gas has a pressure of 70 Nm^{-2} at a temperature of 7°C. If the temperature is increased by 20°C with the volume remaining constant, what is the new pressure?

7 The equation $P^xV^yT^z$ = Constant is Boyle's law if

A $x = 0, y = 0, z = 1$
B $x = 1, y = 0, z = 0$
C $x = 1, y = 1, z = 0$
D $x = 1, y = 1, z = 1$
E $x = 1, y = 1, z = -1$. (JAMB)

8 The equation $P^xV^yT^z$ = Constant is Gay-Lussac's law if

A $x = 1, y = 0, z = -1$
B $x = 1, y = 1, z = 1$
C $x = 0, y = 1, z = 1$
D $x = 1, y = 0, z = 1$
E $x = -1, y = 1, z = 0$

9

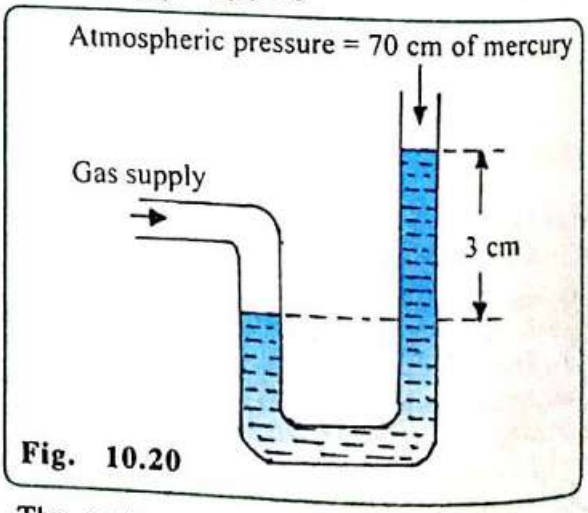


Fig. 10.20

The water manometer in fig. 10.20 is measuring the pressure of the gas supply. If the specific gravity of mercury is 13.6 and the atmospheric pressure is 70 cm of mercury, what is the total pressure of the gas supply in cm of water?

A 67 cm B 73 cm C 949 cm
D 952 cm E 955 cm.

(JAMB)

Scanned with CamScanner

10 Calculate the final centigrade temperature required to change 20 litres of gas at 120°C and 1 atmosphere to 25 litres at 2 atmosphere.

11 Describe how you would use a manometer to measure the pressure of a gas supply. In using such a manometer, the level of mercury in the arm of the V-tube open to the atmosphere is at 7.0 cm mark while the arm attached to the gas supply is at 27.0 cm mark of a metre rule placed between the two arms. If the atmospheric pressure is 760 mm Hg, what is the pressure of the gas?

12 State Boyle's law. A thread of mercury of length 15 cm is used to trap some air in a capillary tube with uniform cross-sectional area and closed at one end. With the tube vertical and the open end uppermost, the length of the trapped air column is 20 cm. Calculate the length of the air column, when the tube is held (i) horizontally (ii) vertically with the open end underneath. (Atmospheric pressure = 76 cm of mercury) (SSCE)

13 Describe an experiment to investigate the relationship between the pressure and the volume of a given mass of air at constant temperature. State three precautions and the conclusions of the experiment.

A glass tube in the shape of a letter *J* has the shorter limb sealed and the longer limb open. Mercury is poured into the tube until the level in either limb is the same when the tube is vertical. In this position, the length of the air column in the sealed limb is 6.3 cm. More mercury is then poured into the tube until the length of the trapped air column is 42 cm. Calculate the difference in the levels of mercury in the limbs if a nearby mercury barometer reads 75.0 cm and the reading of a nearby thermometer has not changed. (SC/GCE)

14 State Charles' law and describe an experiment to verify it. State three precautions necessary to obtain an accurate result.

A fixed mass of a gas occupies 1000 cm^3 at 0°C. If it is heated at constant pressure to 100°C calculate its new volume.

15 State the assumptions of the kinetic molecular theory of gases. How does the law account for Boyle's law, Charles' law, Gay-Lussac's law.

16 What is meant by *absolute zero of temperature; absolute scale of temperature*. Describe an experiment to determine the absolute zero of temperature using a mass of air either at constant volume, or at constant pressure.

17 (i) With the aid of a graph, show the relation between the Celsius and absolute temperature scales. Show on the graph the temperature at which an ideal gas has zero volume.

(ii) Write down the general gas equation. Does the law always hold true for real gases? If not, why?

18 What is meant by s.t.p.? A gas occupies 8 litres at 23°C and 70 cm Hg. What is its volume at s.t.p.?

19 What do you understand by 'the absolute zero of temperature'?

At TC, a tyre gauge records the pressure of a car tyre as 1.6 x 105 Nm^{-2}. Assuming that the volume of the tyre is unchanged, what does the gauge read at 30°C? The reading on a tyre pressure gauge reads the pressure above atmospheric pressure. The atmospheric pressure can be assumed to be 10^5 Nm^{-2}.

20 Write down the ideal gas equation.

A man takes 16 breaths per minute at 18°C and 760 mm Hg. What is his rate of breathing on a mountain-top where the temperature is -5°C and the pressure is 240 mm Hg. Assume that he needs the same mass of air per minute.

21 (a) Using the kinetic theory of matter, explain why

(i) Evaporation causes cooling.

(ii) Boiling water changes to steam without any change in temperature although heat is being supplied to the water.

(b) (i) State Boyles Law.

(ii) With the aid of labeled diagram, describe an experiment to illustrate the relationship between the volume and pressure of given mass of gas at constant temperature.

(iii) State two precautions necessary to obtain accurate result. (SSCE)

22 A uniform capillary tube closed at one end contained dry air trapped by a thread of mercury 8.5 x 10^{-2}m long. When the tube was held horizontally, the length of the air column was 5.0 x 10^{-2}m, when it was held vertically with the closed end downwards the length was 4.5 x 10^{-2}m, Determine the value of the atmospheric pressure (g = 10ms^{-2}; density of mercury 1.36 x 104kg m^{-3}). (SSCE)

23 (a) State difference between boiling and evaporation

(b) A closed inexpansible vessel contains air saturated with water vapour at 77°C. The total pressure in the vessel 1007mm Hg. Calculate the new pressure if the temperature is reduced to 27°C. (S.V.P. water at 77°C and 27°C respectively are 314mmHg and 27mmHg. Treat the air in the vessel as an ideal gas)

(c) Explain why there is no temperature change when a solid being heated changes into liquid at its own melting point. (SSCE)

Scanned with CamScanner

11 WAVES

This chapter will discuss the production and propagation of waves, the different types of waves, the mathematical relationship between the frequency, wavelength, period and velocity of waves. Sound and light will also be treated as wave phenomena. The chapter will also discuss the different properties of waves.

At the end of the chapter the student should be able to:

1 Generate mechanical waves.
2 State the important characteristics of waves.
3 Produce circular and plane waves using a ripple tank.
4 Generate and demonstrate longitudinal and transverse waves using suitable materials.
5 Identify the crest, trough, amplitude, wavelength and points in phase on a given sine waveform.
6 Explain compression, rarefaction, period, cycle and frequency as they apply to waves.
7 Derive and use the relationship between wave velocity, frequency and wavelength.
8 Identify light as electromagnetic waves.
9 Classify waves into longitudinal and transverse waves by using
   a. mode of vibration
   b. direction of propagation.
10 Produce plane and circular waves using a ripple tank.
11 Demonstrate the reflection of sound.
12 Demonstrate stationary waves.
13 Demonstrate the reflection of heat radiation.
14 Demonstrate refraction of water waves and sounds.
15 State and explain four properties of waves.
16 Demonstrate polarization of light waves.

11.1 Wave Motion: Introduction

Wave motion is a means of transferring energy. In such a motion in a material medium, particles of the medium vibrate about their positions of equilibrium. The wave energy is carried forward but not the medium.

When we throw a stone into a pool of water, circular waves form and move outward. Waves will also travel along a cord or rope that is stretched out straight on a table if we vibrate one end back and forth. Water waves and waves on a cord, are two common examples of wave motion. In this chapter we will concentrate on Mechanical waves — waves that travel in a material medium.

All types of waves have as their source a vibration. Water waves for example move with a recognisable velocity, but each particle of the water itself merely oscillates about an equilibrium position. This is easily seen when we observe leaves on a pond as waves pass by. The leaves are not carried forward by the waves, but simply move up and down about an equilibrium point because this is the motion of the water itself. This is a general feature of waves: *Waves can move over large distances but the medium (e.g. the water or the cord) itself has only a limited movement.* A wave consists of oscillations that move without carrying matter with them. Such up-and-down movement or *vibration* or *oscillation* of the medium (e.g. water) which causes waves is an example of *wave motion*.

A **wave motion** is a process of transferring a disturbance (in form of kinetic energy) from one point to another in a medium without any transfer of particles of the medium.

A wave is a disturbance which travels through a medium transferring energy from one point to another without causing any permanent displacement of the medium.

Not all waves, however, require a material medium for their propagation.

Mechanical waves are those waves that require a material medium for their propagation. Examples of such waves are water waves, sound waves, waves on a rope or string.

Electromagnetic waves are waves that do not require a material medium for propagation. Examples are light waves, radio waves, X-rays and gamma-rays.

11.2 Production of Mechanical Waves

The motion of any material object may be considered as a source of waves.

Scanned with CamScanner

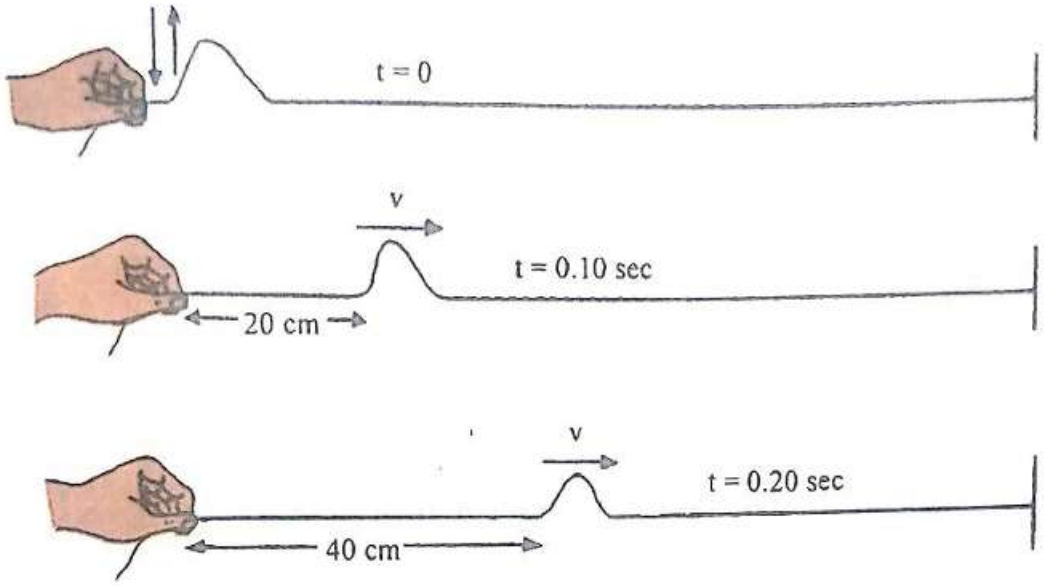


Fig. 11.1 A pulse moving down the rope

Activity 11.1 Waves on a String

1 Hold one end of a tight string or rope as shown in fig. 11.1. The other end of the rope is tied to a nail on a wall. Tie some ribbons all along the rope.
2 Move the end of the rope in your hand up and down rapidly.
3 Observe a set of waves travelling along the rope. Observe the movements of the ribbons.

As we move the end of the rope, a pulse will be sent down the rope (fig 11.1a). A repeating set of pulses sent down the rope is called a *wave*.

The waves have been generated by the vibration of the particles of the rope about their mean position of rest. The ribbons are also observed to move up and down while the wave travels along the rope from the hand to the nail on the wall.

Activity 11.2 Waves in Slinky Spring

1 Obtain a long loose spring called a slinky spring.
2 Support the slinky spring horizontally at one end and move the other end up and down in a vertical direction.
3 Observe the wave as it travels along the spring (fig. 11.2)
4 With the aid of a long pencil compress the free end of the slinky spring as shown in fig 11.3. Notice the successive positions of the coils of the spring after the compression.

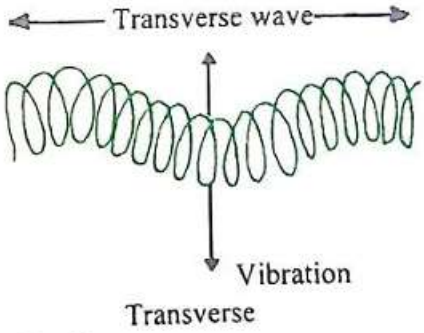


Fig. 11.2 Transverse wave

It will be observed that the compressed part (compression) travels along the spring while one particular turn, indicated by the paper maker, only vibrates back and forth. Every turn of the slinky spring vibrates in the direction in which the compression travels. Thus the wave generated by the compression of the slinky spring travels parallel to the direction of the compression or the disturbance producing the wave.

Scanned with CamScanner

(a) *Positions of coils before compression. The arrow shows the rest position of the turn indicated by the paper marker.*

(b) *The coil is compressed with a pencil. Here the compression has not reached the turn indicated by the marker. Notice that the turns on the left of the pencil are further apart than in (a).*

(c) *The compression has travelled to the right. Notice that the marked turn has also travelled in the same direction.*

(d) *After reflection at the wall, both the compression and the marked turn travel in the opposite direction, i.e. to the left.*

(e) *The marked turn is now back in its rest position. The marked turn, together with the other turns of the coil have merely vibrated back and forth while a wave travelled along the spring.*

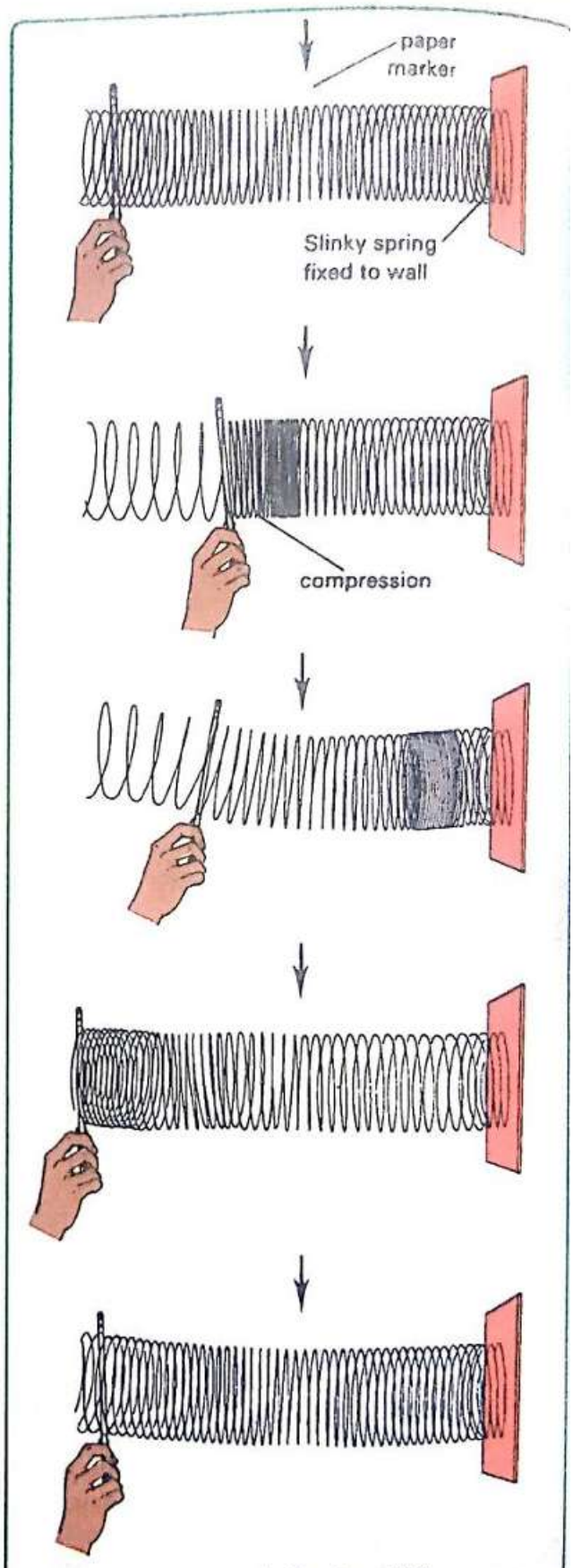


Fig. 11.3 Wave motion

Water waves in ripple tank

If a stone is dropped into a pool of water in a pond or stream, ripples are seen to form, grow quickly and die away. The falling stone creates a disturbance of the calm water surface and produces ripples. Such ripples are called pulses. A pulse consists of a single crest.

A pulse is a sudden increase in magnitude of a physical quantity, shortly followed by a rapid decrease

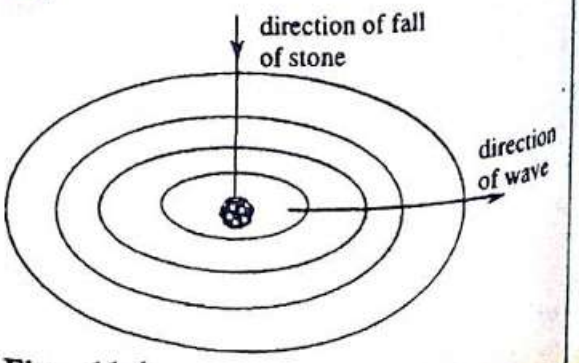


Fig. 11.4

Scanned with CamScanner

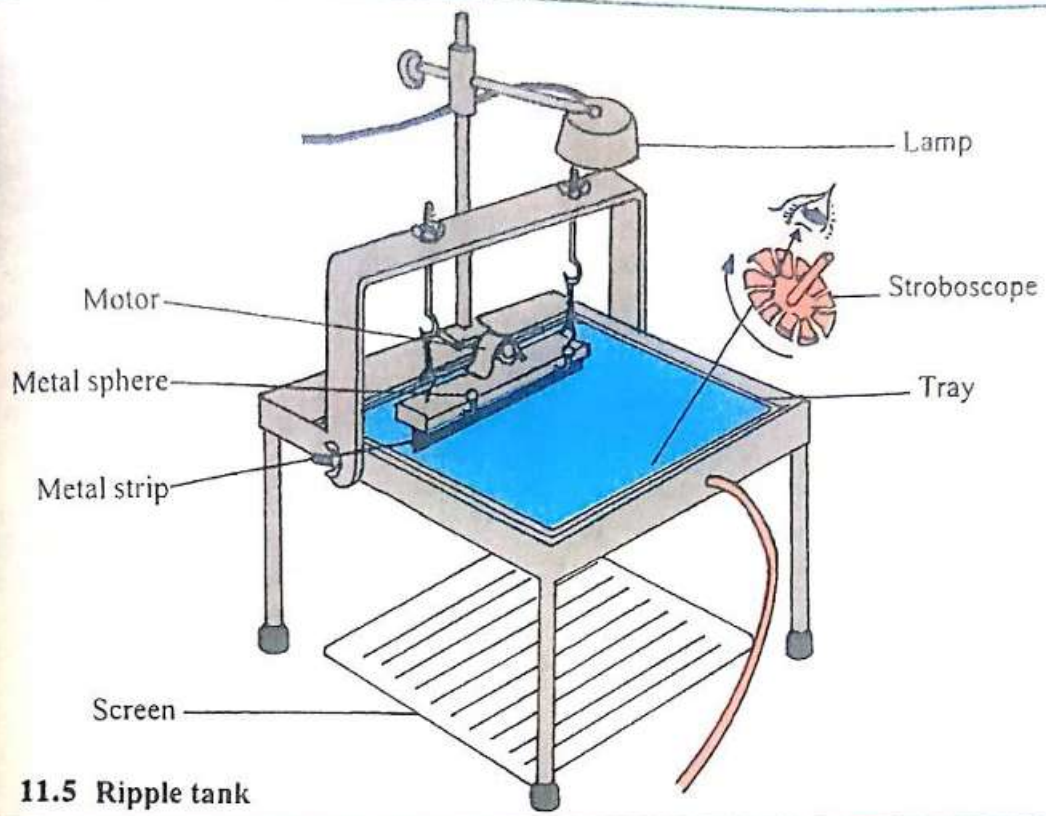


Fig. 11.5 Ripple tank

We can produce continuous water ripples or pulses to form a water wave by disturbing the water surface continuously, e.g. by throwing in stones at regular intervals. In the laboratory water waves can be produced and studied in a **ripple tank.**

The ripple tank consists of

(i) a pool of water in shallow tray with a transparent base which enables us to project the images of the waves on a screen placed below it. The tray is levelled such that it is perfectly horizontal.

(ii) A dipper in the form of a horizontal metal strip or a small sphere, touching the water.

(iii) A vibrator in the form of small electric motor to vibrate the dipper

(iv) A lamp fixed right above the water surface to illuminate it.

The wave is produced by the movement of the dipper which generates continuous pulses. This continuous wave can be made to appear stationary by the use of a *Stroboscope* which consists of a circular wooden disc with a number of equally spaced radial slits and an axle in the centre. The disc can be rotated about its centre and as it rotates the water waves can be viewed through the slits. For a particular speed of rotation, the waves would appear to be stationary. They can then be studied in greater detail.

We can generate two types of waves with the ripple tank system - the *plane wave* and the *spherical or circular wave*. When the dipper of the ripple tank is in the form of a horizontal metal or wooden strip, parallel plane waves are produced.

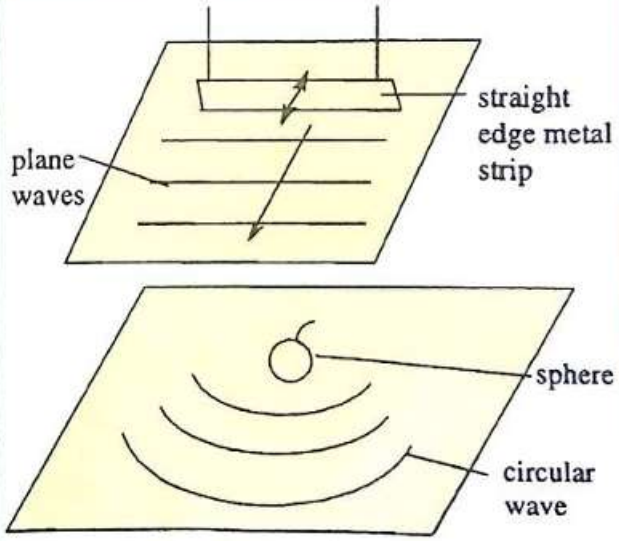


Fig. 11.6 Production of plane and circular waves

When the dipper is in the form of a sphere, spherical or circular waves are produced. Both types of waves are seen to spread out from the source of their production as shown in fig. 11.6.

Wave fronts

A line or surface in the path of an advancing wave on which all the particles are vibrating in step or in phase is called a *wave front.*

Scanned with CamScanner

There are two types of wave fronts – the *plane wave front* and the *spherical or circular wavefront*.

Suppose a stone is thrown into a pool of water. Spherical or circular water waves are generated from S, the point at which the stone hits the water surface. After a time, t, the circular patterns of the wave emerging from the centre of disturbance, S, have travelled a distance vt where v is the speed of the wave.

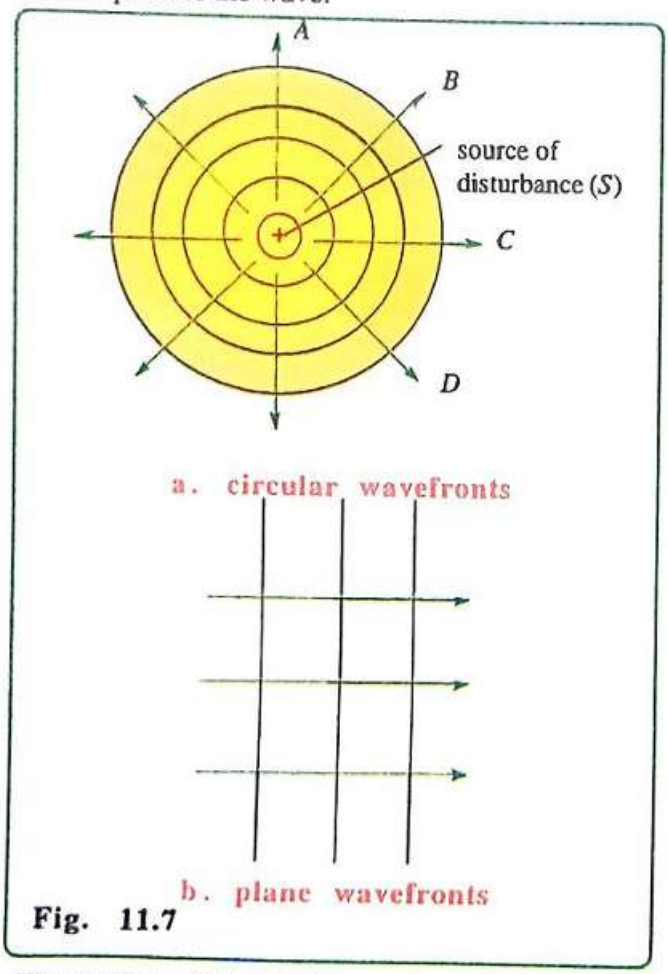


Fig. 11.7

The surface of the circle or sphere containing all particles which are vibrating in *phase* or in step is called a *wavefront*. All points such as A, B, C, D on a wavefront are the same distance from the central point S. All such points are also in the same state of disturbance. At long distances from S the wavefronts are practically straight lines instead of curves. The significance of the wavefront is that it shows how the energy travels from one place in a medium to another.

A line or surface in the path of an advancing wave on which all the particles are in the same phase or state of disturbance is called a **wavefront**.

Waves as a Mode of Energy Transfer

If small pieces of cork float on a water surface and waves are generated for example by throwing a stone into the water, the pieces of cork are seen to bob up and down in their positions as the water wave passes along the water surface. The waves transfer the kinetic energy of the falling stone from the centre of the disturbance outwards. The up and down movement of the cork shows that the water particles themselves merely vibrate up and down but do not move from one place to another along the water surface. The waves carry the energy as they travel. Energy is hence transferred along the direction of propagation of the wave.

Thus with every wave we must distinguish between (i) the vibrations of the material medium which produce the wave and (ii) the wave itself which carries the energy away from the source of disturbance. In this example water is the medium or material in which the energy travels. The particles of the water medium vibrate about a mean position as the wave travels. The vibrations are passed on from one particle of the medium to the next. The direction in which this vibration takes place is significant in classifying the type of wave as we show in the next section.

11.3 Transverse and Longitudinal Waves – Graphical Representation

We have seen that a wave originates from the vibrations of the particles of a medium. Such a wave allows energy to be transferred from one point to another some distance away without any particles of the medium travelling between the two points.

A *Progressive* or *Travelling Wave* is a wave that transfers energy in travelling or moving away from the source of the disturbance. There are two types of progressive waves – the *Transverse waves* and the *Longitudinal waves*. The distinction between the two types of waves is dependent on the direction of the vibrating particles with respect to the direction of travel of the wave.

Transverse waves are waves which travel perpendicularly to the direction of the vibrations producing the waves

Scanned with CamScanner

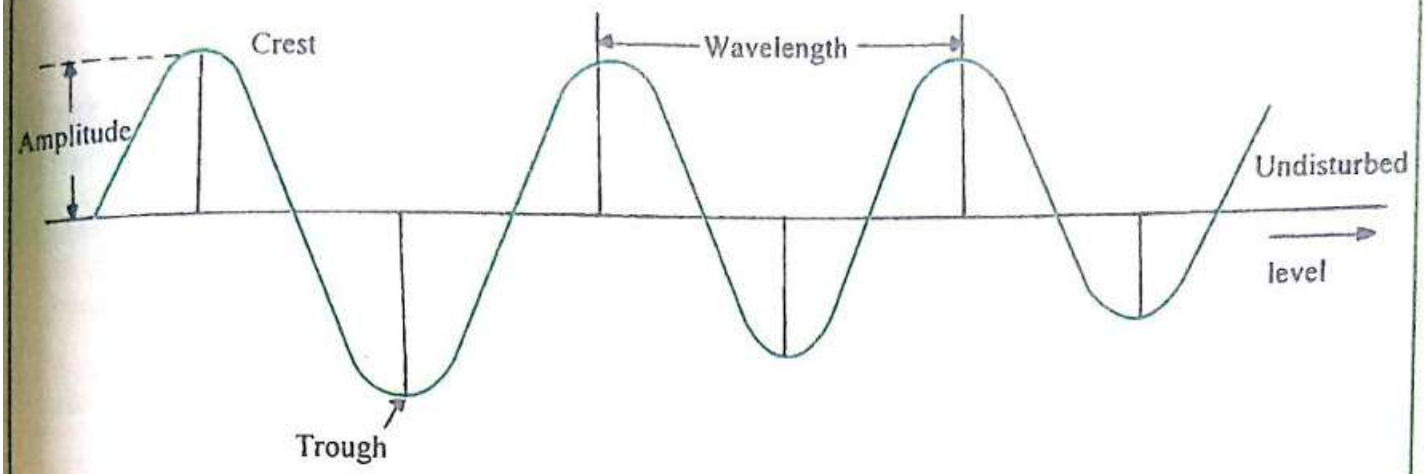


Fig. 11.8 Wave representation

Examples of transverse waves are water waves, light waves, radio waves and waves produced in a rope or string. In all such waves the vibrations are vertical and the direction of the wave motion is horizontal.

A graph of the displacement (y) of the vibrating particles about a mean position against time (t) is as shown in fig 11.8.

The displacement (y) of the various particles will also vary according to their positions or distance (x) from some chosen origin such as the centre of disturbance. The graph shows a series of up and down movements. Some portions of the wave are displaced upwards while some portions are displaced downwards. A displacement above the mean position is taken as positive and that below it is taken as negative.

The region of maximum upward displacement is called a *crest*. The region of maximum downward displacement is called a *trough*.

Longitudinal Waves are waves which travel in a direction parallel to the vibrations of the medium

Longitudinal waves are readily formed on a stretched spring or 'Slinky' by alternatively compressing and expanding one end or by moving the slinky to and fro in the direction of its length (fig. 11.9).

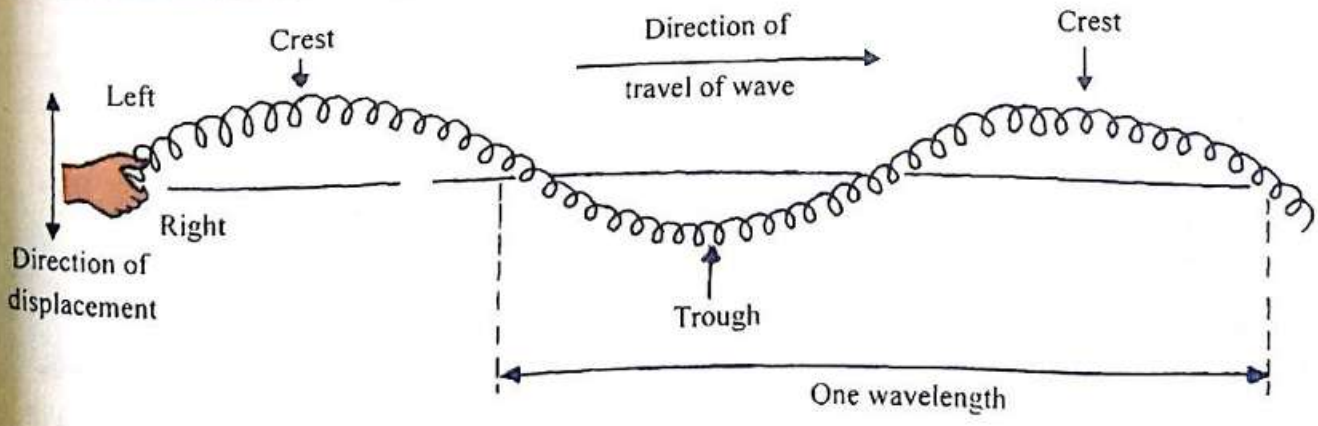


Fig. 11.9 Transverse wave in a spring

Scanned with CamScanner

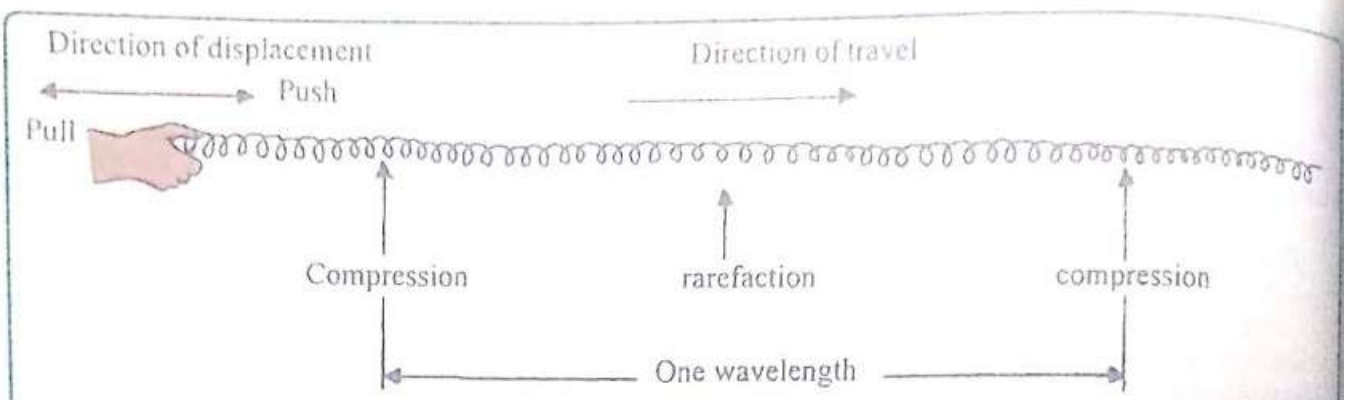


Fig. 11.10 Longitudinal wave in a spring

A series of *compressions* and expansions or *rarefactions* propagate along the slinky or spring. The *compressions* are areas where the coils are momentarily close together, *rarefactions* or *expansions* are areas where the coils are momentarily far apart. Compressions and rarefactions correspond to the crests and troughs respectively of a transverse wave.

Sound waves are an important example of longitudinal waves. A vibrating tunning fork or drum head, for examples, alternatively compresses and rarefies the air and produces longitudinal waves that travel outward in the air as shown in fig. 11.11. Fig 11.11a shows the original undisturbed positions of the air. As the sound wave travels some layers of air molecules crowd together and are known as the *compressions* areas. Here the air density or pressure has a value above the normal. Between these areas are the *rarefaction* areas where the layers are further apart and the air pressure or density is less than normal.

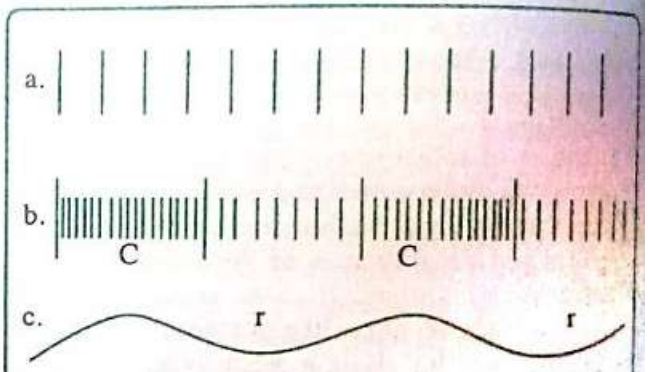


Fig. 11.11

A longitudinal wave can be represented graphically by plotting the density or pressure of air molecules (or coils of a slinky) against position as shown in fig. 11.11C.

Terms Used in Describing Waves

Fig. 11.12 shows a displacement graph of a wave at an instant.

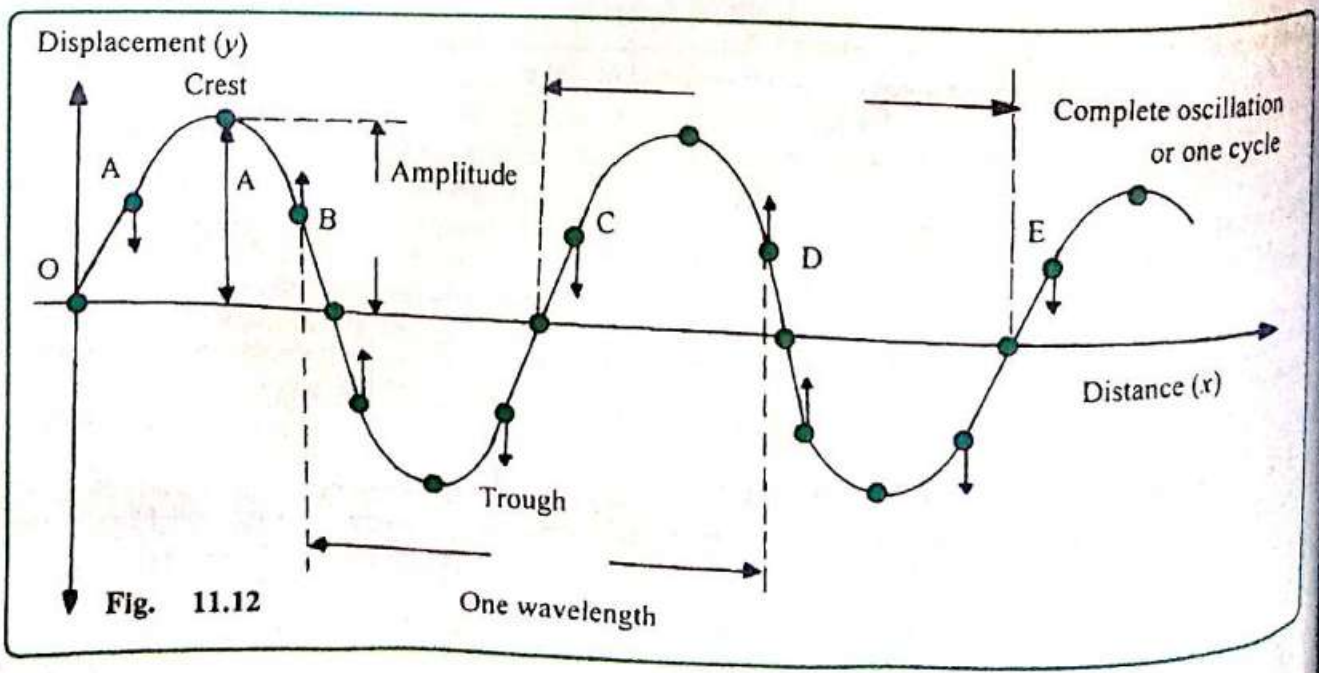


Fig. 11.12

Scanned with CamScanner

The dots on the graph represent the positions of the particles at that instant and the arrows indicate the direction of motion of these particles.

Phase
Particles which are at the same vertical distances from their positions of rest and are moving in the same direction as shown on the graph, are said to be in the same *phase*. Particles A, C and E are in phase.

A Cycle is a complete to-and-fro movement or oscillation of a vibrating particle.

Amplitude
This is the maximum displacement of a particle from its rest position. In fig. 11.12, A is the amplitude of the wave. It is measured in metres.

Wavelength (λ)
This is the distance between two successive crests or successive troughs (or two successive points in phase) of a wave. It is measured in metres.

Frequency (f)
This is the number of complete vibrations or cycles that a particle makes in one second. It is also equivalent to the number of complete waves passing a given point in one second. The S.I. unit of frequency is the Hertz (Hz) which is defined as one cycle or oscillation per second (1 cps).
The frequency of a wave is identical to the frequency of the source that sent it out.

Period (T)
This is the time taken by a wave particle to make one complete oscillation. It is measured in seconds. The period is also the time taken by the wave to travel one wavelength.

The period (T) and frequency (f) are related by

$$f = \frac{1}{T} \qquad 11.1$$

$$\text{or} \quad T = \frac{1}{f}$$

Wave Speed (v)
This is the distance which the wave travels in one second. Its unit is metre-per-second (ms^{-1})

Mathematical Relationships between Frequency (f), Wavelength (λ) and Velocity (v)
There is a very important relation between the frequency (f) wavelength (λ) and velocity (v) of all waves. This is given by:

$$v = f\lambda \qquad 11.2$$

Velocity = Frequency × Wavelength

The above relation is easy to prove. The speed of a wave, v, is given by

$$v = \frac{\text{distance travelled by wave}}{\text{Corresponding time taken}}$$

A wave travels a whole wavelength (λ) in a time given by the period (T)

$$\therefore \quad v = \frac{\lambda}{T}$$

But from equation 11.1, $f = \frac{1}{T}$

Hence $v = \frac{\lambda}{T} = f\lambda$

$$\lambda = vT$$

Example 11.1
The frequency of a vibrating source is 450 Hz and the velocity of the sound it produces in air is $330\,ms^{-1}$. Find how far the sound travels when the source completes 50 vibrations.

Solution
Frequency, $f = 450$ Hz or 450 cycles per sec.
$\therefore$ time for 50 oscillations is given by

$$t = \frac{1}{450} \times 50 = \frac{1}{9} \text{ secs.}$$

Since the velocity of sound in air is $330\ ms^{-1}$, the distance travelled in time $\frac{1}{9}$ secs is given by:

$$s = vt = 330 \times \frac{1}{9}$$
$$= \underline{36.67\ m}$$

Scanned with CamScanner

Example 11.2

A radio station broadcasts at a frequency of 300 kHz. If the speed of the wave is 3×10^8 ms^{-1}, calculate the period and wavelength of the wave.

Solution

Period, $T = \dfrac{1}{\text{Frequency } (f)} = \dfrac{1}{300\times10^3 \text{ Hz}}$

$$T = \frac{1}{3\times10^5} \text{ sec.}$$

$$= \underline{3.3\times10^{-6} \text{ sec}}$$

Wavelength, $\lambda = \dfrac{V}{T} = \dfrac{3\times10^8}{3\times10^5}$ m

$= 1000$ m $= 1$ km.

Example 11.3

A long rope is fixed at one end, and the free end is made to oscillate in one plane at right angles to the rope with a frequency of 4 Hz. The successive crests are 0.6 m apart. Calculate the speed of the waves. For what frequency would the wavelength be 30 cm?

Solution

Distance between two successive crest = Wavelength of wave,

Hence $\lambda = 0.6$ m

$$v = f\lambda$$

$$= 4\times0.6$$

$$= 2.4 \text{ ms}^{-1}$$

When $\lambda = 30$ cm $= 0.3$ m

$$f = \frac{v}{\lambda} = \frac{2.4}{0.3}$$

$$= \underline{8 \text{ Hz}}$$

Mathematical Representation of Wave Motion – Progressive Wave

We can represent a wave mathematically by the equation

$$Y = A\sin\left(\frac{2\pi x}{\lambda}\right) \qquad 11.3$$

where Y = vertical displacement of vibrating particle, P

λ = Wavelength of wave

A = Amplitude of wave

x = horizontal coordinate of the vibrating particle from the origin

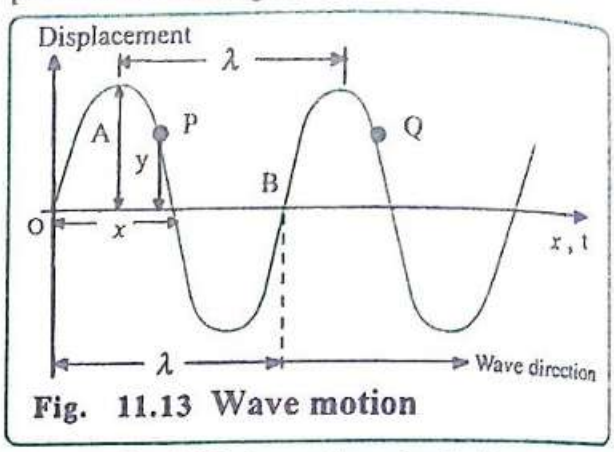


Fig. 11.13 Wave motion

The equation 11.3 above gives the displacement of any particle at a distance x from the origin. The relationship gives a shape that repeats itself every wavelength. It is a sine wave form. The displacement of the particle t seconds after is given by

$$Y = A\sin\frac{2\pi}{\lambda}(x - vt) \qquad 11.4$$

where v is the velocity of the wave. If we put $t = 0$ in equation 11.4, we again obtain equation 11.3. Since $v = f\lambda$, we can write equation 11.4 as

$$Y = A\sin\left(\frac{2\pi x}{\lambda} - \frac{2\pi vt}{\lambda}\right) = A\sin\left(\frac{2\pi x}{\lambda} - \frac{2\pi f\lambda t}{\lambda}\right)$$

$$= A\sin\left(\frac{2\pi x}{\lambda} - 2\pi ft\right) = A\sin\left(\frac{2\pi x}{\lambda} - \omega t\right)$$

where $\omega = 2\pi f$ is known as the angular speed.

$$Y = A\sin\left(\frac{2\pi x}{\lambda} - \omega t\right) \qquad 11.5$$

Example 11.4

A plane progressive wave is represented by the equation $y = 2\sin(2000\pi t - 0.5x)$ where the symbols have their usual meanings. What is the frequency of the wave? (JAMB)

Solution

From equation 11.5, with $\omega = 2\pi f$, we have

$$y = A\sin\left(\frac{2\pi x}{\lambda} - 2\pi ft\right)$$

Comparing the above with the given equation,

$$y = 2\sin(2000\pi t - 0.5x)$$

Scanned with CamScanner

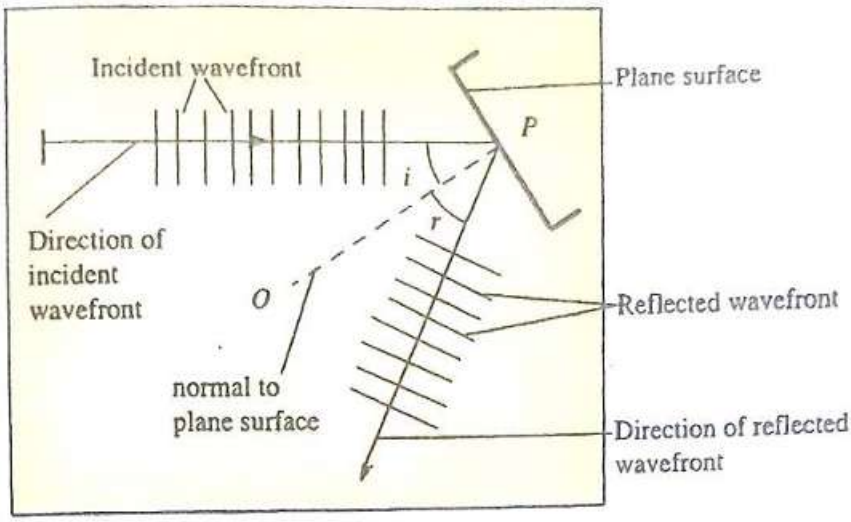


Fig. 11.14 Reflection by a plane surface

we have that:

$2\pi ft = 2000\pi t$

or $2ft = 2000t$

Ans f = 1000 Hz

11.4 Properties of Waves

In this section we shall discuss the following properties of waves: *Reflection*, *Refraction*, *Interference*, *Diffraction*, *Polarization*.

a Reflection of Waves

All waves undergo *reflection*. We can study the reflection of water waves at surfaces using a ripple tank.

A plane metal strip, S, is held vertically in the water in a ripple tank such that parallel plane waves set up in the water surface fall on the metal surface at an angle. The angle which the direction of motion of the plane wavefront makes with the normal to the metal surface is known as *Angle of Incidence (i)* (fig. 11.14).

The plane waves on meeting the plane surface are reflected in a definite direction as another set of plane waves. The angle between the direction of motion of the reflected plane wavefront and the normal OP to the plane surfaces is known as the *Angle of Reflection* (r) Experiments show that in such reflections the angle of incidence is equal to the angle of reflection. This is the law of reflection. The plane metal surface is known as a *plane reflector*.

If circular waves are produced in the ripple tank and made to fall on the plane metal surface held vertically in the water the waves are also reflected and spread out as spherical or circular waves as though they originate from a point I behind the plane metal surface (fig 11.15).

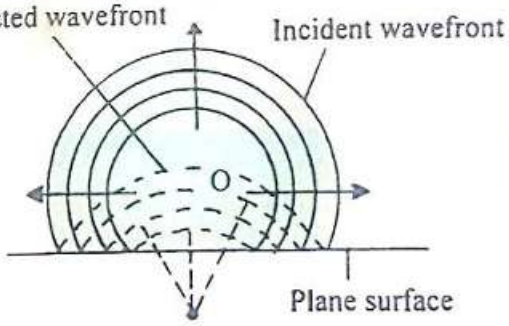


Fig. 11.15 Reflection of circular waves from a plane surface

Experiments show that the point I is as far behind the metal plate as the point O (the source of the wave) is in front. The same effect is observed when a light source is reflected by a plane mirror.

When a plane wave falls on a curved reflector, (a concave or a convex reflecting surface), the reflected waves will be found to be spherical or circular. The reflected circular wavefronts will converge at a point F, the principal focus of the concave reflector (fig. 11.16a) or they will diverge from F behind the convex reflector.

Scanned with CamScanner

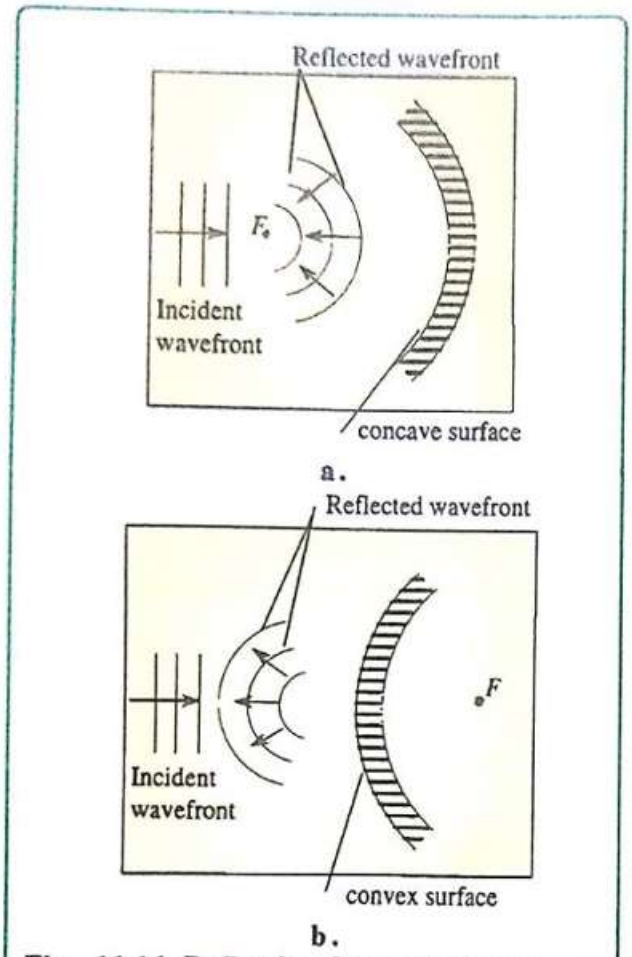


Fig. 11.16 Reflection by curved surfaces

Sound and light waves also undergo reflection at plane surfaces. The reflection of sound wave can be demonstrated using an arrangement as shown in fig 11.17. A ticking watch is placed near the end of one tube touching a hardboard reflecting surface or wall as shown. The ear is placed at the end of the other tube. The tubes are adjusted until the observer hears the loudest sound. At this point it is observed that the angles the tubes make with the normal to the hardboard or wall are equal. Thus the sound wave or reflection obeys the law of reflection – angle of incidence = angle of reflection.

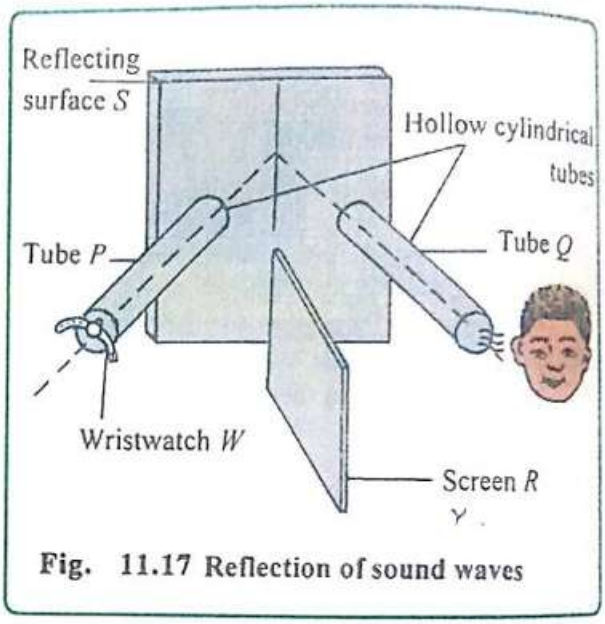


Fig. 11.17 Reflection of sound waves

The sound heard after reflection of sound waves is called an *Echo*. Echo phenomenon is used in underwater exploration for oil and gas and in the mapping of the depths of the sea-bed.

Reflection of Heat Radiation

Heat radiation is much like light and radio waves. All three are examples of electromagnetic radiation. Like other waves, heat radiation can be reflected. This can be demonstrated using the simple set-up as shown in fig 11.18.

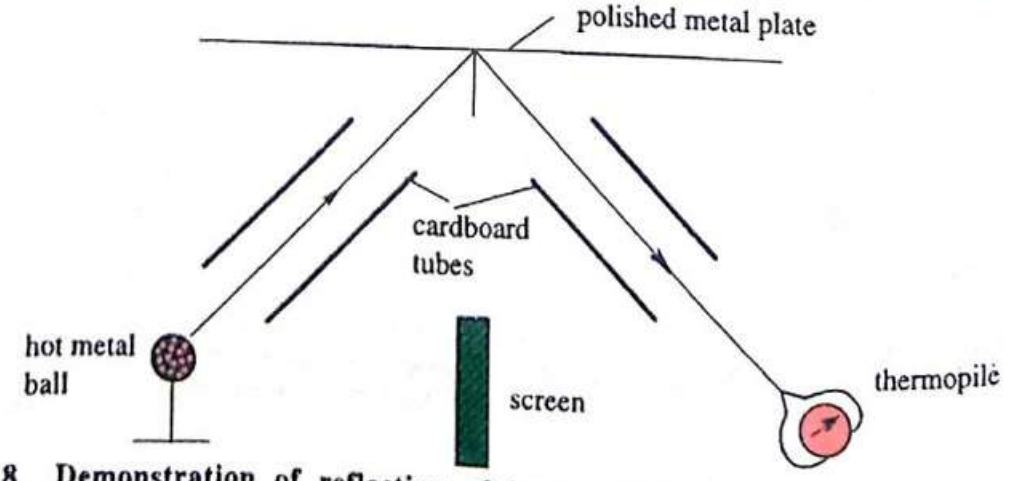


Fig. 11.18 Demonstration of reflection of heat radiation

Scanned with CamScanner

The heat wave originates from the hot metal ball and is directed along a cardboard tube to a polished metal plate which acts as the reflector. The reflected wave is detected by the thermopile, an instrument whose action depends on the electromotive force which appears between the junctions of two different metals at different temperatures. The tubes are adjusted until the thermopile registers the maximum deflection. At this point it is found that the angle of reflection is equal to the angle of incidence. The experiment also demonstrates another law of reflection: the incident and the reflected rays are in the same plane as the normal to the reflecting surface at the point of incidence.

A screen placed between the hot body and the thermopile prevents the later from receiving heat directly from the former.

Refraction of waves at plane boundaries

Consider a plane wavefront travelling from a medium to another medium of different density. Let the wave travel from medium 1 to medium 2. As the wave travelling at an angle to the plane boundary pass from medium 1 to medium 2, there is usually noticed a change in wavelength, speed, and of direction of travel of the waves.

Refraction is the change in the speed and direction of the waves as they cross the boundary between two media of different densities. The angle of incidence (i) is the angle the direction of the incident wavefront makes with the normal at the boundary surface. The angle of refraction (r) is the angle the direction of the refracted ray makes with the normal to the plane boundary. The refractive index, ${}_1n_2$ of medium 1 with respect to medium 2 is related to the speeds of the wave in the two media as follows.

$$\text{Refractive Index, } {}_1n_2 = \frac{\text{Velocity of wave in 1}}{\text{Velocity of wave in 2}}$$

$$= \frac{v_1}{v_2}$$

Since the frequency (f) of the wave remains constant during refraction,

$$v_1 = f\lambda_1,\ v_2 = f\lambda_2$$

where λ_1, λ_2 are the wavelength of the waves in medium 1 and 2 respectively.

Hence

$${}_1n_2 = \frac{v_1}{v_2} = \frac{f\lambda_1}{f\lambda_2} = \frac{\lambda_1}{\lambda_2}$$

For light waves

$${}_1n_2 = \frac{\text{Velocity of light in medium 1}}{\text{Velocity of light in medium 2}} \qquad 11.7$$

For water waves passing from deep to shallow water refractive index is given by

$${}_1n_2 = \frac{\text{Velocity in deep water}}{\text{Velocity in shallow water}}$$

Refractive Index is also given by

$${}_1n_2 = \frac{\sin i}{\sin r}$$

Example 11.5

The wavelength and velocity of a set of plane waves travelling in a medium are 60 cm and 320 cms^{-1} respectively. It meets a plane refracting

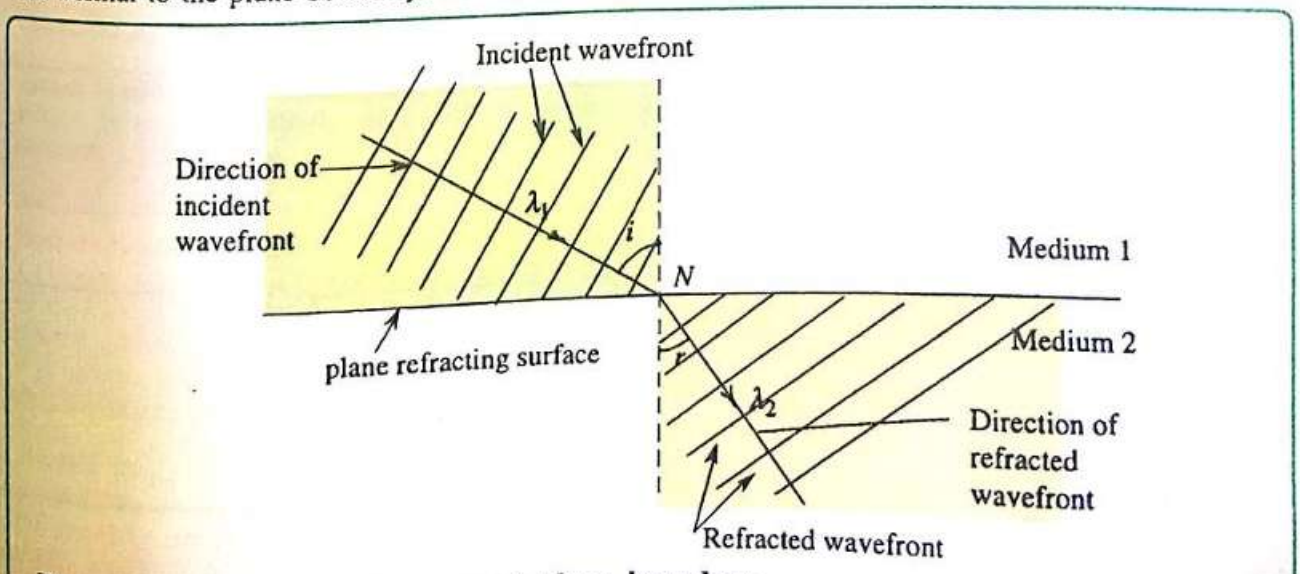


Fig. 11.19 Refraction of waves at plane boundary

Scanned with CamScanner

surface at an angle of incidence of 60°. Its velocity after refraction is 280 cms^{-1}. Calculate the wavelength of the waves in the second medium and also the angle of refraction.

Solution

The refractive index is defined by the following relations:

$$_1n_2 = \frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

$$\frac{\sin 60^\circ}{\sin r} = \frac{320}{280} = \frac{60}{\lambda_2}$$

$$\frac{320}{280} = \frac{60}{\lambda_2}$$

$$\lambda_2 = \frac{280 \times 60}{320}$$

$$= \underline{52.5 \text{ cm}}$$

$$\frac{\sin 60^\circ}{\sin r} = \frac{320}{280} = 1.143$$

$$\sin r = \frac{\sin 60^\circ}{1.143} = 0.7577$$

$$r = \underline{49.26^\circ}$$

Progressive and Standing Waves; Superposition of waves

A wave which continues to spread out transferring energy from the source of disturbance is called a *progressive wave*. Its waveform is as shown in fig 11.13. A *standing* or *stationary* wave is set up when two equal progressive waves of equal amplitude and frequency and travelling in opposite directions are superposed or made to overlap or combine together.

The resultant effect of the superposition of two waves of the same frequency depends on whether the waves arrive at a point in phase (in step). Consider a particle which is in a region where two waves of the same frequency arrive exactly in phase. The displacement-time graphs for such a particle are as shown in fig. 11.20 a-d. Fig 11.20d shows the resultant wave obtained by superposing the two waves. The amplitude of the resultant wave is the algebraic sum of the amplitudes of the two waves. The superposition of two waves of the same frequency and amplitude, which are exactly out of phase results in the creation of a region of zero resultant disturbance [fig. 11.21(d)].

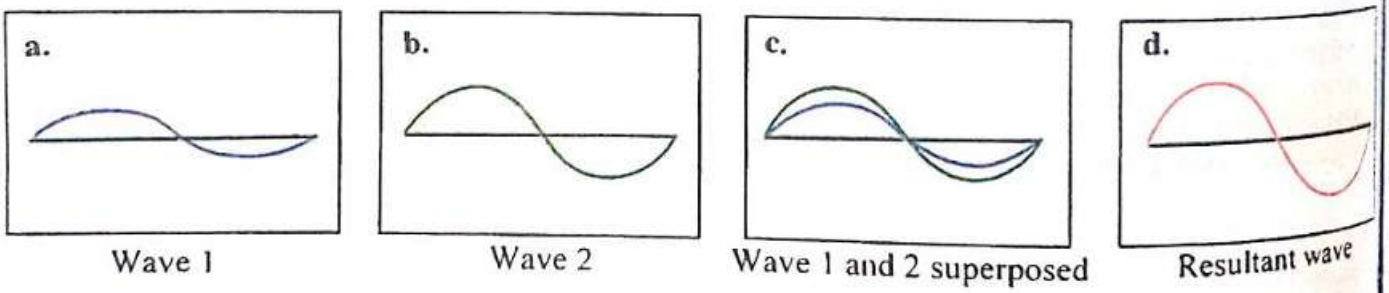


Fig. 11.20

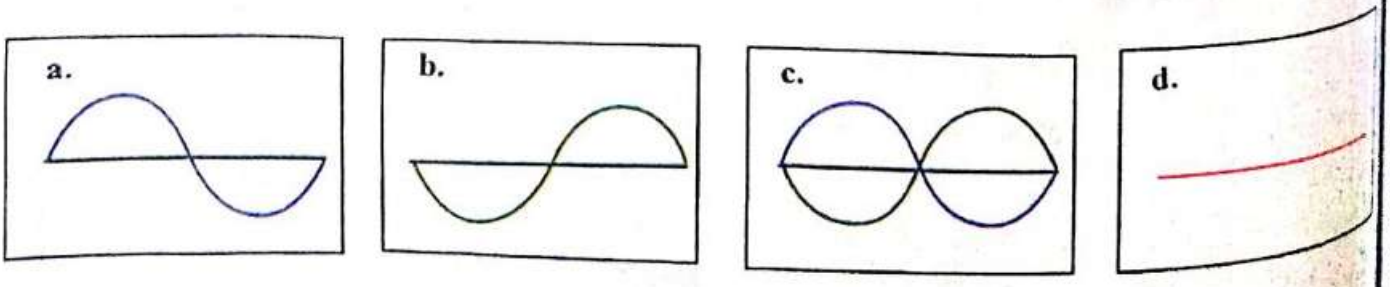


Fig. 11.21 Superposition of two waves of the same frequency which are exactly out of phase

Scanned with CamScanner

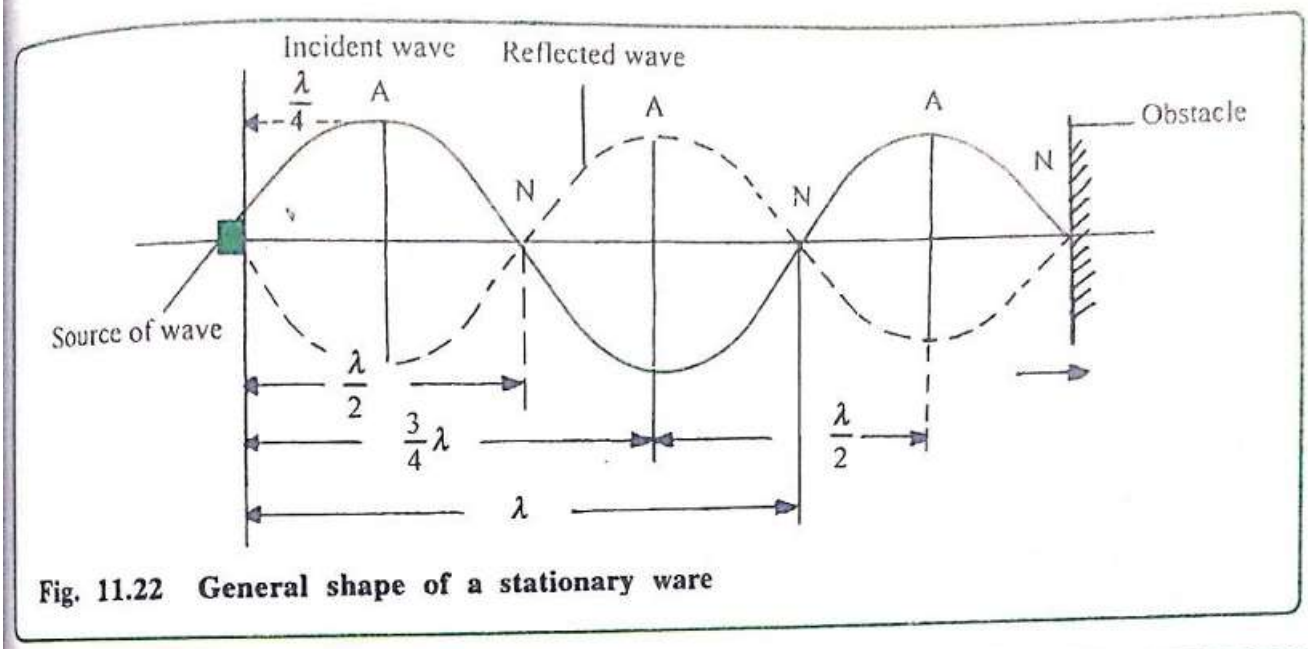


Fig. 11.22 General shape of a stationary ware

Most stationary waves are obtained as a result o the combination or superposition of an inciden wave and its reflected wave. A reflected wave is o the same frequency and amplitude as the inciden wave but both waves are exactly out of phase. The crest of the incident wave coincides with the trough of the reflected wave and vice-versa. The points marked *N* are called *Nodes* of the stationary wave At these points, the displacement is always zero Points marked A at which the maximum displacement occurs are called *Antinodes.*

A **Node** is a point on a stationary wave where there is no movement of the medium.

The nodes are $\lambda/2$ apart,

An **antinode** is a point on a stationary wave where there is maximum displacement of the medium.

Antinodes are $\lambda/2$ apart.

Both longitudinal and transverse waves can produce stationary waves. The vibrations of an air column of a tube can give rise to longitudinal stationary waves. Musical notes produced by wind instruments like flute, trumpet or saxophone are due to longitudinal stationary waves. Stationary waves can also result from the vibrations of a plucked string when both ends of the string are fixed. These are transverse stationary waves. Music produced by stringed instruments, e.g. guitar or piano, results from the stationary waves of plucked strings. From the superposition principle, when two identical waves are brought together, the resultant disturbance is the algebraic sum of the individual disturbances of the two waves.

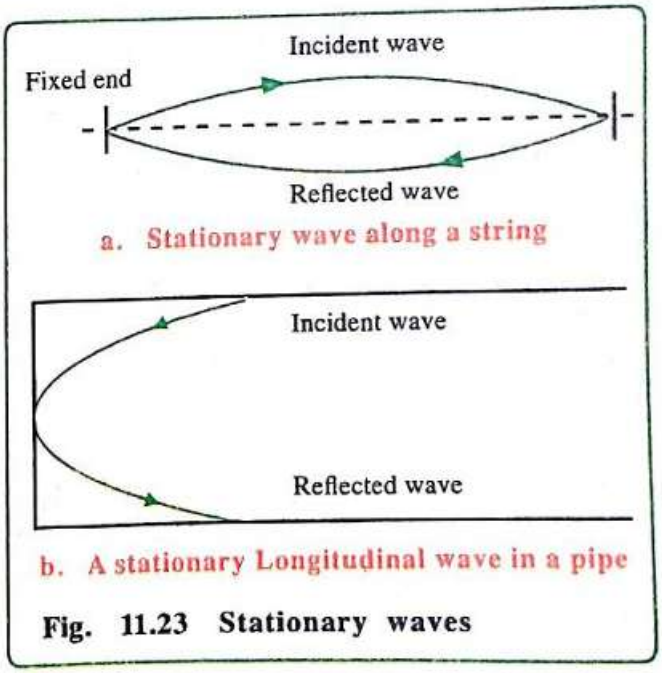


Fig. 11.23 Stationary waves

Let us consider two identical sources of waves vibrating in phase, i.e. they send out wave crest simultaneously and wave troughs simultaneously. Let the waves travel towards the right as shown in fig 11.24.

Let us suppose that (a) the two waves reach a point simultaneously and combine. If the point is the same distance from the two sources of waves, the path difference (Δs) between the two waves is zero, the two waves arrive in phase. The effect is that they reinforce each other, to produce a maximum disturbance at that point. The two waves are said to *interfere constructively*. The same effect is also observed when the path difference of the two waves is a whole number of wavelength. For then the two waves also arrive in phase.

Scanned with CamScanner

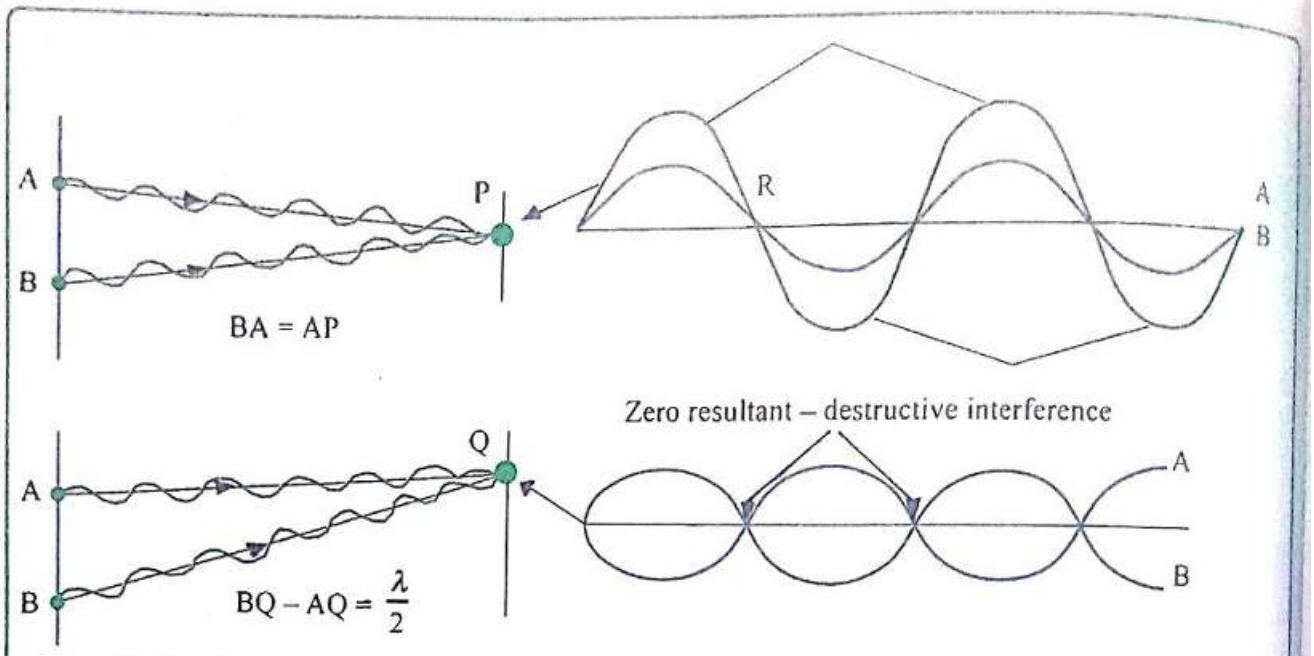


Fig. 11.24 Interference of water waves

Consider another situation where one source is $\frac{1}{2}\lambda$ closer to the point of combination than the other, i.e, path difference = $\frac{1}{2}\lambda$. The two waves no longer arrive in phase but 180° out of phase. The crest from one wave arrives simultaneously with the trough of the second wave. The waves combine to cancel completely the effect of each other. They are said to interfere destructively.

Interference is the effect produced when two waves of the same frequency, amplitude and wavelength travelling in the same direction in a medium are superposed – as they simultaneously pass through a given point.

When the superposition of the two identical waves result in increased disturbance (i.e. they reinforce each other) it is known as *Constructive Interference*. Constructive Interference occurs when the path difference between two identical waves at a point, is

$\Delta s = n\lambda$, where $n = 0, 1, 2 \ldots$

When the resultant effect of the combination of two identical waves result in their annihilation or complete cancellation of the effect of each other (i.e. zero or minimum disturbance), it is called *Destructive Interference*. This occurs when the path difference between the two identical waves at a point is

$$\Delta s = n\lambda + \frac{1}{2}\lambda, \text{ where } n = 0, 1, 2 \ldots$$

Interference can be demonstrated in a ripple tank by vibrating two metal spheres, S_1, S_2 at the same frequency. Circular waves spreading out from S_1, S_2 give rise to wave patterns as shown in fig. 11.25. When the crest of wave from S_1 overlaps the crest of wave from S_2 the resultant effect is a reinforcement of both waves in an increased disturbance. This is known as *Constructive Interference*. It also occurs in the region where troughs from the two waves overlap.

However if a crest from one source is superposed on a trough of equal amplitude from the other source, a region of zero resultant disturbance is created on the water surface. This is *Destructive Interference*.

Two wave sources producing waves that have the same frequency wavelength and amplitude and which are always in phase with each other or have a constant phase difference are said to be *Coherent* sources. Such waves are coherent waves.

Light and sound waves show the interference effect.

Scanned with CamScanner

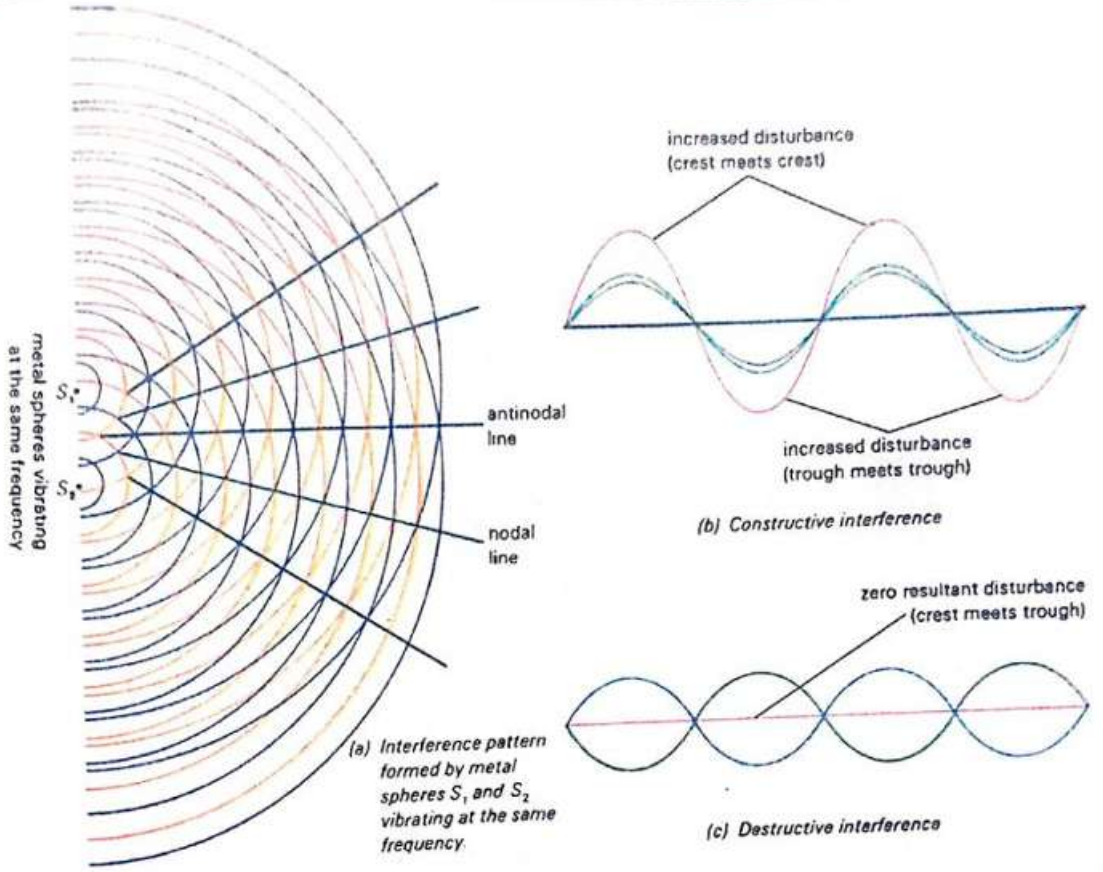


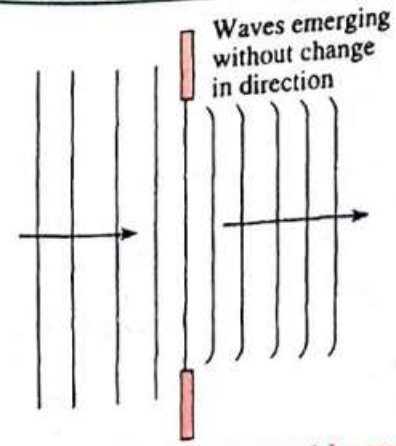
Fig. 11.25 Interference of waves

Diffraction of Waves

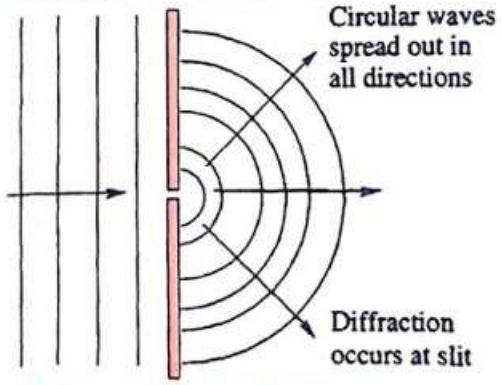
When travelling waves pass through narrow openings or move round obstacles, they tend to spread out in different directions. This phenomenon is known as *Diffraction*. It occurs when the wavelength of the wave is longer than the width of the opening or size of the obstacle.

Diffraction is the ability of waves to bend around obstacles in their path

We can demonstrate diffraction in a ripple tank by the following simple experiment. Two separate strips *A* and *B* are suitably positioned in the ripple tank to provide a small opening for plane waves from a vibrating source to pass through.



a. Plane waves incident on wide opening



b. Plane wave incident on narrow opening

Fig. 11.26

If the opening is fairly wide (about 25 cm) with its width greater than the wavelength of the incident plane waves, the waves emerge through the opening almost unchanged, and in the same direction as the incident wave (fig. 11.26a).

If the opening is made sufficiently narrow (about 1 cm), about the same order as the wavelength of the incident waves, the wave-fronts emerge with a pronounced circular shape and the waves spread out in all directions (fig 11.26b). The waves no longer travel in the same direction from the opening. They are said to be diffracted. Light and sound waves show the diffraction phenomenon.

Scanned with CamScanner

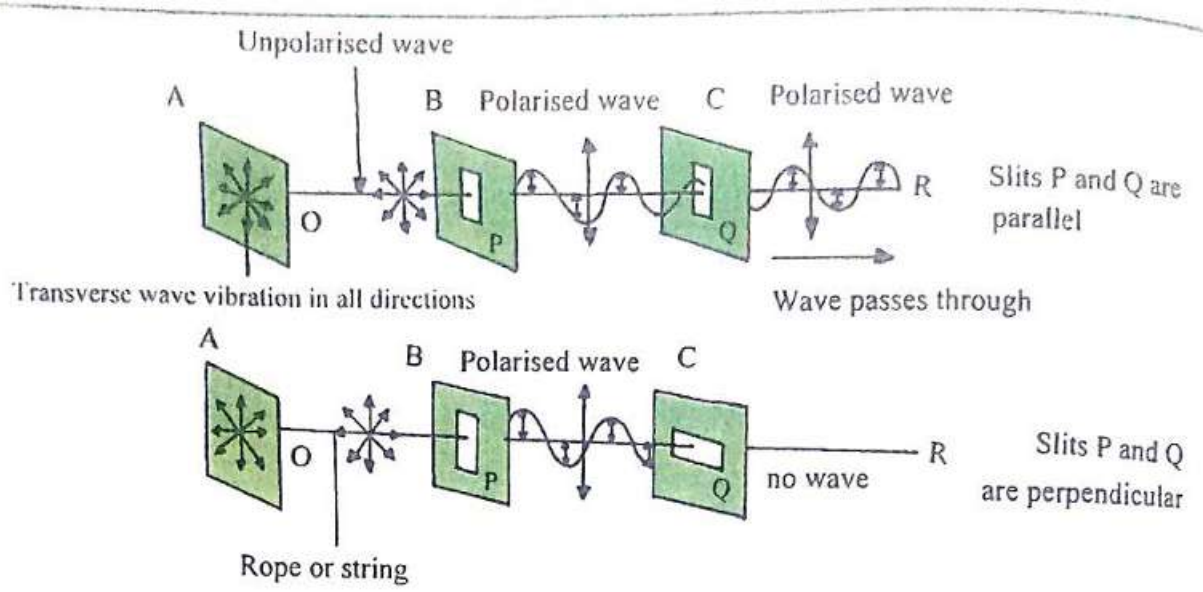


Fig. 11.27 Transverse waves and polarisation

Light waves are diffracted through very narrow openings of about $\frac{1}{5000}$ cm but sound waves are diffracted through wider openings of about 50 cm. These are the order of their wavelengths. The reason why we do not notice the diffraction of light in our daily experience is that the wavelength of light is very small and so the effect is unobservable except when the light passes through very narrow openings.

Polarization of Mechanical Waves

In fig 11.27(a) *B* and *C* are two parallel slits placed along a rope *OR* fixed at *R*. We can produce transverse waves in the rope by shaking it up and down and sideways. The wave moves along the direction *OR* while its vibrations move in different planes perpendicular to *OR* as shown by arrows in the plane of *A*. Such a wave is said to be ***unpolarized*** because its vibrations are in several planes or directions. All these vibrations arrive at *B* but only those parallel to the slit at *B* pass through *B* to *C* and through *C* to *R* as shown in fig 11.27(a).

If the slit at *C* is now put perpendicular to *B* as in fig. 11.27b the vibrations parallel to the slit at *B* pass through to *C* but cannot pass through the slit at *C* to *R*. The wave from *B* to *C* is said to be plane-polarised since it now vibrates only in one plane.

A wave is said to be **plane-polarised** if its vibrations occur only in one plane. This plane is known as the plane of polarisation.

Polarisation is a phenomenon whereby a wave whose vibrations are only in one plane is produced.

Only transverse waves can be plane-polarised. Polarisation therefore occurs with light waves and other electromagnetic waves, e.g. radio waves, x-rays, Infra-red radiation, but not with sound waves because sound waves are longitudinal waves.

Production and Detection of Polarised Light

Polarised light can be produced using polarisers such as Tourmaline crystal, Quartz or a polaroid. Polarisers allow light vibrations of only a particular direction to pass through them. Any other vibrations are absorbed. Thus, to produce a plane-polarised light we allow an unpolarised light (e.g. natural light or candle light) to pass through a polariser, *Q* (fig. 11.28).

To show that the light emerging from *Q* is polarised, another polariser *R* is placed parallel to *Q* and behind it. If *R* is rotated slowly in its vertical plane, about the line of vision, a bright light is seen beyond *R* but the brightness of the light decreases as *R* is rotated becoming darkest when the axis of *Q* and *R* are perpendicular. On further rotation of *R* in the same direction, the emergent light again becomes brighter and brighter and becomes brightest when the axis of *Q* and *R* are

Scanned with CamScanner

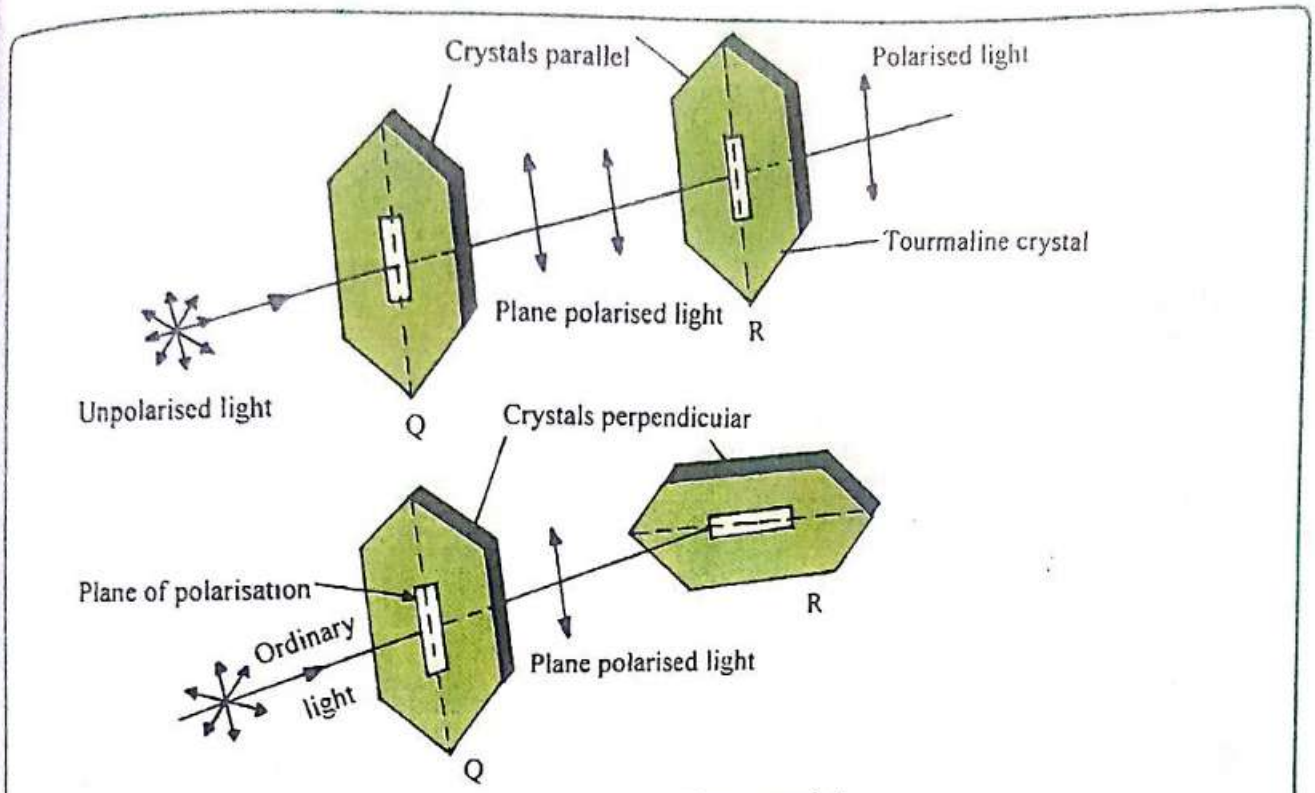


Fig. 11.28 Polarization of light by tourmaline crystal

again parallel. This simple demonstration shows that light waves are transverse waves; otherwise the light emerging from *R* could never be extinguished by simply rotating this crystal.

Thus we can establish whether some light is polarized or not by simply rotating a polarizer (or Polaroid) through which the light is passing. If the light transmitted through the polarizer.

(i) Shows no change in intensity, then the incident light was not polarised.

(ii) Varies slightly in intensity twice per revolution, the incident light was partly plane polarised.

(iii) is cut off completely twice per revolution, the incident light was plane-polarised.

Polarisation by Reflection

Polarisation can also be produced if light is incident at an angle of about 57° on the polished surface of a glass plate. The reflected light is plane-polarised. This is polarisation by reflection.

Practical Application of Polaroids

The polaroid is used in sunglasses to reduce the intensity of incident light and to eliminate reflected light or glare.

Light glare is reflected for example from window panes, glass doors, the road ahead of a driver on a sunny day, or from a polished table top or an open book. This reflected light is polarised light and the glaring can be eliminated by the use of a polaroid.

Summary

1 All waves are due to vibrations. Water and sound waves are due to vibrations of layers of particles in the medium concerned. Light and radio waves which do not require a material medium are due to electromagnetic vibrations.

2 Transverse waves are waves in which the vibrations of the particles are at right angles to the direction of wave motion, or direction of energy flow in the medium. Examples of transverse waves are water waves, light and other electromagnetic waves.

3 Longitudinal waves are waves in which the vibrations of particles are in the same direction as the direction of the wave motion, or direction of energy flow in the medium.

4 The frequency, f, of a wave is the number of complete oscillations per second of the vibrating source producing the wave. It is measured in Hertz.

5 The wavelength, λ, is the distance between successive crests or troughs of the wave.

6 The period T is the time for one complete oscillation. $T = \frac{1}{f}$.

7 For all waves, velocity = frequency × wavelength ($v = f\lambda$).

8 The amplitude of a wave is the maximum displacement of the vibrating particle from its rest position.

9 A wavefront is a line or surface in the path of an advancing wave on which all the particles

Scanned with CamScanner

are in the same phase. The wavefront is always at right angles to the direction of wave motion.

10 The properties of waves are reflection, refraction, diffraction interference and polarisation. Light and other electromagnetic waves can be polarised because they are transverse waves; sound cannot be polarised because it is a longitudinal wave.

11 *Refraction* is due to the change of speed when the wave travels from one medium to another. *Diffraction* is the spreading of waves round corners or around obstacles. It is more noticeable when the wavelength is longer than the size of the obstacle or the width of the opening. Light is diffracted by very narrow openings of the order of the wavelength (about 6×10^{-7} m). Sound waves are diffracted by wider openings (~ 50 cm).

12 Interference effects are produced when two close sources emitting waves of the same frequency and amplitude and wavelength (coherent sources) are superposed. In *Constructive Interference*, the crests of the two waves or their troughs overlap reinforcing each other. In *Destructive Interference*, the crest from one wave overlap the trough from the other equal amplitude, and they cancel the effect of each other.

13 A stationary or standing wave is due to the interference of two waves travelling in opposite direction, e.g a travelling wave and its reflection.

14 A wave is plane-polarised when its vibrations are only in one plane. Polarised light can be produced using tourmaline crystals, quartz or polaroid. Polaroid is used in sunglasses to reduce the sun-glare. Polarisation is the phenomenon of producing transverse vibrations which are only in one plane.

Exercise 11

1 A radio station broadcasts at a frequency of 600 kHz. If the speed of light in air is 3×10^{8} ms^{-1}, calculate the wavelength of the radio wave.

A 22.0×10^{3} m B 5.0×10^{2} m

C 5.0×10^{5} m D 11.8×10^{11} m

2 Which of the following is not a property of longitudinal waves?

A Compression B Reflection
C Refraction D Polarisation
E Diffraction.

3 The maximum displacement of particles of a wave from their equilibrium positions is called

A Wave velocity B Period
C Amplitude D Wavelength
E Frequency

4 What is the distance between a node and an antinode for a transverse wave?

A the wavelength
B twice the wavelength
C three quarters of the wavelength
D one-half of the wavelength
E thrice the wavelength

5 Which of the following statements about wave is/are correct?

I. A wavefront is a line which contains all particles whose vibrations are in phase.
II. The direction of propagation of a wave is the line drawn parallel to the wavefront.
III. A wavefront is a circle which is common to all particles that are in the same state of disturbance.

A I only B II only
C III only D I and II only
E I and III only

6 Which of the following is not a mechanical wave?

A Wave propagated in stretched string
B Waves in closed pipes
C Radio waves
D Water waves
E Sound waves

7 A wave sent out from a source is reflected back to the source in 1.0×10^{-3} s. If the velocity of the wave is 3.0×10^{8} ms^{-1}, the distance of the reflecting surface from the source is

A 1.5×10^{2} m B 3.0×10^{2} m

C 3.0×10^{5} m D 1.5×10^{5} m

E 6.0×10^{5} m.

8 If V is the velocity of a wave, λ its wavelength and T its period, the V, λ and T are related by the expression

A $\lambda = VT$ B $\lambda = \frac{V}{T^2}$

C $T^2 = \lambda V$ D $V = \lambda T$ E $V = \frac{T}{\lambda}$

9 Transverse and longitudinal waves travelling in the same direction in a medium differ essentially in their

A Wavelength B Amplitude
C Frequency
D direction of vibration of the particles of the medium
E Period of vibration of the particles of the medium.

10 Explain the terms, *frequency*, *wavelength*, *velocity*, *period* as applied to wave motion. Sketch a waveform and label the amplitude, wavelength, crest and trough. What is the relationship between wavelength, frequency and velocity?

Scanned with CamScanner

11 Distinguish between transverse and longitudinal waves. Give two examples of each. How would you produce each type of wave using a slinky? Illustrate with diagrams showing the following: crest, trough, compression, rarefaction.

12 Distinguish between progressive and standing waves. How are they produced? Draw a diagram of a stationary wave indicating the node and antinode. What is the distance between two nodes, and between a *node* and an *antinode* in terms of the wavelength of the wave?

13 Explain the terms: *Diffraction*, *Refraction*, *Interference* and *Polarisation*.

Plane waves are incident on a narrow gap. Draw the appearance of the waves or wavefronts before and after they have passed through the gap (i) when the wavelength of the waves is very small (ii) when the wavelength is large. Which of the italicized phenomena above is represented by this effect?

14 What is a plane-polarised wave? How can plane-polarised light be produced and detected?

15 What is a wave-motion?

The equation, $y = A\sin\frac{2\pi}{\lambda}(Vt - X)$ represents a wavetrain in which y is the vertical displacement of a particle at distance X from the origin in the medium through which the wave is travelling. Explain, with the aid of a diagram, what A and λ represent? (SSCE)

16 What do you understand by *interference of waves*? Distinguish between *constructive interference* and *destructive interference*. What are the necessary conditions for two wave trains to have continuous interference so as to produce no effect at certain points? What is meant by *path difference* between two waves? Describe the significance of part difference in the interference of waves.

17 How can plane waves be produced in a ripple tank? How can refraction of waves be demonstrated in such a tank?

In a ripple tank with a 20 Hz vibrator, the wave speed is 0.3 ms^{-1} until the waves reach a shallower region where the speed is 0.2 ms^{-1}.

a. What is the wavelength in each of these parts of the tank?

b. Plane wave-fronts travelling from the deeper part make an angle of 60° with the boundary; at what angle to the boundary do they travel in the shallower part?

18 How can you demonstrate a mechanical analogue of polarisation? Explain the uses of polaroids.

19 Define refractive index for waves passing from one medium to another in terms of (a) angles; (b) velocities.

Water ripples are caused to travel across the surface of a shallow tank by means of a suitable straight vibrator. The distance between successive crests is 4.0 cm and the wave travels 30 cm in 1.5 s. Calculate the *wavelength* and *velocity* of the waves and the *frequency* of the vibrator. Explain the three terms in italics.

20 Fig. 11.29 shows a wave on a cord at a certain instant. The oscillator that sends out the wave is vibrating at 100 Hz. Calculate the following parameters for the wave

A Amplitude B Wavelength
C Period D Speed.

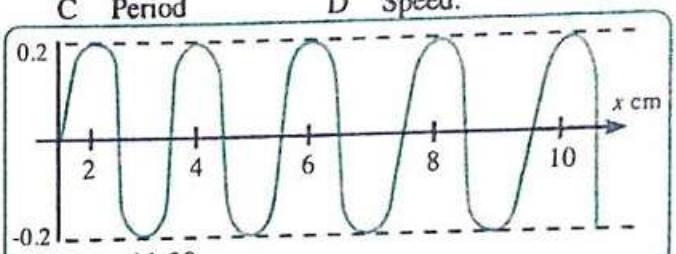


Fig. 11.29

21 List two types of waves apart from light that can be plane polarized. State two application of plane polarized light.

22 (a) Explain (i) Wave Motion (ii) Stationary wave

(b) (i) List four physical properties of a wave.

(ii) Define amplitude and use it to distinguish between the node and antinode of a stationary wave.

(iii) List the factors on which the frequency of vibration in a stretched string depends.

(c) The equation y = 5 Sin (3x - 4), where y is in millimeters, x is in metres and t is in seconds, represents a wave motion. Determine the (i) frequency (ii) period (iii) speed of the wave. (SSCE)

23 (a) Explain (i) refraction of wave (ii) Critical angle

(b) State two conditions necessary for (i) total internal reflection of a wave to occur, (ii) interference wave patterns to be formed.

(c) The distance between two successive crests of a water wave traveling at 3.6ms^{-1} is 0.45m. Calculate the frequency of the wave. (SSCE)

24 The equation $y = a\ sin\ (wt - kx)$ represents a plane wave travelling in a medium along the x- direction , y being the displacement at the point x at time t.

i. Given that x is in metres and t is in seconds, state the units of k and w.

ii What physical quantity does w/k represent? Justify your answer.

iii State whether the wave is travelling in the positive or negative x- direction.

Scanned with CamScanner

12 REFLECTION OF LIGHT WAVES

In chapter 11 we discussed the general properties of mechanical waves including reflection, refraction, diffraction, interference and polarization. Water, light and sound waves were used to illustrate these properties. This chapter will concentrate on the reflection of light waves only.

At the end of the chapter, the student should be able to

1 List different sources of light
2 Determine the angle of reflection for a given angle of incidence
3 Draw ray diagrams to show the formation of images by plane and curved mirrors and explain some practical applications of these mirrors.

12.1 Sources of Light

Light is a form of energy called *luminous energy*. This energy causes a sensation of vision, enabling us to see. There are various sources of light, for example the sun and the stars are natural sources of light. Artificial sources of light are the candle, electric torch, the electric lamp, incandescent and arc lights and fluorescent light.

Self luminous or luminous sources of light are those that generate and emit light by themselves. Examples are the sun, stars, fire-flies and some deep-sea fishes, and the artificial light sources. *Non-luminous* bodies depend on natural or artificial light sources to illuminate them. They are seen only when they reflect the light from a luminous body. For example, light from a car headlamp falling on a road sign in the night, causes the sign to throw back part of this light into the eyes of the car-driver thereby enabling the road sign to be seen. The road sign is a non-luminous body, the head-lamp is a luminous body, an artificial luminous body. Examples of non-luminous bodies are a page of a book, a person's face, a brick, and the moon. The sun's rays illuminate the moon and make it to appear luminous in the night.

12.2 Transmission of Light

Light is an electromagnetic wave. It can pass through a vacuum and through a material medium. If light shines on a body, part of the light is transmitted through the body, the rest is reflected. The amount of light passing through a body depends on the nature of the body. If a large percentage of light falling on a body passes through it, the body is said to be *transparent*. Examples of transparent bodies are glass and water. Because light is easily transmitted through these transparent bodies, we can see objects through them easily. Some objects like frosted glass and tissue paper allow some small amount of light to pass through them. Such objects are called translucent objects. Because the amount of light passing through *translucent* bodies is small, objects cannot be seen clearly through them. Some other bodies do not allow any light to pass through them. These are called *opaque* objects. Examples of opaque objects are wood, bricks, walls, and metal sheets.

Rays and Beams of Light

A light *ray* is the direction or path along which light energy flows. Such rays are indicated in diagrams by thin lines with arrow heads which indicate the direction of travel of the light. A collection of rays is called a *beam*. There are three types of beams.

(i) A *parallel beam* is one in which the light rays are parallel to one another. Search-lights give off parallel beams of light.
(ii) A *convergent beam* is one in which the rays converge or meet at a point. A hand lens can be used to produce such a beam.
(iii) A divergent beam is one in which the light rays all come from a point and spread out or diverge from the source. Lamps produce a divergent beam of light.

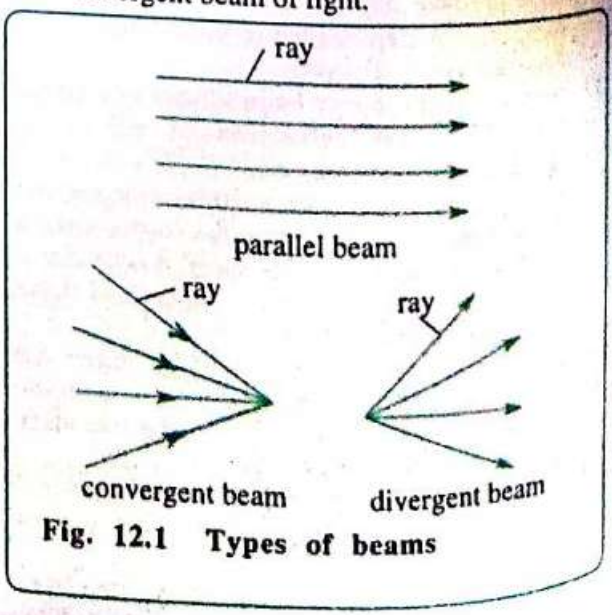


Fig. 12.1 Types of beams

Scanned with CamScanner

The Ray Box

Rays of light are produced in the laboratory by means of a ray box. A simple laboratory ray box consists of a box made of wood or cardboard inside which is a source of light, e.g., a candle or an electric lamp. On one face of the box is a hole or a slit or several slits through which light rays can travel out of the box.

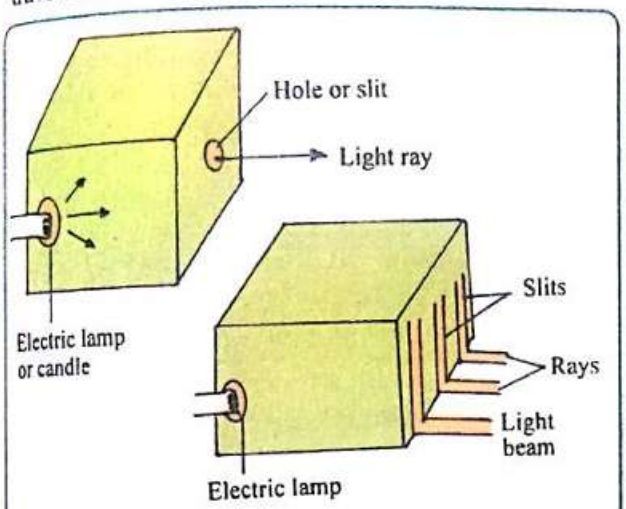


Fig. 12.2 The Ray Box

12.3 Rectilinear Propagation of Light

Light travels in straight lines as can be shown by the following simple demonstration using two cardboard pipes one straight and the other bent. If we place a candle flame on one end of each pipe and view through the other end, light from the candle flame will be seen clearly on the end of the straight pipe but no light will be seen at the end of the bent pipe.

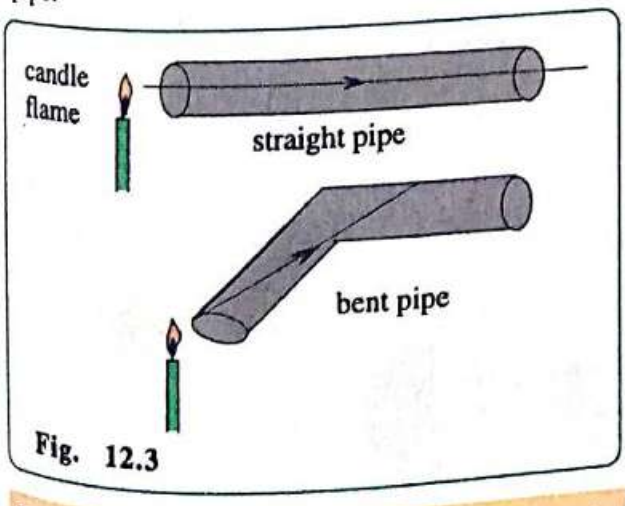


Fig. 12.3

The phenomenon of light travelling in straight lines is known as rectilinear propagation of light

Another demonstration of rectilinear propagation of light is the following simple experiment. We punch a tiny hole in the centre of each of three equally sized cardboards which are then arranged in a straight line in front of each other. A string is passed through the holes and pulled taut to ensure that the holes are aligned in a straight line.

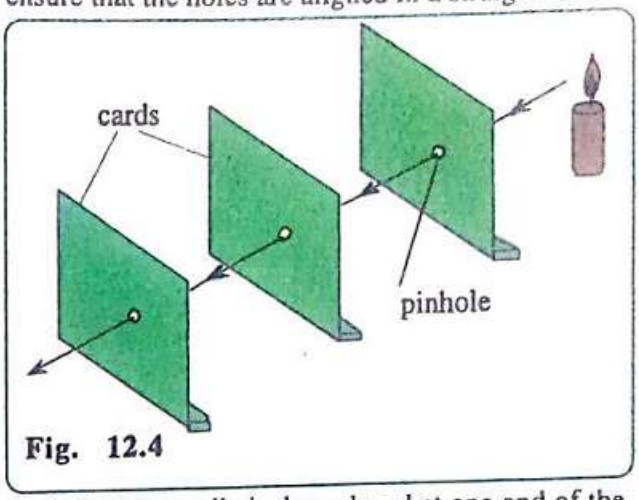


Fig. 12.4

Light from a candle is then placed at one end of the row of cards. When we look through the other end of the row, we will see the light from the candle flame through the holes. If either cardboard is slightly shifted out of the straight line position, the light is cut off.

Two natural effects that result from the rectilinear propagation of light are the formation of *shadows* and *eclipses*. The principle of operation of the pin-hole camera also depends on the fact that light travels in straight lines.

Shadows

A shadow is an area in which light rays from a source cannot reach. A shadow is produced by the obstruction of light by an opaque object. The shade under a tree or a canopy, on a bright sunny day is a shadow produced by the opaque tree or canopy which obstructs light rays from the sun and prevents them from getting to the area of the shade.

Fig. 12.5 shows how shadows are formed. In fig. 12.5a, we use a small light source known as a *point source* of light. If an opaque object e.g. a coin or a solid ball is placed between the point source and a screen, a circular shadow is cast on the screen. The shadow takes the shape of the opaque object producing it. The shadow cast by a point source has sharp edges.

Scanned with CamScanner

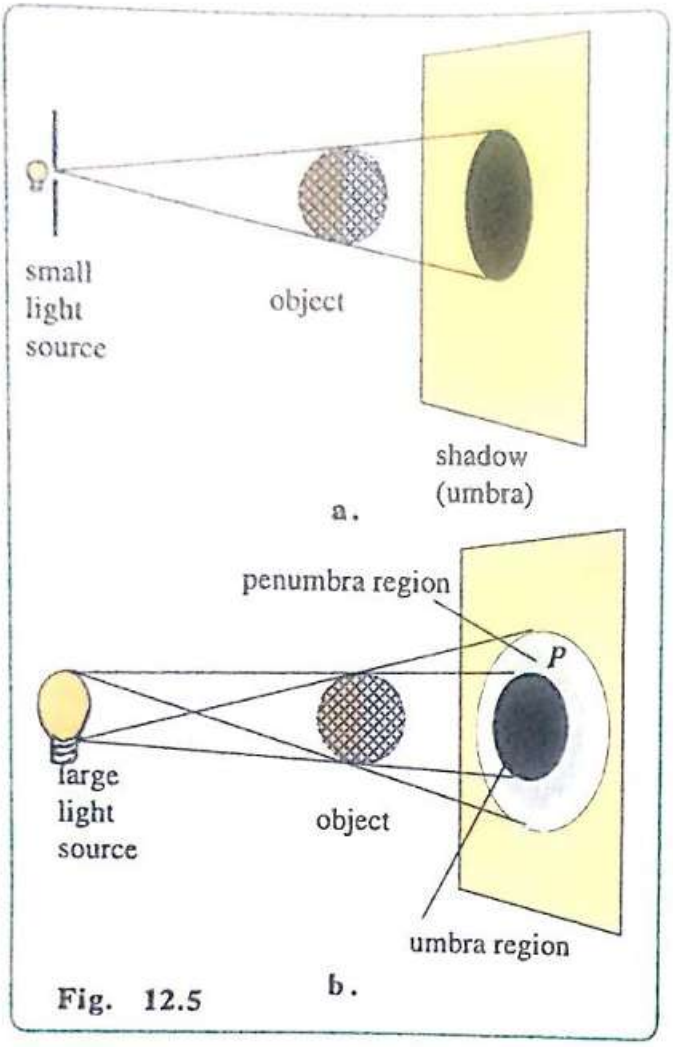


Fig. 12.5

If the light source is large, or extended the shadow formed consists of two parts, a completely dark area known as the *Umbra* and an outer light or grey area known as the *Penumbra* or partial shadow. In the Umbra region, the light from the source is completely blocked by the opaque body. In the Penumbra region the light is partially blocked by the opaque object. The inner region of the shadow receives less light than its outer parts. Thus the penumbra becomes brighter from the umbra end outwards.

Eclipses

An eclipse is a result of a shadow cast by one heavenly body on another. While the Sun is a luminous body, i.e, it produces its own light, the Moon and the Earth are both non-luminous The Moon revolves round the Earth, and the Earth also revolves round the Sun. At certain times during these movements, the three bodies all lie in a straight line. If the Moon is between the Sun and the Earth, the shadow of the Moon will be cast on the Earth's surface. If we are living in the part of the Earth that is in the Moon's shadow, the light from the Sun is cut off from view and we are in darkness. Since we cannot see the Sun, we call this *an Eclipse of the Sun* or *Solar Eclipse*.

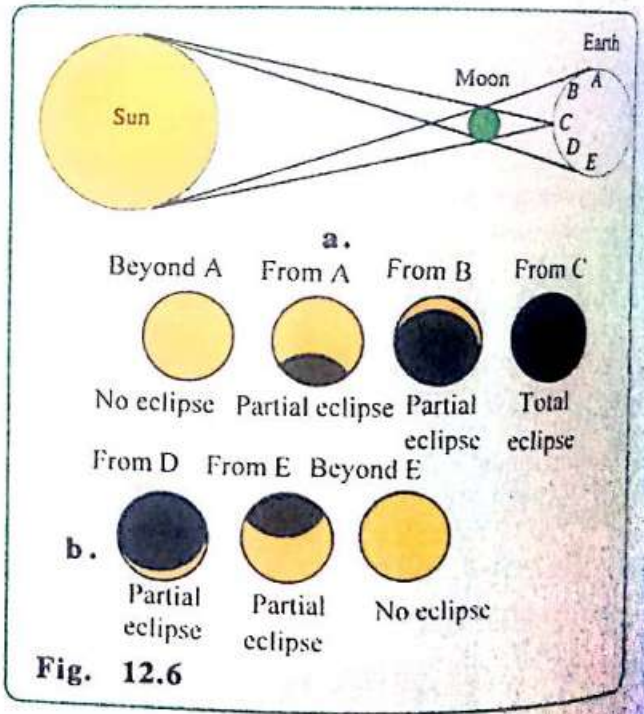


Fig. 12.6

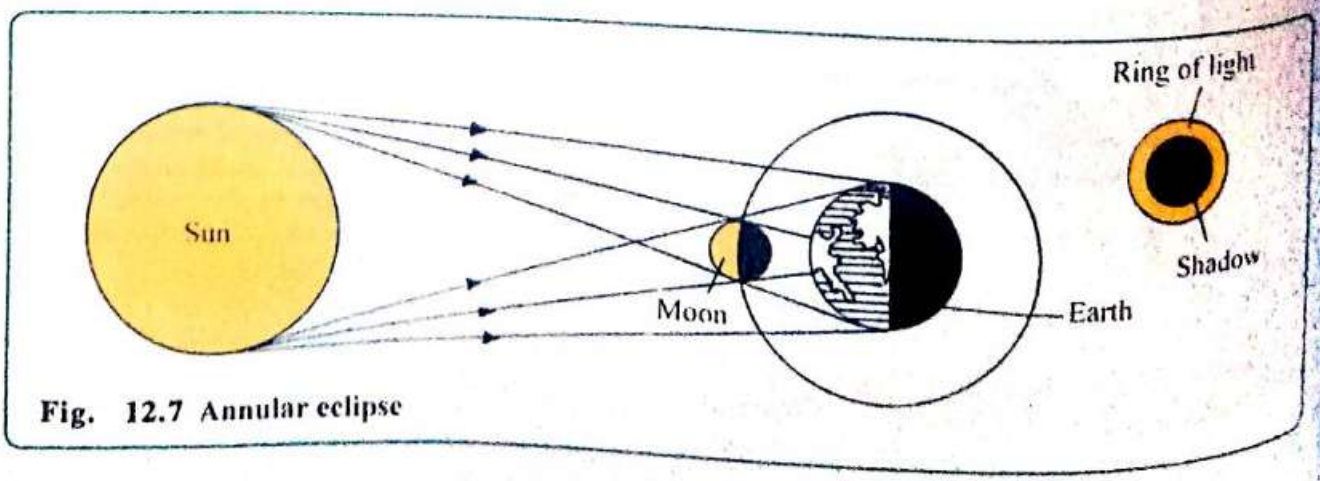


Fig. 12.7 Annular eclipse

Scanned with CamScanner

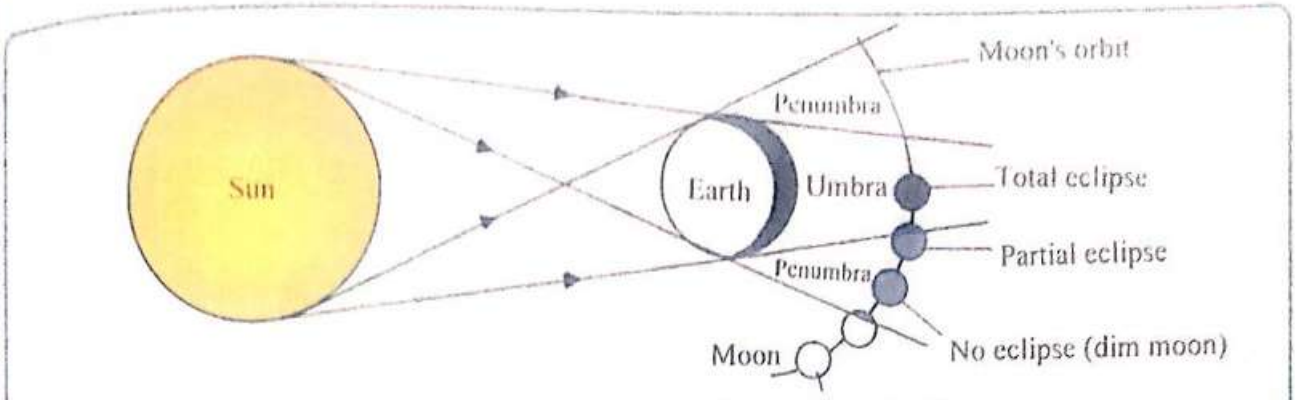


Fig. 12.8 Eclipse of the moon

This usually lasts for just a few minutes. During this period, a portion of the earth is in the shadow cast by the Moon. It is a *total eclipse* for people living in the umbra region of the shadow, and for those in the penumbra regions of shadow, it is a *partial eclipse*. For them the Sun is only partially covered. Fig. 12.6(b) shows the appearance of the Sun as seen from the various regions of the earth.

The Earth and the Moon may be in positions where the extreme rays at the Moon's edge intersect before reaching the Earth. Under such condition, an *annular eclipse* is formed. A ring of light is formed around the shadow of the Moon.

When the Sun, the Earth and the Moon are in line during their movements, with the Earth lying between the Sun and the Moon, the earth being an opaque object casts a shadow on the moon and an *eclipse of the Moon* or lunar Eclipse occurs. In the eclipse of the Moon, the Moon is in the Earth's shadow. As in the solar eclipse, lunar eclipse can be total or partial.

The Pinhole Camera

The Pinhole Camera makes use of the fact that light travels in straight lines. It consists of a light-proof box, one end of which has a small hole made with a pin or needle point. The opposite end has a screen made of tracing paper or ground glass. Light from an object in front of the pinhole passes through it and forms an image on the screen. If the screen is replaced with a photographic paper or film, a picture of the object can be taken with the pinhole camera.

The camera forms an image of an illuminated object AB, in the following way: Each point in the illuminated object AB acts as a point source of light. Light rays from the point A pass through the hole P, and fall on the screen at A'. Rays from B pass through P and fall on the screen at B'.

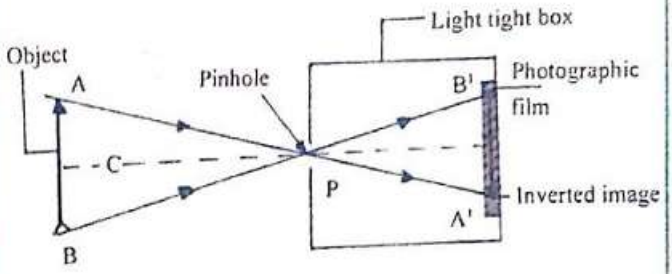


Fig. 12.9 The pinhole camera

Similarly, light rays from all other points on the object will fall on corresponding points between A' and B', forming a complete image of AB. This image is inverted. If the pin-hole is small, the image is sharp, though not very bright since the amount of light entering the camera is small. A brighter image can be obtained by having a larger hole. More light then enters the box to form the bright image but this image is blurred. The wide hole acts like a large number of pin-holes, each forming its own image. These images overlap and are seen as a single blurred image.

If the distance between the pin-hole and the screen is increased the image is enlarged but less bright. The decrease in brightness is as a result of the distribution of the same amount of light over the greater area of the enlarged image.

Scanned with CamScanner

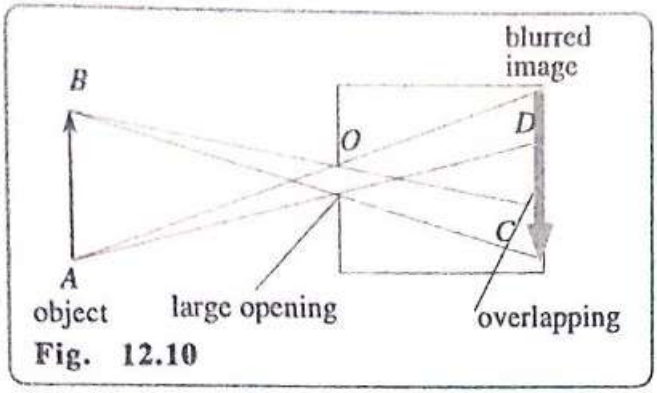


Fig. 12.10

When using the pinhole camera to take pictures of an object, long exposures are necessary to allow sufficient light to enter the box through the pinhole. This long exposure is not good for the film. Snapshots can be taken only with a lens camera (see p. 310) since the larger aperture of the lens admits more light energy per second than a pinhole. A pinhole camera requires no focusing as does the lens camera, but it can take pictures of still objects.

The image formed on the screen of the pinhole camera will be seen more clearly if external light is excluded by covering head and camera with a dark cloth.

Magnification produced by the Pinhole Camera

Magnification is defined as the ratio of the size (or height) of the image to the size (or height) of the object

Consider the ray diagram of the formation of an image by a pinhole camera fig. 12.11.

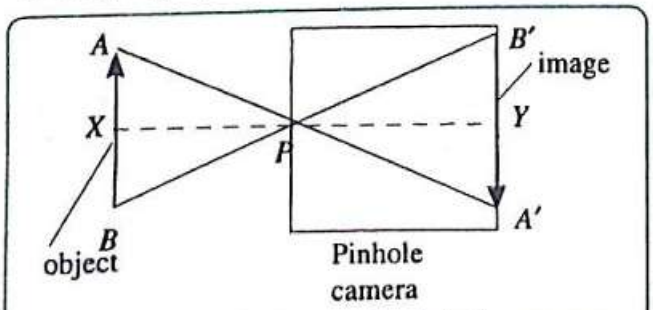


Fig. 12.11 Principle of pinhole camera

Magnification, M,

$$= \frac{\text{Size of image}}{\text{Size of object}} = \frac{B'A'}{BA}$$

The triangles ABP and $A'B'P$ are similar triangles.

Hence we can write

$$\frac{B'A'}{BA} = \frac{B'P}{BP} = \frac{A'P}{AP} \qquad 12.1$$

Also from the similar triangles XPB and YPB', we can write

$$\frac{YP}{XP} = \frac{B'P}{BP} \qquad 12.2$$

$\therefore$ From equations (12.1) and (12.2) we have

$$\frac{YP}{XP} = \frac{B'P}{BP} = \frac{B'A}{BA}$$

YP = distance of image from pinhole

XP = distance of object from pinhole

$\therefore$ Magnification, $M = \dfrac{\text{Size of image}}{\text{Size of object}}$

$$= \frac{\text{height of image}}{\text{height of object}}$$

$$= \frac{\text{distance of image from pinhole}}{\text{distance of object from pinhole}}$$

$$= \frac{\text{length of camera}}{\text{distance of object from pinhole}}$$

Example 12.1

The height of an object placed at a distance of 60 cm from the hole of a pinhole camera is 15 cm. If the length of the camera is 10 cm, calculate the magnification produced by the camera and the height of the image.

Solution

$$M = \frac{\text{length of camera}}{\text{distance of object from pinhole}}$$

$$= \frac{10}{60} = \frac{1}{6} = 0.1666$$

$$\text{Also } M = \frac{\text{height of image } (h')}{\text{height of object } (h)}$$

$$\frac{1}{6} = \frac{h'}{15 \text{ cm}}$$

$$h' = \frac{15}{6} = \underline{2.5 \text{ cm}}$$

Example 12.2

What is the width of the image of an object which subtends an angle of 8° at the pinhole of a camera 0.5 m long? How wide is the image of the sun in this camera. Take the mean distance of the Sun as 1.5×10^{11} m, and diameter of sun as 1.4×10^{9} m.

Solution

Let width of image = h (m)

$$\tan 8^\circ = \frac{h}{0.5 \text{ m}}$$

$$h = 0.5 \times \tan 8^\circ = 0.5 \times 0.1405 \text{ m}$$

$$= \underline{7 \text{ cm}}$$

Scanned with CamScanner

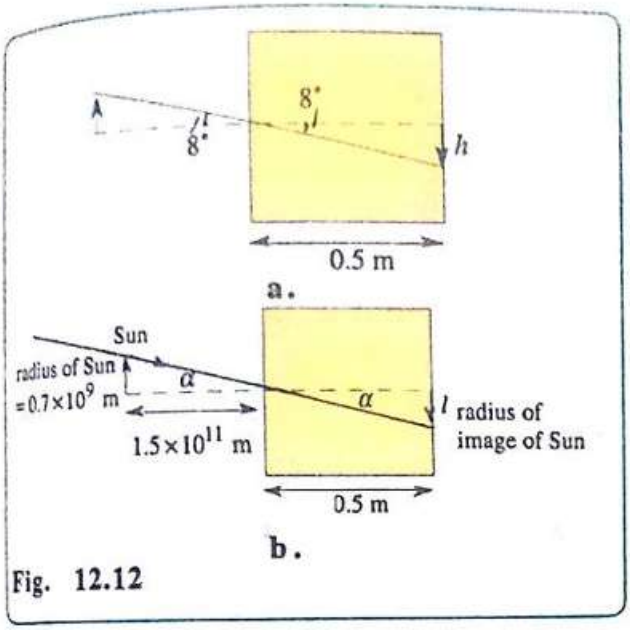


Fig. 12.12

Let radius of image of the sun be l (m)

$$\frac{l}{0.5} = \frac{0.7\times10^{9}}{1.5\times10^{11}} = \tan\alpha$$

$$l = \frac{0.5\times0.7\times10^{9}}{1.5\times10^{11}} = 0.233\times10^{-2} \text{ m}$$

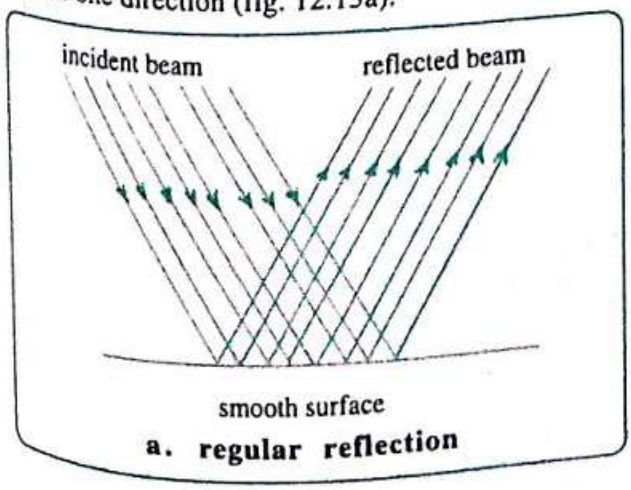
$$= \underline{0.23 \text{ cm}}$$

$\therefore$ diameter of image $= \underline{0.46 \text{ cm}}$

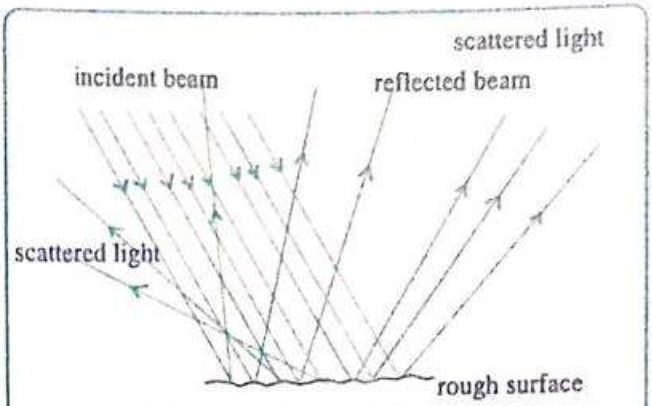
12.4 Reflection of Light at Plane-surfaces

Non-luminous objects are seen when they reflect or throw back light from a luminous object like a candle flame or the sun. When light falls on such surfaces, it is either absorbed, transmitted or reflected. Sometimes a combination of the above processes may occur.

There are two types of reflection - *Regular reflection* and *Scattered or diffuse reflection*. In regular reflection, parallel rays of light incident on a smooth or polished surface are reflected as parallel rays in one direction (fig. 12.13a).



a. regular reflection



b. irregular reflection

Fig. 12.13 Regular and diffuse reflection

In diffused or irregular reflection, parallel rays of light incident on a rough or irregular surface are reflected in various directions.

Mirrors produce regular reflection, but a page of a book or the surface of cloth give rise to scattered or diffuse reflection.

Laws of Reflection

A plane mirror is produced by coating one side of a plane glass plate with silver. The silvered surface prevents light rays that strike the glass from passing through the glass and serves to reflect the light. We can investigate the laws of reflection by the following experiment:

Activity 12.1

1 Fix a sheet of paper firmly on a drawing board. Mount a plane mirror vertically along a line AB on the paper (fig. 12.14). Draw a line ON to bisect AO at right angles. Draw lines on the paper making angles such as 20°, 30°, 40°, 60°, 80°, to ON. The line ON that is perpendicular to the mirror at N is called the Normal to the mirror.

2 Direct a narrow pencil of light with a ray box, on to the mirror along the line making an angle of 20° with ON. This is the *incident ray*, CN. Mark the reflected ray ND with two dots. Draw a line to join the two dots. This line represents the *reflected ray*,

3 Repeat the experiment for other angles. The angle (i) between the incident ray and the normal is called the *angle of incidence*. The angle between the reflected ray and the normal is called the *angle of reflection* (r).

By measuring this set of angles for each experiment, we observe that the angle of incidence is always equal to the angle of reflection.

Scanned with CamScanner

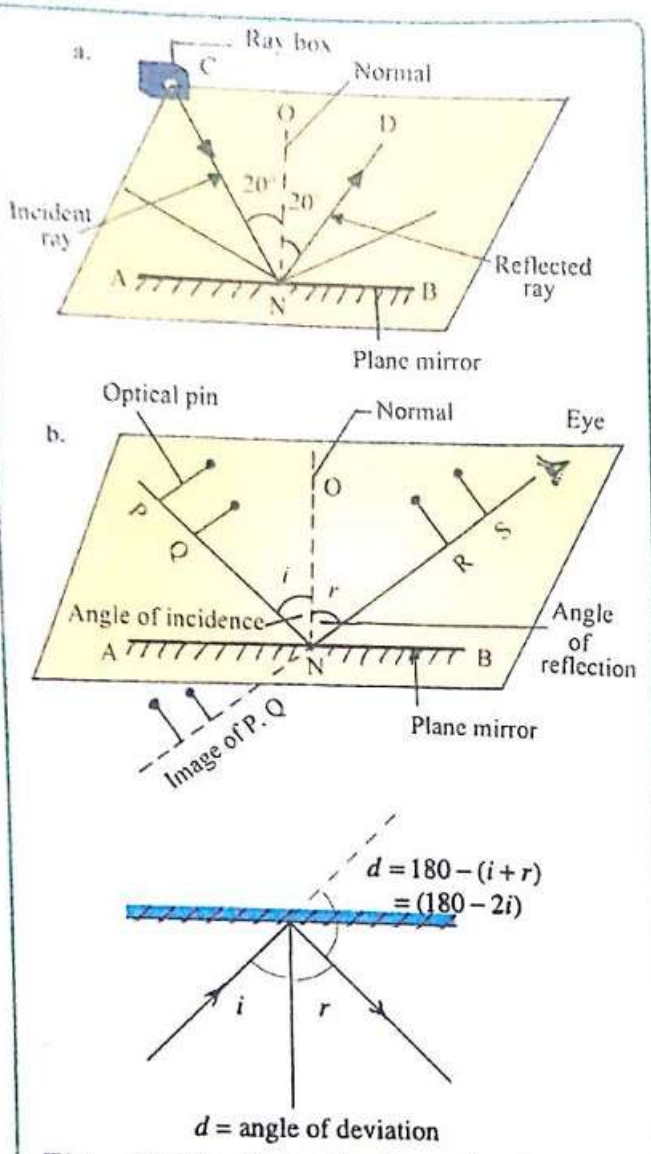


Fig. 12.14c Investigating the laws of reflection

The above experiment can also be performed using optical pins instead of a ray box. Two optical pins P and Q are fixed in front of the mirror along the line making an angle of 20° to ON. Two other pins R, S are used to fix the two points in front of the mirror where the eye sees them in line with the images of P and Q in the mirror. The line RS represents the reflected ray. The angles of incidence and reflection are measured as before.

The Laws of Reflection

1. The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.
2. The angle of incidence(i) is equal to the angle of reflection (r).

From fig. 12.14(c) we see that when an incident ray is reflected by the mirror, the ray is deviated from its original path through an angle, $d = 180 - (i + r)$. Since from the law of reflection, $i = r$, the angle through which the incident ray is deviated is $180 - 2i$.

Principle of Reversibility of Light

There is a general principle in optics known as the *principle of reversibility of light*. This principle states that the path of a light ray is reversible. This is easily seen when we consider the reflection by a plane mirror shown in fig 12.15.

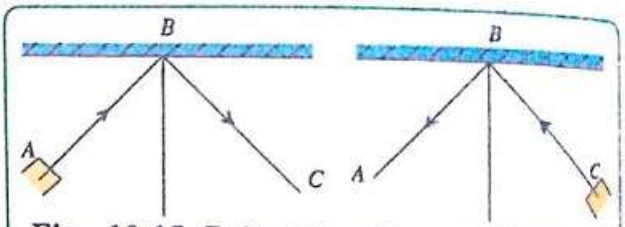


Fig. 12.15 Principle of reversibility of light

When a light ray is incident on the plane mirror at B along AB, it will be reflected along BC in accordance with the laws of reflection. Similarly if the light ray is incident along CB, it will be reflected along BA. Thus the path of the light rays are the same and hence the incident and reflected rays can be interchanged.

Image Formation by a Plane Mirror

Consider a candle flame placed in front of a plane-mirror in fig. 12.16.

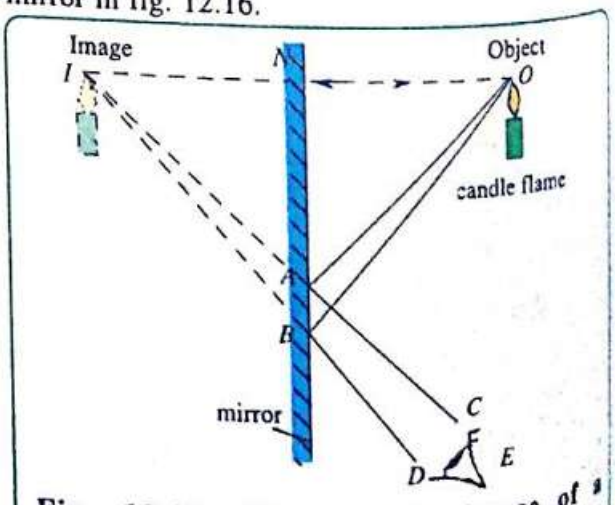


Fig. 12.16 Viewing the image of a candle in a plane mirror

An eye at E sees the upright image of the candle flame as if it were located at I behind the mirror. Two incident rays OA and OB from the tip O, of

Scanned with CamScanner

the candle flame strike the mirror at A and B and are reflected as AC and BD respectively. After reflection the rays appear to the eye at E as coming from I, the tip of the image of the candle flame in the mirror. I is the point of intersection of the two reflected rays when produced backwards. A ray ON normal or perpendicular to the mirror is reflected back along NO, but ON produced meets the other two rays at I such that $ON = NI$.

As no rays actually come from I since AC and BD are actually diverging from each other, the image in the mirror is described as a *Virtual image*. Such an image appears to be at I but it is not actually there since it cannot be caught on a screen placed at I. This type of image is to be distinguished from the image formed by a pin-hole camera which is a *real image* and can be seen on a screen. A virtual image has no real existence.

A **virtual image** is one that cannot be caught on a screen. It is one through which rays of light do not actually pass but which is nevertheless visible to the eye

A **real image** is one that can be caught on a screen. Light rays actually pass through a real image.

Activity 12.2: To locate the image of an object behind a plane mirror

1 Stand a plane mirror on a flat board and fix an optical pin, O, to stand erect in front of the mirror. Observe the image of O in the mirror.
2 Stand two other pins P and Q in line with the image I seen in the mirror. Repeat with two other pins R and S in a different position.

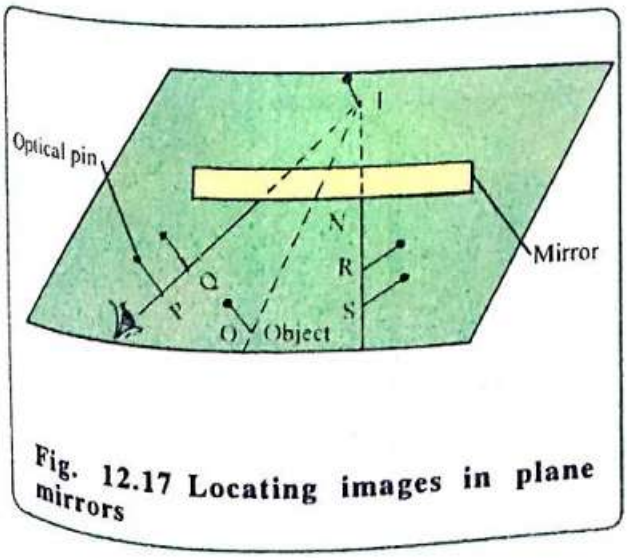


Fig. 12.17 Locating images in plane mirrors

3 Remove the pins on the mirror and produce the lines PQ and RS. Their point of intersection behind the mirror is the position of I. (fig. 12.17)
4 Measure ON and NI.

We can also locate the image of the object by the method of *non-parallax*.

1 Stand the plane mirror on the flat board with the reflecting surface of the mirror vertical.
2 Fix the object pin in front of the mirror.
3 Place a second pin known as the *search pin* or *image pin* behind the mirror so that it appears to be in the same position as the image of the object pin. At this position, there is no relative motion between the top of the image or search pin and the bottom part of the image of the object pin as the observer moves his eyes from side to side.

When there is no relative motion between two pins or two objects as the eyes are moved, we say there is *no parallax* between them. But if the objects appear to move relative to one another as the observer moves his eyes, we say there is parallax between them.

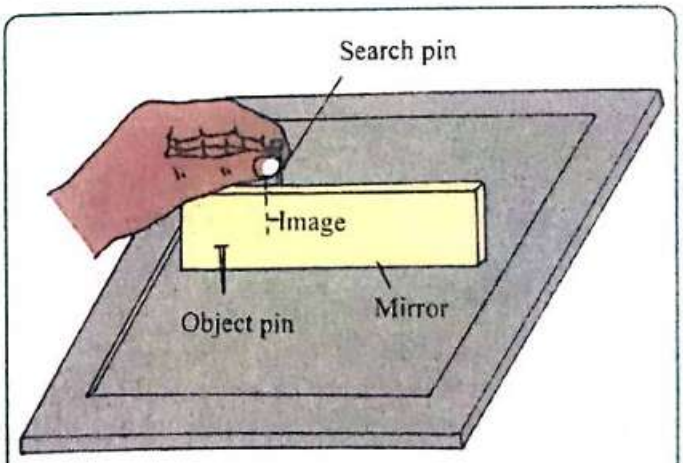


Fig. 12.18 Finding image position by non-parallax

Parallax is the apparent relative movements of two aligned objects when they are not at the same distance from an observer.

From both experiments, it will be seen that the object distance in front of the mirror equals the image distance behind the mirror i.e. $ON = NI$.

Scanned with CamScanner

Lateral Inversion

If we stand in front of a plane mirror e.g. a dressing mirror and raise our right palm, the image in the mirror appears like a left palm. The student should try this out. He should also try to 'shake hands' with this image hand in the mirror and the effect will be most glaring.

Also if we write some alphabets (e.g. *MAP*) on a piece of paper and place the paper in front of the mirror as shown in fig. 12.19, the letters appear to be turned around.

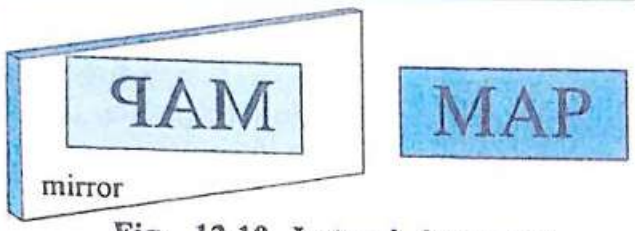


Fig. 12.19 Lateral inversion

This effect of the plane mirror on objects placed in front of it whereby the appearance of the image looks like a reversal of the object is known as *lateral inversion*. Lateral inversion is due to the fact that an object is perpendicularly opposite its image behind the mirror.

As a summary we note the following characteristics of the image formed by a plane mirror:

1 It is the same size as the object
2 It is virtual
3 It is laterally inverted
4 It is as far behind the mirror as the object is in front of the mirror
5 It is upright.

Images formed by Inclined Mirrors

When two mirrors are placed at an angle to each other, an observer sees several images of an object placed in front of the mirrors. The number of images formed depends on the angle between the two mirrors according to the formula.

$$n = \frac{360}{\theta} - 1 \qquad 12.4$$

where n is the number of images, and θ is the angle between the two mirrors. When $\theta = 90°$, $n = \frac{360}{90} - 1 = 3$ images. This case is illustrated in fig. 12.20. The ray diagram for the formation of the three images is as shown in fig 12.21.

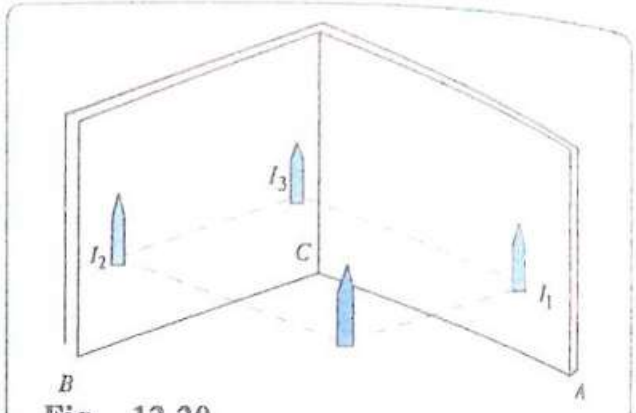


Fig. 12.20

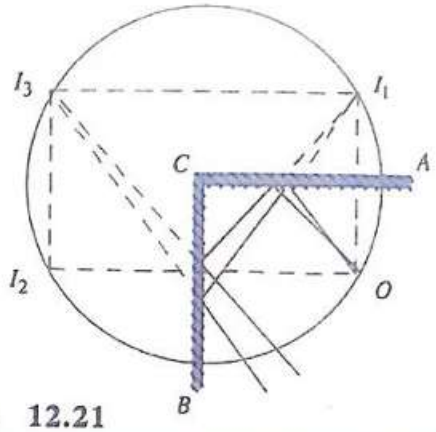


Fig. 12.21

The image I_1 is due to the reflection at the mirror AC and the image I_2 is due to the reflection at BC. The image I_3 is due to the reflection at AC and then at BC i.e. a combination of the images of I_1 and I_2 formed by both mirrors.

When $\theta = 180$, the two mirrors will act as a single mirror and therefore form only one image. When $\theta = 0$, i.e. the two mirrors are placed parallel to each other, and an object is placed between them, the number of images formed is infinite (because $\frac{360}{0} = \infty$) These images all lie in a straight line through the object, and perpendicular to the mirrors.

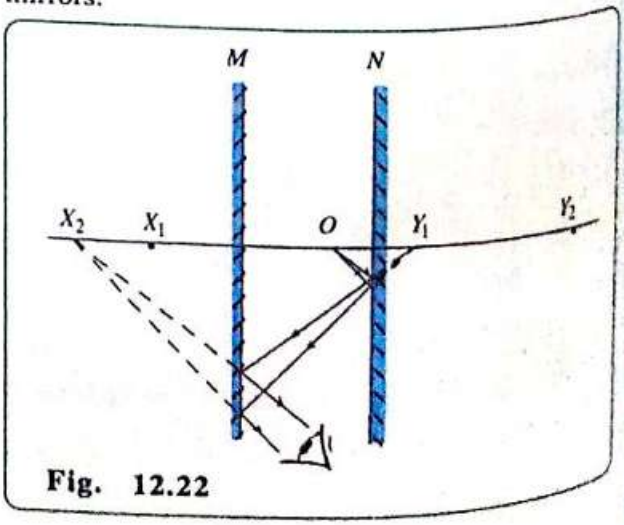


Fig. 12.22

Scanned with CamScanner

The *Kaleidoscope* is a toy which uses the above principle to produce multiple images. It consists of a tube in which two mirrors are fixed, inclined at an angle of 60°.

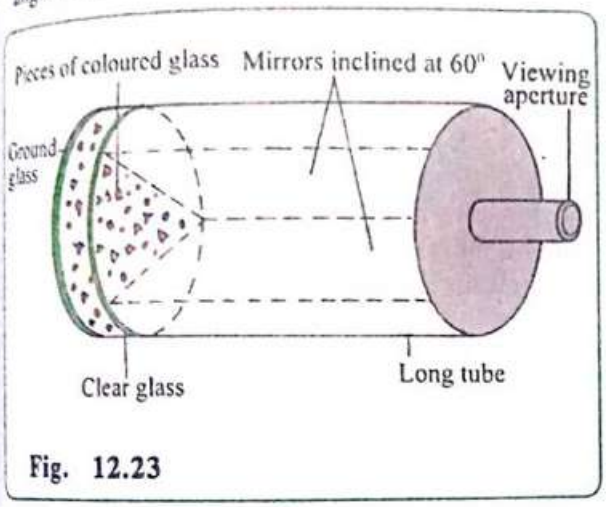


Fig. 12.23

At one end of the tube is a window or an opening for viewing. The other end of the tube is sealed by ground glass, on which are placed several loose pieces of coloured glass or plastic chips. The images of the pieces of coloured glass when viewed through the opening are seen symmetrical in a circle. The tube can be shaken to rearrange the glass pieces and obtain various patterns.

The Periscope

The Periscope is a practical application of reflection by plane mirrors. It is an instrument used for looking over barriers or obstacles. In particular, they are used in armoured tanks by soldiers to see out above the armour plating.

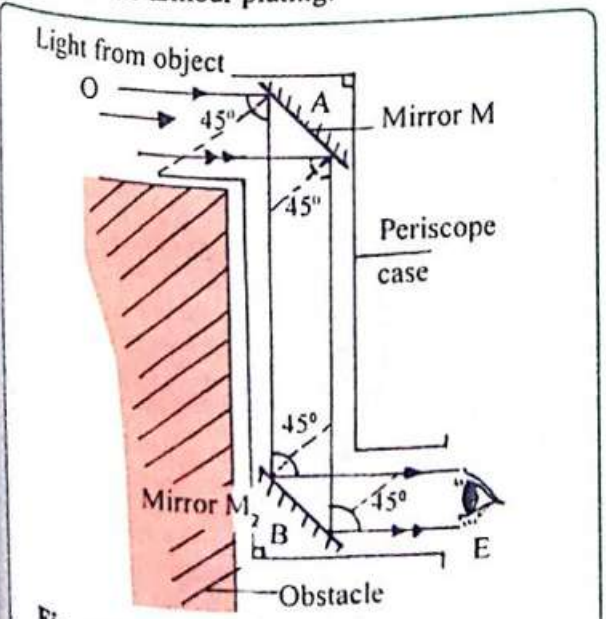


Fig. 12.24 A periscope

A simple periscope consists of two parallel plane mirrors fixed facing one another and inclined at 45° to the walls of the tube in which the mirrors are fixed. A horizontal ray of light from the object O falls on the mirror M_1 at an angle of 45° to the normal. As the angle of incidence is equal to the angle of reflection, the reflected ray AB from M_1 is at right angles to OA. This ray strikes the mirror M_2 also at 45° to the normal and is similarly reflected along BE at right angles to AB into the eye at E.

Thus by the reflection of two plane mirrors, an object hidden from view can be seen by an observer.

The major disadvantage of the periscope is that it often produces multiple images.

Effect of mirror rotation on a reflected ray. Mirror Galvanometer

If the direction of an incident ray on a mirror is kept constant and the mirror is rotated through an angle, it will be found that the reflected ray will be rotated through twice that angle. This is an important fact which is made use of in *mirror galvanometers* used for the measurement of very small electric currents, and in the *navigator's sextant*.

We can prove this important observation using the laws of reflection.

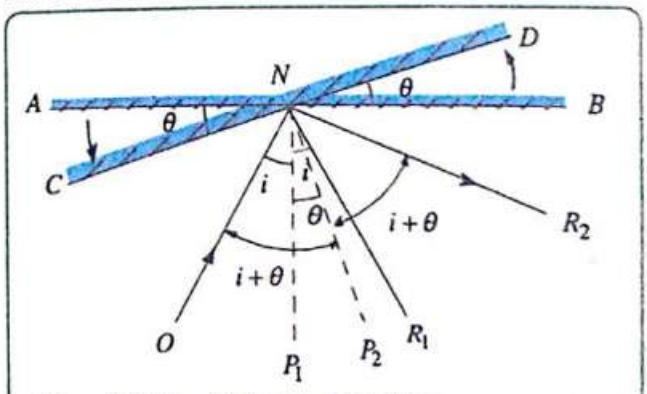


Fig. 12.25 Mirror rotation

Let ON be a ray incident at an angle i on the mirror AB at N. P_1N is the normal to the mirror at N. The reflected ray NR_1 is also inclined at angle r $(= i)$ to the normal. If the mirror is rotated to a new position at CD through an angle θ, the new reflected ray is NR_2. If the direction of the incident ray ON is kept constant and the normal to the mirror at CD is P_2N, then the new angle of incidence is ONP_2 which is equal to $(i + \theta)$. Note that the normal P_1N is also rotated through θ to

Scanned with CamScanner

P_2N. By the law of reflection, the angle of reflection in this new position is also equal to $(i + \theta)$. Hence angle $P_2NR_2 = (i + \theta)$. The angle between the incident ray ON and the new reflected ray NR_2 equals angle ONR_2 which is equal to $(i+\theta) + (i + \theta) = 2(i + \theta)$.

The angle of rotation of the reflected ray when the mirror is turned through θ is $2i + 2\theta - 2i = 2\theta$. Hence we have shown that

If the incident ray has a constant direction and the mirror is rotated through an angle θ, the reflected ray is rotated through an angle 2θ.

Example 12.3

The reflection of a narrow beam of light, incident normally on a plane mirror falls on a meter rule parallel to the mirror and at a distance of 1 m. Calculate the angle of rotation of the mirror if the reflected beam is displaced 21.26 cm along the metre rule when the mirror is rotated.

Solution

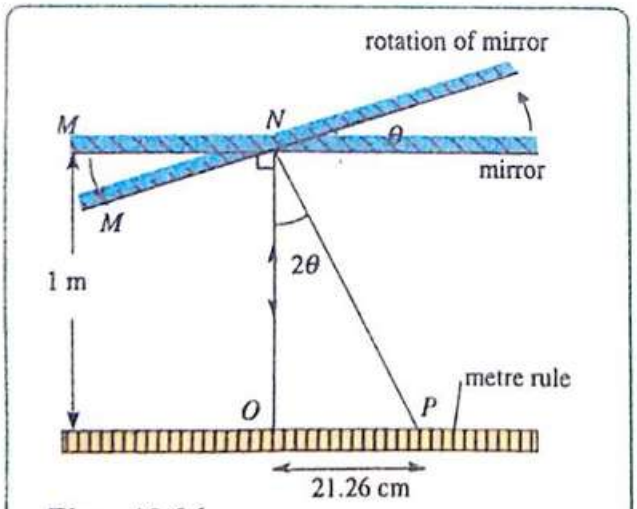


Fig. 12.26

Suppose a beam of light ON is incident normally on the mirror at N. This beam is reflected straight back along NO to the point O on the metre scale. Incident ray is ON. Reflected ray after rotation is NP. Let the mirror be rotated through θ°. Then the reflected ray is rotated through $2\theta^\circ$,

i.e. angle $ONP = 2\theta^\circ$

$$\tan 2\theta = \frac{21.26}{100 \text{ cm}} = 0.2126$$

$$2\theta = \tan^{-1} 0.2126 = 12.0^\circ$$

$$\theta = 6^\circ$$

Angle of rotation of mirror is $\underline{6^\circ}$

12.5 Reflection of Light by Curved or Spherical Mirrors

Apart from plane mirrors, curved mirrors also reflect light incident on them. In this section we shall consider only those curved mirrors which form parts of the surfaces of spheres. Such mirrors are therefore called *spherical mirrors*, and they have a large number of scientific and practical applications. Shaving mirrors, car driving mirrors, parabolic mirrors of telescopes, mirror reflectors of searchlights are such applications.

There are two types of curved mirrors – *Convex and Concave* mirrors. They are made by silvering a glass surface which is part of a sphere. If the inside surface of this spherical part is silvered and the outside surface is the reflecting part, the resulting mirror is called a *convex* or *diverging* mirror. If the outside surface is silvered and the inside surface is the reflecting part, the resulting mirror is known as a *concave* or *converging* mirror (fig 12.27)

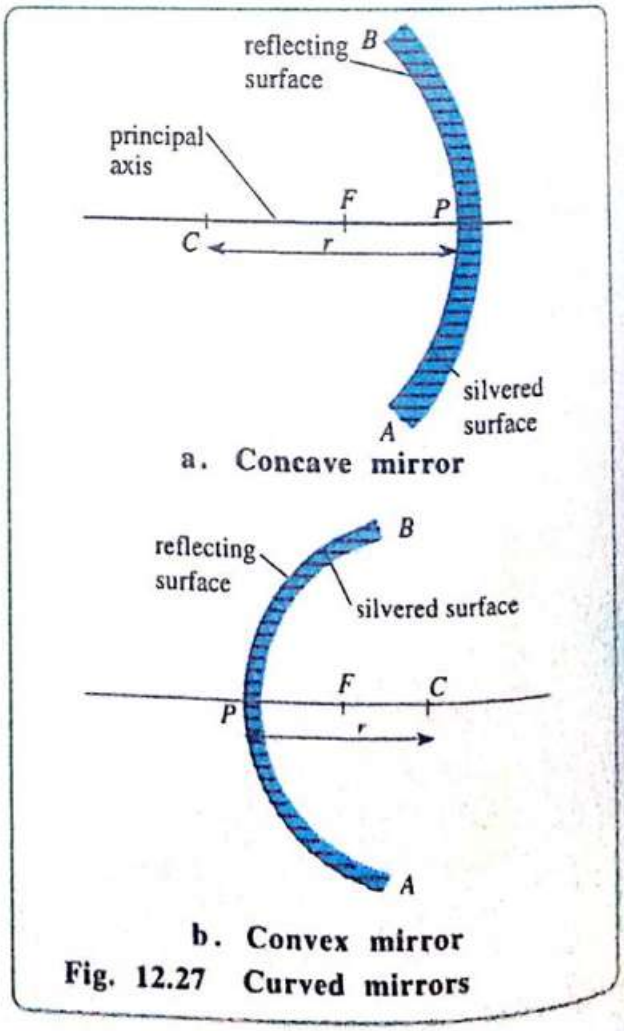


Fig. 12.27 Curved mirrors

Scanned with CamScanner

The essential parts of spherical mirrors are: the *Aperture*, *the pole*, the *centre of curvature*, the *principal axis*, the *radius of curvature*. These are illustrated in fig. 12.27.

The *Aperture* is the width (AB) of the mirror.

The *Pole* (P) is the centre of the reflecting surface of the curved mirror.

The *Centre of curvature* (C) is the centre of the sphere of which the mirror forms a part.

The *Radius of Curvature* (r) is the distance CP. It is the radius of the sphere of which the mirror forms a part.

The *Principal Axis* is the line PC from the pole to the centre of Curvature.

When parallel rays of light close to and parallel to the principal axis are incident on a concave mirror, they *converge* or come together after reflection to a point F on the principal axis called the *principal focus*. By contrast, when such rays are incident on a convex mirror, they *diverge* from the surface and appear to come from the point F on the principal axis behind the mirror.

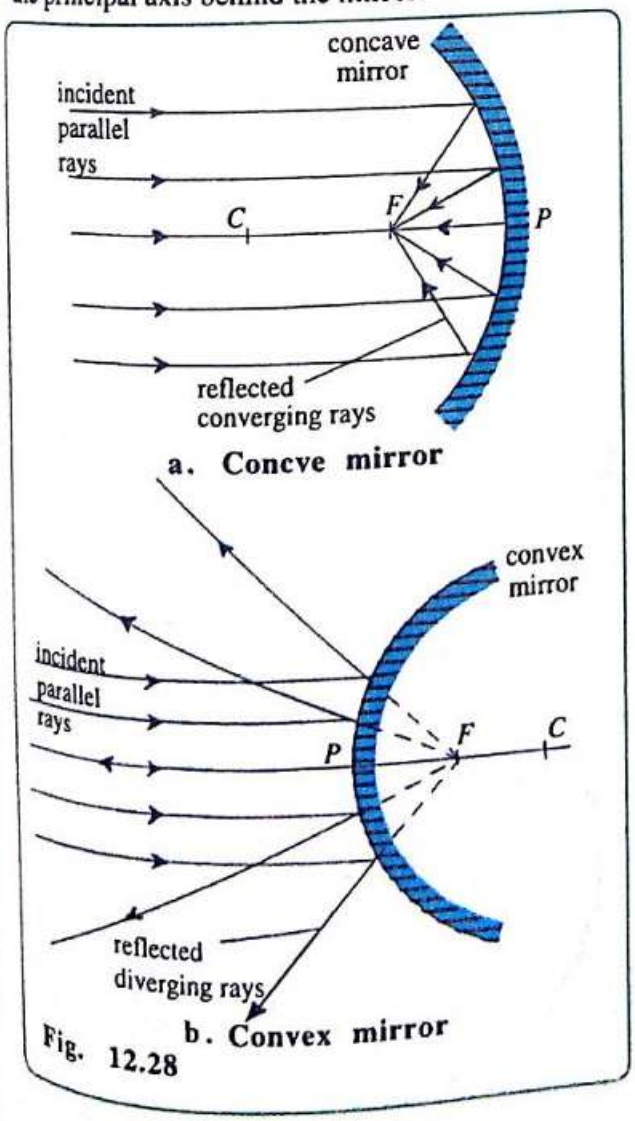


Fig. 12.28

The **Principal focus** (F) of a curved mirror is that point on the principal axis to which incident rays parallel and close to the principal axis converge or from which they appear to diverge after reflection.

The principal focus of a concave mirror is said to be a *real focus* because the reflected rays actually pass through it. This point of convergence can actually be obtained on a screen placed in front of the mirror as a bright spot of light. On the contrary the principal focus of a convex mirror is a *virtual focus* as the reflected rays do not actually pass through it. If we place a screen behind the convex mirror, no bright spot of light or image of an object is seen on the screen.

The focal length (f) of a curved mirror in each case is the *distance* (FP) *from the principal focus to the pole*. It is found, experimentally and also by the geometry of the mirror, to be equal to half the *radius of curvature* for both types of mirrors.

$$f = \frac{r}{2} \qquad 12.5$$

Thus the principal focus (F) is midway between the pole (P) and the centre of curvature (C). In fig. 12.28, $PF = FC$.

Formation of images by Curved Mirrors

The size, nature and position of an image formed by a curved mirror depends on the position of the object from the pole of the mirror. We can study these by the use of an illuminated object, a concave mirror and a screen.

The concave mirror is placed on a stand and light from the illuminated object is made to fall centrally on it by the adjustment of the height of the mirror. The mirror is slightly turned to one side so as to form an image where possible, on the screen.

First, we place the object a long way from the mirror. The distance of the screen is then adjusted until a sharp image is obtained. This image is found to be diminished, real and inverted, and located between the centre of curvature and the principal focus of the mirror. When the object is moved closer to the mirror, the image becomes larger but it is still real and inverted.

Scanned with CamScanner

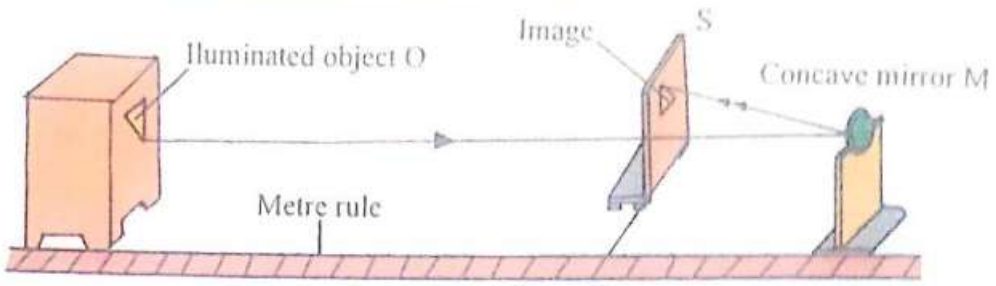


Fig. 12.29 Investigating images formed by a concave mirror

When the object is at *C*, the centre of curvature, the image is found to be located also at the centre of curvature, real, inverted and the same size as the object. If the object is between the centre of curvature and the principal focus, the image appears larger than the object but still remains real and inverted. It is located beyond the centre of curvature. When the object is placed at the principal focus, no image is seen on the screen. Its image is at infinity When the object is between the principal focus and the pole, the image is larger than the object, but it is erect and virtual. Thus it is not seen on the screen, as it appears behind the mirror.

If a convex mirror is used, an erect and virtual image smaller than the object results for any position of the object. We can use the results above to distinguish between a convex and a concave mirror by merely looking at our face in the mirror. If you look at a convex mirror, your face looks small and erect whatever the distance between it and the mirror. With a concave mirror, a large and erect image of your face is seen when the face is close to the mirror. Hence concave mirrors are used as shaving mirrors. If the concave mirror is held far away from the face, an inverted image is observed.

Ray diagrams of images

Fig. 12.30 (a)–(e) illustrates graphically the nature and location of the images formed in the experiment above for a concave mirror.

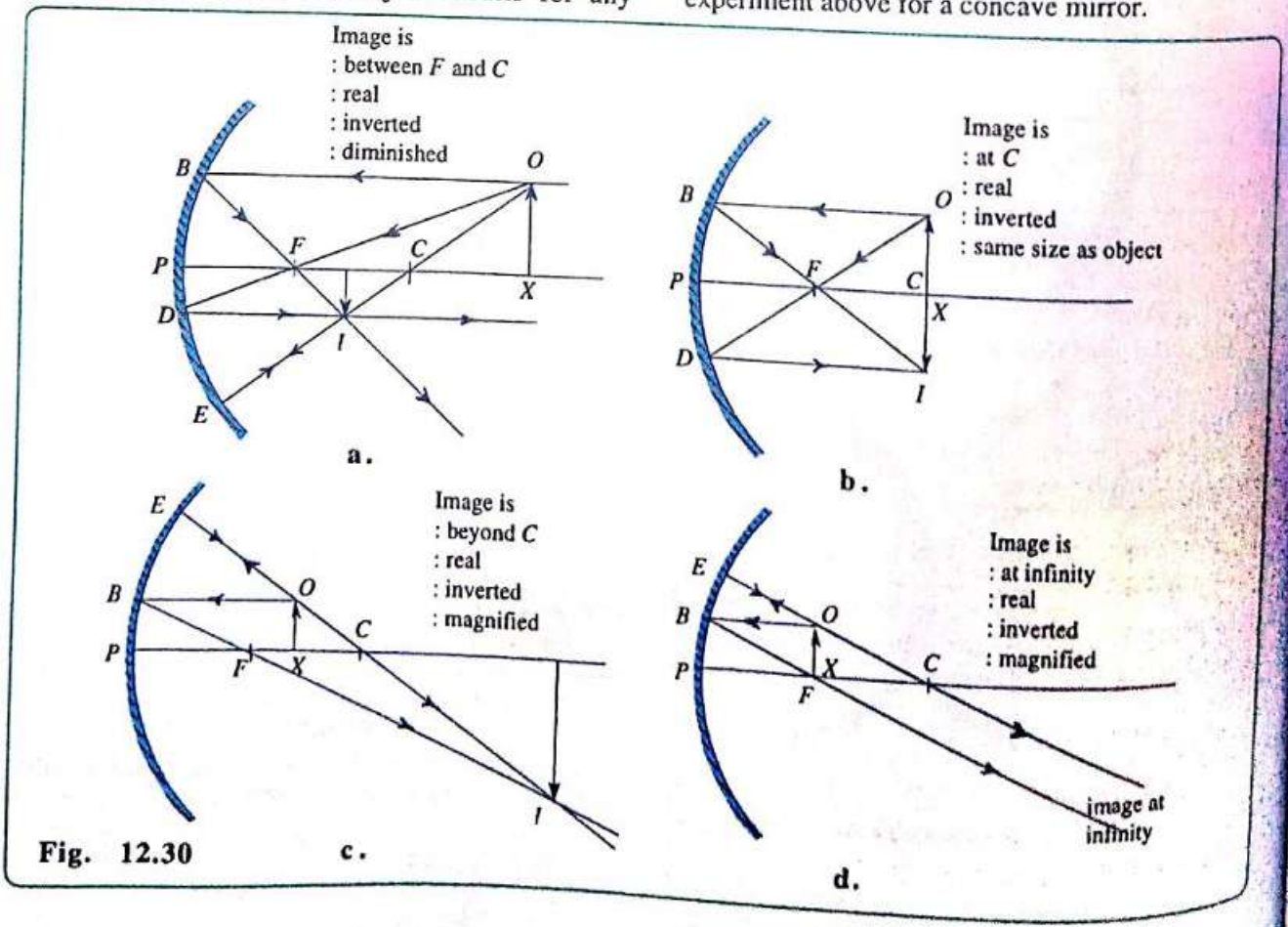


Fig. 12.30

Scanned with CamScanner

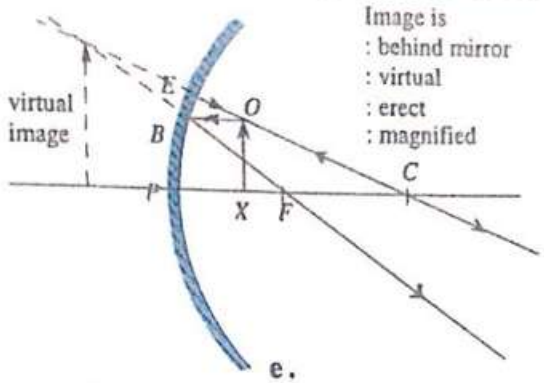


Fig. 12.30 Ray diagrams of images in concave mirrors

Fig 12.31 shows the situation for a convex mirror. The images obtained in mirrors can always be drawn to scale by means of a ray diagram. To do this we make use of the following facts.

(i) A ray OB parallel and close to the principal axis passes through the principal focus(F) after reflection.

(ii) A ray OC through the centre of curvature, strikes the mirror normally at E and is reflected back along the same line or path.

(iii) As a corollary to (i) any ray from O which passes through the principal focus strikes the mirror at D and is reflected parallel to the principal axis.

The point I to which these three reflected rays intersect represent the required image of O, the top of the object. The image of the foot of the object lies on the axis, since the ray XCP strikes the mirror normally at P and is reflected straight back.

In fig 12.30 (a)-(d) OB is the ray parallel and close to the principal axis which after reflection passes through F, the principal focus. Ray OC passes through the centre of curvature C and is reflected back on its path as indicated by the double arrows. In fig 12.30(e) where the object is between F and P, we note that the rays appear to intersect at a point behind the mirror. The broken lines indicate virtual rays and virtual images.

A convex mirror forms only virtual erect and diminished images as shown in fig 12.31. The image I is formed at the intersection of the reflected rays produced backwards (in broken lines) and is formed behind the mirror. This is the case for all positions of the object in front of the mirror.

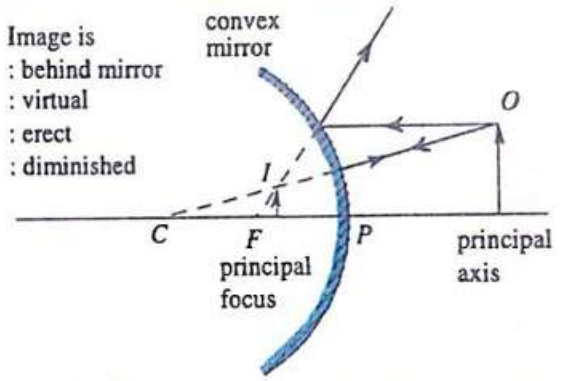


Fig. 12.31 Image formed by a convex mirror

Experimental Determination of the focal length of curved mirrors

1 Quick but approximate method

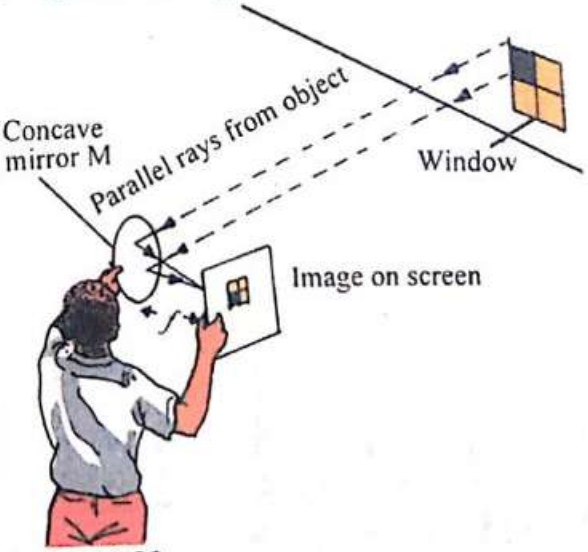


Fig. 12.32

A quick method of finding an approximate value of the focal length of a concave mirror is to focus the image of a distant object on a screen. A concave mirror is held with its reflecting surface facing a distant object (e.g

Scanned with CamScanner

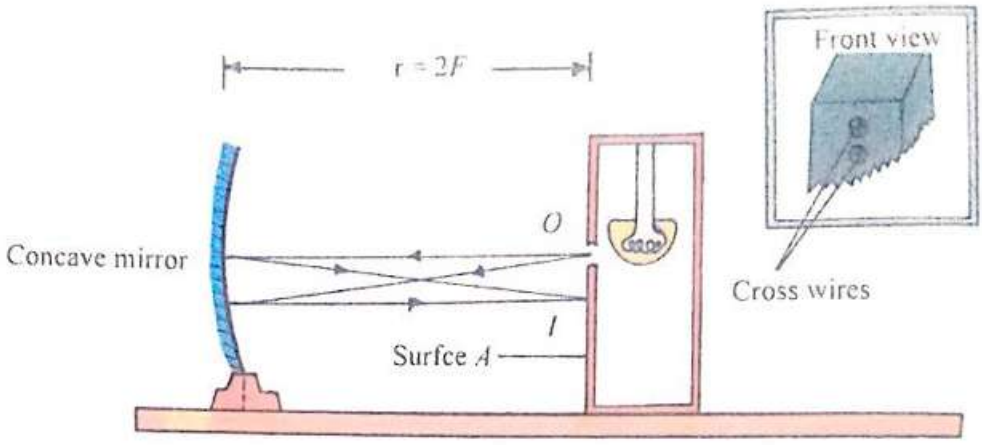


Fig. 12.33 Finding radius of curvature of concave mirror

window panes) at an angle. A screen in the form of a sheet of paper is moved in front of the mirror until the sharpest image of the window forms on it. The distance between the screen and the mirror is measured and this gives an approximate value of the focal length of the mirror.

As the window is far from the mirror, the rays from it may be considered as parallel rays when they strike the mirror. They will therefore come to a focus after reflection at the principal focus.

2 Focal length from measurement of the radius of curvature

An illuminated object is placed in front of a concave mirror mounted on a mirror holder. The position of the mirror is adjusted until a sharp image of the object is obtained on the surface A at the same position as the object. The distance between the mirror and the illuminated object is measured. This distance gives the radius of curvature, r, of the mirror. The experiment is repeated two more times. Then focal length f is given by

$$f = \frac{r}{2}$$

3 By the Use of Search pin and the method of no-parallax

An object pin is stuck into a cork and attached to a stand. The concave mirror and the optical pin are arranged as shown in fig 12.34

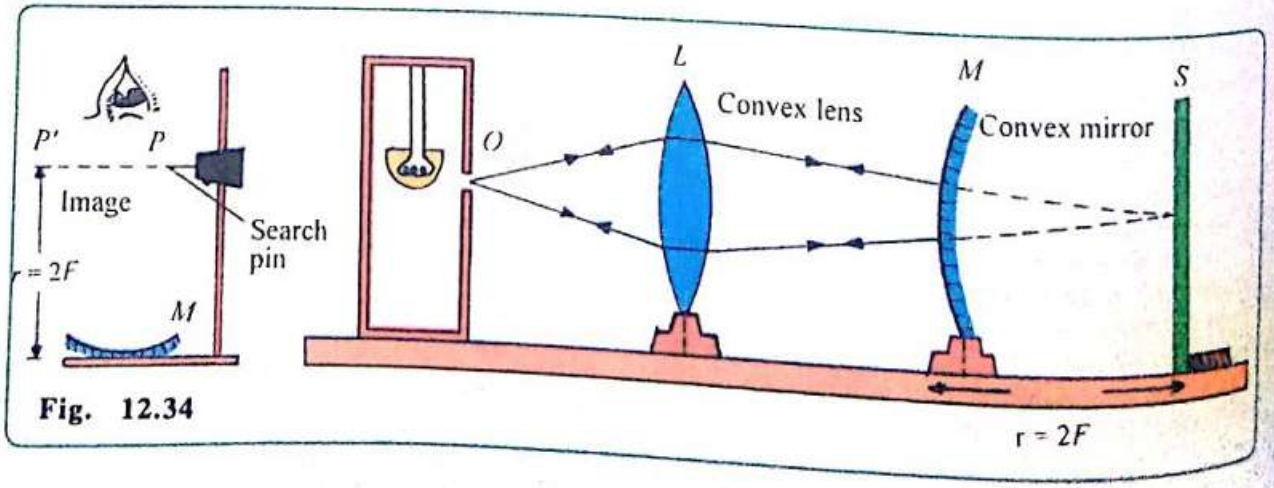


Fig. 12.34

Scanned with CamScanner

The pin is moved up and down until it coincides with its image P when viewed from above. At this position there is no parallax between the pin and its inverted image as seen in the mirror. The distance between the pin and the mirror gives the value of r. The focal length f is obtained from the relation

$$f = \frac{r}{2}$$

4 By the use of the mirror equation

The object distance, u, the image distance, v, and the focal length, f, of spherical mirror are related by the equation

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \qquad 12.6$$

The equation can be used to obtain the focal length of a concave mirror. An approximate focal length value is first obtained as already described. An illuminated object (e.g. the cross wire of a ray box) is placed at a distance of about $\frac{3}{2}f$ in front of the mirror.

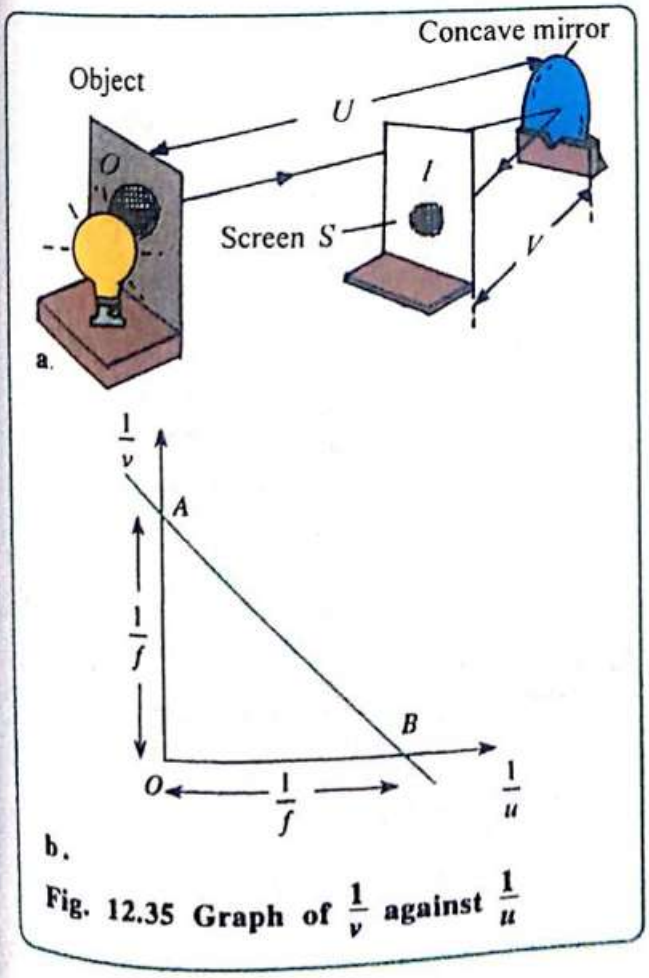


Fig. 12.35 Graph of $\frac{1}{v}$ against $\frac{1}{u}$

The exact distance is measured and recorded as u. The screen position is adjusted until a sharp image is obtained. The distance from the mirror to the screen is measured and recorded as v. The experiment is repeated for various object distances, u, and the corresponding values of v are obtained. The value of f can be obtained by substituting the values of u and v to the mirror formula (equation 12.6) and finding the average value of f or by plotting the value of $\frac{1}{u}$ against $\frac{1}{v}$. The intercept on either axis is equal to $\frac{1}{f}$. The reciprocal of this intercept gives the value of the focal length of the mirror.

Sign convention

When the mirror formula (equation 12.6) is used in solving practical problems, it is necessary to add a positive (+) or a negative (–) sign to each of the distances according to a sign rule or convention. Two commonly used sign conventions are given below:

	Real is Positive	New Cartesian
1	Real objects and real images are considered to be at a positive distance from the mirror	Distances measured to the right of the mirror are positive
2	Virtual images are at a negative distance from the mirror	Distances measured to the left of the mirror are negative
3	Focal length of a concave mirror is positive	Focal length of a concave mirror is negative
4	Focal length of a convex mirror is negative	Focal length of a convex mirror is positive

Linear Magnification (M) produced by a mirror is given by

$$M = \frac{v}{u} = \frac{\text{height of image}}{\text{height of object}} \qquad 12.7$$

Scanned with CamScanner

Example 12.4

An object is placed 10 cm in front of a concave mirror whose radius of curvature is 12 cm. Calculate the position, nature and magnification of the image produced.

Solution

Real-is-positive

$f = +\frac{r}{2} = +\frac{12}{2}$

$= +6$ cm

$u = +10$ cm

Using $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$= \frac{1}{6} - \frac{1}{10} = \frac{1}{15}$

$v = +15$ cm

Since v is +ve, image is real

$M = \frac{v}{u} = \frac{15}{10} = 1.5$

New Cartesian

$f = \frac{-12}{2} = -6$ cm

$u = -10$ cm

Using $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$= \frac{1}{-6} - \frac{1}{-10}$

$= \frac{1}{-6} + \frac{1}{10}$

$= \frac{1}{10} - \frac{1}{6} = \frac{1}{-15}$

$v = -15$ cm

Image is 15 cm from mirror on the same side as the object. So image is real

$M = \frac{v}{u} = \frac{-15}{-10} = +1.5$

Thus we obtain the same result whatever the convention we use.

Example 12.5

An object is placed 15 cm in front of a convex mirror and an image is produced 5 cm behind the mirror. Calculate the focal length of the mirror.

Solution

Use Real-is-positive convention

$u = +15$ cm, $v = -5$ cm (–ve because a convex mirror always produces a virtual image)

Using $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

$\frac{1}{15} + \frac{1}{-5} = \frac{1}{f}$

$\frac{1}{15} - \frac{1}{5} = \frac{1}{f}$

$\frac{1}{f} = \frac{-2}{15 \text{ cm}}$

$f = \underline{-7.5 \text{ cm}}$

Example 12.6

A concave mirror of radius of curvature 20 cm produces an inverted image 3 times the size of an object placed on and perpendicular to the axis. Calculate the positions of the object and image

Solution

$r = 20$ cm

$f = \frac{20}{2} = 10$ cm

$M = \frac{v}{u} = 3$

$\therefore \quad v = 3u$

Using $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{10} = \frac{1}{u} + \frac{1}{3u} = \frac{4}{3u}$

$u = \frac{40}{3} = \underline{13.3 \text{ cm}}$

$v = 3u = \underline{40 \text{ cm}}$

Practical applications of Curved mirrors

1 Concave mirrors are used as shaving mirrors because when a man places his face near the mirror (i.e. between the principal focus and the pole), he sees an enlarged, erect and virtual image of the face.

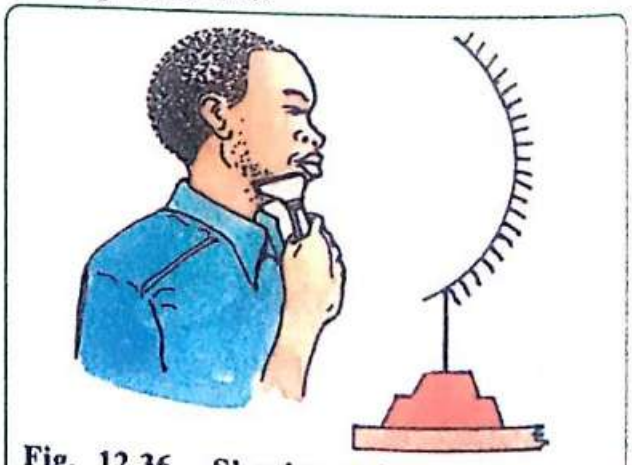


Fig. 12.36 Shaving mirror

2 Concave mirrors are used as reflectors in reflecting telescopes and microscopes.

3 Convex mirrors are used as driving mirrors of motor cars because they give an erect image of an object behind the driver. In addition, such convex mirrors provide a wide field of view. Therefore, objects within a large angle can be seen with the help of the mirror.

Convex mirror has some disadvantages when used as driving mirror. These are (i) The image is always smaller than the object. (ii) It gives a false impression of the distance as image seems further away.

Scanned with CamScanner

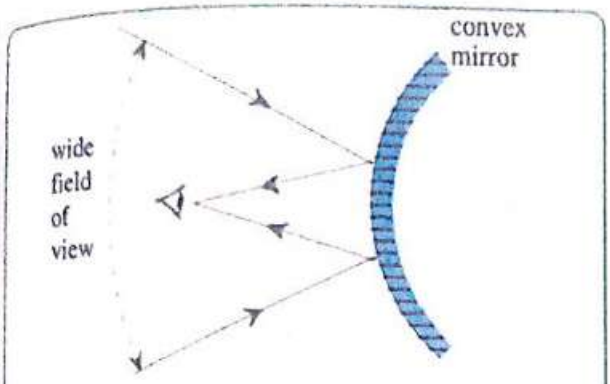


Fig. 12.37 Use of convex mirror for wide field of view

4 Use of parabolic mirrors in car head lamps and searchlights.

A parabolic mirror is a special type of concave mirror which has the shape of a parabola. The parabolic mirror produces a wide parallel beam of light of constant intensity when a small light source is placed at its focus (fig. 12.38).

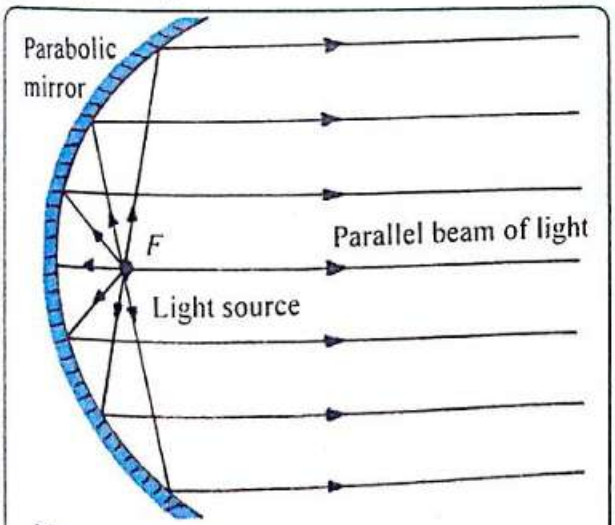


Fig. 12.38 Parabolic mirror

Parabolic mirrors are used in car headlamps and as searchlights. Spherical concave mirrors are not used in these devices because they do not provide a parallel beam of constant intensity.

Summary

1 Luminous sources of light are those that generate and emit light by themselves. Non-luminous sources are depended on natural or artificial light sources to illuminate them.

2 A ray is the direction or path along which light energy flows.

3 A beam is a collection of rays.

4 Beams of light may be parallel, convergent or divergent.

5 Light travels in straight lines and this is responsible for the formation of shadows, eclipses and the formation of inverted images by Pinhole cameras.

6 A shadow produced by a large light source may consist of the umbra region (total darkness) and the penumbra region (partial darkness).

7 An *eclipse of the sun* occurs when the sun casts the shadow of the moon on the earth's surface.

8 An eclipse of the moon occurs when the moon lies in the shadow of the earth.

9 In *regular reflection*, parallel rays of light incident on a smooth surface are reflected in the same direction. In diffuse or scattered reflection, parallel rays of light incident on a rough surface are reflected in various directions.

10 Laws of Reflection:
   (i) The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.
   (ii) The angle of incidence is equal to the angle of reflection.

11 The paths of light rays are reversible.

12 The image in a plane mirror is virtual, upright, the same size as the object, laterally inverted and as far behind the mirror as the object is in front.

13 A real image is one that can be caught on a screen. In a real image, light rays actually travel from all points of the image to the eye.

14 A virtual image cannot be caught on a screen. Light rays only appear to come from the image.

15 *Parallax* is the apparent relative movement of objects caused by the actual movement of the observer. *Non parallax* occurs when both the object and its image move together in the same direction when an observer moves his eyes from side to side.

16 When a mirror is rotated through an angle θ, the reflected ray is rotated through an angle 2θ.

17 The Periscope is one practical application of plane mirrors. It consists of two parallel plane mirrors inclined at 45° to the horizontal.

18 A concave mirror converges incident parallel rays. A convex mirror diverges incident parallel rays.

19 A concave mirror has a real focus, a convex mirror has a virtual focus.

20 The images formed by a concave mirror are real and inverted unless the object is nearer the

Scanned with CamScanner

mirror than its focal length, when the image is then upright, virtual and magnified. If the object is at the centre of curvature of the mirror, the image is the same size as the object, it is also real and inverted.

21 A convex mirror always forms a virtual, diminished, and erect image, irrespective of the position of the object from the mirror.

22 Mirror formula: $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$; $M=\frac{v}{u}$

23 Some practical applications of
a. Concave mirrors
(i) As shaving mirrors
(ii) A Parabolic mirrors used in searchlights and car headlamps. A parabolic mirror produces a wide parallel beam of light of constant intensity
b. Convex mirrors are used as driving mirrors. They have a wide field of view.

Exercise 12

1 Which of the following expressions about the image formed by a plane mirror is/are correct?
I virtual II magnified III erect
IV farther from the mirror than the object
A I only B I and II only
C I and III only D I and IV only
E III and IV only.

2 Which of the following processes will increase the size of the image in a pinhole camera?
A Enlarging the size of the pinhole
B Decreasing the size of the pinhole
C Moving the object away from the camera
D Moving the object nearer to the camera
E Shortening the length of the camera.

3 If the angle of incidence of a ray of light on a plane mirror is 40°, the deviation of the incident ray after reflection is
A 40° B 100° C 80°
D 140° E 160°

4 The image of an object placed at 2f from a concave mirror is
A real, diminished and inverted
B real, magnified and inverted
C virtual, same size and inverted
D real, same size and inverted
E real, same size and erect.

5 An image which can be formed on a screen is said to be
A virtual B blurred C inverted
D erect E real.

6

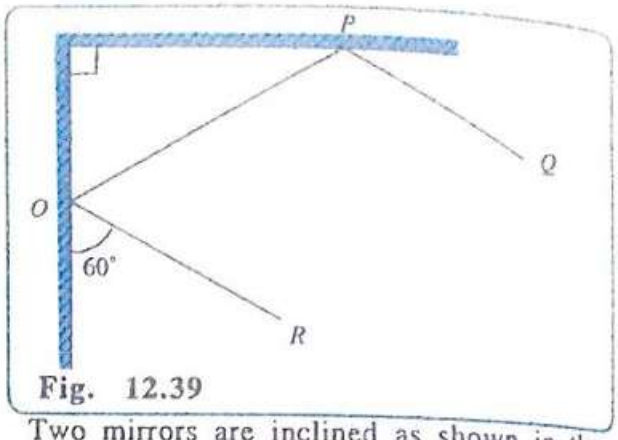


Fig. 12.39

Two mirrors are inclined as shown in the diagram above. A ray of light RO strikes the arrangement at O and emerges along PQ. The emergent ray has been deviated through
A 240° B 180° C 120°
D 60° E 30°

7 For a concave mirror to form a real diminished image, the object must be placed
A behind the mirror
B between the mirror and its focus
C between the focus and the centre of curvature
D at the centre of curvature
E at a distance greater than the radius of curvature.

8 Draw ray diagrams to show how (i) a plane mirror (ii) a convex mirror (iii) a concave mirror, can form an image. Describe a practical application of each type of mirror.

9 (i) Describe the construction and mode of image formation by a pinhole camera. What is the effect on the image produced by the camera when
a. the diameter of the pinhole is increased
b. the object distance is doubled
c. the length of the camera is increased.
(ii) The screen of a pin-hole camera is a square, of side 0.16 m and it is 0.15 m behind the pinhole. The camera is placed 11 m from a flag staff and positioned so that the image of the flag staff is formed centrally on the screen. The image occupies three-quarters of the height of screen. What is the height of the flag staff? What are the major disadvantages of a pinhole camera compared to a lens camera.

10 Explain the terms *radius of curvature* and *principal focus* as applied to a concave mirror, illustrating your answer by diagrams. What are the disadvantages and advantages of spherical

Scanned with CamScanner

and parabolic reflectors, used in optical systems?

An object, 0.04 m high, is placed in front of and at right angles to the axis of a concave mirror which has radius of curvature 0.15 m. If a real image 0.12 m high is obtained, determine by scale drawing or otherwise, the distance of the object from the mirror.

How far, and in what direction, must the object be moved in order to obtain a virtual image 0.12 m high.

11 What is meant by (a) the *radius of curvature* (b) the *focal length* of a spherical mirror? What is the relationship between the two? Describe an experiment to measure the focal length of a concave mirror. A concave mirror of radius of curvature 40 cm is to be used to obtain a real image of an object. The image is to be one-third as large as the object. Where should the object be placed, and where is the image to be found?

12 Distinguish between *real* and *virtual* images. With the aid of diagrams, show the formation of real and virtual images by a concave mirror.

An object 7 m high is placed 15 cm from a convex spherical mirror of radius 45 cm. Describe its image.

13 With the aid of diagrams describe the formation of solar eclipse, lunar eclipse, annular eclipse. Distinguish between a total eclipse and a partial eclipse of the Sun.

14 Show by means of sketch diagrams the arrangement of mirrors in a (i) periscope (ii) kaleidoscope.

15 Draw a diagram to illustrate the reflection at a plane surface; label the incident ray, the reflected ray, the normal, the angle of incidence, the angle of reflection.

State the laws of reflection and describe an experiment to verify the law relating the angle of incidence and the angle of reflection.

16 Explain the term parallax as used in optics. Describe the method of non-parallax to locate the position of an image in a plane mirror.

How is the distance of object in front of a plane mirror related to the distance of image from the mirror.

An optician uses a plane mirror to help him. Suppose a patient sits in a chair 2.5 m away from him. He views the image of a chart which faces the mirror and is located 30 cm behind him. How far away is the chart as viewed from his eyes.

17 (i) Explain using a ray diagram, how an image is formed by a convex mirror.

(ii) State one advantage and one disadvantage of using a convex mirror as a driving mirror.

Scanned with CamScanner

13 REFRACTION OF LIGHT WAVES

In this chapter, we shall discuss the change in the direction of light waves as they travel from one medium to another, e.g. from air to glass or vice versa. The chapter will discuss the refraction of light through a rectangular glass block, triangular prism, converging and diverging lenses and also the dispersion of white light by a triangular glass prism.

At the end of the chapter, the student should be able to:

1. Explain how the direction of light changes as it travels from one medium to another.
2. Deduce a value of refractive index of a given material from measurements of angles of incidence and refraction.
3. Explain the meanings of critical angle and total internal reflection, stating the conditions under which they occur.
4. Establish the relationship between critical angle and refractive index and apply it to the solution of simple problems.
5. Trace light rays through a triangular prism and obtain graphically the value of angle of minimum deviation.
6. Obtain the spectrum of white light.
7. Trace rays of light through converging and diverging lenses and obtain images due to the rays.
8. Derive and use the lens formula to solve simple numerical problems on lenses.

13.1 Introduction: Definitions

When a ray of light travels from one transparent medium to another of different density, its direction is abruptly changed at the surface separating the two media. This is known as the *refraction* of the light ray. Thus a light ray appears to bend as it crosses the boundary of two different media. Refraction is due to the difference in the speed of light in the different media.

Refraction is the bending of a light ray as it crosses the boundary between two media of different densities, thus causing a change in its direction.

The phenomenon of refraction is responsible for the following common observations: (i) The bottom of a clear river or pond appears shallower than it really is. (ii) A rod or spoon appears bent or broken when it is partially immersed in water or any liquid. (iii) Letters in print seem to be nearer when we place a thick block of glass over them.

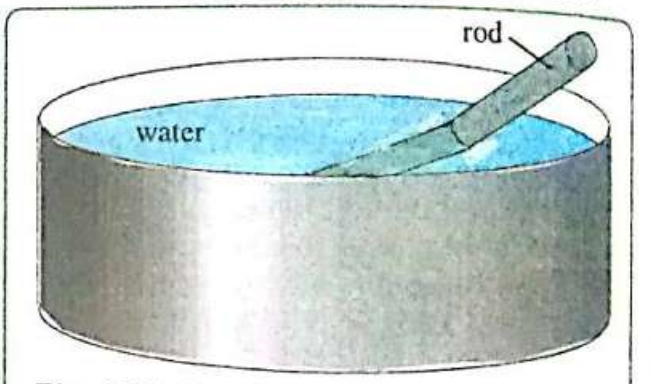


Fig. 13.1 A rod appears bent in water

Let us consider a ray of light travelling from a medium A to another medium B as shown in fig. 13.2. The incident ray PO is the path along which the light travels in the first medium.

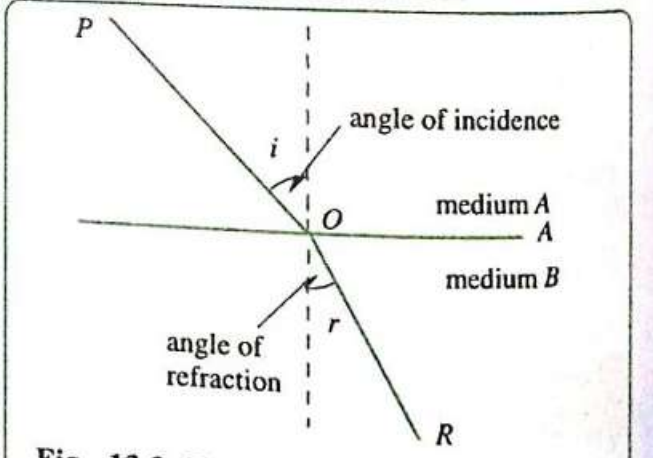


Fig. 13.2 The bending of light ray

The angle of incidence (i) is the angle which the incident ray makes with the normal to the surface at O. The incident ray is refracted as it enters the second medium, and it travels along a path OR. OR is known as the refracted ray.

The angle of refraction (r) is the angle between the refracted ray and the normal to the surface at O.

Observations show that when the light travels from a medium A to a more dense medium B, the refracted ray bends towards the normal and angle of refraction (r) is therefore less than the angle of incidence. Conversely if light travels from a denser to a less dense medium, the refracted ray bends

Scanned with CamScanner

away from the normal and the angle of refraction is more than the angle of incidence.

13.2 Refraction through rectangular glass block

We can study the refraction of light at air-glass boundary using a rectangular block. A thick rectangular glass block, $ABCD$, is placed horizontally on a thick sheet of paper and the outline of the glass block is drawn with a pencil. A light ray from a ray-box is shone onto the glass surface AB along a line PO inclined at an acute angle to the normal ON. The ray is refracted in the glass slab and emerges from the side CD, along a line RS. The glass block is then removed and the line OR representing the path of the light ray in the glass is drawn. The line RS is found to be parallel to the line PO.

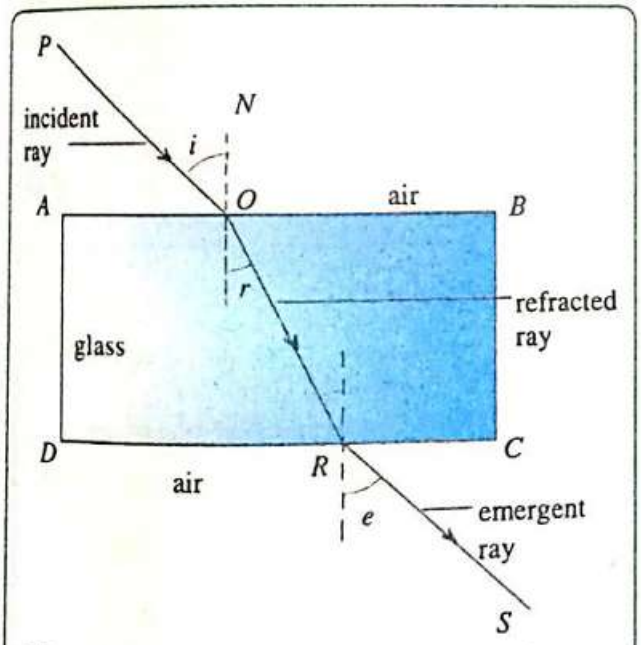


Fig. 13.3 Refraction through a rectangular glass block

In fig 13.3, PO is known as the incident ray, OR is the refracted ray and RS is the emergent ray. The angle (i) between the incident ray and the normal at O is the angle of incidence. The angle (r) between the refracted ray OR and the normal at O is the angle of refraction. The angle (e) between the emergent ray RS and the normal at R is the angle of emergence.

If the incident ray is along the normal NO i.e. at an angle of 90° to the side AB, ($i = 0$) it passes straight through the glass emerging on the face CD without change of direction.

Laws of refraction

There are two laws governing the refraction of light.

(i) The incident ray, the refracted ray and the normal at the point of incidence all lie on the same plane
(ii) The ratio of the Sine of the angle of incidence to the Sine of the angle of refraction is a constant for a given pair of media

The second law is known as *Snell's law*:

$$\frac{\sin i}{\sin r} = \mathbf{n}, \quad \text{a constant for a given pair of media} \qquad 13.1$$

The constant **n**, is known as the *refractive index* of the second medium with respect to the first medium. It is a number which gives a measure of refraction or bending of light as it travels from one medium to another. If light is travelling from air to glass, the refractive index of glass is given by

$$_aN_g = \frac{\text{sine of angle of incidence in air}}{\text{sine of angle of refraction in glass}}$$

If light travels from glass to air then the refractive index $_gN_a$ is given by

$$_gN_a = \frac{\text{sine of angle of incidence in glass}}{\text{sine of angle of refraction in air}}$$

From the principle of the reversibility of light we have that:

$$_aN_g = \frac{1}{_gN_a} \qquad 13.2$$

Since refraction is due to the change in the speed of light as it travels from one medium to another, the refractive index is also given by

$$_aN_g = \frac{\textbf{speed of light in air (vacuum)}}{\textbf{speed of light in glass}}$$

To Verify Snell's Law of refraction

We can verify Snell's law using a rectangular glass block. The block is placed on a sheet of paper and its outline is marked by a pencil. Lines such as P O are drawn making angles of 30°, 40°, 50°, 60° and 70° with the normal ON. A single ray of light

Scanned with CamScanner

from a ray box is directed along PO and the emergent ray ST is marked with a pencil.

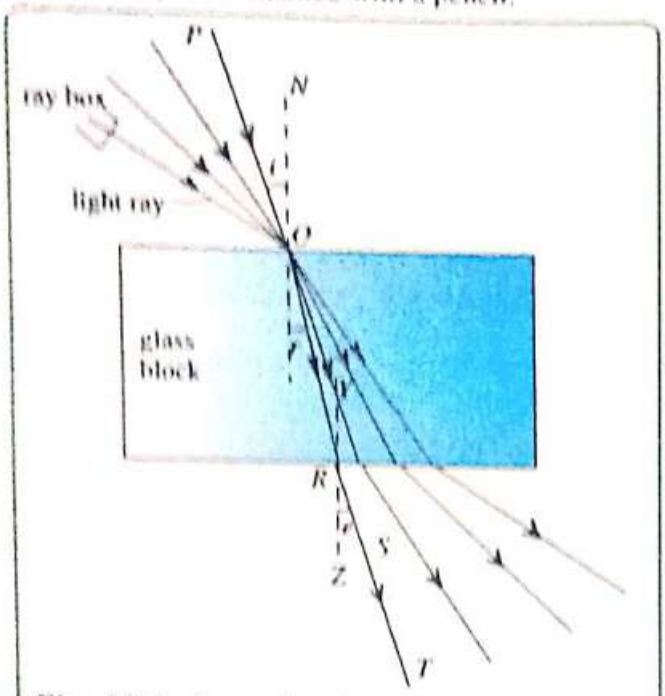


Fig. 13.4 Investigation of the laws of refraction

The line TSR is drawn to cut the glass block at R. The block is removed and the points O and R are joined. The normal YRZ is drawn. The line OR represents the refracted ray in the glass. The angle of incidence i, and the angle of refraction r are measured and recorded. The experiment is repeated for other angles of incidence and the corresponding angles of refraction are measured and recorded as shown in the table below

Table 13.1

i	r	sin i	sin r	

When sin i is plotted against sin r, a straight line graph passing through the origin is obtained as shown in fig. 13.5. Thus proves that the ratio $\frac{\sin i}{\sin r}$ = Constant, which verifies Snell's law. The slope of the graph gives the refractive index for glass.

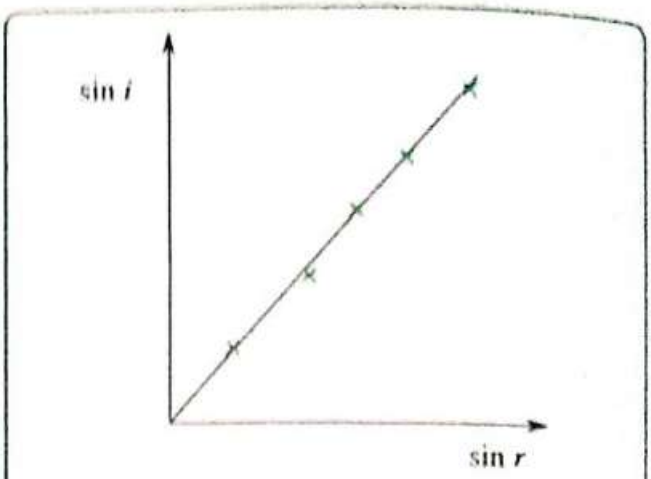


Fig. 13.5 Graph of sin i against sin r

In the absence of a ray box, we can use optical pins to define the incident ray and sighting pins are used to locate the emergent ray. Two pins P_1, P_2 are fixed on the line PO which represents the incident ray. A sighting Pin Q_1 is moved along CO until it appears to be in the same straight line as P_1P_2 when viewed through the glass block. A second pin Q_2 is fixed so that Q_1 and Q_2 appear coincident with the image of P_1 and P_2 in the glass block.

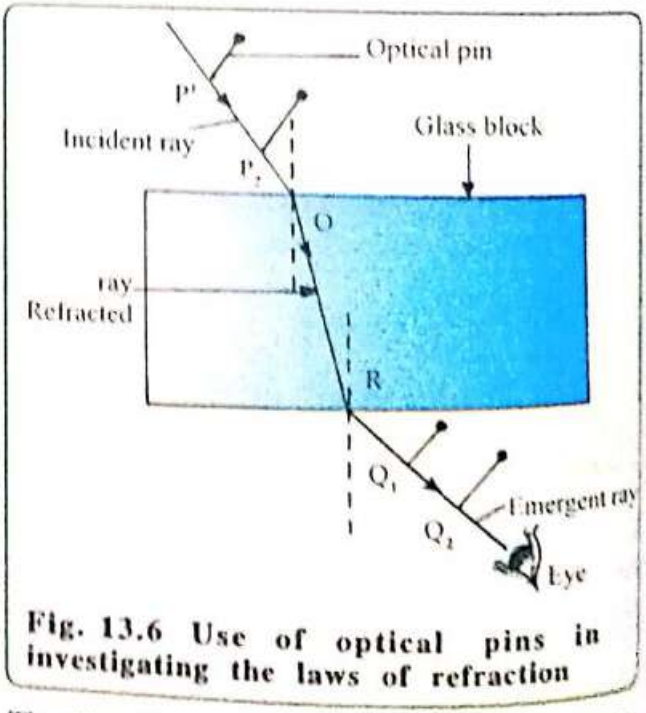


Fig. 13.6 Use of optical pins in investigating the laws of refraction

The glass block is removed and the incident ray is drawn through P_1 and P_2 while the emergent ray is drawn through Q_1 and Q_2. The refracted ray is drawn by joining the point, O, at which the line P_1P_2 meets the glass surface to the point R where the emergent ray leaves the glass surface. The

Scanned with CamScanner

angles of incidence and refraction are measured as before.

Real and Apparent Depth

The depth of a river or a swimming pool always appears shallower than it actually is. When a glass block is placed on top of an object, e.g. a pin or a mark on a piece of paper, the object when viewed from directly above, appears nearer the top. This apparent depth is caused by refraction.

Theory shows that the real depth, the apparent depth, and the refractive index are related by the formula

$$n = \frac{\text{Real Depth}}{\text{Apparent Depth}} \qquad 13.3$$

Thus if the real depth of a river bed is 10 metres but its apparent depth is 7.5 metres then the refractive index of the river water is given by

$$n = \frac{10}{7.5} = 1.33$$

Experiment: To find the refractive index of glass by the real and apparent depth method

A glass block is placed vertically over an object, e.g. a pin or a straight line drawn on paper (fig. 13.7)

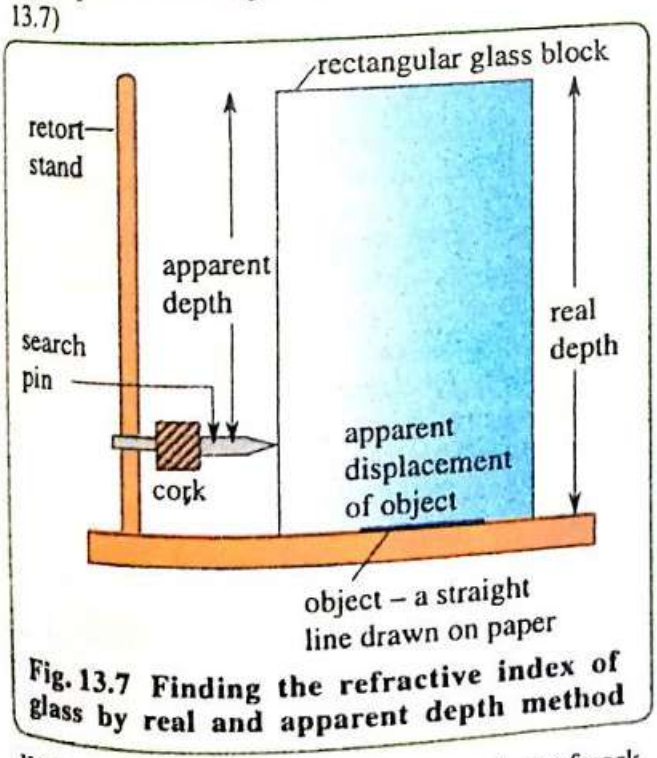


Fig. 13.7 Finding the refractive index of glass by real and apparent depth method

With the aid of a search pin fixed to a piece of cork held on a clamp, that can slide up and down, the apparent depth of the object is found by the method of non-parallax. The search pin is moved up and down until there is no parallax between it and the image of the object as seen through the glass slab from above.

The true or real depth and the apparent depth are measured as shown in the diagram and the refractive index of glass is calculated from

$$n = \frac{\text{real depth}}{\text{apparent depth}} = \frac{D}{d}$$

To find the refractive index of a liquid e.g. water

We can also find the refractive index of a liquid by the real and apparent depth method.

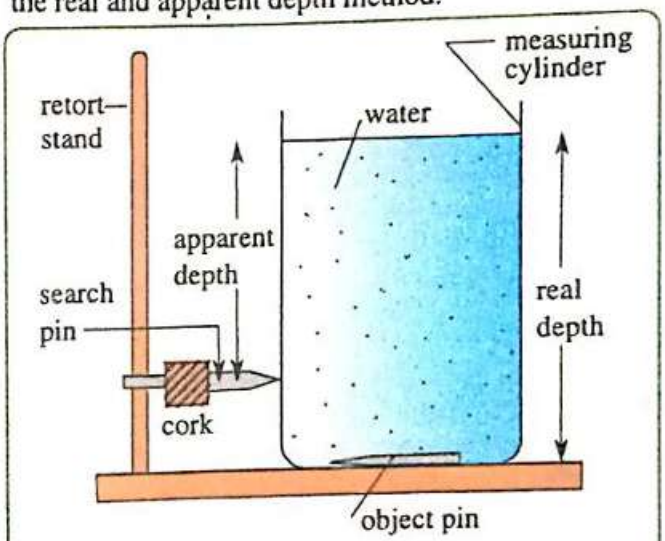


Fig. 13.8 Finding the refractive index of a liquid

An optical pin is placed at the bottom of a beaker. Water is poured into the beaker until it is about three-quarters full. A search pin attached horizontally to a sliding cork is moved up and down until it coincides with the apparent position of the pin as seen from above the liquid, as judged by the no-parallax method. The real and apparent depths of the pin from the surface of the water are measured and the refractive index of the liquid is calculated from

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$

Example 13.1

What is the real depth of a swimming pool which appears to be 10 m deep when viewed directly from above its surface? (Refractive index of water = $\frac{4}{3}$).

Scanned with CamScanner

Solution

$$n = \frac{4}{3} = \frac{\text{real depth}}{\text{apparent depth}}$$

$$\frac{4}{3} = \frac{\text{real depth}}{10}$$

$$\text{real depth} = \frac{40}{3} = \underline{\underline{13.13 \text{ m}}}$$

Explanation of effects of refraction by ray diagrams

1 Apparent depth of a swimming pool

Consider an object, e.g. a piece of stone, at the bottom of a swimming pool fig. 13.9.

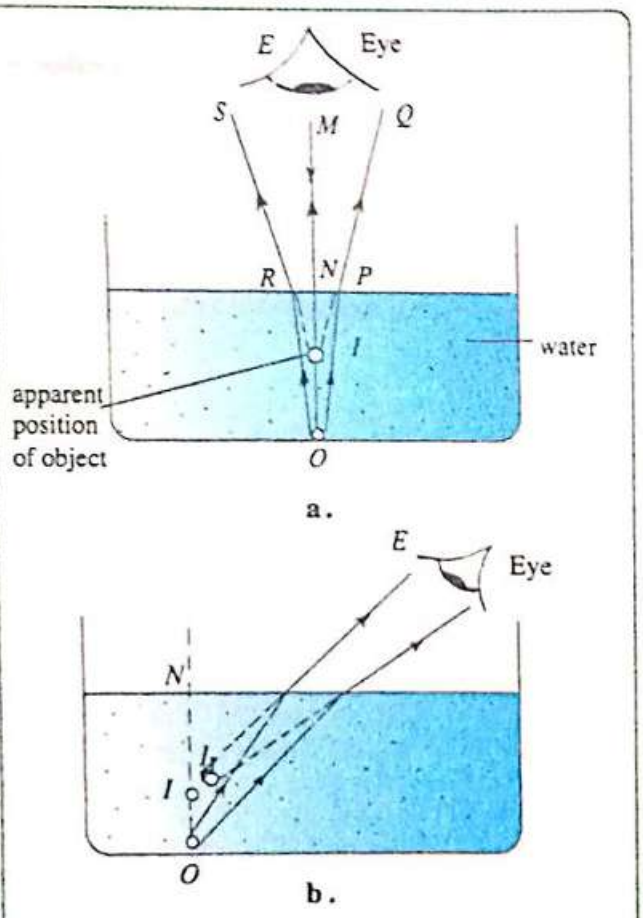


Fig. 13.9 Apparent depth of object in water

For an eye at E directly above the object at O, the object appears to be at I which appears nearer the water surface than O. This arises as follows: A ray ON normal to the water surface passes into the air along NM in the same direction. A ray OR slightly inclined to ON is refracted away from the normal on leaving the water surface and emerges into the air, travelling along RS. Another ray OP is refracted along PQ. When RS and PQ are produced backwards, they meet at I, so that to an observer's eye at E, the two emerging rays RS and PQ appear to originate from I, their point of intersection. Thus the object appears to the observer to be at I. Since I is nearer the water surface than O, the bottom of the pool appears to be raised upwards to the level of I. The pool therefore appears shallower than it actually is.

To another observer looking at the object in the water from the side it appears to be at I_1 which is slightly higher than I and to the side of it as shown in fig 13.9b.

2 Apparent bending of a stick partly immersed in water

A pencil or a stick partially immersed in water, appears bent upwards or broken at the water surface. This is due to the fact that rays from different points of the object below the water surface are refracted away from the normal as they emerge from water to air.

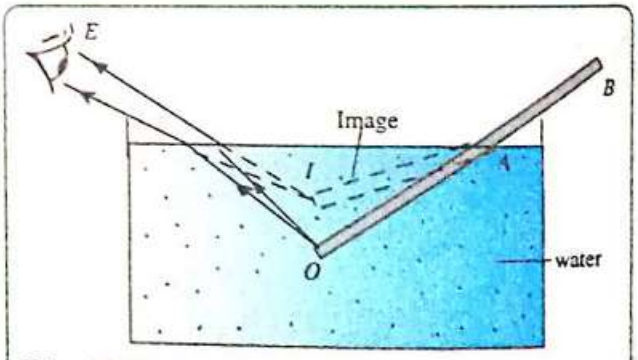


Fig. 13.10 Apparent bending of a rod in water

For example, rays from the point O after refraction at the water-air interface appear to an observer's eye at E as coming from I. Similarly all points along the stick between O and A appear higher or nearer the water surface than they actually are. The image of OA is therefore IA which is not in the same straight line as BA. Thus the stick appears bent upwards.

3 Bringing objects into view

An object e.g. a piece of coin, placed in a bowl so that it is not visible to an observer at E, can be brought to his view by pouring sufficient water to the bowl. This is due to refraction at the water-air interface.

Scanned with CamScanner

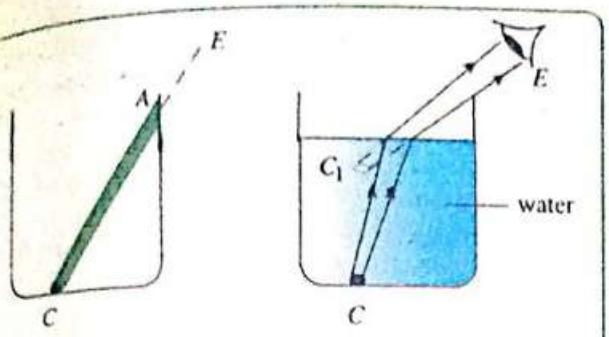


Fig. 13.11 A coin brought into view by refraction

When there is no water in the bowl rays from the coin at C just pass below the edge of the vessel at A, and thus the coin is hidden from the eye at E. When water is poured into the bowl to a sufficient level, refraction of rays from C at the water surface causes the coin to be seen, as such rays now enter the observer's eye. The coin now appears to be at C_1.

13.3 Total Internal Reflection

An interesting and useful effect can sometime occur when light travels from an optically more dense medium to a less dense medium, and the angle of incidence is greater than a certain critical value.

To study this phenomenon we place a semicircular glass block on a sheet of white paper and draw its outline with a pencil. The midpoint A of its plane side is marked on the paper. Using a ray box we direct an incident ray OA towards A. This ray will be refracted along AB away from the normal since glass is more dense than air. Part of the incident ray is also reflected back into the glass by the bottom surface PQ of the glass block. This reflected ray AD is weak and is not as bright as the refracted ray AB.

As the angle of incidence is gradually increased, the angle of refraction also increases and the refracted ray is further bent away from the normal. At each value of the angle of incidence, a strong refracted ray and a weak reflected ray are obtained. As we continue to increase the angle of incidence of the light ray in the glass block, a point is finally reached where the refracted ray does not emerge from the glass but lies along the path PQ on the glass-air interface. This particular angle of incidence at which the angle of refraction is 90° and the refracted ray lies along the glass air surface is known as the *critical angle*, Its value depends on the nature of the two media, or their relative refractive indices.

The critical angle is the angle of incidence in the denser medium when the angle of refraction in the less dense medium is 90°.

When the angle of incidence slightly exceeds this critical angle, the refracted ray disappears and a strong reflected ray is obtained. The incident ray is then said to be totally reflected back into the denser medium along PQ. Such a reflection is known as *total internal reflection*.

Total Internal Reflection is the reflection of an incident ray of light at the interface between the medium of incidence and another medium of lower refractive index when the angle of incidence in the denser medium exceeds the critical angle.

Conditions for Total Internal Reflection

For total internal reflection to occur, two conditions must be fulfilled:

1 Light must be travelling from an optically more dense to an optically less dense medium.

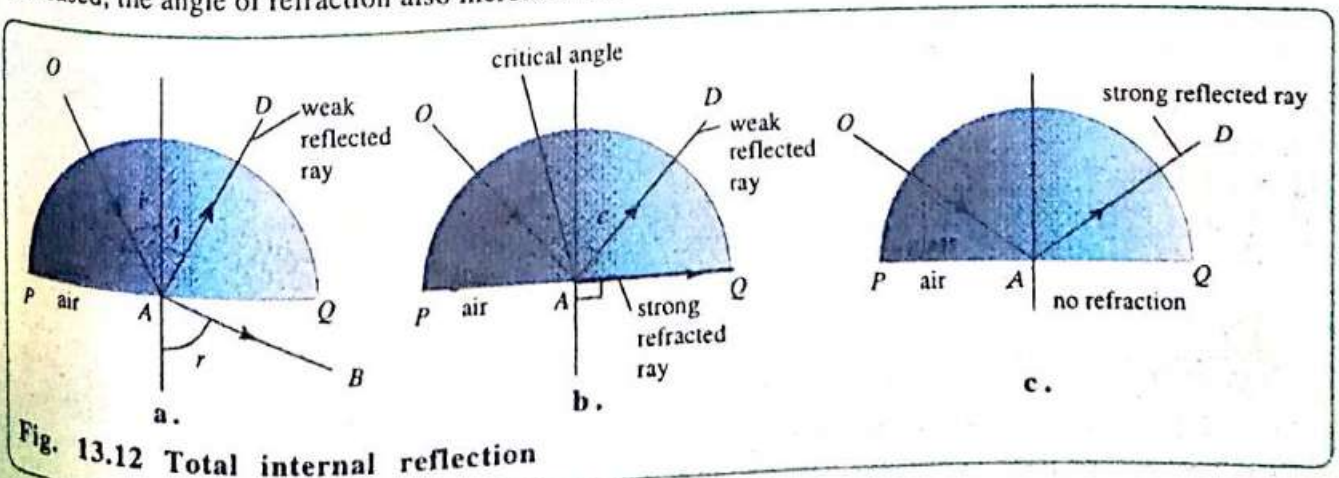


Fig. 13.12 Total internal reflection

Scanned with CamScanner

2 The angle of incidence in the denser medium must be greater than the critical angle.

Critical angle and Refractive Index

Consider fig. 13.13 where light travels from glass to less dense medium, air, and the angle of incidence equals the critical angle.

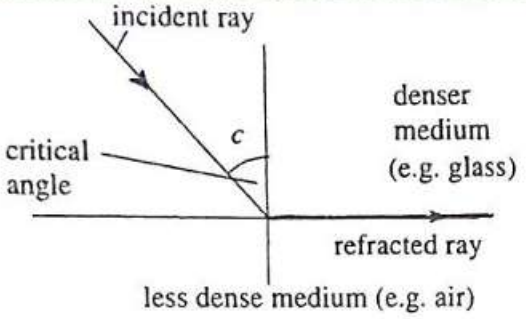


Fig. 13.13 The relation between critical angle and refractive index

From Snell's laws we have that

$$\frac{\sin i}{\sin r} = \frac{\sin C}{\sin 90^\circ} = {}_gn_a = \frac{1}{{}_ag} = \frac{1}{n} \qquad 13.4$$

where ${}_gn_a$ is the refractive index for a ray passing from glass to air and ${}_an_g$ is the refractive index for a ray passing from air to glass, n is the refractive index of glass by definition.

Since the refractive index (n) of glass is 1.5 we have that

$$\sin C = \frac{1}{1.5} = \frac{2}{3} = 0.6667.$$

Hence C is approximately equal to 42°. So for an angle of incidence in glass greater than 42 , there is total internal reflection.

Generally we can write that

$$\text{Refractive index } (n) = \frac{1}{\sin C} \qquad 13.5$$

This follows from equation 13.4 where $\sin 90^\circ = 1$.

Example 13.2

What is the critical angle for light travelling from water to air. Refractive index of water $= \frac{4}{3}$.

Solution

$$n = \frac{1}{\sin C}$$

$$\frac{4}{3} = \frac{1}{\sin C} \quad \text{or}$$

$$\sin C = \frac{3}{4} = 0.75$$

$$C = \underline{48^\circ\ 36'}$$

Total Internal Reflection in Nature

1 **Mirages** A mirage is a natural effect due to total internal reflection occurring at the boundary of a hot and cooler air. We often notice it as we travel along a tarmac road on a hot day, when we seem to see simmering pools of water ahead of us. The road appears wet and shiny as if after rain. These pools of water disappear as we approach them because they are mere optical illusion.

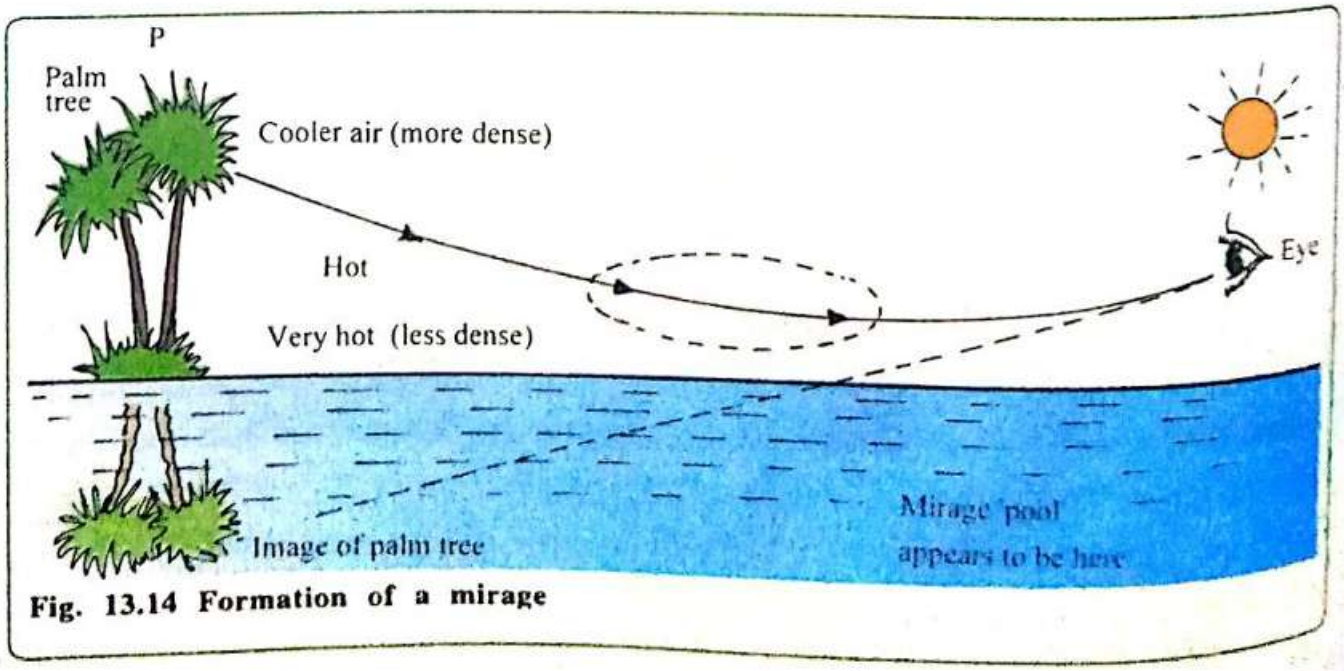


Fig. 13.14 Formation of a mirage

Scanned with CamScanner

This phenomenon is explained by the progressive refractive bending of light from the patch of a sky or a top of a tree, as it passes through successive layers of optically less dense air. Because of solar heating, the layer of air nearer the ground is hotter than those above. The air density decreases from ground towards the sky. As the refractive index of air depends on its density, the lower or hotter layer of air has a lower refractive index than the layer above.

Rays from a patch of the sky or the top of a tree proceeding downwards travel from air layers of higher refractive index to layers of lower refractive index. The rays therefore bend gradually from the incident direction until they enter a certain layer near the ground where the angle of incidence exceeds the critical angle and total internal reflection occurs. The rays are now reflected upwards and reach the observers eye. The observer sees the distant tree as well as its inverted image. The patch of the sky is similarly reflected and appears as a pool of water.

2 Field of View of a fish under water

A fish under water or a swimmer moving on his back under water, can have a full view of everything above the water surface, even the sun in the horizon, as long as the water surface is not ruffled.

Rays of light such as PL and QR reach the eye E of the fish after refraction along LE and RE. The most extreme rays from air reaching the eye of the fish E are AB and DC. These rays just graze the air-water boundary and their angle of incidence is 90°.

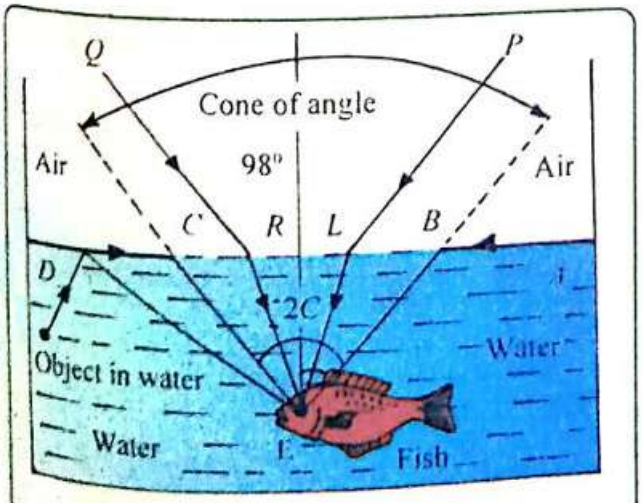


Fig. 13.15 Fish eye-view under water

The angle of refraction r is given by

$$\frac{\sin 90}{\sin r} = \frac{4}{3};\quad \sin r = \frac{3}{4} = 0.75$$

and hence r is approximately equal to 49°. All light from outside the water entering the eye of the fish in the water is concentrated into a cone of half angle 49°. So a fish can see everything in the air above the water within a cone BEC of total angle $2 \times 49° = 98°$.

The fish is also able to see objects in the water due to total internal reflection. The water surface acts as a mirror in forming such images of objects in the water and on the bottom.

13.4 Refraction of Light through Triangular Prisms

When light rays pass through a rectangular glass block, the incident and the emergent rays are parallel to each other but the emergent ray is displaced to one side. There is no deviation or change in the direction of the emergent ray when compared with the incident ray.

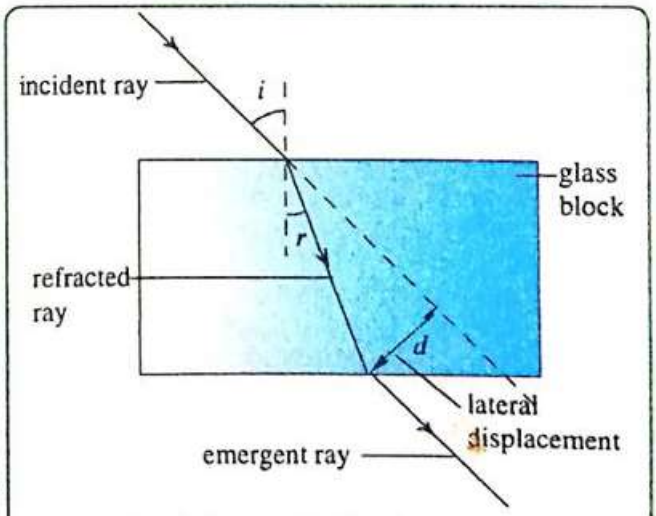


Fig. 13.16 Lateral displacement

Let us consider the passage of light rays through a triangular glass prism, fig 13.17. Consider one such ray PQ incident on the prism side AB. The incident, the refracted, and the emergent rays are as shown in the diagram. We observe that the emergent ray is not parallel to the incident ray, but is at an angle to it.

Scanned with CamScanner

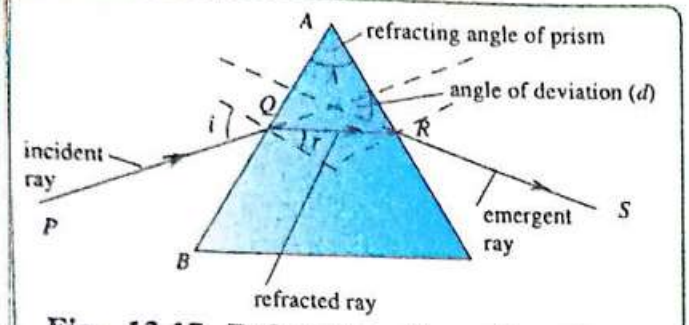


Fig. 13.17 Refraction through prism

The prism deviates the incident ray through an angle known as *angle of deviation*. This is the angle between the incident ray and the emergent ray of light passing through a prism. The amount of deviation is determined by (a) the angle of incidence (b) the refracting angle (A) of the prism and (c) the refractive index of the material of the prism.

Experiments show that the angle of deviation (d) varies with the angle of incidence (i) as shown in fig 13.18b.

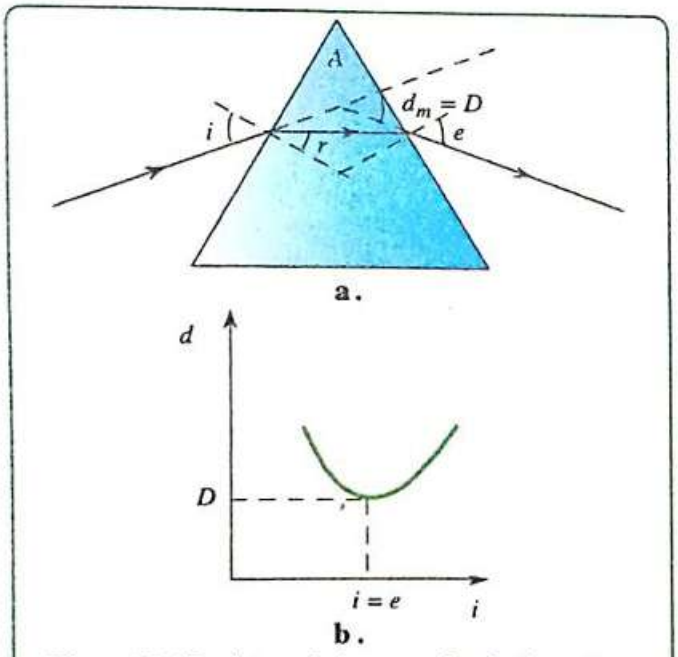


Fig. 13.18 At minimum deviation $i = e$

There is thus a minimum angle of deviation (D) and this occurs when the incident and emergent rays are equally inclined to their respective surfaces (fig. 13.18b). The ray passes symmetrically through the prism and the angle of incidence equals angle of emergence ($i = e$).

It can be shown that at minimum deviation, the refractive index (n) of the glass, the refracting angle (A) of the prism and the angle of minimum deviation (D) are related by the equation.

$$n = \frac{\sin \frac{1}{2}(A + D)}{\sin \frac{1}{2}A} \qquad 13.6$$

Thus by measuring the angle of minimum deviation and the refracting angle of the prism, we can calculate the refractive index of the material of the prism.

Activity 13.1 To find the refractive index of glass by the method of minimum deviation of light through a glass prism

1 Place a glass prism on a tracing paper and draw its outline ABC. Remove the prism, measure and record the value of A.

2 Draw normal to the line AB at N.

3 Erect two pins at P_1 and P_2. Replace the prism and fix two other pins at P_3 and P_4 such that the pins appear to be in a straight line with the images of the pins P_1 and P_2 when viewed from the side AC of the prism. Remove the prism. Join P_3 and P_4 producing it to meet line TN at Z. Draw the normal XY. Measure and record the angle of emergence e and the angle of deviation d.

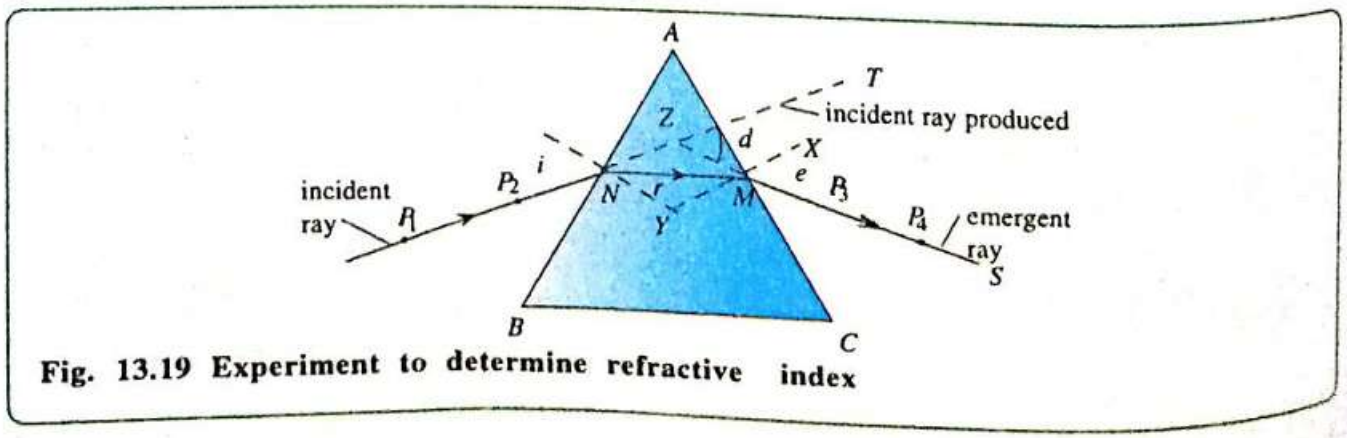


Fig. 13.19 Experiment to determine refractive index

Scanned with CamScanner

4 Repeat the experiment with $i = 45^\circ, 50^\circ, 55^\circ, 60^\circ, 65^\circ$. In each case measure and record the corresponding angle e, and d. Tabulate your readings.

5 Plot a graph with d on the vertical axis and i on the horizontal axis starting both axes from the origin.

6 Obtain the minimum deviation D from your graph.

7 Calculate the refractive index of glass from the relation

$$n = \frac{\sin \frac{1}{2}(A + D)}{\sin \frac{1}{2} A}$$

8 If a ray of light is incident normally on the face AB of the prism, draw a ray diagram to show the path of the ray as it passes through the prism (Take refractive index of glass = 1.5)

The graph of the above experiment shows that as the angle of incidence increases, the angle of deviation, d, decreases until at a certain minimum value, the angle of deviation begins to increase for any further increase in the angle of incidence. The graph is thus a curve as in fig. 13.18b. If the angle of incidence on face AB of the prism is such that the refracted ray NM meets the face AC at angle of incidence greater than the critical angle for glass-air interface (about 42°) then the light is totally internally reflected and emerges from the face BC of the prism.

Example 13.3

A ray of light experiences a minimum deviation when passing symmetrically through an equilateral triangular prism. Calculate the angle of incidence of the ray. (Refractive index of glass = 1.5)

Solution

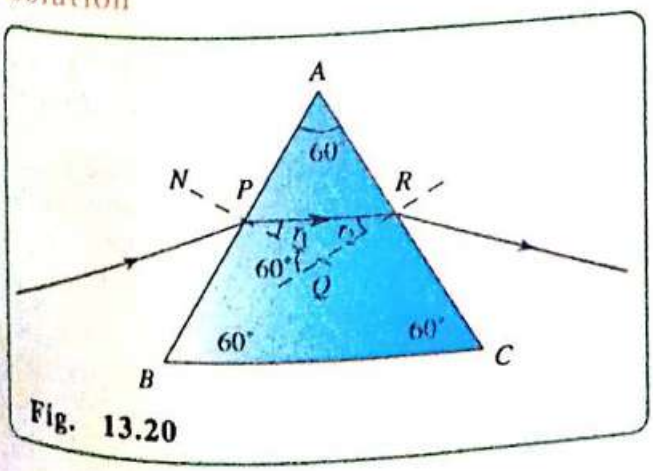


Fig. 13.20

When the ray passes symmetrically through the prism, the angle of incidence i = angle of emergence, e and $r_1 = r_2 = r$ the angle of refraction in the glass. Also PR, the refracted ray is parallel to the base BC of the prism when the ray passes symmetrically through the prism.

As PR is parallel to BC, angle $BPR = 180^\circ - 60^\circ = 120^\circ$. But NPQ is normal to AB.

Hence angle $QPB = 90^\circ$.

Hence angle $r_1 = 120^\circ - 90^\circ = 30^\circ$

(Also the geometry of the prism can be used to show that $r_1 = 30^\circ$ or $\frac{A}{2}$)

Using Snell's law, we have

$$\frac{\sin i}{\sin r} = n$$

$$\frac{\sin i}{\sin 30^\circ} = 1.5$$

$$\sin i = 1.5 \times \sin 30^\circ = 0.75$$

$$i = \sin^{-1} 0.75 = \underline{\underline{48.6^\circ}}$$

Totally Reflecting Prisms

In Chapter 12, we showed the construction of periscopes using two plane mirrors arranged parallel to themselves and at 45° to the axis of the tube. Plane mirrors however produce multiple images due to multiple reflection from the mirror surface. Therefore plane mirrors are not used in high quality periscopes. Instead glass prisms with angles of 90°, 45°, 45° are used as reflectors. These right angle isosceles prisms have the advantage that only one image is produced.

Consider such a right angle isosceles prism as shown in fig 13.21. A ray of light incident normally on the side BC will pass straight through it and strike the hypothenus BA at an angle of 45°. As this angle is greater than, the critical angle for glass ($c = 42^\circ$), the ray will be totally reflected with an angle of reflection equal to 45°. The reflected ray YZ strikes the glass face AC at right angles and passes straight through. Such a right angled isosceles prism can therefore be used as a total reflecting prism.

A right-angled isosceles prism can be arranged so that a ray of light incident on it is deviated through 90° (fig 13.21b) or through 180° (fig. 13.21c).

Scanned with CamScanner

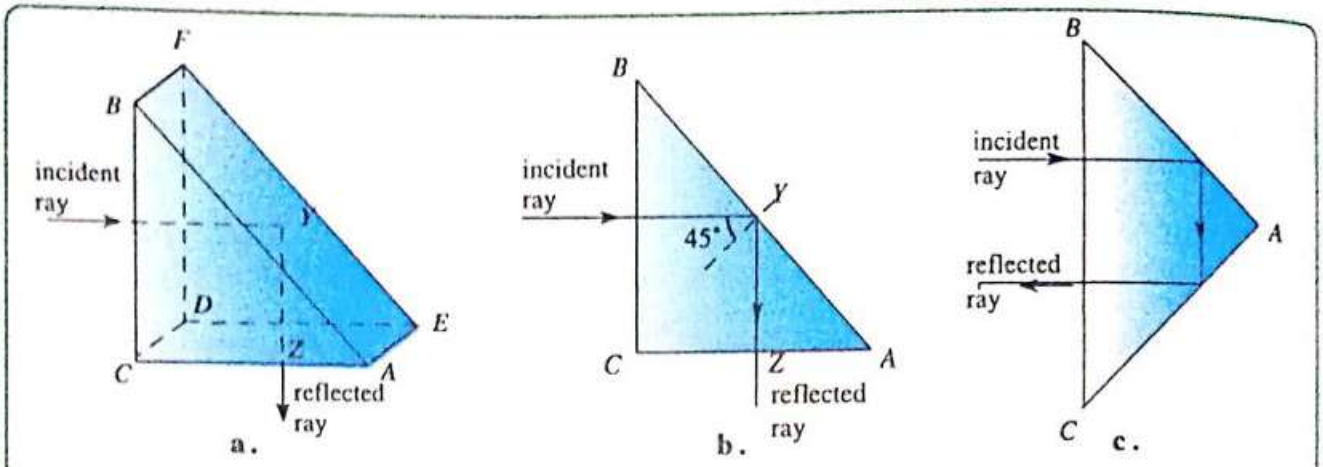


Fig. 13.21 Totally reflecting prism

For a *periscope*, two such right angled isosceles prisms are arranged as shown in fig. 13.22.

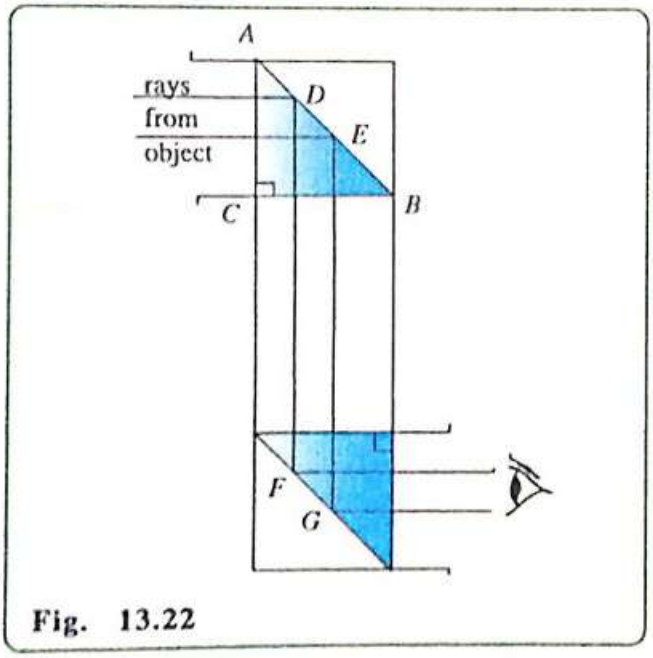


Fig. 13.22

A second prism is placed below the first. Light from the first prism falls on the second one and behaves in exactly the same way as already described above. Total internal reflection takes place at D and F, also at E and G. The rays emerge along a line parallel to the initial incident ray direction. An eye at Y sees a bright image of the object.

A right-angled isosceles prism can also be used in a prism binocular to produce upright images that would otherwise appear inverted to the viewer (fig. 13.23).

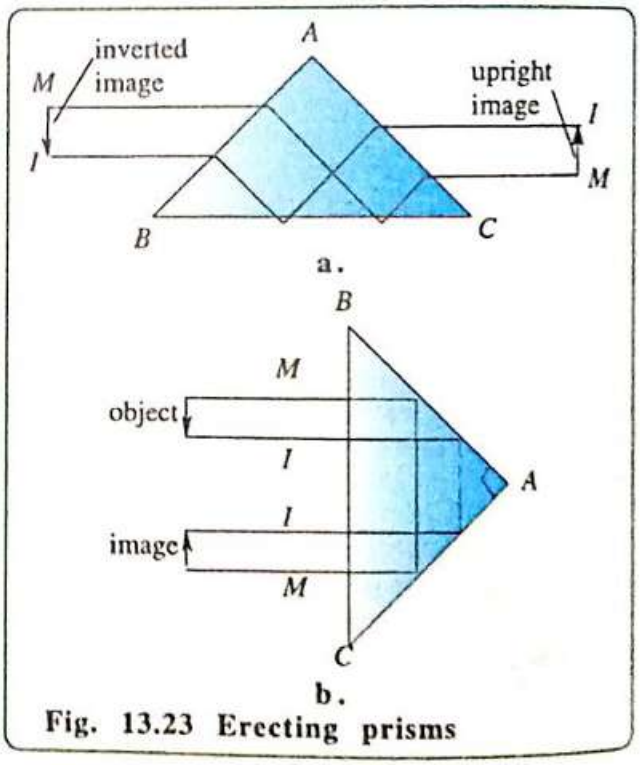


Fig. 13.23 Erecting prisms

A prism binocular uses a combination of right-angled isosceles prisms to achieve a high magnification without being as long as a telescope (fig 13.24).

Prisms have the advantage that they do not tarnish and deteriorate as mirrors do. Also mirrors absorb more light and give a fainter image.

Scanned with CamScanner

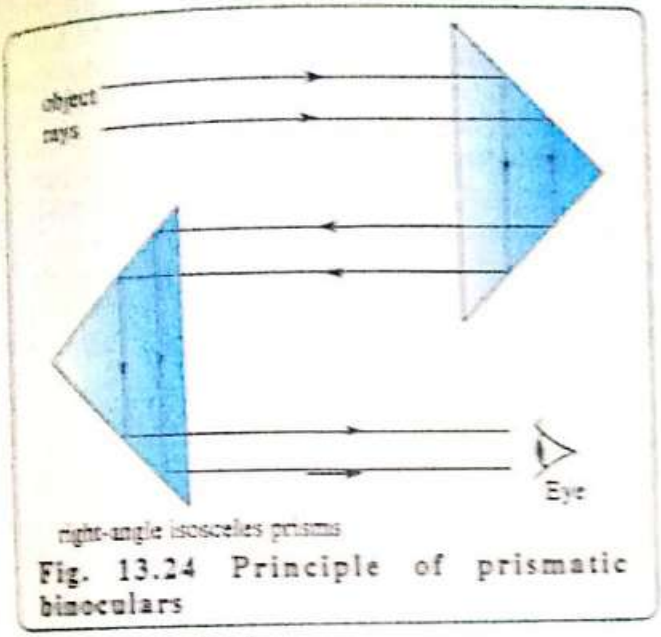


Fig. 13.24 Principle of prismatic binoculars

Example 13.4

The angle of incidence of a ray of light striking an equilateral triangular prisms ABC of refracting angle 60° is 40°. Calculate

(i) the angle of refraction at the first face

(ii) the angle of emergence

(iii) If the light is incident normally on the first face AB, describe with the aid of a diagram what is the fate of this light as it passes through the prisms.

Solution

(i)

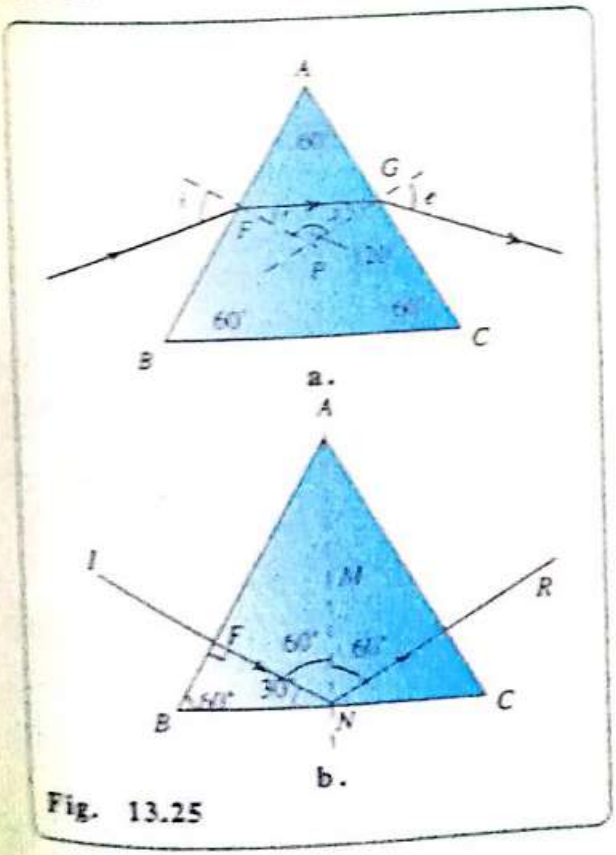


Fig. 13.25

Let r be the angle of refraction at the first face AB.

From Snell's law

$$n = \frac{\sin i}{\sin r}$$

$$1.5 = \frac{\sin 40^\circ}{\sin r}$$

$$\sin r = \frac{\sin 40^\circ}{1.5} = \frac{0.6428}{1.5} = 0.4285.$$

$\therefore$ $r = 25.37^\circ = 25^\circ\ 22'$

(ii) Let the angle of emergence be e. Consider the rectangle $AFPG$.

the angle AFP + angle $AGP = 180^\circ$

Angle FAG + angle $GPF = 1B0$

Since angle $FAG = 60^\circ$, angle $GPF = 180^\circ - 60^\circ = 120^\circ$

$\therefore$ $r + x + 120^\circ = 180^\circ$

$r + x = 60^\circ$

But $r = 25.37^\circ$; hence $x = 34.63^\circ$

x° is the angle of incidence on the glass-air boundary on the face AC.

$\therefore$ $\frac{\sin x}{\sin e} = \frac{1}{1.5}$ (since light is now travelling from glass to air)

$$\frac{\sin 34.63}{\sin e} = \frac{1}{1.5}$$

$\sin e = 1.5 \times \sin 34.63^\circ$

$= 0.8524$

$e = 58.47^\circ = 58^\circ\ 30'$

(iii) In fig 13.25(b) the incident ray IF is normal to the face AB. It therefore passes through undeviated and strikes the face BC at N. Here we see that the angle of incidence $FNM = 60^\circ$. But from $\sin C = \frac{1}{n} = \frac{1}{1.5}$ we see that the critical angle for glass is about 42°.

$\therefore$ Since $60^\circ > 42^\circ$, the angle of incidence is greater than the critical angle. The ray IN is hence totally internally reflected at N as shown. There is therefore no emergent ray.

What happens to this reflected ray NR as it strikes the face AC?

13.5 Dispersion of White Light

In an experiment, first performed by Isaac Newton, it was observed that when white light is passed through a prism, an elongated coloured patch of

Scanned with CamScanner

light is obtained on a screen placed behind the prism.

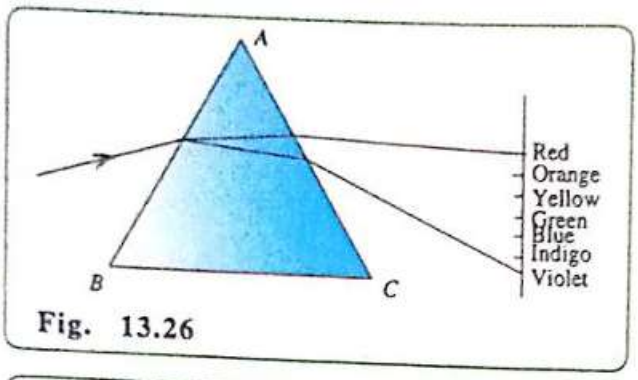


Fig. 13.26

fact that the different colours of white light travel at different speeds through the glass. Each colour is therefore refracted in a slightly different direction or angle through the glass. Careful observation of the spectrum shows that the red light is deviated least and the violet light is deviated most. Since red is deviated least from its original direction in air, it follows that its change of speed on entering the glass prism is least. Therefore red light travels in glass with the greatest speed. Similar reasoning shows that the violet light travels in glass with the least speed. In a vacuum however all the colours travel with the same speed.

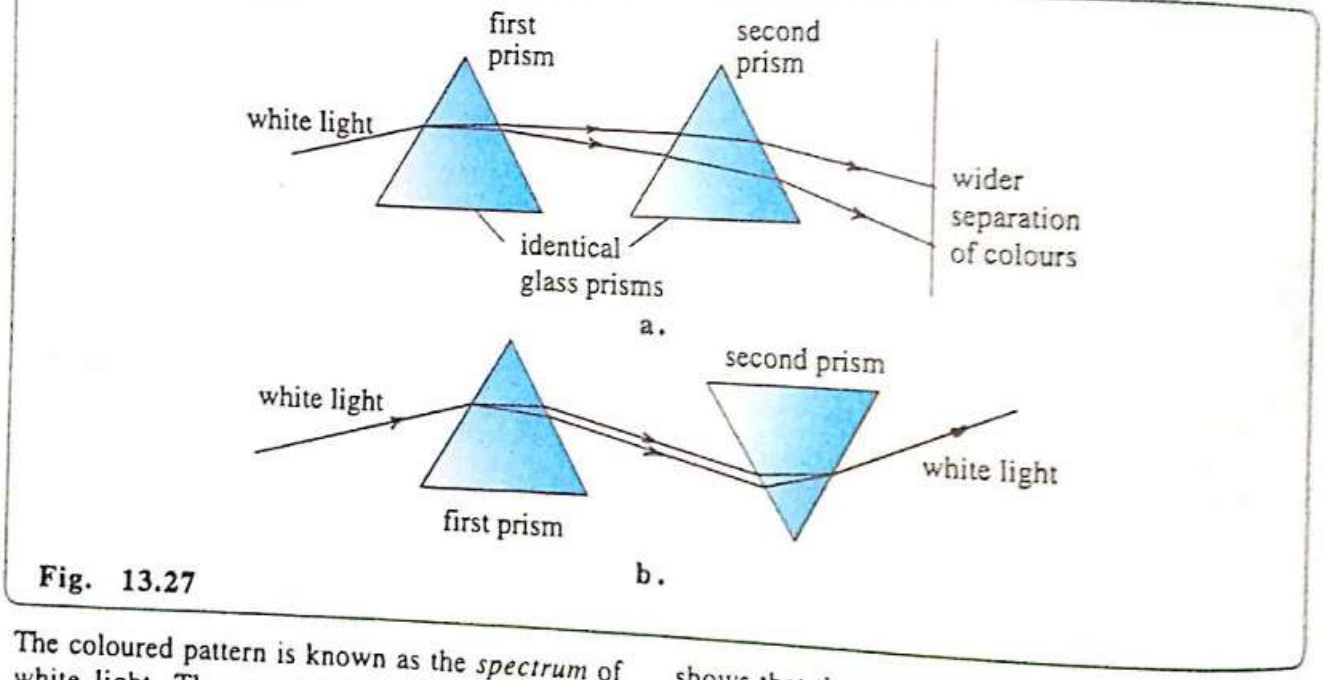


Fig. 13.27

The coloured pattern is known as the *spectrum* of white light. The spectrum consists of the Red, Orange, Yellow, Green, Blue, Indigo, Violet (ROYGBIV), in that order from the apex side *A* of the prism.

When another identical prism is placed to intercept the refracted rays in the manner shown in fig.13.27a, the same arrangement of colours emerge on the screen, but this time it is found that the colours are more widely separated.

If the second prism is inverted as shown in fig. 13.27b, the colours are seen to disappear and only a patch of white light will be visible on the screen. The disappearance of the colours was due to their recombination to reproduce white light.

Dispersion

This is the separation of white light into its component colours of Red, Orange, Yellow, Green, Blue, Indigo and Violet. Dispersion is due to the

The rainbow is formed by the dispersion of sunlight by spherical raindrops floating in the air.

Monochromatic Light

This is light of one wavelength. When such light of one colour or wavelength is passed through a prism, refraction occurs without dispersion.

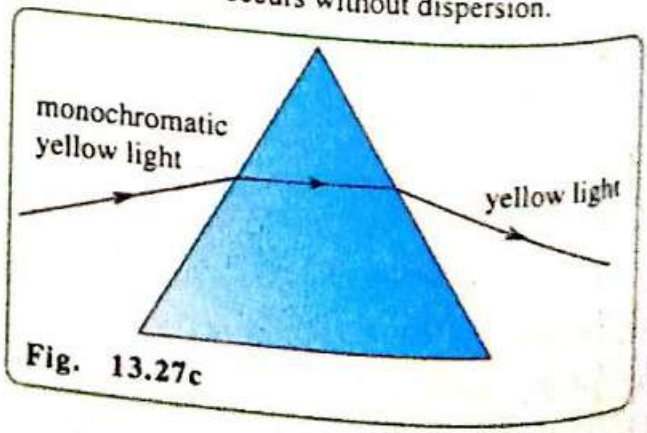


Fig. 13.27c

Scanned with CamScanner

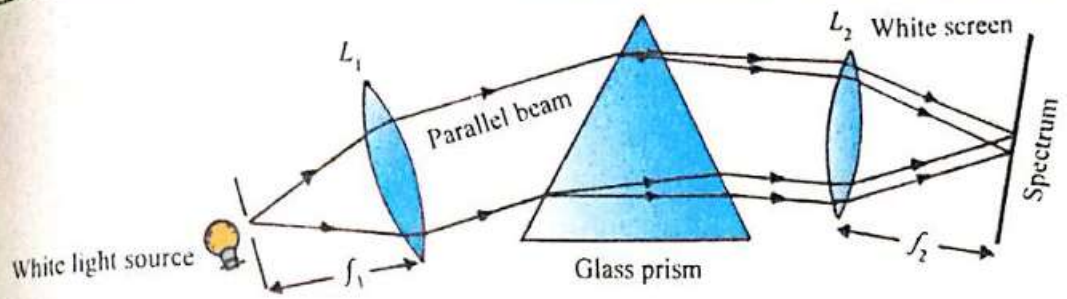


Fig. 13.28 Lens set-up to obtain pure spectrum

Monochromatic light of a yellow colour is conveniently obtained from a *sodium lamp*, a special lamp containing sodium vapour.

Production of a pure spectrum

The spectrum produced by a prism is an impure spectrum because the different colours overlap. A *pure spectrum* is that in which the colours are clearly separated or distinct from each other. To produce such a pure spectrum we use two converging lenses and a prism in an arrangement as shown in fig. 13.28.

The apparatus consists of the following components:

a. A narrow slit S illuminated by a bright source of white light. The slit produces a series of narrow coloured images which minimizes the chances of overlapping colours.

b. A converging lens L_1 produces the parallel rays that are incident on the prism. The slit is placed at the principal focus of the lens to ensure that parallel rays are produced by the lens.

c. A 60° prism that disperses the white light into beams of coloured light. Each beam of coloured light is in the form of parallel rays but at an angle to the other beams.

d. Another converging lens L_2 which collects the parallel beams of different colours and brings each to a separate focus in the focal plane of the lens.

e. A screen at the focal plane of the lens L_2, on which the coloured bands of the spectrum appear well defined.

To obtain a clear spectrum we use a very fine slit and a bright light source.

If the second lens L_2 in fig 13.28 is not used, we can obtain a fairly pure spectrum.

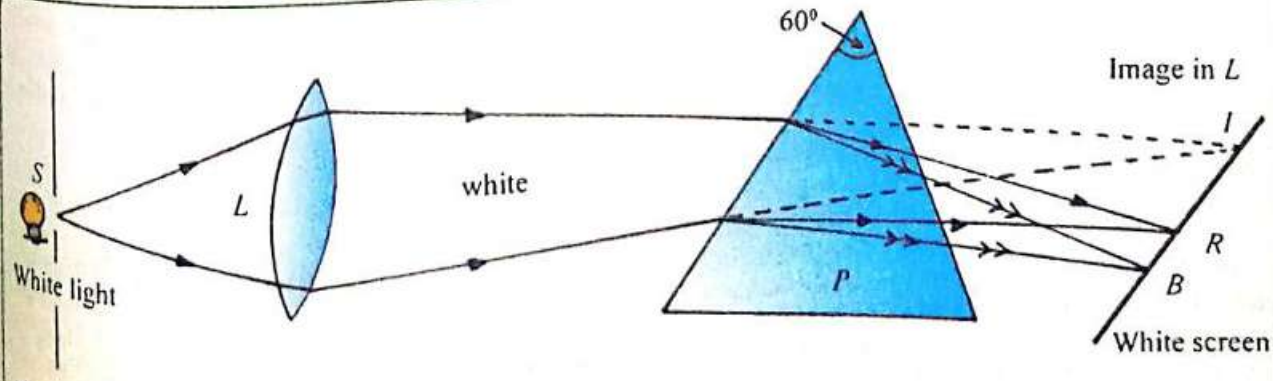
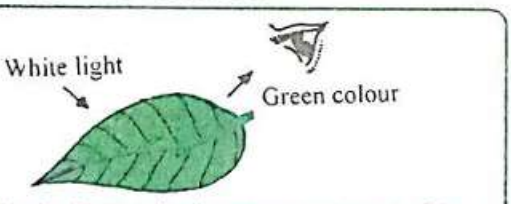


Fig. 13.29

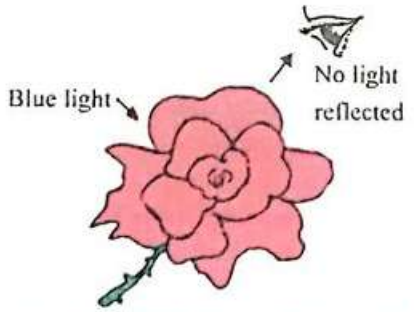
Scanned with CamScanner

Colours of Objects

The colour of an object depends on the colours in the light incident or falling on it, and also on the absorption and reflection of this light by the object. A white object appears white in daylight because it reflects equally all the colours of the spectrum. If a red light is incident on it, it appears red. A black object in daylight absorbs all the colours of the spectrum and reflects none at all. So if it is illuminated by any colour e.g. blue light, it still appears black. A green leaf, appears green in white light (or daylight) because it reflects green but absorbs all the other colours of white light. The green leaf looks dark or black if it is illuminated in a dark room by blue light. A red rose flower that appears red in daylight is also red only in red light but appears black in any light which it does not reflect. Hence if the red flower is illuminated by a blue light in a dark room, the red flower will appear black. This is because it absorbs all the blue light, leaving no colour to be reflected.



a. 'Green' leaf reflects only the green component of the white light, therefore it appears green



b. Red rose absorbs the blue light and appears black.

Fig. 13.30 Colour appearance of an object

Additive mixing of Colours

We can obtain a variety of colours by mixing the different colours of the spectrum. We cannot however, obtain Red, Green and Blue colours by mixing other colours. These three colours are therefore called the *primary colours*. The adding of primary colours to produce other colours is known as *additive colour mixing* or *additive combination* of colours. *Secondary colours* are the colours we obtain by mixing any two of the primary colours.

Examples of such mixing are:

Red + Green = Yellow

Blue + Green = Cyan (turquoise or blue-green colour)

Blue + Red = Magenta (a purple colour)

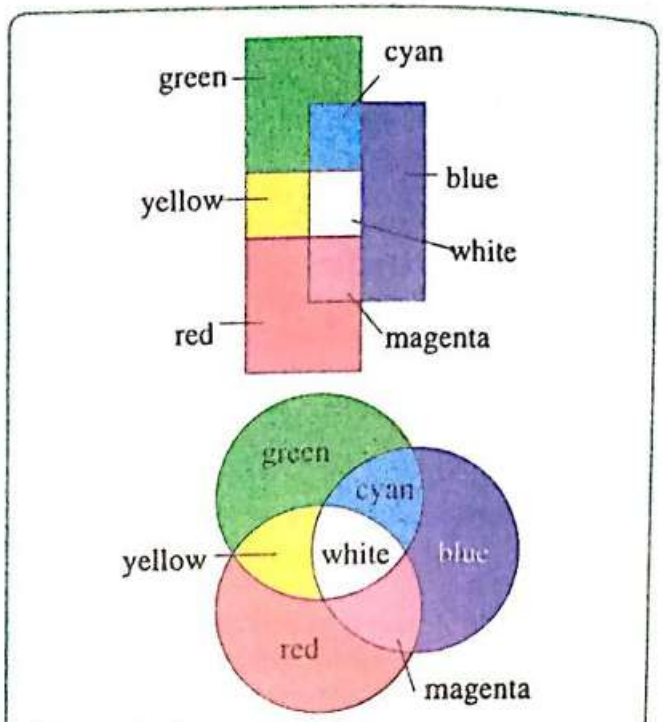


Fig. 13.31

Yellow, Cyan, and Magenta are the secondary colours. All the three primary colours combine to produce white

Red + Green + Blue = White.

A useful way of remembering the addition of colours is by the colour triangle (fig. 13.32). Each primary colour is placed at the corner of an equilateral triangle *RGB*. Each secondary colour is at the midpoint of the sides *RB*, *BG* and *GR*, and represents the colour obtained by mixing or adding equal quantities of the primaries at the corners. The centre *O* of the triangle represents white light obtained by the addition of equal quantities of the three primary colours. A primary colour when mixed with an appropriate secondary colour, that is, the colour directly opposite to it in the colour triangle, will produce white. For example:

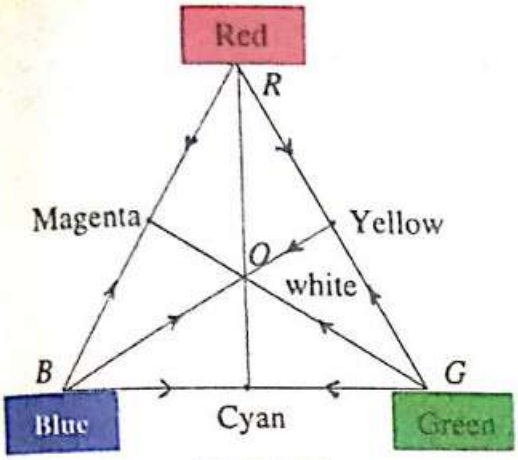
Green + Magenta = White

Blue + Yellow = White

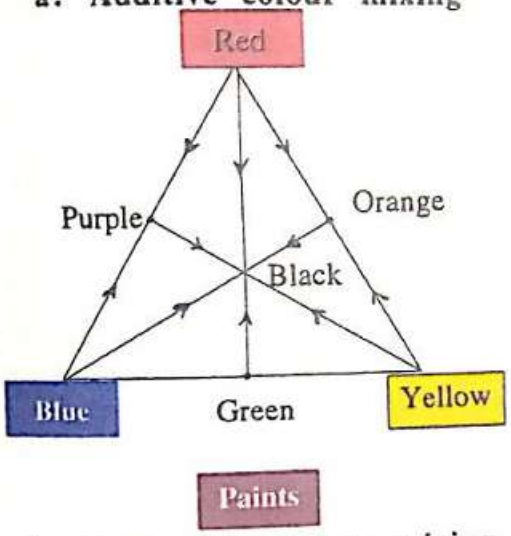
Red + Cyan = White

Such colours which produce white light when mixed together in suitable proportions are called *complementary colours*. Thus Green and Magenta are complementary colours. So also are Blue and Yellow, Red and Cyan.

Scanned with CamScanner



a. Additive colour mixing



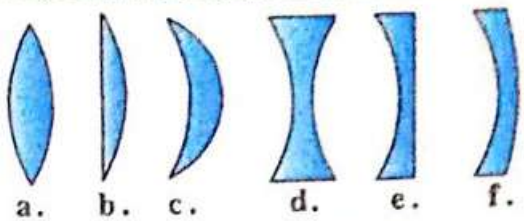
b. Subtractive colour mixing

Fig. 13.32 Colour triangle

13.6 Refraction of Light through lenses

Refraction through lenses involves the same change in direction of light rays, as discussed for rectangular glass blocks and triangular prisms. A lens is any transparent material with two faces, of which at least one is curved. It causes a beam of light passing through it to either converge or diverge. There are therefore two kinds of lenses - (i) the *converging* or *convex lens* and (ii) the *diverging* or *concave lens*.

Converging (convex) lens makes a parallel beam of light to converge to a point. Such lenses are thicker at the centre than at the edges.



a. biconvex
b. plano-convex
c. concavo-convex
d. biconcave
e. plano-concave
f. convexo-concave

Fig. 13.33

A diverging (concave) lens makes a parallel beam of light to appear to diverge from a point. Such lenses are thinner at the centre than at the edges. The various forms of convex and concave lenses are shown in fig 13.33. We shall be dealing mainly with *biconvex* and *biconcave* lens which are the most common in school laboratory experiments. Plano-convex and plano-concave lenses are used in optical instruments. The concavo-convex (or converging meniscus) and the convexo-concave (or diverging meniscus) are used as contact lenses to correct defective vision because they fit the curvature of the eyeball.

Most lenses are made from glass. A few are made with quartz or plastic. The eye has a crystalline biconvex lens.

Parameters of a Lens

We define the following important parameters of a lens: *optical centre*, *principal axis*, *principal focus*, *focal length* and the *power* of the lens.

The *optical centre* (C) of a lens is the point through which rays of light pass without being deviated by the lens.

The line passing through the optical centre of the lens and joining the centres of curvature of its surfaces is termed the principal axis.

The *principal focus* (F) of a converging or convex lens is the point to which all rays parallel and close to the principal axis converge after refraction through the lens.

The *principal focus* of a diverging or concave lens is the point from which all rays parallel and close to the principal axis appear to diverge after refraction through the lens. A lens has two principal foci since light may pass through a lens in either direction.

Scanned with CamScanner

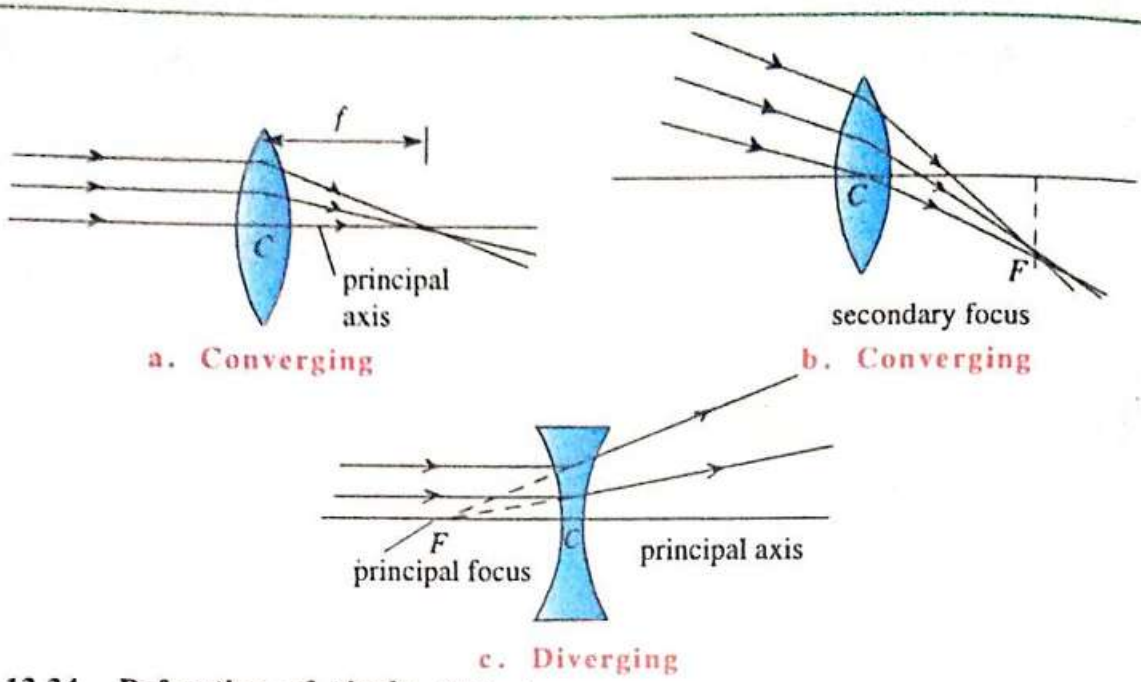


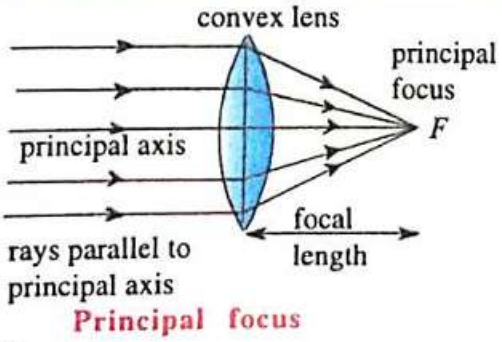
Fig. 13.34 Refraction of single rays

The *focal length* (f) is the distance between the optical centre and the principal focus of the lens.

The power (P) of a lens is equal to the reciprocal of the focal length and is measured in *dioptres* when f is in metres.

$$P = \frac{1}{f}$$

Because of the principle of reversibility of light, rays originating from F and passing through the convex lens will emerge parallel to the principal axis (fig. 13.35).



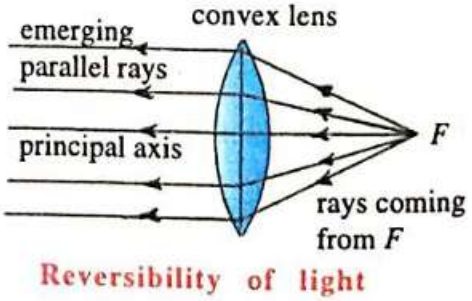
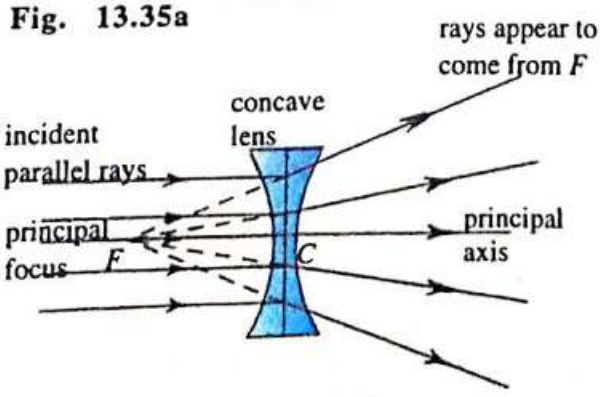


Fig. 13.35a



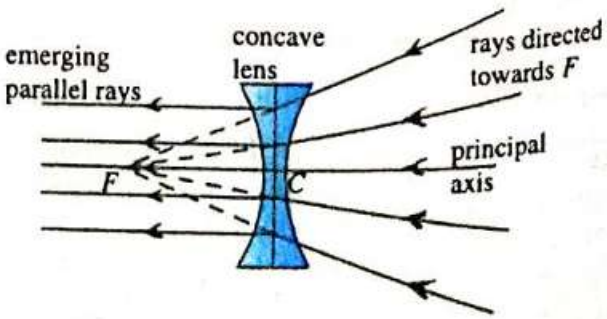


Fig. 13.35b

Scanned with CamScanner

From the ray diagrams of fig. 13.35 we note that
(i) the principal focus of a converging lens is real
(ii) the principal focus of a diverging lens is virtual.

Formation of images by lenses: Construction of ray diagrams

Light rays incident on a converging lens converge to form a real image after passing through a converging or convex lens. For a diverging lens, the rays from an object diverge to form a virtual image of the object. If we know the distance of the object from the lens and the focal length of the lens we can obtain the position and nature of the image by the construction of ray diagrams as outlined below. Three classes of rays are used to obtain the position and nature of the image:

a. A ray from the object, parallel to the principal axis which refracts through the lens and passes through the principal focus.

b. A ray from the object passing through the optical centre of the lens which passes through undeviated.
The intersection of the two rays corresponds to the position of the top of the image of the object.

c. Sometimes, a third ray may be used. A ray from the object through the principal focus which emerges parallel to the principal axis after refraction. The three emergent rays intersect at a point, the image position.

So, to find the position and nature of the image we construct the following:

1 Choose a suitable scale for the construction.

2 Draw a straight vertical line MN to represent the position of the lens and mark the position of the focus F according to the scale. CF represents the focal length.

3 Draw an arrowed line OA perpendicular to OC to represent the object so that OC represents the distance of the object from the lens and OA its height according to the scale.

4 From the top A of the object, draw a ray AP parallel to the principal axis. AP is refracted through F'. Draw another ray AC through the centre of the lens C. This ray passes straight through without changing direction. The point of intersection B of the two emerging rays PB and CB represents the top of the image. Complete the image by drawing BI normal to the principal axis.

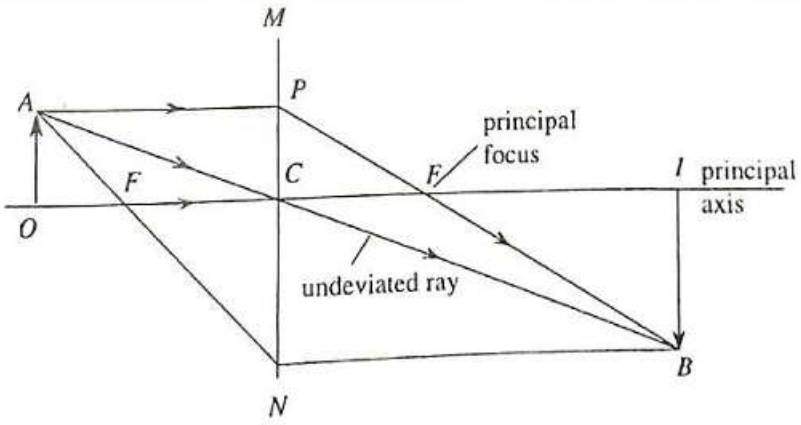


Fig. 13.36 Construction of image location

Table 13.2

Position / Object	Image	Nature	Size
At infinity	At F	Real; inverted	Diminished (fig. 13.27)
Beyond $2F$	Between F and $2F$	Real; inverted	Diminished (fig. 13.38)
At $2F$	At $2F$	Real; inverted	Same size (fig. 13.39)
Between F and $2F$	Beyond $2F$	Real; inverted	Magnified (fig. 13.40)
At F	At infinity	Real; inverted	Magnified (fig. 13.41)
Between F and C	Beyond the object	Virtual; erect	Magnified (fig. 13.42)

Scanned with CamScanner

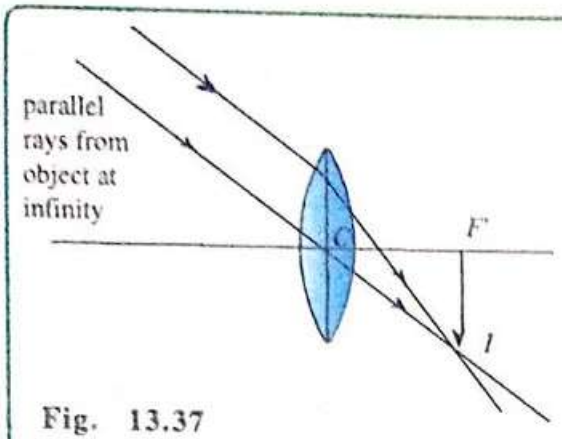


Fig. 13.37

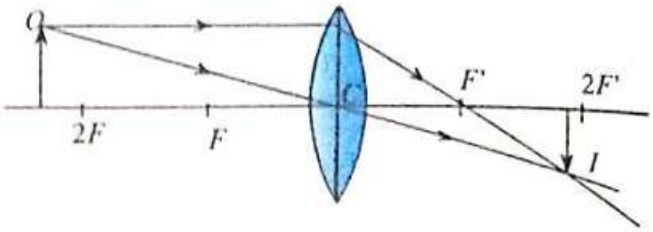


Fig. 13.38

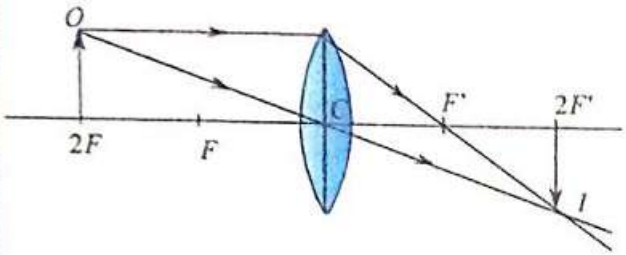


Fig. 13.39

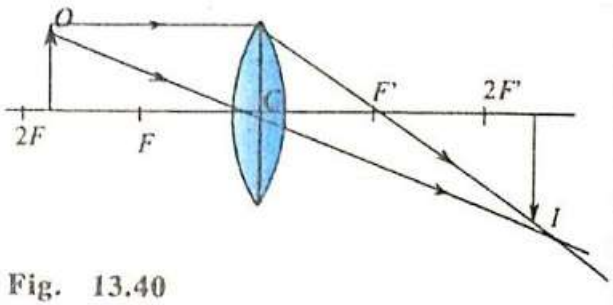


Fig. 13.40

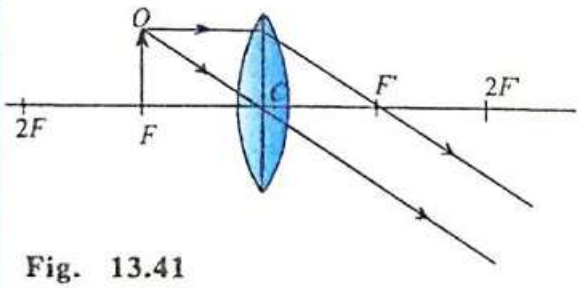


Fig. 13.41

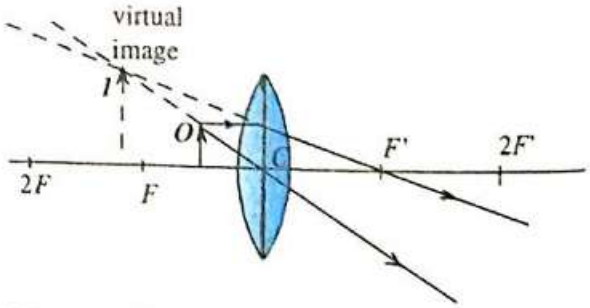


Fig. 13.42

The position and nature of images formed by convex lenses depend on the position of the object from the lens. We summarise in table 13.1 such position, size and nature of the image in relation to the object position. The relevant ray diagrams are as shown in fig. 13.37 – 13.42.

Images formed by Diverging or Concave Lens

For all positions of the object, the image is virtual, erect and diminished. Fig. 13.43 shows the path taken by a ray travelling parallel with the principal axis and another ray passing through the optical centre, *C*, of the lens. The image formed is on the same side as the object and always between the object and the lens.

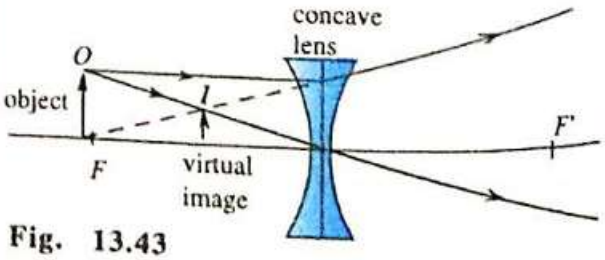


Fig. 13.43

The Lens Formula and Sign Convention

Experiments show that there is a relation between the object distance, u, the image distance, v, and the focal length, f, of a lens. The relation is given by the lens formula.

Scanned with CamScanner